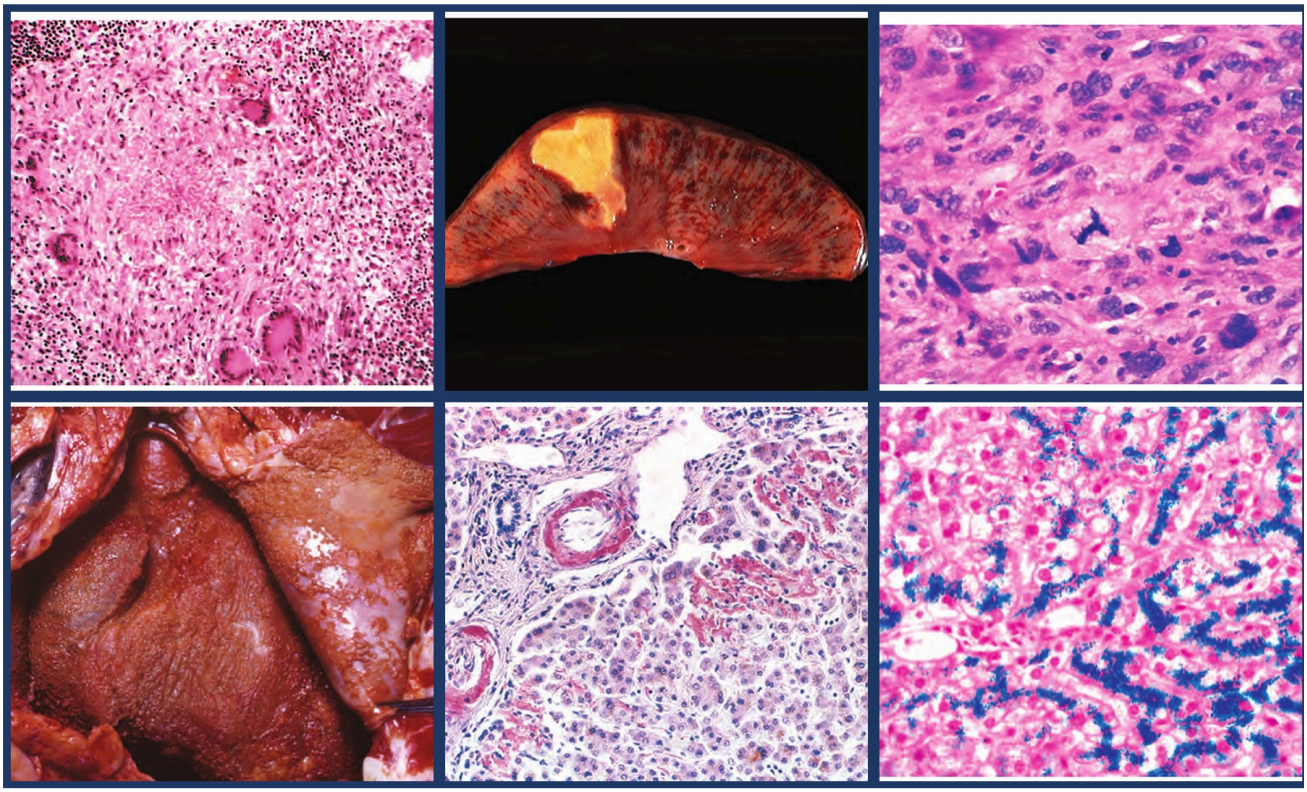
MINISTRY OF HEALTH OF THE REPUBLIC OF UZBEKISTAN
TASHKENT STATE MEDICAL UNIVERSITY
Department of Pathological Anatomy

PATHOLOGICAL ANATOMY
PART 1

complete study manual

A study manual for practical classes



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This manual is based on the 11 practical class topics of the 5th-semester calendar-thematic plan of the course “Pathological Anatomy” and covers general pathological anatomy: the response of the cell to injurious agents, cell injury and cell death, degenerations, inflammation and repair, hemodynamic disorders, immunopathological processes and amyloidosis, tumors, infectious diseases, and the pathology of newborns and children.

At the end of each chapter there are algorithms for describing gross specimens and microscopic slides, review questions, test questions and clinical (clinicopathological) cases. The answers to the test questions are given at the end of the book.

The manual is based on the textbook “Pathological Anatomy” by M.S. Abdullakhodjaeva, the textbook “Pathological Anatomy” by A.I. Strukov and V.V. Serov, the study manual “General Pathological Anatomy” of the Tashkent Medical Academy, and Robbins and Cotran “Pathologic Basis of Disease”, Pathoma and the Kaplan USMLE review books. The figures are taken from these sources, and the source is given under each figure. Labels inside some figures remain in the language of the original source.

The text and structure of the manual were prepared with the help of the artificial intelligence Claude (Anthropic) on the basis of the textbooks listed. The compiler selected the material, defined its structure and checked it. Before clinical information is applied in practice, it is recommended to compare it with the main textbooks.

The manual is intended for 3rd-year students of the faculties of general medicine, pediatrics and preventive medicine of higher medical education institutions.

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Preface

Pathological anatomy is a fundamental science that studies the material basis of disease, that is, the structural changes that occur in cells, tissues and organs during disease. It links theoretical medicine with the clinic: it is impossible to understand the signs of disease, make a diagnosis and choose the right treatment without knowing the essence of pathological processes. Today the pathologist is the specialist who makes the final diagnosis on biopsy and surgical material, and treatment tactics are often based on the pathologist's conclusion.

This first part of the manual is devoted to **general pathological anatomy** and follows the order of the 11 practical classes in the 5th-semester calendar-thematic plan of the course. Each chapter corresponds to one practical class and has the same structure:

- **Learning objectives** — what the student should know and be able to do.
- **Theoretical part** — the definition, causes, mechanisms of development, gross and microscopic appearance and outcomes of the pathological process.
- **Clinicopathological section** — the significance of the process studied in clinical practice, clinical examples.
- **Practical skills** — the description of the gross specimens and microscopic slides studied in the class, with conclusions.
- **Review questions, test questions and clinical cases** — for checking independent preparation. The answers to the clinical cases are given within the chapter, and the answers to the test questions at the end of the book.
- **Quick review** — the key definitions, numbers and rules, to look over in a few minutes before an exam.

Four kinds of colored box appear in the text:

> **REMEMBER**
> Yellow box — a rule, number or definition that must be memorized.

> **CLINICAL SIGNIFICANCE**
> Blue box — the significance of the knowledge in clinical practice.

> **MEMORY AID**
> Green box — a technique or short rule that makes memorizing easier.

> **CAUTION**
> Orange box — common mistakes, concepts that are easily confused, and discrepancies between textbooks.

The main sources of the manual are the textbook “Pathological Anatomy” by M.S. Abdullakhodjaeva and the textbook “Pathological Anatomy” by A.I. Strukov and V.V. Serov; modern molecular mechanisms and clinical examples have been added from Robbins and Cotran's “Pathologic Basis of Disease”, Pathoma and Kaplan. The descriptions of gross specimens and microscopic slides rely on the study manual “General Pathological Anatomy” of the Tashkent Medical Academy. The figures are taken from these sources, and the source is given under each figure. Where the textbooks differ in classification or terminology, both points of view are given.

I hope that the manual will help students prepare for practical classes, describe gross specimens and microscopic slides, and prepare for midterm and final assessments.

TOPIC 1

Introduction to pathology. The cell as the structural unit of health and disease. Cellular responses to injurious agents and stress

Learning objectives

The student should know:

- the content, tasks, objects and research methods of pathological anatomy (autopsy, biopsy, experiment);
- the history of pathological anatomy and the Uzbek school of pathologists;
- the basic concepts of disease: etiology, pathogenesis, morphogenesis, pathomorphosis, complication, outcome;
- the main structures of the cell and their significance in pathology;
- the sequence of cellular responses to stress and injurious agents;
- the types of cellular adaptation: hypertrophy, hyperplasia, atrophy, metaplasia — definition, causes, mechanism, morphology and outcomes.

The student should be able to:

- apply the scheme for describing gross specimens and microscopic slides;
- identify and describe gross specimens of cardiac hypertrophy, urinary bladder hypertrophy, endometrial hyperplasia, brown atrophy of the liver and heart, hydronephrosis and hydrocephalus;
- identify and describe microscopic slides of bronchial epithelial metaplasia, glandular hyperplasia of the endometrium and brown atrophy of the liver;
- distinguish physiological from pathological adaptation, hypertrophy from hyperplasia, and atrophy from hypoplasia;
- explain the clinical significance of adaptive processes.

1.1. Content and tasks of pathological anatomy

Pathology (Greek pathos — suffering, disease; logos — study) is the science that studies the causes and mechanisms of disease and the resulting structural and functional changes in cells, tissues and organs. **Pathological anatomy** is the anatomy of the diseased organism: it studies the material (structural) basis of disease and serves as a bridge between theoretical medicine and the clinic.

Pathological anatomy consists of two parts:

- **General pathological anatomy** — the general pathological processes that occur in any disease: cell injury and death, degenerations, circulatory disorders, inflammation, immunopathological processes, regeneration, adaptation and tumors. This manual is devoted to this part.
- **Systemic (special) pathological anatomy** — the morphology of individual diseases (nosological entities): atherosclerosis, pneumonia, hepatitis, glomerulonephritis and others.

Tasks of pathological anatomy (after M.S. Abdullakhodjaeva):

1. To study the morphologic changes that occur in organs, tissues, cells and intracellular structures in diseases and their complications.
2. To establish the causes and conditions of the development of disease — its **etiology**.

3. To study the mechanism of development of disease — its **pathogenesis**.
4. To establish the sequence in which structural changes appear during the disease — its **morphogenesis**.
5. To study the structural basis of protective, compensatory and adaptive processes.
6. To analyze changes in the picture of a disease under the influence of treatment or changing living conditions — **pathomorphosis**.

REMEMBER
In pathological anatomy, changes are studied at **four levels**: **the organism and the organ** (gross), **the tissue and the cell** (microscopic — light microscope), **intracellular structures** (ultrastructural — electron microscope), and **molecules** (molecular pathology). For example, in sickle cell anemia the change begins at the molecular level: glutamic acid at position 6 of the hemoglobin beta chain is replaced by valine.

Research methods of pathological anatomy

1. Autopsy (Greek autopsia — seeing with one's own eyes; synonyms: necropsy, postmortem examination, dissection of the body) — examination of the dead body by dissection to identify the structural changes caused by disease. The significance of autopsy:

- comparison of the clinical and pathological diagnoses and detection of diagnostic and treatment errors (clinicopathological conferences);
- determination of the cause of death and its thanatogenesis (mechanism of death);
- study of new and rare diseases and of pathomorphosis;
- timely detection of infectious diseases and prevention of epidemics;
- collection of statistical data on the structure of morbidity and mortality;
- medical education and continuing training of physicians. **2. Biopsy** (Greek bios — life, opsis — vision) — removal of a piece of tissue from a living patient and its microscopic examination for diagnostic purposes. Types of biopsy:

Table 1.1. *Types of biopsy*

Type	Method of obtaining the material	Example
Incisional (cutting out a piece)	A piece is cut from the altered area	Skin; a piece from the edge of a breast tumor
Excisional	The pathological focus is removed entirely	Small nodule, lymph node
Needle core (trephine)	A core of tissue is taken with a special needle	Liver, kidney, prostate, bone marrow
Endoscopic	Taken with forceps through an endoscope	Gastro-, colono-, bronchoscopy
Aspiration	Aspirated with a fine needle (cytology)	Thyroid nodule
Surgical specimen	Organ removed at surgery	Appendix, gallbladder, uterus
Urgent (intraoperative)	Frozen section during surgery, result in 15–20 minutes	Determining whether a tumor is malignant or benign and the extent of surgery

Tasks of biopsy: to confirm or clarify the clinical diagnosis; to detect the early stage of a disease; to distinguish tumors from hyperplastic (inflammatory) processes; to establish the histologic type, grade

and margins of a tumor (radicality of surgery); to assess the effect of treatment.

3. Experiment — artificial reproduction of a disease in animals to study its etiology, pathogenesis and morphogenesis step by step.

Modern morphologic methods: histochemistry (detection of substances with special stains — PAS reaction, Congo red, Sudan III), immunohistochemistry (detection of antigens in tissue with labeled antibodies — the main method in tumor diagnosis), electron microscopy (transmission and scanning), fluorescence microscopy, in situ hybridization (FISH), polymerase chain reaction (PCR) and next-generation sequencing (molecular pathology).

> **CLINICAL SIGNIFICANCE**
> Today the pathologist is not a “doctor of the dead” but **a treating physician of the living patient**: biopsies and surgical specimens make up most of the work of pathology departments. A cancer diagnosis is not made without histologic examination, and chemotherapy or the extent of surgery is often chosen according to the pathologist's report (tumor type, grade, receptor status, presence or absence of tumor at the margins).

Brief history

Three periods are distinguished in the development of pathological anatomy:

1. **Anatomic (macroscopic) period** — until the middle of the 19th century. **Avicenna (Ibn Sina)** in “The Canon of Medicine” (1020) linked the signs of disease with changes in the organs. The Italian anatomist **G.B. Morgagni**, in his work “On the Seats and Causes of Diseases Investigated by Anatomy” (1761), compared the results of about 700 autopsies with clinical signs — he is regarded as **the founder of pathological anatomy**. M. Bichat (pathology of tissues) and K. Rokitansky (more than 30,000 autopsies, the humoral theory of pathology) were major figures of this period.
2. **Microscopic period** — from the middle of the 19th to the middle of the 20th century. The German pathologist **R. Virchow** (1821–1902) created the theory of **cellular pathology** in 1855–1858: “every cell from a cell” (omnis cellula e cellula), and disease is the result of injury to cells. Experimental pathology developed.
3. **Ultramicroscopic and molecular period** — from the middle of the 20th century to the present: electron microscopy, histochemistry, immunomorphology, molecular genetics. This period is a direct continuation of Virchow's cellular pathology. In Russia, the first department of pathological anatomy was opened in 1849 at Moscow University (A.I. Polunin). Major representatives of the Moscow school were A.I. Abrikosov, I.V. Davydovsky, A.I. Strukov, A.P. Avtsyn and V.V. Serov; of the St. Petersburg school — M.M. Rudnev and N.N. Anichkov.

In Uzbekistan, pathological anatomy took shape as an independent discipline in 1919 with the opening of the medical faculty at Turkestan State University. The first head of the department was Professor **V.V. Vasilevsky** (1920–1930). **The founder of the Uzbek school of pathologists** is Professor **G.N. Terekhov** (head of the department at Tashkent State Medical Institute in 1930–1966). Later the department was headed by Associate Professor P.J. Tulaganov and Professor **M.S. Abdullakhodjaeva** (1969–1990). Scientists of the department made a major contribution to the study of sprue (chillashir), visceral leishmaniasis, pellagra, brucellosis, ulcerative colitis, hemorrhagic fever and geographic pathology. Departments of pathological anatomy were opened in Samarkand in 1932 (S.F. Tatarenchik) and at the Central Asian Medical Pediatric Institute in 1972 (V.A. Alimov). Today regional pathology bureaus and the Republican Center of Pathological Anatomy operate in the country.

1.2. General doctrine of disease

Disease is a disruption of normal life activity resulting from damage to the structures of the organism and impairment of its functions under the influence of external and internal injurious (pathogenic) factors. In the words of I.V. Davydovsky, disease is the organism's adaptation, of a particular form and degree, to its living conditions: along with damage, it also displays protective, compensatory and adaptive processes.

Table 1.2. *Basic concepts of disease*

Concept	Definition	Example (myocardial infarction)
Etiology	The cause of the disease and the conditions that bring it about	Atherosclerosis and thrombosis of a coronary artery
Pathogenesis	The mechanism of development of the disease — the chain of events from the cause to the signs	Ischemia → decrease in ATP → death of cardiomyocytes
Morphogenesis	The structural expression of pathogenesis — the sequence of morphologic changes	Ischemic stage → necrosis → granulation tissue → scar
Pathomorphosis	A persistent change in the clinical and morphologic picture of a disease: natural (spontaneous) or induced (by treatment)	Reduction of the infarct size due to thrombolysis and stenting
Complication	A new process that is pathogenetically linked to the underlying disease but differs from it	Cardiac aneurysm, cardiac rupture, cardiogenic shock
Outcome	The end of the disease: recovery (complete, incomplete), transition to a chronic course, death	Postinfarction cardiosclerosis

Three stages are distinguished in the development of a disease (D.S. Sarkisov): **1)** formation of the disease (predisease, latent period) — structural changes are balanced by compensatory reactions, and there are no clinical signs; **2)** the full-blown (clinical) period of the disease — compensation fails and signs appear; **3)** recovery or another outcome.

> **REMEMBER**
> **Etiology** answers the question “**why?**”, **pathogenesis** — “**by what mechanism?**”, and **morphogenesis** — “**through what structural changes?**”. A single etiologic agent can cause different diseases through different pathogenetic pathways (for example, streptococcus — tonsillitis, rheumatic fever, glomerulonephritis), whereas different agents can cause the same morphologic change (for example, fatty liver — alcohol, obesity, diabetes, hypoxia).

Etiologic factors are divided into two groups: **genetic** (mutations, chromosomal abnormalities, hereditary predisposition) and **acquired** (infection, nutrition, chemical and physical factors). Most common diseases (atherosclerosis, hypertension, diabetes mellitus) are **multifactorial**, resulting from the combined effect of hereditary predisposition and the environment.

1.3. The cell as the structural unit of health and disease

According to Virchow's doctrine, the smallest living unit of the organism is the cell, and disease begins with injury to cells. The cell organelles work in connection with one another, and each of them can be a “target” in a particular pathological process.

Table 1.3. *Cell structures and their significance in pathology*

Structure	Main function	Significance in pathology
Plasma membrane	Barrier, transport (Na+/K+-ATPase), receptors	Failure of the ion pump — cellular swelling; membrane disruption — a sign of irreversible injury
Nucleus	Genetic information, DNA synthesis and repair	Pyknosis, karyorrhexis, karyolysis — signs of necrosis; mutations — tumors
Mitochondria	Oxidative phosphorylation — ATP production; regulation of apoptosis (cytochrome c)	Injured first in hypoxia; swelling and amorphous densities — a sign of irreversible injury
Endoplasmic reticulum (ER)	Rough ER — protein synthesis; smooth ER — lipid synthesis, drug detoxification (cytochrome P-450)	Detachment of ribosomes — reduced protein synthesis; misfolded proteins — “ER stress” and apoptosis; hypertrophy of the smooth ER under the influence of drugs
Golgi complex	Modification of proteins and their preparation for secretion	Impaired secretory function
Lysosomes	Digestion by hydrolytic enzymes (autophagy, heterophagy)	Lysosomal storage diseases (Gaucher, Niemann–Pick); the aging pigment lipofuscin; release of enzymes in necrosis
Peroxisomes	Oxidation of long-chain fatty acids, breakdown of H2O2	Injury related to free radicals
Cytoskeleton	Cell shape, movement, transport of organelles	Mallory bodies (alcoholic hepatitis), neurofibrillary tangles (Alzheimer disease), effect of colchicine
Extracellular matrix	Support, storage of growth factors, signaling	Fibrosis, sclerosis, amyloidosis

Proliferative capacity of tissues

According to their capacity to divide, cells are divided into three groups. This determines their response to injury (hyperplasia, regeneration or scar).

Table 1.4. *Groups of cells by proliferative activity*

Group	Characteristic	Examples
Labile (continuously dividing)	Proliferate continuously throughout life; lost cells are rapidly replaced	Epithelium of the skin, mouth, vagina and intestine; hematopoietic cells
Stable (quiescent)	Normally remain in the G0 phase and enter division when injured	Parenchyma of the liver, kidney and pancreas; fibroblasts, smooth muscle, endothelium
Permanent (nondividing)	Do not divide after birth; when lost, they are replaced by a scar	Neurons, cardiac and skeletal muscle cells

MEMORY AID

“Labile — Lining, Stable — liver, Permanent — Pump and brain”: labile tissues border the external environment (skin, mucous membranes) and are constantly renewed; permanent tissues (heart, brain) cannot be restored once lost — which is why myocardial infarction and stroke heal with a scar.

1.4. Cellular responses to stress and injurious agents

A normal cell functions within a limited range of function and structure — in a state of **homeostasis**. Depending on the strength and duration of the load or injurious agent acting on it, the cell gives one

of the following responses:

1. **Adaptation** — in response to physiological load or some pathological stimuli, the cell reaches a new steady state and preserves its viability: hypertrophy, hyperplasia, atrophy, metaplasia.
2. **Reversible injury** — when the limits of adaptation are exceeded or the stimulus is mild, functional and morphologic changes appear in the cell, but the cell recovers if the agent is removed.
3. **Irreversible injury and cell death** — if the stimulus is strong or prolonged, the cell passes the "point of no return" and dies by **necrosis** or **apoptosis** (Topic 2). In addition, chronic mild stimuli lead to intracellular **accumulation** of various substances (degenerations — Topics 3 and 4), **pathologic calcification** and **cellular aging**.

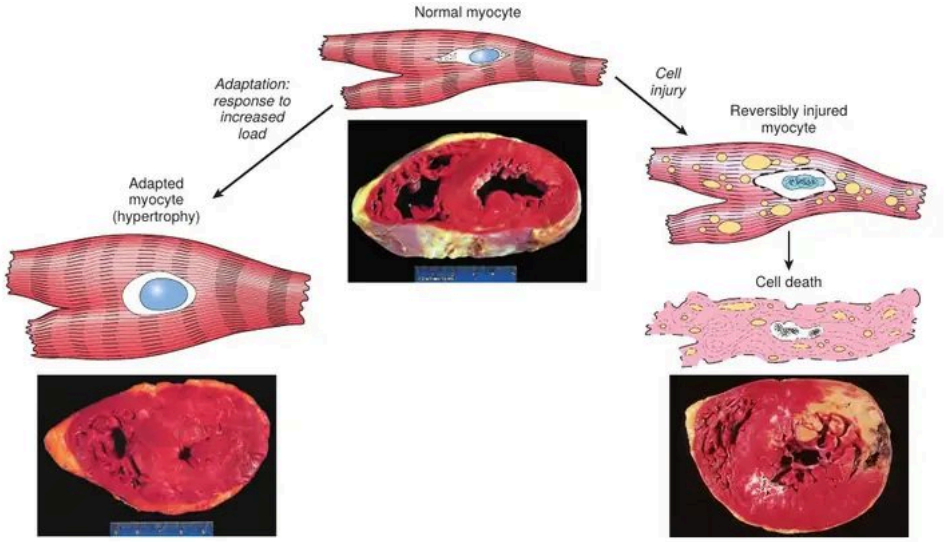


Figure 1.1. *Relationship between normal, adapted, reversibly injured and irreversibly injured myocardial cells. Left — thickening of the left ventricular wall due to pressure overload (hypertrophy); lower right — myocardial infarction (with triphenyltetrazolium chloride staining, viable myocardium is dark red and the dead area is pale) (Robbins, 9th ed., Fig. 2-2)*

CAUTION

In the Uzbek curriculum title, the expression "ruhiy zo'riqish" is a rendering of the English word **stress**. In pathology, **cellular stress** does not mean a psychological state but any excessive load or injurious influence on the cell (hypoxia, heat, toxin, hormonal signal, etc.). The cell responds by synthesizing "stress proteins" (heat shock proteins, chaperones) and by adaptive reactions.

Table 1.5. *Type of stimulus and cellular response (after Robbins)*

Nature of the stimulus	Cellular response
Altered physiological stimuli, some nonlethal injurious stimuli	Adaptation
— increased demand or stimulation (growth factors, hormones)	Hyperplasia, hypertrophy
— decreased nutrients or stimulation	Atrophy
— chronic irritation (physical or chemical)	Metaplasia
Oxygen deprivation, chemical injury, microbial infection	Cell injury
— acute and transient	Reversible injury: cellular swelling, fatty change
— severe and progressive (including DNA damage)	Irreversible injury → necrosis or apoptosis

Nature of the stimulus	Cellular response
Metabolic disorders (inherited or acquired), chronic injury	Intracellular accumulations, calcification
Sublethal injuries accumulated over a lifetime	Cellular aging

Causes of cell injury

1. **Oxygen deprivation (hypoxia)** is the most common cause. **Ischemia** (reduced arterial blood flow) is the most dangerous form of hypoxia: it stops not only the delivery of oxygen but also the delivery of nutrients and the removal of metabolites. Other causes of hypoxia are respiratory failure, anemia and carbon monoxide poisoning.
2. **Physical agents** — mechanical trauma, extremes of temperature, sudden changes in atmospheric pressure, ionizing radiation, electric current.
3. **Chemical agents and drugs** — poisons (arsenic, cyanide, mercury salts), alcohol, narcotic drugs, air pollutants, even high concentrations of physiological substances (glucose, table salt, oxygen), and toxic effects of drugs.
4. **Infectious agents** — prions, viruses, rickettsiae, bacteria, fungi, parasites.
5. **Immunologic reactions** — autoimmune diseases, allergic reactions, excessive inflammation against microbes.
6. **Genetic derangements** — congenital malformations (Down syndrome), enzyme deficiencies (storage diseases), misfolded proteins.
7. **Nutritional imbalances** — protein-energy malnutrition, vitamin deficiencies and, conversely, overnutrition (obesity, atherosclerosis).
8. **Aging** — decline of the replicative and reparative capacity of the cell.

> **MEMORY AID**
> Causes of cell injury — **“GAIN CHIP”**: **G**enetic, **A**ging, **I**mmune, **N**utritional, **C**hemical, **H**ypoxia, **I**nfection, **P**hysical.

1.5. Hypertrophy

Hypertrophy (Greek hyper — excess, trophe — nourishment) is an increase in the size of a tissue or organ due to **enlargement of its cells**. No new cells appear in a hypertrophied organ; the existing ones enlarge. Cell enlargement is due to increased synthesis of structural proteins and organelles (myofibrils, mitochondria).

> **REMEMBER**
> In **hypertrophy** the cells **enlarge**; in **hyperplasia** the **number of cells increases**. In organs made of nondividing cells (heart, skeletal muscle), an increased load produces only **hypertrophy**. In cells capable of division (uterus, prostate, thyroid gland), hypertrophy and hyperplasia often occur **together**.

Mechanism. Mechanical stretch (load), agonists (alpha-adrenergic agonists, angiotensin) and growth factors (IGF-1) act on membrane receptors and activate intracellular signaling pathways (PI3K/AKT, G protein–coupled receptors). As a result, transcription factors (GATA4, NFAT, MEF2) activate the genes of structural proteins. In hypertrophied cardiomyocytes, the synthesis of **fetal-type** proteins (for example, the beta isoform of the myosin heavy chain, natriuretic peptide) resumes.

Types of hypertrophy

Table 1.6. *Types of hypertrophy (after M.S. Abdullakhodjaeva)*

Type	Cause and essence	Examples
Physiological (work)	Increased functional load in a healthy organism	Athlete's heart, skeletal muscles of weightlifters; uterine myometrium in pregnancy (hypertrophy + hyperplasia under the influence of estrogen)
Compensatory (pathological work)	Increased load on an organ due to a pathological process; the cause acts continuously	Left ventricular hypertrophy in arterial hypertension and aortic stenosis, right ventricular hypertrophy in mitral stenosis; hypertrophy of the gastric wall in pyloric stenosis, of the bladder wall in prostatic hyperplasia
Vicarious (substitutive)	When one of a pair of organs is removed or destroyed, the remaining one enlarges	Hypertrophy of the second kidney, lung or testis after removal of the first
Neurohumoral (hormonal, correlative)	Dysfunction of the endocrine glands	Acromegaly (excess of somatotropic hormone), gynecomastia in men, endometrial hyperplasia in hyperestrogenemia
Hypertrophic growths	Have no compensatory significance; related to chronic inflammation or impaired lymph flow	Mucosal polyps, condylomata acuminata, elephantiasis

Three stages are distinguished in the morphogenesis of **compensatory hypertrophy of the heart** (F.Z. Meerson):

1. **Emergency (injury) stage** — the load increases suddenly, the myocardium works under strain, and dystrophic changes and energy deficiency appear in cardiomyocytes.
2. **Compensation (stable hyperfunction) stage** — the number of myofibrils and mitochondria increases and the cells enlarge; the heart wall thickens, while its cavity narrows or stays unchanged — **concentric hypertrophy**. Function is adequately preserved.
3. **Decompensation (exhaustion) stage** — the capillaries can no longer supply the enlarged cells with enough blood; hypoxia, degeneration, death of cardiomyocytes and sclerosis develop. The cardiac cavities dilate (**myogenic dilatation**) — **eccentric hypertrophy**; heart failure begins.

REMEMBER

The normal weight of the heart is about **250–350 g**, and the thickness of the left ventricular wall is **0.8–1.2 cm**. In severe hypertrophy the weight of the heart can increase **2–3 times or more**; a heart weighing over 1000 g is called a “**bovine heart**” (cor bovinum).

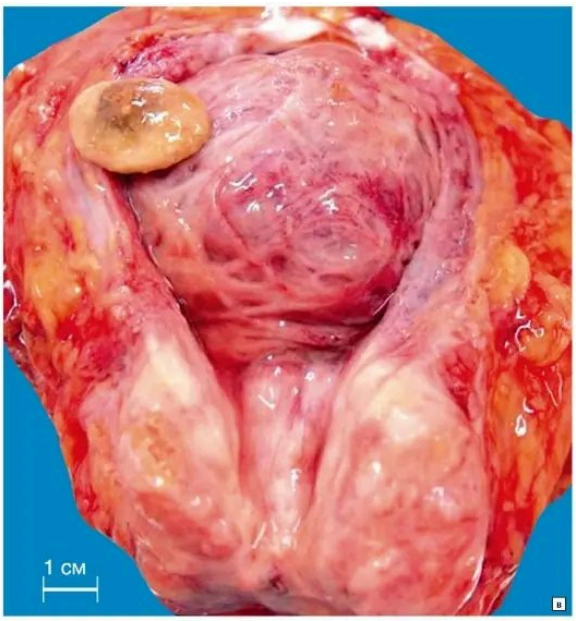


Figure 1.3. *Hypertrophy of the bladder wall and prostatic hyperplasia: the wall is thickened, and muscle bundles bulge beneath the mucosa (trabeculated bladder) (TMA study guide, 2018)*

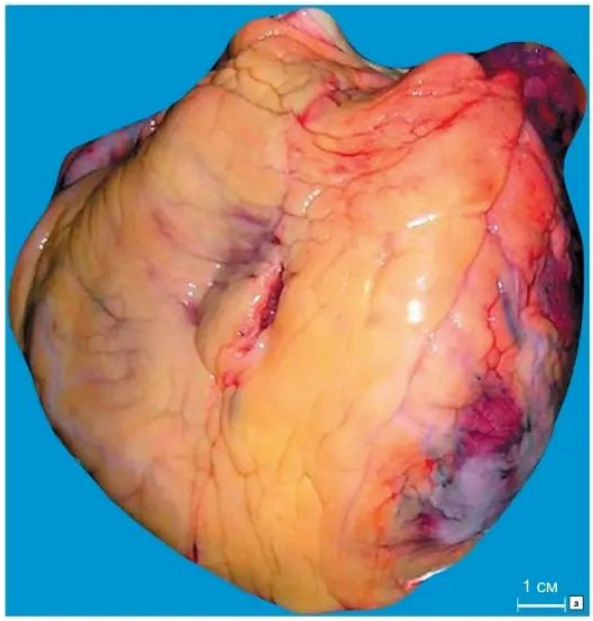


Figure 1.2. *Cardiac hypertrophy (decompensation stage): the heart is enlarged and flabby, with a rounded apex (TMA study guide, 2018)*

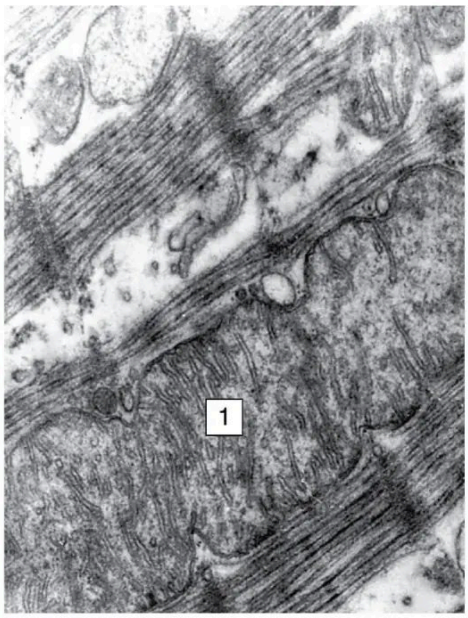


Figure 1.4. *Myocardial hypertrophy in the compensation stage (electron microscopy): mitochondria (1) and myofibrils are numerous and large (TMA study guide, 2018)*

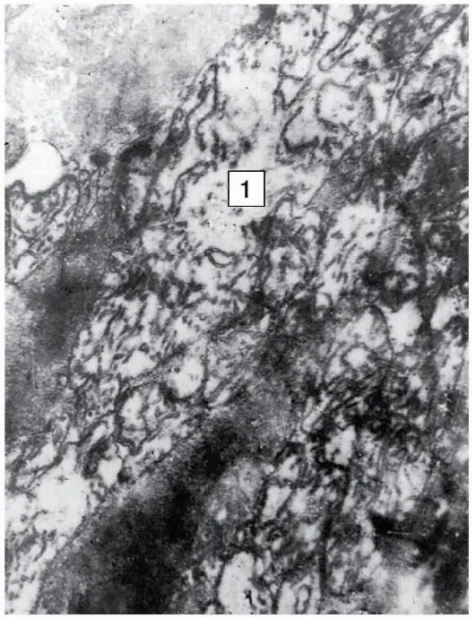


Figure 1.5. *Myocardial hypertrophy in the decompensation stage (electron microscopy): destructive changes in mitochondria (1) (TMA study guide, 2018)*

CAUTION

True hypertrophy must be distinguished from **false hypertrophy**. In false hypertrophy the organ enlarges not because of the parenchyma but because of the growth of fat, connective tissue or bone, while the parenchyma undergoes atrophy and function **declines** (for example, fatty infiltration of the heart, “pseudohypertrophy” of the calf muscles in Duchenne muscular dystrophy). Congenital enlargement of organs (gigantism) is not considered hypertrophy either.

1.6. Hyperplasia

Hyperplasia (Greek plasis — formation) is an **increase in the number of cells** in a tissue or organ. It occurs only in tissues made of cells capable of division. Hyperplasia develops through growth factor–

driven proliferation of mature cells and, sometimes, through the formation of new cells from tissue stem cells.

Physiological hyperplasia:

- **hormonal** — the glandular epithelium of the breast at puberty and in pregnancy, the uterus in pregnancy;
- **compensatory** — after removal of part of the liver (for example, in living-donor transplantation), the remaining part grows back to its original size. The daily regeneration of Prometheus's liver in Greek myth is cited as an allusion to this phenomenon.

Pathologic hyperplasia is most often the result of excessive action of hormones or growth factors:

- **glandular hyperplasia of the endometrium** — when the balance between estrogen and progesterone is disturbed and estrogen predominates; clinically it manifests as acyclic uterine bleeding;
- **benign prostatic hyperplasia** — under the influence of androgens (dihydrotestosterone);
- **hyperplasia of the thyroid gland** — in Graves disease (stimulating antibodies to TSH receptors);
- hyperplasia of fibroblasts and blood vessels in **wound healing** (under the influence of growth factors);
- **viral hyperplasia** — warts and mucosal papillomas caused by papillomaviruses.

CLINICAL SIGNIFICANCE

Pathologic hyperplasia is a **controlled** process: it stops when the hormonal or other stimulus disappears. This is its main difference from a tumor. However, because there is much cell division in the area of hyperplasia, the likelihood of mutations increases, so it can become **fertile soil for tumor development**. Women with atypical endometrial hyperplasia have a much higher risk of endometrial cancer; hyperplasias caused by papillomaviruses are also associated with cancer. Benign prostatic hyperplasia, however, is not considered a precancerous condition.

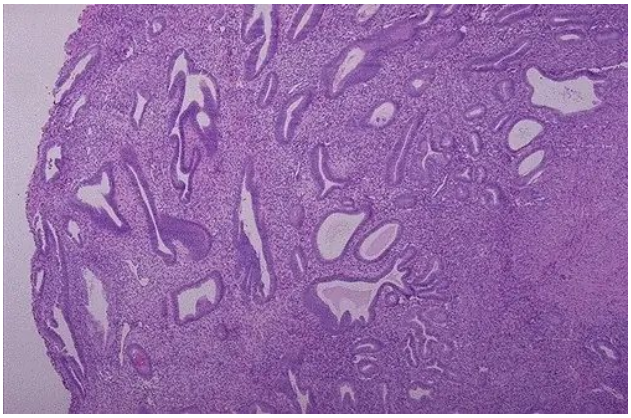


Figure 1.7. *Glandular hyperplasia of the endometrium (microscopic slide, hematoxylin and eosin): the glands are numerous and elongated, some are cystically dilated, and the stroma is cellular (TMA study guide, 2018)*



Figure 1.6. *Glandular hyperplasia of the endometrium (gross specimen): the uterus is enlarged, and the mucosa is unevenly thickened (TMA study guide, 2018)*

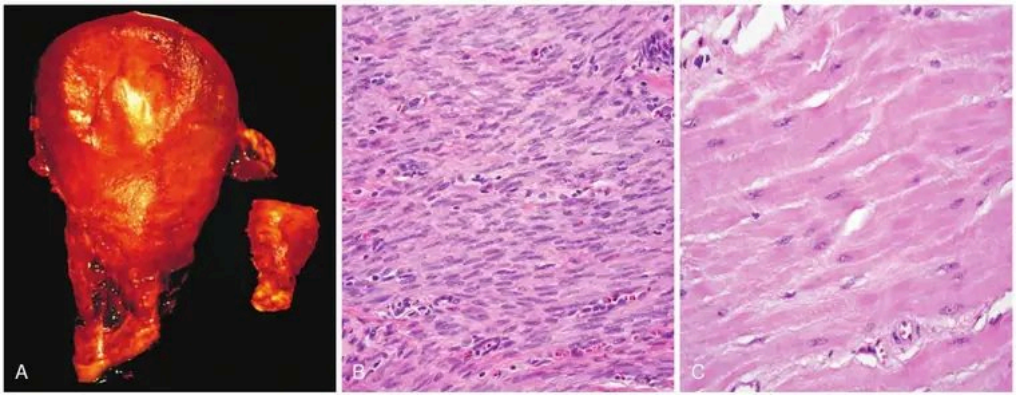


Figure 1.8. *Physiological hypertrophy of the uterus in pregnancy: A — normal uterus (right) and gravid uterus (left); B — small spindle-shaped smooth muscle cells of the normal uterus; C — large, plump smooth muscle cells of the gravid uterus (at the same magnification) (Robbins, 9th ed., Fig. 2-3)*

1.7. Atrophy

Atrophy (Greek a — negation, trophe — nourishment) is a reduction in the size of cells, tissues and organs **during life**, with weakening or cessation of their function. Atrophic cells have fewer structural proteins, mitochondria, myofilaments and less ER; the cell is smaller but alive. Atrophy can also be accompanied by a decrease in the number of cells.

Mechanism: decreased protein synthesis and increased protein degradation. Degradation occurs mainly through the **ubiquitin–proteasome system**: in nutrient deficiency and disuse, cellular proteins are “tagged” with ubiquitin and degraded in proteasomes. Atrophy is often accompanied by increased **autophagy** (“self-eating”): the cell digests its own organelles in autophagic vacuoles to obtain energy for survival. Undigested residues accumulate as the pigment **lipofuscin** — the organ becomes **brown** (**brown atrophy**).

> **CAUTION**
> Atrophy must be distinguished from **congenital malformations**: **agenesis** — the organ or its primordium is completely absent; **aplasia** — the organ remains rudimentary and lacks its typical structure; **hypoplasia** — the organ is small and underdeveloped from birth. Atrophy, in contrast, is the **subsequent** shrinkage of a **normally developed** organ.

Types of atrophy

Physiological atrophy occurs throughout life: obliteration of the umbilical arteries and the ductus arteriosus (Botallo's duct), involution of the thymus after puberty, and atrophy of the gonads, skin, bones (osteoporosis) and brain in old age (senile atrophy).

Pathologic atrophy occurs at any age and can be reversible or irreversible. By extent, it is **generalized** or **local**.

Generalized atrophy — cachexia (exhaustion, wasting). Its types:

1. **Alimentary** — starvation, impaired absorption of nutrients. In starvation the fat depots (subcutaneous tissue, omentum) atrophy first, then the skeletal muscles and internal organs (liver, heart), and the brain last of all — which is why a person dying of starvation remains conscious until the final moments.
2. **Cancer cachexia** — especially in tumors of the digestive tract; caused by tumor intoxication and cytokines (tumor necrosis factor — TNF).
3. **Dyshormonal** — pituitary insufficiency (Simmonds disease — pituitary cachexia), myxedema.

4. **Cerebral** — damage to the hypothalamus (tumor, inflammation).
5. In **chronic infections** (tuberculosis, brucellosis, chronic dysentery) and other severe chronic diseases. In cachexia the subcutaneous fat layer almost disappears, and the remaining adipose tissue becomes orange-brown due to the accumulation of **lipochrome**; the fat of the epicardium and bone marrow is soaked with serous fluid and becomes mucoid (**serous atrophy of adipose tissue**); the heart and liver become small and brown (**brown atrophy**).

Table 1.7. *Types of local atrophy*

Type	Mechanism	Examples
Dysfunctional (disuse)	Weakening or cessation of organ function	Atrophy of muscles under a cast after a bone fracture; atrophy of the alveolar process of the jaw after tooth extraction; atrophy of the optic nerve after removal of the eye
Due to inadequate blood supply	Arterial narrowing — chronic ischemia	Shrinkage of the kidney in renal artery atherosclerosis (atherosclerotic contracted kidney); atrophy of the cerebral cortex in cerebral atherosclerosis — senile dementia
Due to pressure	Prolonged compression of tissue, ischemia	Hydronephrosis (obstruction of urine outflow dilates the renal pelvis and thins the renal parenchyma); hydrocephalus (cerebrospinal fluid accumulates in the ventricles of the brain and the brain tissue atrophies); usuration of vertebrae and ribs under the pressure of an aortic aneurysm
Neurotic (loss of innervation)	Cessation of nerve impulses and trophic influence	Atrophy of skeletal muscles in poliomyelitis, nerve transection or trauma
Due to physical and chemical agents	Radiation, drugs, hormones	Atrophy of the bone marrow and gonads after irradiation; atrophy of the adrenal cortex with prolonged glucocorticoid therapy; atrophy of the thyroid gland under the influence of iodine preparations
Due to loss of endocrine stimulation	Cessation of hormonal influence	Atrophy of the endometrium, vaginal epithelium and breast in menopause

REMEMBER

In appearance, the surface of an atrophic organ is **smooth** (for example, in most atrophies of the liver and kidney) or **granular** (for example, the primary contracted kidney). Atrophy can be a **reversible** process: if the cause is removed early, the structure and function of the organ are restored. In severe atrophy (especially when sclerosis is added), reversal is not possible.

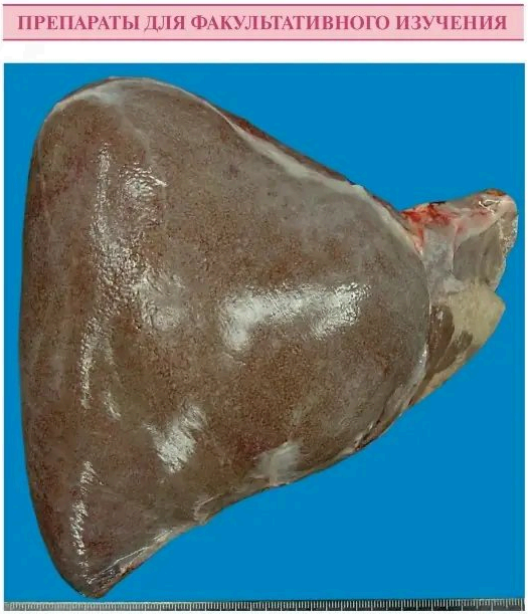


Figure 1.9. *Brown atrophy of the liver: the organ is small, the capsule is wrinkled, and the cut surface is brown (TMA study guide, 2018)*



Figure 1.10. *Brown atrophy of the heart: the heart is small, the epicardium is wrinkled, and the coronary arteries are tortuous (TMA study guide, 2018)*

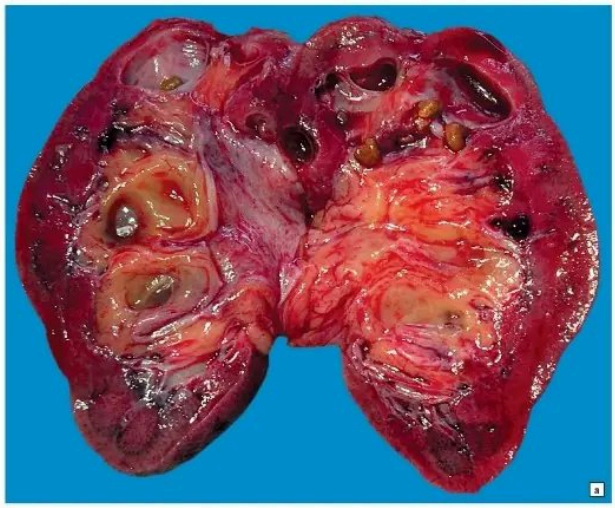


Figure 1.11. *Hydronephrosis: the kidney is enlarged, the pelvis and calyces are markedly dilated, and the parenchyma is thinned (pressure atrophy) (TMA study guide, 2018)*

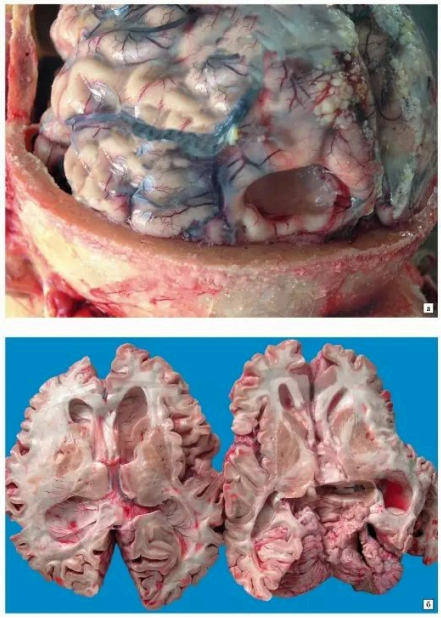


Figure 1.12. *Internal hydrocephalus: the lateral ventricles of the brain are dilated, and the brain tissue is atrophic (TMA study guide, 2018)*

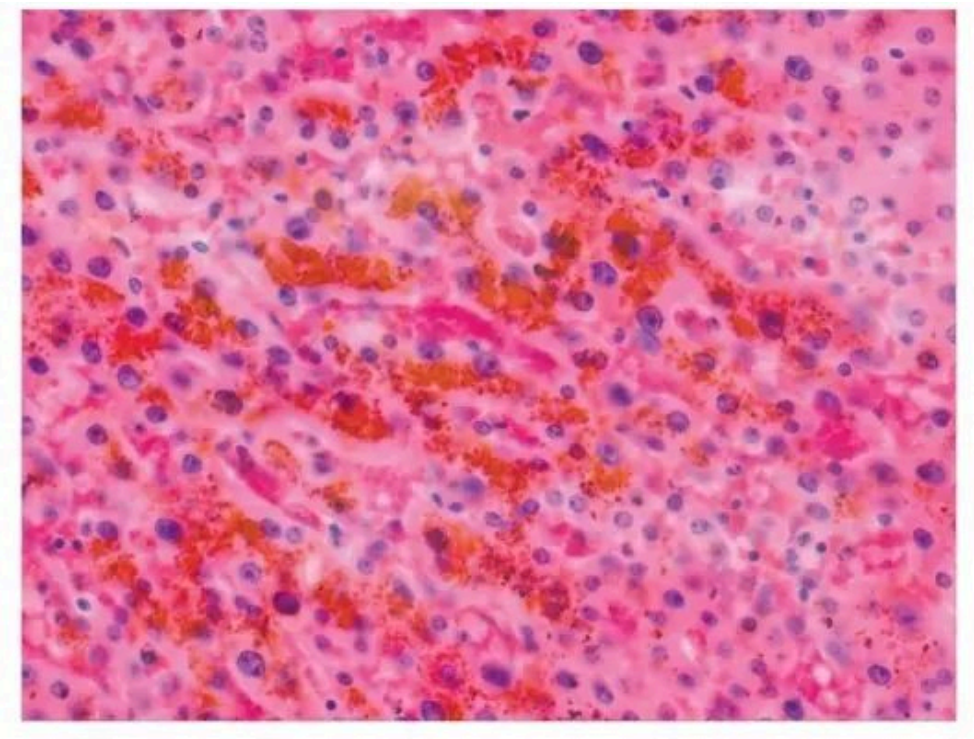


Figure 1.13. *Brown atrophy of the liver (microscopic slide, hematoxylin and eosin): hepatocytes and their nuclei are small, the liver cell plates are thinned, and fine brown lipofuscin granules are seen in the cytoplasm (especially in the center of the lobule) (TMA study guide, 2018)*

1.8. Metaplasia

Metaplasia (Greek metaplasso — I form differently) is the **replacement of one type of mature (differentiated) cell by another type of mature cell**. It is a reversible process that represents adaptation by replacing a cell with one better able to withstand adverse conditions.

Mechanism. A mature cell does not directly turn into another type. Metaplasia results from the **reprogramming of the stem (cambial) cells** of the tissue: under the influence of cytokines, growth factors and extracellular matrix components, stem cells differentiate along a different pathway. Certain transcription factors and retinoic acid (vitamin A) play an important role in this process. Metaplasia occurs **only within one tissue type**: epithelium turns into epithelium, and connective tissue into another type of connective tissue.

Table 1.8. *Main types of metaplasia*

Type	Normal epithelium → new epithelium	Cause and site
Squamous	Ciliated columnar → stratified squamous	Bronchi of smokers (most common); endocervical epithelium of the cervix (transformation zone); vitamin A deficiency; stones in the biliary and urinary tracts
Glandular (columnar, intestinal) metaplasia	Stratified squamous → glandular columnar (with goblet cells)	Barrett esophagus — in the lower esophagus in gastroesophageal reflux
Intestinal metaplasia	Gastric epithelium → intestinal epithelium	Chronic atrophic gastritis (Helicobacter pylori)
Connective tissue metaplasia	Fibroblasts → osteoblasts, chondroblasts	Myositis ossificans — bone formation in muscle after trauma; ossification in old scars and atherosclerotic plaques

CLINICAL SIGNIFICANCE

Although metaplasia is an **adaptation**, it comes at a price: the squamous epithelium in a smoker's bronchus is resistant, but it **does not secrete mucus and has no cilia** — mucociliary clearance is impaired and infections become more frequent. Most importantly, if the stimulus that caused metaplasia persists, **dysplasia and cancer** develop: squamous metaplasia in the bronchus is the soil for **squamous cell lung carcinoma**, Barrett esophagus for **esophageal adenocarcinoma**, and intestinal metaplasia for **gastric adenocarcinoma**. Therefore metaplasia is considered a **precancerous (background) process**.

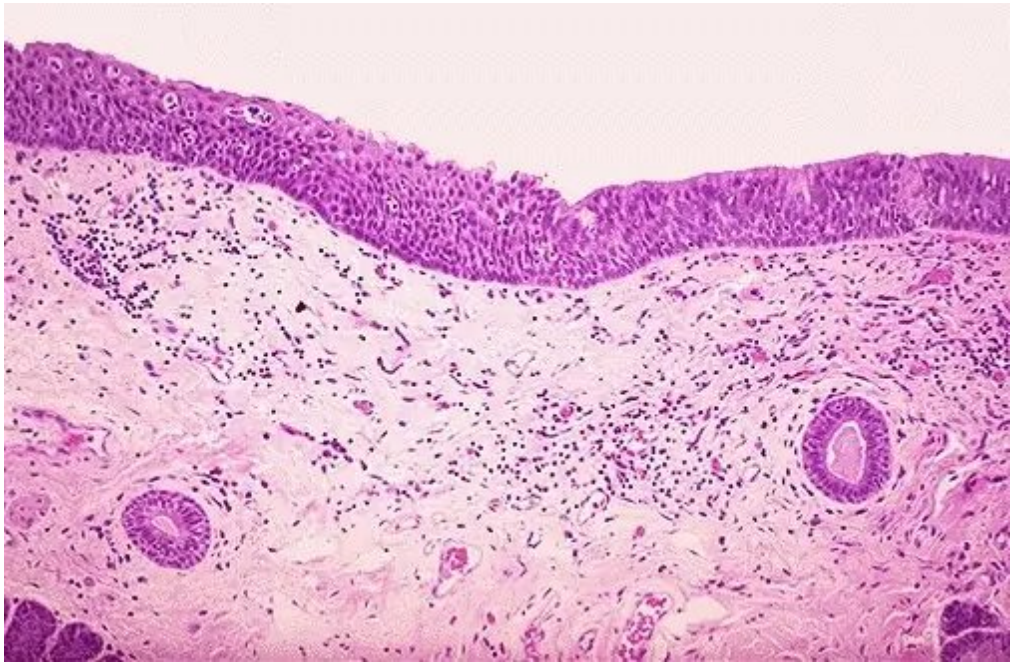


Figure 1.14. *Squamous metaplasia of the bronchial epithelium (microscopic slide, hematoxylin and eosin): left — normal pseudostratified ciliated epithelium, right — stratified squamous epithelium; inflammatory infiltrate in the wall (TMA study guide, 2018)*

MEMORY AID

To remember the direction of metaplasia: **“Bronchus goes Squamous, Barrett goes Glandular”**. The airways put on a **shield** (squamous epithelium) against smoke, while the esophagus, like the stomach and intestine, produces **glands** (mucus) against acid. Both pave the way to cancer: in the bronchus — squamous cell carcinoma, in the esophagus — adenocarcinoma.

1.9. Dysplasia

Dysplasia (Greek dys — disorder, plasis — formation) is a disturbance of the proliferation and differentiation of epithelial cells, manifested by cellular atypia and disruption of tissue structure (histoarchitecture).

Signs of dysplasia:

- variation in the shape and size of cells and nuclei (**pleomorphism**);
- enlargement and hyperchromasia of nuclei, increased nuclear-to-cytoplasmic ratio;
- increased number of mitoses, which also occur in the upper layers of the epithelium;
- loss of cell polarity and of layering. By grade, dysplasia is **mild (I), moderate (II) and severe (III)**. Current classifications often use two grades: **low-grade** and **high-grade** intraepithelial neoplasia. When severe dysplasia involves the full thickness of the epithelium but does not cross the basement membrane, it is called **carcinoma in situ**.

CAUTION

Some textbooks (M.S. Abdullakhodjaeva) discuss dysplasia in the chapter on adaptive processes, whereas the Robbins textbook places it in the chapter on **tumors** (neoplasia). Indeed, dysplasia is **not an adaptation**: it is a **precancerous process**. Mild dysplasia can regress if the cause is removed, whereas severe dysplasia often progresses to invasive cancer. Dysplasia is discussed in detail in Topic 9.

1.10. Clinicopathological section

Adaptation — when is it beneficial and when harmful?

Adaptive processes allow the organism to survive under new conditions, but each of them turns into disease beyond a certain limit:

- **Hypertrophy.** In a patient with arterial hypertension, left ventricular hypertrophy preserves cardiac function for years. However, the capillaries cannot adequately supply the thickened myocardium — the risk of decompensation, arrhythmias and sudden cardiac death increases. Left ventricular hypertrophy detected by ECG and echocardiography is an independent risk factor for cardiovascular complications.
- **Hyperplasia.** In an elderly man with prostatic hyperplasia, urination becomes difficult, the bladder wall undergoes compensatory hypertrophy (trabeculated bladder), and then diverticula, urinary retention, cystitis and hydronephrosis develop.
- **Atrophy.** A patient confined to bed for a long time develops muscle atrophy and osteoporosis; therefore early mobilization and rehabilitation are necessary.
- **Metaplasia.** If squamous metaplasia is found in a bronchial biopsy of a long-term smoker, this is a strong argument for quitting: if smoking stops, the metaplasia can regress; if it continues, dysplasia and cancer develop.

Thanatogenesis and determination of the cause of death

At autopsy, the pathologist lists the diseases that led to death in the following order: **underlying disease** (the nosological entity that caused death directly or through its complications), **complications of the underlying disease**, **concomitant diseases** and **cause of death**. The clinical and pathological diagnoses are compared (the diagnoses agree or disagree), and the reasons for disagreement are discussed at a clinicopathological conference.

CLINICAL SIGNIFICANCE

At the autopsy of a 62-year-old patient with essential hypertension: heart weight 520 g, left ventricular wall 2.2 cm, a fresh infarct focus in the anterior wall of the left ventricle. Structure of the pathological diagnosis: **underlying disease** — myocardial infarction; **background disease** — essential hypertension (concentric hypertrophy of the left ventricle); **complication** — acute heart failure. Here left ventricular hypertrophy is recorded as the **result** of adaptation that has passed into decompensation.

Practical skills

Algorithm for describing a gross specimen

1. Identify which organ the specimen is (the organ or part of it).
2. **Size** (dimensions, weight) — enlarged, reduced, normal.
3. **Shape** — preserved or altered.
4. State of the **capsule (serous membrane)** — smooth, shiny, dull, thickened, wrinkled.

5. **Consistency** — soft, firm, hard (the description is based on the figure and explanations).
6. **Color of the surface and cut surface**, and whether the structure on the cut surface (pattern, lobules) is preserved.
7. If there is a pathological focus — its location, number, shape, size, color and borders.
8. Conclusion: name of the pathological process.

Algorithm for describing a microscopic slide

1. Identify the staining method (hematoxylin and eosin, van Gieson, Sudan III, Congo red, etc.).
2. At low magnification, identify the organ and its part.
3. Where the normal structures are preserved and where they are altered.
4. At high magnification, describe the changes in cells, nuclei, cytoplasm and intercellular substance.
5. State of the blood vessels and stroma; presence or absence of an inflammatory infiltrate.
6. Conclusion: name of the pathological process.

Skill 1. Gross specimens

1. **“Cardiac hypertrophy (myocardial hypertrophy)”.** The heart is enlarged and its weight increased; the left ventricular wall is 2 cm or thicker (normally 0.8–1.2 cm), and the trabecular and papillary muscles are thickened. In the compensation stage the cavity is narrowed (concentric hypertrophy); in decompensation the heart is flabby, the apex is rounded and the cavity is dilated (eccentric hypertrophy, myogenic dilatation). Conclusion: compensatory (work) hypertrophy.
2. **“Bladder hypertrophy, prostatic hyperplasia”.** The prostate is enlarged and nodular; the bladder wall is thickened, and muscle bundles bulge beneath the mucosa in a network pattern (trabeculated bladder), sometimes with false diverticula. Conclusion: compensatory hypertrophy of the bladder wall.
3. **“Glandular hyperplasia of the endometrium”.** The uterus is enlarged; in its cavity the endometrium is unevenly thickened, with polypoid growths. Conclusion: hormonal (hyperestrogenic) hyperplasia.
4. **“Brown atrophy of the liver”.** The liver is small, with a sharp edge, a wrinkled capsule, firm, and brown on the cut surface. Conclusion: a manifestation of generalized atrophy (cachexia) — brown atrophy.
5. **“Brown atrophy of the heart”.** The heart is small, there is little subepicardial fat, the coronary arteries are tortuous, and the myocardium is brown. Conclusion: brown atrophy (accumulation of lipofuscin).
6. **“Hydronephrosis”.** The kidney is enlarged; the pelvis and calyces are dilated, forming cavities; the parenchyma is thinned, and the cortex and medulla are indistinguishable. Conclusion: pressure atrophy.
7. **“Internal hydrocephalus”.** The ventricles of the brain are markedly dilated, and the brain tissue is thinned. Conclusion: pressure atrophy.
8. **“Scar after myocardial infarction”.** Whitish, dense scar tissue in the left ventricular wall; the surrounding myocardium is thickened. Conclusion: regenerative hypertrophy (the myocardium around the scar).

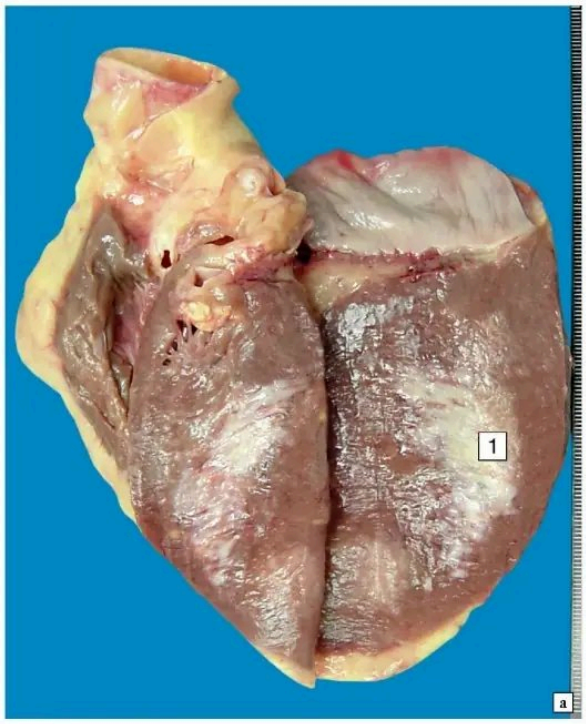


Figure 1.15. *Scar after myocardial infarction (1): a wide, dense, whitish scar in the left ventricular wall; the remaining myocardium shows regenerative hypertrophy (TMA study guide, 2018)*

Skill 2. Microscopic slides

1. **“Metaplasia of the bronchial epithelium”** (hematoxylin and eosin). Normal pseudostratified ciliated epithelium in one part of the bronchial mucosa, stratified squamous epithelium in another; lymphocytic infiltration in the wall. Conclusion: squamous metaplasia — a precancerous background process.
2. **“Glandular hyperplasia of the endometrium”** (hematoxylin and eosin). The mucosa is thickened; the glands are numerous, elongated and convoluted, some cystically dilated, with a “gland within a gland” appearance; the epithelium is proliferating and the stroma is cellular. Conclusion: glandular (glandular-cystic) hyperplasia.
3. **“Brown atrophy of the liver”** (hematoxylin and eosin). Hepatocytes and nuclei are small, the liver cell plates are thin, and the sinusoids between them are dilated; fine yellowish-brown lipofuscin granules are seen in the hepatocytes of the lobule center. Conclusion: brown atrophy.
4. **“Myocardial hypertrophy”** (hematoxylin and eosin). Cardiomyocytes are thickened, and their nuclei are enlarged, hyperchromatic and “rectangular” (box-shaped). Conclusion: compensatory hypertrophy.

Review questions

1. State the content and tasks of pathological anatomy.
2. What research methods are used in pathological anatomy? Explain the significance of autopsy.
3. List the types of biopsy and its tasks.
4. Describe the three periods of development of pathological anatomy. What was R. Virchow's contribution?
5. Who is the founder of the Uzbek school of pathologists? Briefly tell the history of the departments.

6. Explain the concepts of etiology, pathogenesis, morphogenesis and pathomorphosis with examples.
7. State the significance of cell organelles in pathology.
8. Give examples of labile, stable and permanent tissues.
9. State the sequence of cellular responses to stress and injurious agents.
10. List the main causes of cell injury.
11. Definition, types and mechanism of hypertrophy. Stages of compensatory hypertrophy of the heart.
12. What is the difference between hyperplasia and hypertrophy? Give examples of physiological and pathologic hyperplasia.
13. Describe the types of atrophy. How does brown atrophy develop?
14. What is metaplasia? Its types and clinical significance.
15. Signs of dysplasia and how it differs from metaplasia.

Test questions

1. The founder of pathological anatomy is:
   A) G.B. Morgagni
   B) A.I. Abrikosov
   C) K. Rokitansky
   D) M. Bichat
   E) R. Virchow
2. The author of the theory of cellular pathology:
   A) A.I. Strukov
   B) Avicenna (Ibn Sina)
   C) I.V. Davydovsky
   D) R. Virchow
   E) G.B. Morgagni
3. The founder of the Uzbek school of pathologists:
   A) M.S. Abdullakhodjaeva
   B) V.A. Alimov
   C) S.F. Tatarenchik
   D) V.V. Vasilevsky
   E) G.N. Terekhov
4. Performed during surgery to determine the nature of a tumor and the extent of surgery:
   A) Aspiration biopsy
   B) Autopsy
   C) Routine histologic examination
   D) Electron microscopy
   E) Urgent (intraoperative) biopsy

5. The branch that studies the mechanism of development of disease:
   A) Thanatogenesis
   B) Pathomorphosis
   C) Pathogenesis
   D) Morphogenesis
   E) Etiology

6. Permanent (nondividing) cells include:
   A) Cardiomyocytes
   B) Endothelium
   C) Intestinal epithelium
   D) Fibroblasts
   E) Hepatocytes

7. Thickening of the left ventricular wall in arterial hypertension:
   A) False hypertrophy
   B) Compensatory (work) hypertrophy
   C) Neurohumoral hypertrophy
   D) Vicarious hypertrophy
   E) Physiological hypertrophy

8. Enlargement of one kidney after removal of the other:
   A) Hyperplasia
   B) Hormonal hypertrophy
   C) Vicarious hypertrophy
   D) Hypertrophic growth
   E) False hypertrophy

9. Characteristic of the decompensation stage of cardiac hypertrophy:
   A) Complete preservation of cardiac function
   B) Myogenic dilatation and heart failure
   C) Increase in the number of mitochondria
   D) Narrowing of the cavities
   E) Increase in the number of myofibrils

10. The main cause of glandular hyperplasia of the endometrium:
    A) Deficiency of thyroid-stimulating hormone
    B) Excess of somatotropic hormone
    C) Androgen deficiency
    D) Excess of estrogen (hyperestrogenemia)
    E) Excess of progesterone

11. The pigment that gives tissue a brown color in brown atrophy:
    A) Bilirubin
    B) Lipofuscin
    C) Hematoidin
    D) Hemosiderin
    E) Melanin

12. Hydronephrosis is an example of which type of atrophy?
    A) Physiological
    B) Dysfunctional
    C) Due to inadequate blood supply
    D) Due to pressure
    E) Neurotic

13. The most common metaplasia in the bronchi of smokers:
    A) Cartilaginous metaplasia
    B) Osseous metaplasia
    C) Intestinal metaplasia
    D) Glandular metaplasia
    E) Squamous metaplasia

14. Barrett esophagus increases the risk of which tumor?
    A) Adenocarcinoma
    B) Lymphoma
    C) Melanoma
    D) Squamous cell carcinoma
    E) Leiomyosarcoma

15. An organ that is small and underdeveloped from birth:
    A) Cachexia
    B) Aplasia
    C) Hypoplasia
    D) Atrophy
    E) Agenesis

Clinical cases

Case 1

A 58-year-old man had suffered from arterial hypertension for 15 years and was not treated regularly. In the last year he became short of breath when climbing stairs, and edema appeared in his legs. He died of chronic heart failure. At autopsy: heart weight 610 g, left ventricular wall 2.4 cm, the cavities are dilated, the apex of the heart is rounded, and the myocardium is flabby.

Questions: 1) What process was observed in the heart? 2) What is its type and stage? 3) State the mechanism and microscopic signs of hypertrophy.

Answer: 1) Hypertrophy of the heart (mainly of the left ventricle). 2) Compensatory (work) hypertrophy caused by arterial hypertension (increased peripheral resistance). Dilatation of the cavities, rounding of the apex and heart failure indicate the decompensation stage (eccentric hypertrophy, myogenic dilatation). 3) Mechanical stretch and hormonal factors (angiotensin II, adrenergic agonists) activate signaling pathways and increase the synthesis of structural proteins and organelles; cardiomyocytes do not divide, they only enlarge. Microscopically, the cardiomyocytes are thickened, with large, hyperchromatic, box-shaped nuclei; in decompensation — degeneration, death of cardiomyocytes and interstitial sclerosis.

Case 2

A 72-year-old woman was treated for a long time for gastric cancer and died. At autopsy the subcutaneous fat layer is almost absent, and the remaining adipose tissue is orange. The heart weighs 180 g and is small, the subepicardial coronary arteries are tortuous, and the myocardium is brown. The liver weighs 950 g, its capsule is wrinkled, and the cut surface is brown.

Questions: 1) What generalized process was observed? 2) What are the changes in the heart and liver called? 3) What causes the brown color of the organs and the orange color of the adipose tissue?

Answer: 1) Cachexia — generalized atrophy, in this case cancer cachexia (tumor intoxication, TNF and other cytokines, impaired absorption of nutrients). 2) Brown atrophy of the heart and liver. 3) The brown color is due to the accumulation of lipofuscin pigment, formed from undigested residues of autophagy in atrophic cells. The orange color of the adipose tissue is due to the increased concentration of lipochrome pigment.

Case 3

A 45-year-old man has smoked a pack of cigarettes a day for 25 years. Because of chronic bronchitis, bronchoscopy was performed and a biopsy was taken from the bronchial mucosa. Histologic examination: one part of the bronchus is lined by pseudostratified ciliated epithelium and another part by stratified squamous epithelium; there is no cellular atypia.

Questions: 1) Formulate the histologic conclusion. 2) Explain the mechanism of this process. 3) What are its clinical significance and prognosis?

Answer: 1) Squamous metaplasia of the bronchial epithelium without atypia. 2) Under the influence of tobacco smoke, the stem (basal) cells of the bronchial epithelium are reprogrammed and differentiate into the more resistant stratified squamous epithelium instead of ciliated epithelium. 3) In metaplasia, mucus secretion and ciliary function are lost — mucociliary clearance is impaired, and bronchitis and infections become more frequent. Metaplasia is a precancerous background process: if smoking continues, dysplasia and squamous cell lung carcinoma may develop. If smoking stops, the metaplasia can regress, so the patient is strongly advised to quit smoking.

Quick review

Table 1.9. *Key facts on Topic 1*

Question	Answer
Founder of pathological anatomy	G.B. Morgagni (1761)
Author of cellular pathology	R. Virchow (1858)
Founder of the Uzbek school	Prof. G.N. Terekhov
Main methods	Autopsy, biopsy, experiment
Etiology / pathogenesis / morphogenesis	Cause / mechanism / sequence of structural changes
Pathomorphosis	A persistent change in the picture of a disease (natural or due to treatment)
Permanent cells	Neurons, cardiomyocytes, skeletal muscle cells
Sequence of cellular responses	Adaptation → reversible injury → irreversible injury → necrosis/apoptosis
The most common cause of injury	Hypoxia (ischemia)
Hypertrophy	Enlargement of cells

Question	Answer
Hyperplasia	Increase in the number of cells
Stages of cardiac hypertrophy	Emergency → compensation (concentric) → decompensation (eccentric, myogenic dilatation)
Normal left ventricular wall	0.8–1.2 cm
Vicarious hypertrophy	When one of a pair of organs is lost, the other enlarges
Mechanism of atrophy	Ubiquitin–proteasome degradation + autophagy
Pigment of brown atrophy	Lipofuscin
Pressure atrophy	Hydronephrosis, hydrocephalus
Metaplasia	Replacement of one mature cell type by another (reprogramming of stem cells)
Bronchial metaplasia	Squamous → risk of squamous cell carcinoma
Barrett esophagus	Glandular (intestinal) metaplasia → risk of adenocarcinoma
Dysplasia	Atypia + disrupted architecture; a precancerous process; the severe grade → carcinoma in situ

Glossary of terms

- **Agenesis** — complete absence of an organ or its primordium as a result of disturbed ontogenesis.
- **Aplasia** — persistence of an organ only as an early primordium, without its typical structure.
- **Atrophy** (Greek a — negation, trophe — nourishment) — reduction in the size of cells, tissues and organs during life, with weakening of their function.
- **Heterotopia** — presence or displacement of tissue in a site that is not typical for it.
- **Hypertrophy** (Greek hyper — over, excess, trophe — nourishment) — increase in the size of a cell, tissue or organ.
- **Hypoplasia** — incomplete (subnormal) development of an organ.
- **Dysontogenesis** (disontogenesis) — disturbance of ontogenesis (individual development).
- **Dysplasia** — a marked disturbance of epithelial proliferation and differentiation, with cellular atypia and disruption of histoarchitecture.
- **Metaplasia** (Greek metaplasso — I form differently) — transformation of one type of tissue into another, related type of tissue.

TOPIC 2

Reversible and irreversible cell injury and cell death: causes, mechanisms and morphology. Apoptosis. Necrosis

Learning objectives

The student should know:

- the general laws of cell injury and the difference between reversible and irreversible injury;
- the main biochemical mechanisms of cell injury: ATP depletion, mitochondrial damage, calcium influx, free radicals, membrane and DNA damage;
- the features of ischemic, ischemia–reperfusion and toxic injury;
- the morphologic signs of reversible injury (cellular swelling, fatty change);
- the definition, stages, microscopic signs, etiologic and clinicomorphologic types and outcomes of necrosis;
- the definition, causes, mechanisms and morphology of apoptosis and how it differs from necrosis;
- the concept of other pathways of cell death (necroptosis, pyroptosis, ferroptosis) and of autophagy;
- death of the organism and postmortem changes.

The student should be able to:

- identify the signs of necrosis in a microscopic slide (changes in the nucleus and cytoplasm);
- describe gross specimens of ischemic infarction of the spleen, hemorrhagic infarction of the lung, gangrene of the leg, gangrenous appendicitis, caseous necrosis of a lymph node and of the kidney, and calcified foci (petrifications) in the lung;
- describe microscopic slides of necrosis of the renal tubular epithelium, caseous necrosis and Councilman bodies;
- distinguish necrosis from apoptosis and relate the types of necrosis to clinical situations.

2.1. General laws of cell injury

When an injurious agent acting on a cell exceeds its adaptive capacity, **cell injury** begins. Its outcome is determined by four factors:

1. **The type, duration and severity of the injurious agent.** A small dose of toxin or brief ischemia causes reversible injury; a large dose or prolonged ischemia leads to cell death.
2. **The type, state and adaptability of the cell.** Skeletal muscle tolerates complete ischemia for 2–3 hours, a cardiomyocyte for 20–30 minutes, and a neuron for only 3–5 minutes.
3. **The genetic features of the cell.** For example, polymorphism of cytochrome P-450 enzymes determines different responses to the same drug dose in different people.
4. **Several interdependent systems of the cell are injured at the same time:** membranes, aerobic respiration (ATP production), protein synthesis, the cytoskeleton and the genetic apparatus.

REMEMBER

Reversible injury — if the injurious agent is removed, the cell returns to normal. Its two main morphologic signs are **cellular swelling** and **fatty change**. The signs of **irreversible injury** are **severe mitochondrial dysfunction** (complete loss of the ability to produce ATP, amorphous densities in the matrix) and **profound disruption** of **membranes** (plasma and lysosomal membranes). These two phenomena are the cell's "point of no return".

Biochemical changes in the cell appear **before** morphologic changes: function is impaired within a few minutes, ultrastructural changes appear after several tens of minutes, changes visible by light microscopy after hours, and gross changes much later. For example, in myocardial ischemia a cardiomyocyte is irreversibly injured in 20–60 minutes, the light-microscopic signs of necrosis are detected after 4–12 hours, and the gross signs after 12–24 hours.

CLINICAL SIGNIFICANCE

This is why a patient who died suddenly may have no gross or even microscopic signs of myocardial infarction. In such cases the pathologist uses histochemical methods: **loss of glycogen** in injured cardiomyocytes (PAS reaction) and failure to stain with **triphenyltetrazolium chloride** (loss of dehydrogenase activity) are the earliest signs of ischemia. Clinically, **troponins** and creatine kinase-MB, which are released into the blood when the cardiomyocyte membrane is disrupted, are the main markers of myocardial necrosis.

2.2. Mechanisms of cell injury

ATP depletion

ATP (adenosine triphosphate) is produced mainly in mitochondria by oxidative phosphorylation. **Hypoxia and ischemia** and mitochondrial poisons (cyanide) rapidly deplete ATP stores. A fall in ATP to 5–10% leads to the following consequences:

- the **Na^+/K^+-ATPase** (sodium pump) slows down — Na+ and water enter the cell and K+ leaves: **cellular swelling**, dilatation of the ER;
- **anaerobic glycolysis** increases — glycogen stores are exhausted, lactic acid accumulates and the **pH falls**: nuclear chromatin clumps (condensation) and enzyme activity changes;
- the **Ca^{2+} pump** fails — calcium increases in the cytoplasm;
- **ribosomes** detach from the rough ER and polysomes break down — **protein synthesis decreases**;
- misfolded proteins accumulate and trigger "ER stress" (the unfolded protein response) and apoptosis.

Mitochondrial damage

Mitochondria are damaged by increased cytosolic Ca2+, free radicals and hypoxia. A high-conductance channel opens in the inner membrane — the **mitochondrial permeability transition pore** — the membrane potential is lost and oxidative phosphorylation stops. In addition, **cytochrome c** and other proteins leak from the intermembrane space into the cytoplasm and initiate apoptosis. Thus mitochondrial damage leads the cell to death by one of two pathways: ATP exhaustion — **necrosis**, release of cytochrome c — **apoptosis**.

Disturbance of calcium homeostasis

Normally the concentration of free cytosolic Ca^{2+} is 10,000 times lower than outside the cell. Under the influence of ischemia and toxins, Ca^{2+} enters the cytoplasm from outside the cell and from intracellular stores (mitochondria, ER). Excess Ca^{2+} activates the following enzymes:

- **phospholipases** — degrade membranes;
- **proteases** — degrade membrane and cytoskeletal proteins;
- **endonucleases** — degrade DNA and chromatin;
- **ATPases** — accelerate ATP depletion. In addition, Ca^{2+} opens the mitochondrial permeability pore and initiates apoptosis.

Free radicals (oxidative stress)

Free radicals are highly reactive molecules that have an unpaired electron in their outer orbit. The most important in cell injury are **reactive oxygen species (ROS)**: the superoxide anion (O_2^-), hydrogen peroxide (H_2O_2) and the most reactive — the hydroxyl radical (•OH).

Sources of free radicals:

- normal metabolism — incomplete reduction of oxygen in mitochondria;
- ionizing radiation (hydrolyzes water to form •OH);
- the “respiratory burst” of neutrophils and macrophages in inflammation (NADPH oxidase);
- metabolism of drugs and chemicals (CCl_4, paracetamol (acetaminophen));
- transition metals — iron and copper (Fenton reaction: $Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + \bullet OH + OH^-$);
- nitric oxide (NO) — forms peroxynitrite.

Harmful effects of free radicals:

1. **Lipid peroxidation in membranes** — a chain reaction with unsaturated fatty acids.
2. **Oxidation of proteins** — protein folding is disrupted, enzyme activity is lost and degradation in proteasomes increases.
3. **DNA damage** — single-strand breaks, mutations; contributes to cellular aging and malignant transformation. **The antioxidant defense system:** antioxidant vitamins (vitamins E, A, C, beta-carotene, glutathione), iron- and copper-binding proteins (transferrin, ferritin, lactoferrin, ceruloplasmin) and enzymes — **catalase** (breaks down H_2O_2), **superoxide dismutase** (converts O_2^- to H_2O_2), **glutathione peroxidase**.

> **MEMORY AID**
> Enzymes that protect against free radicals — **“CSG”**: **C**atalase, **S**uperoxide dismutase, **G**lutathione peroxidase. Remember the formation of the hydroxyl radical (•OH) as “**iron + peroxide = Fenton**”: this is why liver and heart cells are injured in hemochromatosis (iron overload).

Membrane damage and DNA damage

Membrane damage is the final stage in most types of injury. Its causes are ROS (lipid peroxidation), decreased phospholipid synthesis, activation of phospholipases and cytoskeletal disruption. As a result:

- damage to the **mitochondrial membrane** — ATP production stops and apoptotic factors are released;
- damage to the **plasma membrane** — osmotic balance is lost and intracellular contents leak out;

- damage to the **lysosomal membrane** — enzymes (RNase, DNase, proteases, phosphatases) are released into the cytoplasm and cause the cell to **"digest itself"** — this is the basis of the morphology of necrosis. **DNA damage** (by radiation, chemotherapy drugs, ROS) activates the **p53** protein: it arrests the cell cycle in the G1 phase and gives time for DNA repair. If the damage cannot be repaired, p53 initiates **apoptosis**.

Ischemic and hypoxic injury

In **hypoxia** (reduced oxygen with preserved blood flow), glycolysis continues and injury develops more slowly. In **ischemia** (cessation of blood flow), glycolytic substrates are not delivered either and metabolites (lactic acid) are not removed — therefore injury is faster and more severe.

Ischemia–reperfusion injury. Sometimes, when blood flow is restored to ischemic but still viable tissue (thrombolysis, stenting, surgery), injury paradoxically **increases**. The causes:

- in the presence of returning oxygen, large amounts of **free radicals** are generated in damaged mitochondria and leukocytes;
- **Ca^{2+} overload** increases;
- **inflammation** — leukocytes and plasma proteins enter the injured area and cause additional injury;
- the **complement** system is activated (IgM antibodies are deposited in ischemic tissue).

Chemical (toxic) injury

Chemicals injure the cell in two ways:

1. **Direct action** — by binding to molecular components: mercuric chloride binds to the sulfhydryl groups of membrane proteins; cyanide blocks mitochondrial cytochrome oxidase; chemotherapy drugs are directly cytotoxic.
2. **Conversion to toxic metabolites** — mainly by the **cytochrome P-450** enzymes of the smooth ER of the liver. For example, **carbon tetrachloride (CCl_4)** is converted in the liver to the highly reactive free radical CCl3: lipids undergo peroxidation, the ER swells, protein synthesis (including lipoproteins) decreases, fat accumulates in hepatocytes and necrosis develops. In large doses, the toxic metabolite of **paracetamol** (acetaminophen) depletes glutathione stores and causes centrilobular necrosis of the liver.

> **CLINICAL SIGNIFICANCE**
> **N-acetylcysteine** is used as an antidote in paracetamol poisoning — it restores glutathione stores. In people who consume alcohol, cytochrome P-450 enzymes are induced and glutathione is low, so even an ordinary dose of paracetamol can damage their liver.

2.3. Morphology of reversible injury

Reversible injury has two main morphologic manifestations:

1. Cellular swelling (hydropic change, vacuolar degeneration) — entry of water into the cell as a result of failure of the ion pumps. It is the **first** sign of almost all forms of injury.

- Grossly: the organ is slightly enlarged and heavier, its color is pale, and the cut surface is dull and looks "boiled" (in older literature — "cloudy swelling", granular degeneration).
- Microscopically: the cells are enlarged, with small clear vacuoles in the cytoplasm (distended segments of the ER) — **vacuolar (hydropic) degeneration**. This condition is discussed in detail in Topic 3. **2. Fatty change (fatty degeneration)** — small and large lipid vacuoles in the

cytoplasm. It is seen most often in cells involved in lipid metabolism — hepatocytes and cardiomyocytes (Topic 3).

Ultrastructural changes of reversible injury:

1. changes in the **plasma membrane** — formation of blebs (blebbing), loss of microvilli, loosening of intercellular attachments.
2. changes in **mitochondria** — swelling, appearance of small amorphous densities.
3. **dilatation of the ER**, detachment of ribosomes and breakdown of polysomes.
4. changes in the **nucleus** — clumping of chromatin granules.
5. **myelin figures** in the cytoplasm — layered phospholipid structures derived from damaged membranes.

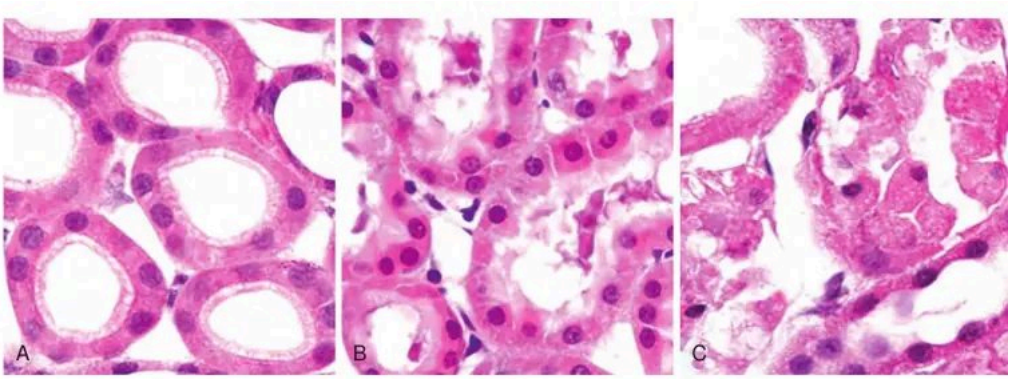


Figure 2.1. *Reversible and irreversible injury of the renal tubular epithelium: A — normal tubules, epithelial cells viable; B — early (reversible) ischemic injury: blebs on the cell surface, increased cytoplasmic eosinophilia, some cells swollen; C — necrosis (irreversible injury): nuclei lost, cells fragmented, contents leaked out (Robbins, 9th ed., Fig. 2-9)*

2.4. Necrosis

Necrosis (Greek nekros — dead) is the death of cells, tissue or part of an organ **in a living organism**, with complete cessation of their vital activity. Necrosis is the end result of irreversible injury and develops as a result of **denaturation** of cellular components and **enzymatic digestion** of the dead cell. The enzymes that digest the cell come from the lysosomes of the dead cell itself (**autolysis**) and from leukocytes attracted by inflammation (**heterolysis**).

> **REMEMBER**
> Necrosis is always a **pathologic** process: it involves **a group of cells**, membrane integrity is lost, intracellular contents leak out and **provoke inflammation**. Apoptosis, in contrast, is often **physiological**, occurs in individual cells, the membrane is preserved and it **does not provoke** inflammation.

Autolysis must be distinguished from necrosis. Autolysis (self-digestion) is the breakdown of cells in the dead body after death under the action of their own enzymes; there is **no inflammatory reaction**. Autolysis develops fastest in organs rich in enzymes (pancreas, gastric mucosa, adrenal medulla).

Stages of necrosis

1. **Paranecrosis** — reversible changes (resembling necrosis but still able to regress).
2. **Necrobiosis** — irreversible degenerative changes; catabolic reactions predominate over anabolic ones.
3. **Cell death** — its exact time is difficult to establish.

4. **Autolysis (and heterolysis)** — breakdown of the dead cell's substrate by hydrolytic enzymes and inflammatory cells.

Microscopic signs of necrosis

Table 2.1. *Microscopic signs of necrosis*

Structure	Change	Essence
Nucleus	Karyopyknosis	The nucleus shrinks, the chromatin condenses and stains darkly (basophilia)
	Karyorrhexis	The pyknotic nucleus breaks into fragments
	Karyolysis	The nucleus dissolves and disappears — chromatin is degraded by DNases, and the nucleus does not stain (“ghost cell”)
Cytoplasm	Eosinophilia	Loss of RNA and denatured proteins bind more eosin — the cytoplasm is deep pink
	Plasmocoagulation	Proteins are denatured, and the cytoplasm becomes uniform (homogeneous) and dense
	Plasmorrhexis	The cytoplasm breaks into fragments
	Plasmolysis	The cytoplasm dissolves
Intercellular substance	Swelling, imbibition, fibrinoid change and dissolution of collagen fibers	Necrosis of connective tissue

MEMORY AID
Order of nuclear changes — **“Pyk-Rhex-Lys”**: **Pyk**nosis (shrinks) → karyo**rrhex**is (fragments) → karyo**lys**is (dissolves). Likewise for the cytoplasm: **coagulation → rhexis → lysis**.

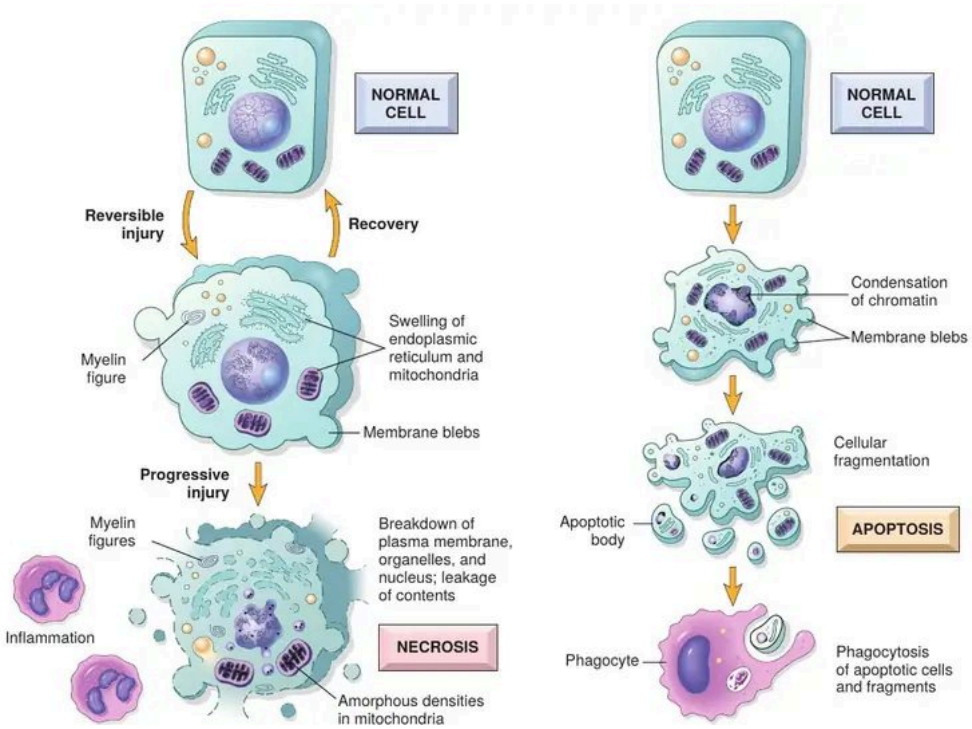


Figure 2.2. *Diagram of the morphologic changes of cell injury. Left column — reversible injury (swelling of the cell and organelles, membrane blebs, myelin figures) and its progression to necrosis (breakdown of the plasma membrane, organelles and nucleus, leakage of contents, amorphous densities in mitochondria, inflammation). Right column — apoptosis (chromatin condensation, membrane blebs, fragmentation of the cell into apoptotic bodies and their phagocytosis) (Robbins, 9th ed., Fig. 2-8)*

Etiologic types of necrosis

Table 2.2. *Etiologic and pathogenetic types of necrosis*

Type	Cause	Mode of action	Example
Traumatic	Physical and chemical agents directly	Direct	Trauma, burns, frostbite, action of acids and alkalis
Toxic	Bacterial toxins, chemical poisons	Direct	Myocardial necrosis caused by diphtheria exotoxin, mercuric chloride nephrosis
Trophoneurotic	Damage to the nervous system, impaired tissue trophism	Indirect	Bedsores, trophic ulcers after spinal cord injury
Allergic	Immune reactions	Indirect	Arthus phenomenon, fibrinoid necrosis in autoimmune diseases
Vascular (ischemic)	Cessation of arterial blood flow	Indirect	Infarction — of the myocardium, brain, kidney, spleen

Clinicomorphologic forms of necrosis

1. Coagulative (dry) necrosis is the most common form. The dead tissue **preserves its structural outlines** for several days: proteins and enzymes are denatured and proteolysis stops. The cells have no nuclei and their cytoplasm is eosinophilic, but the outlines of cells and tissue remain visible as “ghosts”. It occurs in organs rich in protein and poor in water — the heart, kidney, spleen and liver. In all organs except the brain, **ischemia** leads to coagulative necrosis.

Grossly: the focus is dry, firm, gray-yellow or whitish, and separated from the surroundings by a red rim (demarcation zone).

Special types of coagulative necrosis:

- **ischemic infarction** (myocardium, kidney, spleen);
- **waxy (Zenker) necrosis** — in the rectus abdominis and thigh adductor muscles in typhoid fever and other severe infections; the muscle is whitish and looks like wax;
- **caseous (cheese-like) necrosis**;
- **fibrinoid necrosis** (discussed separately below).

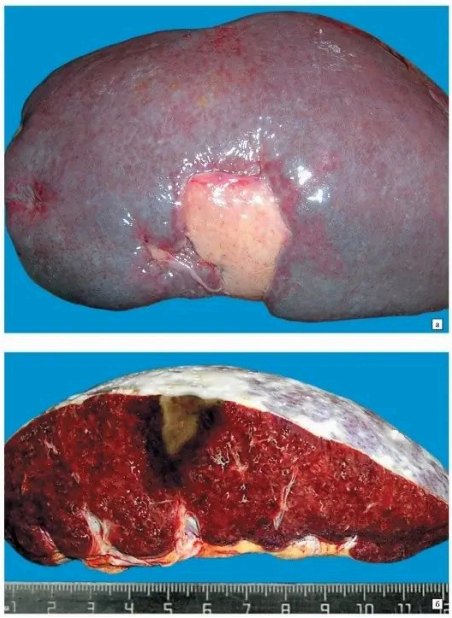


Figure 2.4. *Ischemic infarction of the spleen: a whitish, firm, triangular focus beneath the capsule; on the cut surface its base faces the capsule (TMA study guide, 2018)*

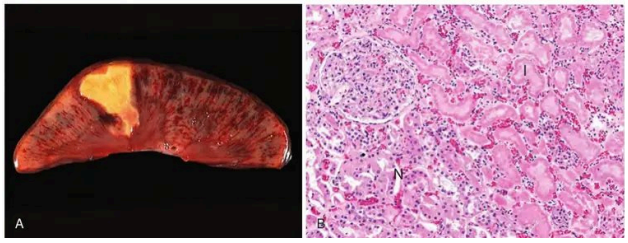


Figure 2.3. *Coagulative necrosis of the kidney: A — wedge-shaped renal infarct (yellow); B — border of the infarct: normal kidney on the left (N), necrotic cells on the right (I) — cell outlines are preserved, nuclei are absent; an inflammatory infiltrate between them (Robbins, 9th ed., Fig. 2-11)*

2. Liquefactive (wet) necrosis — the dead tissue **liquefies and dissolves**. It occurs in tissues rich in water and enzymes and poor in protein. Main examples:

- in the **brain** after hypoxia and ischemia (ischemic stroke) — the focus is called **“gray softening”**, and later a **cyst** forms in its place;
- in **bacterial (sometimes fungal) infections** — leukocyte enzymes dissolve the tissue and form **pus** (abscess).

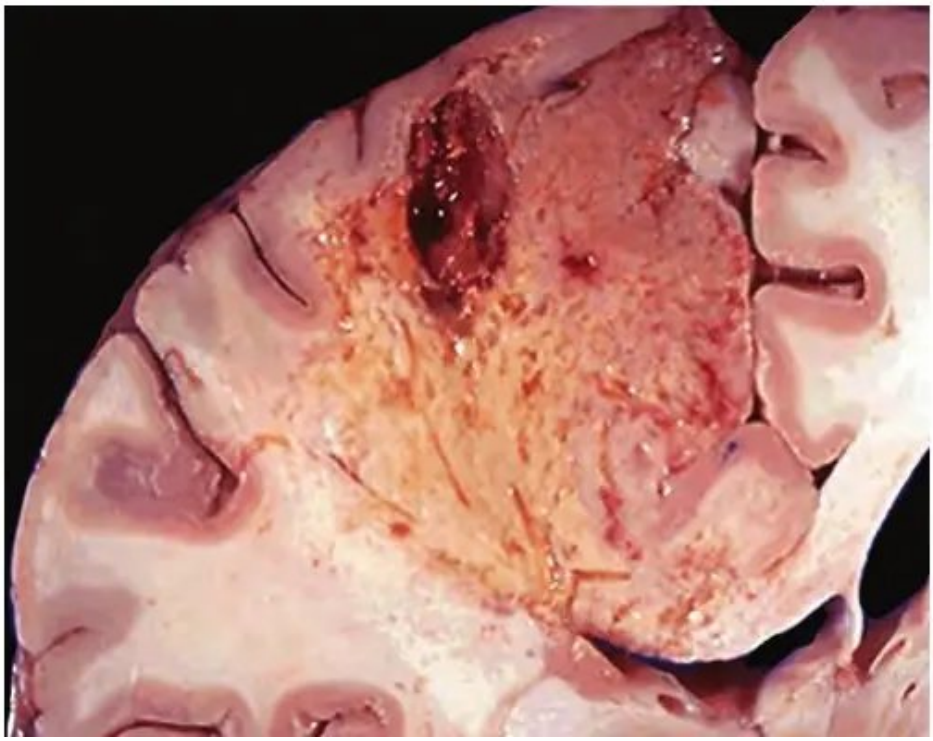


Figure 2.5. *Liquefactive necrosis of the brain: in the area of cerebral infarction the tissue softens and dissolves (Robbins, 9th ed., Fig. 2-12)*

3. Caseous (cheese-like) necrosis is a special type of coagulative necrosis. Grossly it is a **white-yellow, dry, crumbly, cheese-like** mass. It occurs mainly in **tuberculosis**, rarely in fungal infections (histoplasmosis), syphilis (gumma) and lymphogranulomatosis. Microscopically, the caseous focus appears as a **structureless, granular, pink mass**: the cell outlines are **not preserved** (unlike coagulative necrosis), and it is surrounded by **granulomatous inflammation** (epithelioid cells, Pirogov–Langhans giant cells, lymphocytes).



Figure 2.6. *Caseous necrosis of the lung (tuberculosis): a large focus filled with a white-yellow, cheese-like mass in the lung tissue (Robbins, 9th ed., Fig. 2-13)*

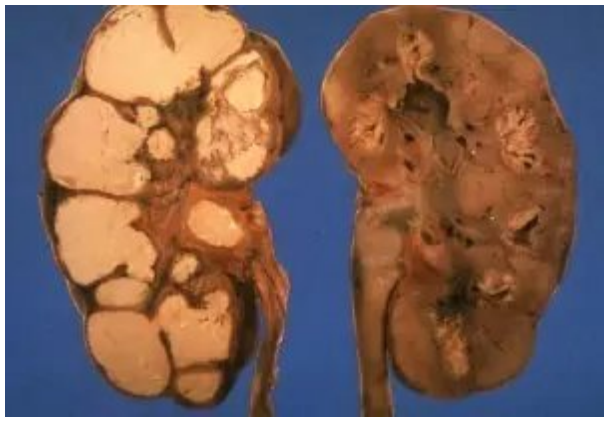


Figure 2.7. *Renal tuberculosis — caseous necrosis: the kidney structure is destroyed, with cavities (caverns) filled with a cheese-like mass (TMA study guide, 2018)*

4. Fat necrosis (steatonecrosis) — focal destruction of adipose tissue. It is seen mainly in **acute pancreatitis**: activated pancreatic lipases escape and break down the triglycerides of the adipose tissue of the peritoneum, omentum and mesentery. The released fatty acids combine with calcium to form **calcium soaps** (**saponification**) — grossly, **chalky white spots** are seen. Microscopically: the

shadowy outlines of fat cells, basophilic calcium deposits between them and inflammation. **Traumatic fat necrosis** of the breast may be clinically mistaken for cancer.

5. Fibrinoid necrosis — necrosis of connective tissue and of the vessel wall. It usually occurs in **immune reactions** (antigen–antibody complexes deposited in the vessel wall together with fibrin) — vasculitis, rheumatic diseases, and also in **malignant hypertension**. With hematoxylin and eosin, a **bright pink, homogeneous, fibrin-like** mass is seen in the vessel wall. This process is also discussed in Topic 3 as the irreversible stage of fibrinoid swelling.

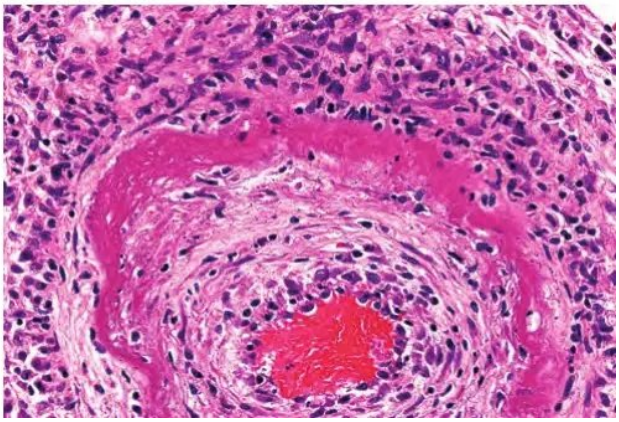


Figure 2.9. *Fibrinoid necrosis of an artery: the vessel wall is circumferentially replaced by a bright pink necrotic mass, with inflammation (neutrophils) around it (Robbins, 9th ed., Fig. 2-15)*

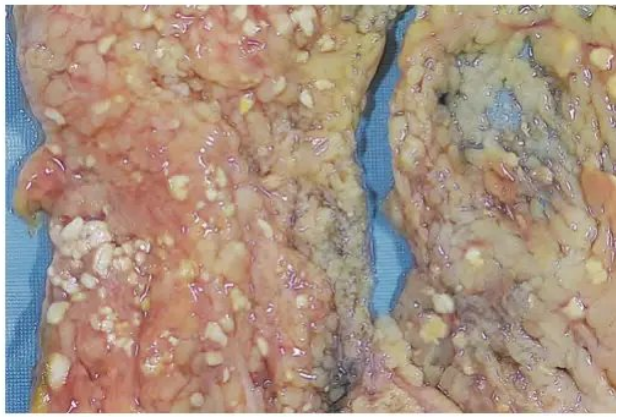


Figure 2.8. *Fat necrosis (acute pancreatitis): chalky white spots in the mesentery — deposits of calcium soaps (Robbins, 9th ed., Fig. 2-14)*

6. Gangrene (Greek gangraina — “gnawing sore”) is necrosis of tissues **in contact with the external environment** (skin, intestine, lung, limbs, uterus). Because iron sulfide (a breakdown product of hemoglobin — a compound of hemoglobin iron and hydrogen sulfide from the air) forms in the necrotic focus, the tissue becomes **black or gray-black**.

Table 2.3. *Types of gangrene*

Type	Condition	Morphology	Example
Dry gangrene	Arterial blood flow stops, the tissue loses water, and no infection is added	The tissue is dry, wrinkled, hard, black, “mummified”; it is separated from healthy tissue by a clear demarcation line	Atherosclerosis, obliterating endarteritis, toes in frostbite
Wet gangrene	Putrefactive bacteria are added to the necrosis; the tissue contains much water (venous congestion, edema)	The tissue is swollen, soft, wet, dirty green or black, with a foul odor; there is no demarcation line; intoxication is severe	Gangrene of the intestine (intestinal obstruction, mesenteric thrombosis), gangrene of the lung, gangrene of the foot in diabetes mellitus, noma (gangrene of the facial soft tissues in debilitated children)
Gas (anaerobic) gangrene	Deep wounds infected with clostridia (Clostridium perfringens)	The tissue is swollen and crackles on palpation (crepitation) — gas bubbles	War wounds, contaminated wounds
Bedsore (decubitus)	Prolonged pressure + trophic disturbance	Necrosis of the skin and soft tissues over bony prominences (sacrum, scapula, heel)	Weak patients confined to bed for a long time



Figure 2.10. *Wet gangrene of the foot (diabetes mellitus): the tissues are black-bluish and swollen (TMA study guide, 2018)*



Figure 2.11. *Gangrenous appendicitis: the vermiform appendix is thickened and dirty green, with fibrinous deposits on its serosa (TMA study guide, 2018)*

7. Sequestrum — a part of necrotic tissue lying freely among living tissues and not undergoing autolysis. It occurs most often in **bone** — in suppurative osteomyelitis. A **sequestral cavity** forms around the sequestrum; it is filled with pus, which drains outside through sinus tracts (fistulas).

8. Infarction is **vascular (ischemic) necrosis**, i.e., necrosis resulting from cessation of the blood supply to part of an organ. The forms of infarction (white, red, white with a red rim) and its features are discussed in detail in Topic 7.

Outcomes of necrosis

Demarcation inflammation develops around the necrotic focus: the vessels dilate (hyperemia), and there is edema and leukocyte infiltration. This zone separates the dead tissue from the living tissue. Further outcomes:

- **Favorable outcomes:**

– **organization** — the necrotic focus is replaced by granulation tissue, and a **scar** forms (for example, postinfarction cardiosclerosis); – **encapsulation** — the focus is surrounded by a connective tissue capsule; – **petrification** (calcification) — calcium salts are deposited in the necrotic focus (dystrophic calcification, Topic 4); for example, the **Ghon focus** remaining from the primary tuberculous focus; – **ossification** — bone tissue forms in the focus; – **cyst** formation — in place of liquefactive necrosis (for example, in the brain); – **mutilation** — separation of the necrotic part (in dry gangrene of a finger). • **Unfavorable outcomes: purulent dissolution** — suppurative breakdown of the necrotic focus, which can lead to sepsis; loss of function of the organ containing the necrotic focus (cardiac rupture, paralysis after stroke).

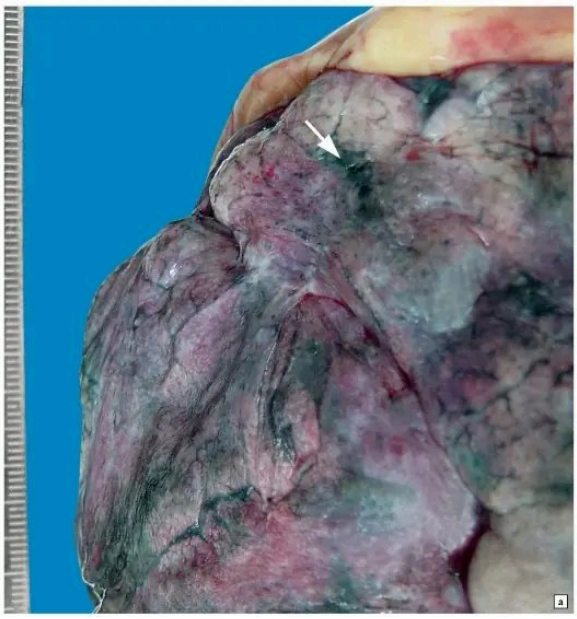


Figure 2.12. *Petrifications in the lung: stone-hard, round foci in the lung tissue (pointer) — calcification of necrosis (TMA study guide, 2018)*

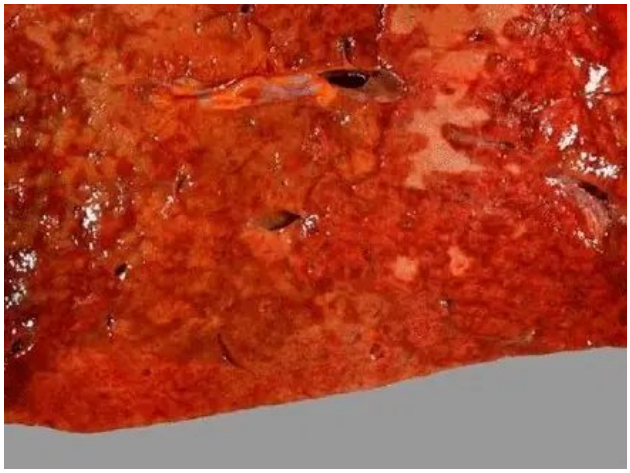


Figure 2.13. *Infarcts of the liver: irregular, firm, gray foci on the cut surface (TMA study guide, 2018)*

CLINICAL SIGNIFICANCE

The significance of necrosis depends on its location and size. In vital organs (heart, brain), even a small necrosis can cause death. Substances released from the necrotic focus are **detected in the blood** and used in diagnosis: in myocardial necrosis — **troponin I and T**, creatine kinase-MB; in liver necrosis — **ALT and AST** (transaminases); in pancreatitis — **amylase and lipase**; in skeletal muscle necrosis (rhabdomyolysis) — creatine kinase and myoglobin. In addition, extensive necrosis leads to a **resorptive syndrome** — fever, leukocytosis and an increased ESR.

2.5. Apoptosis

Apoptosis (Greek apoptosis — “falling off of leaves”) is **programmed cell death**: the cell destroys itself by degrading its own DNA and nuclear and cytoplasmic proteins with special intracellular enzymes (caspases). The membrane of the apoptotic cell **remains intact**, and the cell fragments (apoptotic bodies) are rapidly engulfed by macrophages, so **no inflammation develops**. Apoptosis was described as a distinct type of death in 1972 by J. Kerr, A. Wyllie and A. Currie.

Causes of apoptosis

Physiological apoptosis — elimination of unneeded, aged or dangerous cells:

- programmed death of cells during embryogenesis (implantation, organogenesis, disappearance of the tissue between the fingers);
- involution of hormone-dependent tissues when the hormone decreases (endometrium in the menstrual cycle, ovary in menopause, breast after cessation of lactation);
- maintenance of cell numbers in continuously renewing cell populations (intestinal crypts, lymphocytes);
- elimination of inflammatory cells (neutrophils, lymphocytes) that have completed their task;
- elimination in the thymus of **self-reactive lymphocytes** directed against self antigens (immune tolerance);
- death of virus-infected and tumor cells under the action of cytotoxic T lymphocytes.

Pathologic apoptosis:

- irreparable **DNA damage** (radiation, chemotherapy drugs) — via p53;
- **accumulation of misfolded proteins** (ER stress) — neurodegenerative diseases (Alzheimer and Parkinson diseases);
- some **infections** — viral hepatitis (Councilman bodies), HIV infection;
- **atrophy of organs** after duct obstruction (obstruction of the pancreatic, salivary gland or renal ducts).

Morphology of apoptosis

- **Cell shrinkage** — the cytoplasm becomes dense, and the organelles are tightly packed (but keep their normal structure).
- **Chromatin condensation** — the most characteristic feature: chromatin aggregates densely beneath the nuclear membrane, and the nucleus breaks into fragments (**karyorrhexis**).
- **Membrane blebs** and fragmentation of the cell into membrane-bound **apoptotic bodies** containing cytoplasm, organelles and nuclear fragments.
- **Phagocytosis of apoptotic bodies by macrophages**. In a hematoxylin and eosin–stained slide, an apoptotic cell is a **single** cell separated from its surroundings by a clear space, **round or oval, with deeply eosinophilic cytoplasm** and dense nuclear fragments. Because there is no inflammatory reaction and the process is completed rapidly (within a few hours), apoptosis is difficult to see in histologic slides.

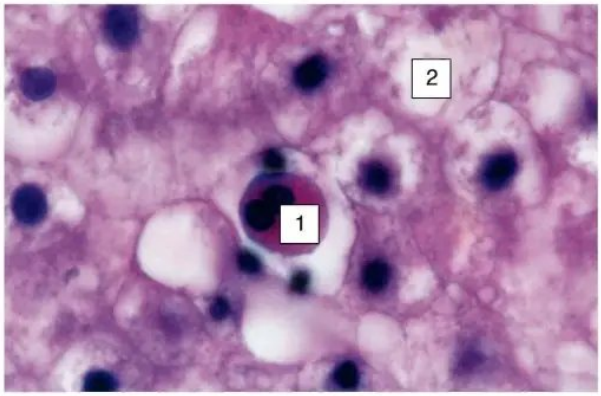


Figure 2.14. *Apoptotic bodies in viral hepatitis (Councilman bodies) — microscopic slide: a round, eosinophilic, dense body (1) between the hepatocyte plates (TMA study guide, 2018)*

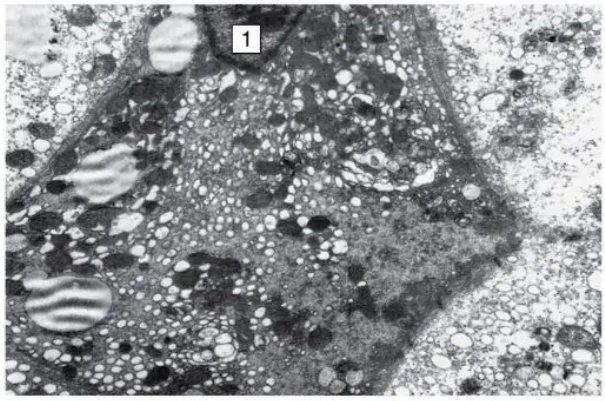


Figure 2.15. *Apoptosis (electron microscopy): the cytoplasm is condensed, the organelles are tightly packed, and the chromatin is condensed (1) (TMA study guide, 2018)*

Mechanisms of apoptosis

Apoptosis consists of two phases: **initiation** — activation of caspases, and **execution** — breakdown of the cell. **Caspases** are proteases with cysteine in their active site that cleave proteins after aspartic acid (cysteine-aspartic proteases). There are two pathways for initiating apoptosis:

1. The intrinsic (mitochondrial) pathway is the main mechanism of apoptosis. The permeability of the outer mitochondrial membrane is regulated by the **BCL2 family of proteins**:

- **anti-apoptotic** proteins — **BCL2**, BCL-XL, MCL1: keep the mitochondrial membrane intact; they are produced in response to growth factors and survival signals;
- **pro-apoptotic** (effector) proteins — **BAX and BAK**: oligomerize and form channels in the mitochondrial membrane;

- **regulated “sensor” proteins of apoptosis** — BIM, BID, BAD, PUMA, NOXA (BH3-only proteins): when stress is sensed (DNA damage, absence of growth factors, misfolded proteins), they activate BAX/BAK and block BCL2. If the cell loses survival signals or is injured, **cytochrome c** leaks into the cytoplasm through BAX/BAK channels, binds the APAF-1 protein to form the **apoptosome** and activates **caspase-9**.

2. The extrinsic (death receptor) pathway. The **death receptors** on the cell surface belong to the TNF receptor family: **Fas (CD95)** and TNFR1. **Fas ligand (FasL)** on the surface of cytotoxic T lymphocytes binds to the Fas receptor and brings together three receptor molecules; their cytoplasmic “death domains” activate **caspase-8** through the adapter protein FADD. This pathway is important in the elimination of self-reactive lymphocytes and in killing by cytotoxic T lymphocytes.

3. The execution phase. Caspase-9 and caspase-8 activate the **executioner caspases (caspase-3 and caspase-6)**. They activate a DNase that cleaves DNA between nucleosomes (into fragments of 180–200 base pairs — a “ladder” pattern on electrophoresis) and degrade the proteins of the nuclear matrix and cytoskeleton.

4. Removal of apoptotic cells. Phosphatidylserine, normally located in the inner leaflet of the membrane, flips to the outer leaflet in an apoptotic cell — this is an “**eat me**” signal for macrophages. Apoptotic bodies are rapidly phagocytosed without inflammation.

MEMORY AID

To remember the caspases of apoptosis: **“Intrinsic — 9, Extrinsic — 8, Executioners — 3 and 6”**. Imagine the intrinsic pathway as a **“mitochondrial gate”**: BCL2 is the gatekeeper (keeps it closed), BAX/BAK are the gate breakers, and cytochrome c is the “messenger of death” that comes out through the gate.

CLINICAL SIGNIFICANCE

Both **decreased** and **increased** apoptosis lead to disease. **Too little apoptosis:** in follicular lymphoma the t(14;18) translocation causes **overproduction of BCL2** — lymphocytes do not die and accumulate; with TP53 mutation, damaged cells are not eliminated — tumors; with defects of Fas or FasL, self-reactive lymphocytes survive — autoimmune diseases. **Too much apoptosis:** neurodegenerative diseases, ischemic injury, viral infections. A drug that blocks BCL2 — **venetoclax** — is used to treat chronic lymphocytic leukemia.

Table 2.4. *Differences between necrosis and apoptosis*

Feature	Necrosis	Apoptosis
Nature	Always pathologic	Often physiological, sometimes pathologic
Cell size	Enlarged (swelling)	Reduced (shrinkage)
Extent	Groups of cells, an area of tissue	Individual (single) cells
Nucleus	Pyknosis → karyorrhexis → karyolysis	Chromatin condensation and fragmentation
Plasma membrane	Disrupted	Intact, with altered structure (phosphatidylserine on the outside)
Cellular contents	Enzymatically digested, leak out	Retained within apoptotic bodies
Inflammation	Present (usually)	Absent
Energy requirement	Does not require ATP (ATP depletion is the cause)	An active, ATP-dependent process
DNA breakdown	Random (a “smear” on electrophoresis)	Internucleosomal (“ladder”)

2.6. Other pathways of cell death and autophagy

- **Necroptosis** — programmed (regulated) necrosis: morphologically it resembles necrosis (cell swelling, membrane rupture, inflammation) but is genetically programmed. It is initiated via TNFR1, is caspase-independent and involves the **RIPK1 and RIPK3 kinases** and the MLKL protein. It is seen in pancreatitis, reperfusion injury, neurodegenerative diseases and some viral infections (a defense against viruses that block caspases).
- **Pyroptosis** — an inflammatory, fever-inducing cell death associated with **inflammasome** activation, caspase-1 and IL-1 production in intracellular microbial infection (Greek pyr — fire).
- **Ferroptosis** — death resulting from iron-dependent lipid peroxidation; it develops when the glutathione system is insufficient.
- **Autophagy** — digestion by the cell of its own organelles and cytoplasm in lysosomes. It is primarily a **survival mechanism** (a source of energy in starvation, clearance of damaged organelles), but excessive autophagy can lead to cell death. Defects in autophagy genes are associated with Crohn disease, neurodegenerative diseases and tumors.

> **CAUTION**
> The textbooks of Strukov and Abdullakhodjaeva discuss two main types of cell death — necrosis and apoptosis. Recent editions of Robbins add **necroptosis, pyroptosis and ferroptosis** to them. When "types of cell death" are asked at the exam, it is advisable first to describe necrosis and apoptosis fully and then briefly mention the modern types.

2.7. Death of the organism and postmortem changes

Death is the irreversible cessation of the vital activity of the organism. By cause, death is divided into **natural** (physiological — from old age), **violent** (trauma, poisoning, homicide, suicide) and death **from disease**. Sudden, unexpected death is called **sudden death** (most often in cardiovascular diseases).

Stages of death:

1. **Agony** — the state before death: disordered fading of functions.
2. **Clinical death** — breathing and cardiac activity have stopped, but there are no irreversible changes in the tissues yet. At normal temperature it lasts **5–6 minutes** (the time the cerebral cortex tolerates hypoxia); during this period resuscitation can restore the organism.
3. **Biological death** — irreversible changes begin in the tissues, first of all in the brain; resuscitation is ineffective. In modern medicine **brain death** (irreversible cessation of all functions of the brain, including the brainstem) is recognized as the death of the person.

Table 2.5. *Postmortem (cadaveric) changes*

Change	Essence and timing
Cooling of the body (algor mortis)	Heat production stops, and body temperature gradually falls to that of the environment
Postmortem rigidity (rigor mortis)	Because of ATP depletion, actin–myosin cross-bridges in the muscles do not detach; it begins 2–4 hours after death in the masticatory muscles and spreads from top to bottom; it disappears in the same order after 24–48 hours (autolysis)
Postmortem lividity (livor mortis)	Blood flows by gravity to the lowest parts of the body: first the hypostasis stage (the spot disappears on pressure), then hemolyzed blood soaks into the tissue — the imbibition stage (does not disappear on pressure)

Change	Essence and timing
Drying of the body	Loss of moisture by the skin, mucous membranes and cornea (“Larcher spots” in the eye)
Postmortem clotting of blood	After death, blood clots form in the vessels and cardiac chambers; they are smooth, shiny, elastic and not attached to the wall (unlike a thrombus)
Autolysis and putrefaction	Breakdown of cells by their own enzymes and later by intestinal bacteria; in putrefaction gases form and the body turns green

CAUTION

At autopsy it is important to distinguish a **postmortem blood clot** from a **thrombus** formed during life: a thrombus is **dull, dry, friable, layered** and **attached** to the vessel wall; a postmortem clot is **smooth, shiny, elastic**, not attached to the wall, follows the shape of the vessel and separates into layers resembling “chicken fat” (yellow, plasma part) and “currant jelly” (red, red cell part) (Topic 7).

2.8. Clinicopathological section

Ischemia–reperfusion: is a quickly opened vessel not always beneficial?

In a patient presenting with myocardial infarction, the vessel is opened in the first hours (stenting or thrombolysis). This greatly reduces the necrotic focus — cardiomyocytes that are still reversibly injured are saved. However, as soon as blood flow is restored, some cells die because of **reperfusion injury** (free radicals, Ca2+ overload, leukocytes), and reperfusion arrhythmias appear. Microscopically, reperfused myocardium shows **hemorrhages** and **contraction band necrosis** (hypercontracted cardiomyocytes with transverse “bands”).

Types of necrosis in the clinic

- **Coagulative necrosis:** infarcts of the heart, kidney and spleen — clinically pain and impaired organ function.
- **Liquefactive necrosis:** ischemic stroke — a brain cyst after several weeks; abscesses — removal of pus (drainage) is necessary.
- **Caseous necrosis:** tuberculosis — the caseous mass is discharged through a bronchus, forming a **cavity**, and the patient spreads mycobacteria to others.
- **Fat necrosis:** acute pancreatitis — blood calcium falls (it is used up in soap formation), which is a sign of a severe course.
- **Gangrene:** diabetic foot — if it progresses to wet gangrene, amputation and prevention of sepsis are necessary.

CLINICAL SIGNIFICANCE

Chemical liver necrosis. In a patient who has taken a large dose of paracetamol (for example, 10–15 g), there are few symptoms on the first day, but on days 2–3 jaundice, a sharp rise in ALT and AST (several thousand units) and liver failure develop. Liver biopsy shows **central (centrilobular) necrosis** — because the cytochrome P-450 enzymes that form the toxic metabolite are located mainly in the center of the lobule, and this area is the least supplied with oxygen.

Practical skills

Skill 1. Gross specimens

1. **“Ischemic infarction of the spleen”.** A whitish-yellow, firm, triangular (wedge-shaped) focus beneath the splenic capsule; its base faces the capsule and its apex faces the hilum of the spleen; a red rim (demarcation zone) around the focus; fibrin deposits on the capsule. Conclusion: coagulative necrosis — white infarct.
2. **“Hemorrhagic (red) infarction of the lung”.** A dark red, firm, wedge-shaped focus in the lung tissue; its base faces the pleura, the pleura is dull, with fibrinous deposits. Conclusion: red infarct (Topic 7).
3. **“Gangrene of the leg”.** The foot and toes are black-bluish; in dry gangrene — wrinkled, hard, separated from healthy tissue by a demarcation line; in wet gangrene — swollen, soft, without demarcation. Conclusion: gangrene (specify the type).
4. **“Gangrenous appendicitis”.** The vermiform appendix is thickened, dirty green or black, its serosa is dull, with fibrinopurulent deposits. Conclusion: a form of intestinal gangrene (wet gangrene).
5. **“Caseous necrosis of a lymph node”.** The lymph node is enlarged and firm; on the cut surface — a white-yellow, dry, crumbly, cheese-like mass. Conclusion: caseous necrosis (tuberculosis).
6. **“Caseous necrosis of the kidney (renal tuberculosis)”.** The kidney structure is destroyed, the cortex and medulla are indistinguishable, with cavities (caverns) filled with a cheese-like mass. Conclusion: caseous necrosis.
7. **“Petrifications in the lung”.** Small, stone-hard, whitish, round foci in the lung tissue; the knife grates when cutting. Conclusion: an outcome of necrosis — petrification (for example, the Ghon focus).
8. **“Gray softening of the brain (ischemic infarction)”.** A soft, mushy, gray focus with indistinct borders in the brain tissue. Conclusion: liquefactive necrosis.

Skill 2. Microscopic slides

1. **“Necrosis of the renal tubular epithelium”** (hematoxylin and eosin; for example, in mercuric chloride poisoning or shock). In the convoluted tubules of the renal cortex the epithelial cells lack nuclei (**karyolysis**), in some cells the nucleus is shrunken (pyknosis) or fragmented (karyorrhexis); the cytoplasm is homogeneous and eosinophilic (plasmocoagulation) or broken into fragments (plasmorrhexis); cell debris in the tubular lumen. The glomeruli are preserved. Conclusion: necrotic nephrosis (acute tubular necrosis).
2. **“Caseous necrosis of a lymph node”** (hematoxylin and eosin). The structure of the lymph node is lost, and most of it is occupied by a structureless, granular, pink mass; around it are epithelioid cells, Pirogov–Langhans giant cells and lymphocytes. Conclusion: caseous necrosis (tuberculosis).
3. **“Necrosis of the tonsil in leukemia (necrotic tonsillitis)”** (hematoxylin and eosin). Most of the tonsillar epithelium and lymphoid tissue has undergone necrosis — a structureless eosinophilic mass (1); a leukocytic infiltrate at its border (2). Conclusion: necrotic tonsillitis.
4. **“Apoptotic bodies in viral hepatitis (Councilman bodies)”** (hematoxylin and eosin). Single round, shrunken cells with deeply eosinophilic cytoplasm and a condensed or absent nucleus between the liver cell plates; no inflammatory reaction around them. Conclusion: apoptosis of hepatocytes.

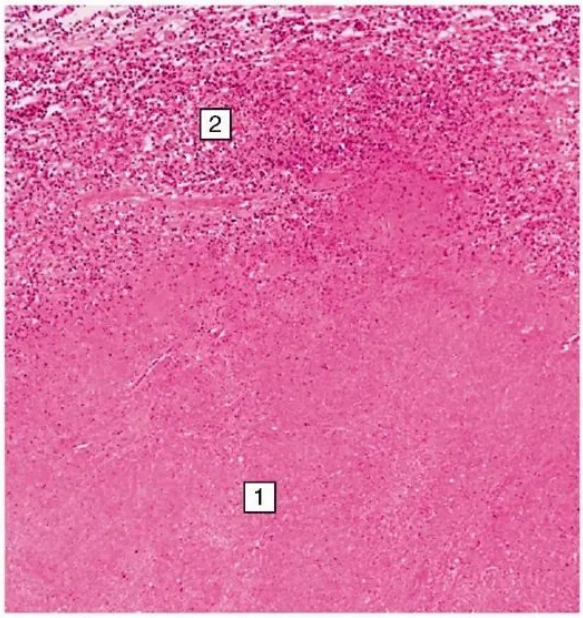


Figure 2.17. *Caseous necrosis of a lymph node in tuberculosis — microscopic slide: structureless pink caseous mass (1) surrounded by a granulomatous reaction (2) (TMA study guide, 2018)*

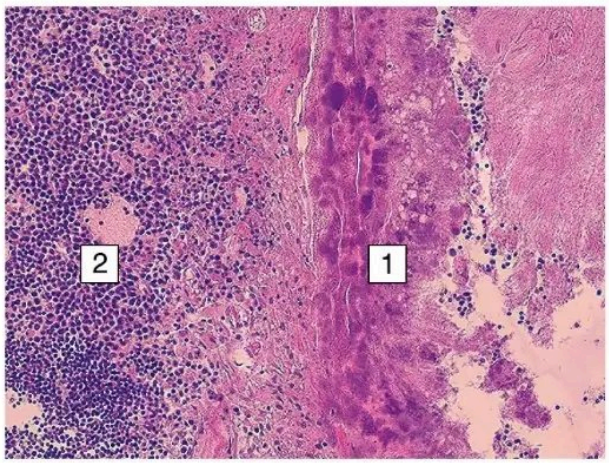


Figure 2.16. *Necrosis of the tonsil in leukemia — microscopic slide: the area of necrosis (1) is bordered by a leukocytic infiltrate (2) (TMA study guide, 2018)*

Review questions

1. What factors determine the outcome of cell injury?
2. State the main morphologic signs of reversible and irreversible injury.
3. What are the consequences of ATP depletion for the cell?
4. How do calcium ions and free radicals injure the cell? List the components of the antioxidant defense system.
5. Explain the mechanism of ischemia–reperfusion injury.
6. What is the mechanism of the toxic action of carbon tetrachloride and paracetamol?
7. What is necrosis? State the stages of necrosis. How does necrosis differ from autolysis?
8. List the microscopic signs of necrosis (changes in the nucleus and cytoplasm).
9. State the etiologic types of necrosis with examples.
10. What is the difference between coagulative and liquefactive necrosis? In which organs do they occur?
11. Describe caseous, fat and fibrinoid necrosis.
12. State the types of gangrene and the differences between dry and wet gangrene.
13. List the outcomes of necrosis.
14. What is apoptosis? Give examples of physiological and pathologic apoptosis.
15. Explain the intrinsic and extrinsic pathways of apoptosis.
16. What are the differences between necrosis and apoptosis?
17. What is the difference between clinical and biological death? List the postmortem changes.

Test questions

1. The main sign of irreversible cell injury:
   A) Fatty change
   B) Amorphous densities in mitochondria and profound disruption of membranes
   C) Dilatation of the ER
   D) Detachment of ribosomes
   E) Cellular swelling

2. The main cause of cellular swelling:
   A) Activation of caspases
   B) Increased protein synthesis
   C) Reduced activity of the Na+/K+-ATPase
   D) DNA breakdown
   E) Rupture of lysosomes

3. Participates in the formation of the hydroxyl radical in the Fenton reaction:
   A) Magnesium
   B) Calcium
   C) Potassium
   D) Iron
   E) Sodium

4. Dissolution and disappearance of the nucleus is called:
   A) Karyolysis
   B) Karyopyknosis
   C) Plasmocoagulation
   D) Karyorrhexis
   E) Plasmorrhexis

5. Necrosis that develops in the brain as a result of ischemia:
   A) Fat necrosis
   B) Caseous
   C) Fibrinoid
   D) Coagulative
   E) Liquefactive

6. Formation of chalky white spots in the omentum in acute pancreatitis:
   A) Fat necrosis (saponification)
   B) Fibrinoid necrosis
   C) Caseous necrosis
   D) Sequestrum
   E) Waxy necrosis

7. The disease in which caseous necrosis occurs most often:
   A) Typhoid fever
   B) Tuberculosis
   C) Rheumatic fever
   D) Acute pancreatitis
   E) Myocardial infarction

8. A feature characteristic of wet gangrene:
   A) Occurs only with cessation of arterial blood flow
   B) No intoxication
   C) Putrefactive infection is added, the tissue is swollen and foul-smelling
   D) A clear demarcation line
   E) The tissue is dry and wrinkled

9. Necrotic tissue lying freely among living tissues in bone:
   A) Infarct
   B) Cyst
   C) Petrification
   D) Mutilation
   E) Sequestrum

10. Necrosis that occurs in the rectus abdominis muscles in typhoid fever:
    A) Waxy (Zenker)
    B) Liquefactive
    C) Fibrinoid
    D) Fat
    E) Caseous

11. Not characteristic of apoptosis:
    A) Translocation of phosphatidylserine to the outer leaflet of the membrane
    B) Formation of apoptotic bodies
    C) A strong inflammatory reaction
    D) Cell shrinkage
    E) Chromatin condensation

12. The substance that is released into the cytoplasm and forms the apoptosome in the intrinsic (mitochondrial) pathway of apoptosis:
    A) TNF
    B) Phosphatidylserine
    C) BCL2
    D) Cytochrome c
    E) Fas ligand

13. A protein with an anti-apoptotic effect:
    A) Caspase-3
    B) BCL2
    C) BAK
    D) BAX
    E) p53

14. The morphologic manifestation of hepatocyte apoptosis in viral hepatitis:
    A) Russell bodies
    B) Aschoff granulomas
    C) Mallory bodies
    D) Pirogov–Langhans cells
    E) Councilman bodies

15. A feature distinguishing a postmortem blood clot from a thrombus:
    A) Occurs only in arteries
    B) Layered structure
    C) Dull, dry, friable
    D) Smooth, shiny, elastic, not attached to the wall
    E) Attached to the vessel wall

Clinical cases

Case 1

A 64-year-old man has suffered for many years from diabetes mellitus and atherosclerosis of the leg arteries. A month ago the big toe of his right foot became black, dry and wrinkled and was clearly demarcated from healthy tissue. Over the last 3 days the foot has become swollen and bluish-black, a foul odor has appeared, and his temperature is 39 °C.

Questions: 1) What process was present a month ago? 2) What is the current condition called, and why did it change? 3) State the differences between the two conditions and their outcomes.

Answer: 1) Dry gangrene — because of atherosclerosis, arterial blood flow gradually stopped and the tissue lost water; no infection was added; there is a clear demarcation line. 2) Wet gangrene — putrefactive bacteria have been added to the necrotic focus; in diabetes, tissue edema and weakened immunity predispose to this. 3) In dry gangrene the tissue is hard and mummified, demarcation is clear, there is no intoxication, and the outcome may be mutilation (spontaneous separation). In wet gangrene the tissue is swollen, soft and foul-smelling, there is no demarcation and intoxication is severe; urgent amputation is necessary because of the risk of sepsis.

Case 2

A 55-year-old woman developed severe, band-like pain in the upper abdomen after heavy drinking. Blood amylase and lipase were sharply increased, and the calcium level was decreased. The woman died. At autopsy the pancreas is swollen, with hemorrhages; there are numerous chalky white spots in the omentum and mesentery.

Questions: 1) Your diagnosis? 2) What process are the chalky white spots? Explain its mechanism. 3) Why is blood calcium decreased?

Answer: 1) Acute (hemorrhagic) pancreatitis (pancreatic necrosis). 2) Fat necrosis (steatonecrosis): activated pancreatic lipases break down the membranes of fat cells and the triglycerides, and the released fatty acids combine with calcium to form calcium soaps (saponification) — these are the chalky white spots. Microscopically: shadowy outlines of fat cells, basophilic calcium deposits, inflammation. 3) Blood calcium decreases because it is used up in soap formation; hypocalcemia is a sign of a severe course of pancreatitis.

Case 3

A 32-year-old man has chronic viral hepatitis B. Liver biopsy: single round, shrunken cells with deeply eosinophilic cytoplasm and dense nuclear fragments between the liver cell plates; there is no inflammatory reaction around them. Lymphocytic infiltration in the portal tracts of the lobule.

Questions: 1) What are the described cells called? 2) What type of cell death is this? Which of its mechanisms is involved? 3) State the differences between this type of death and necrosis.

Answer: 1) Councilman (apoptotic) bodies. 2) Apoptosis. Cytotoxic T lymphocytes eliminate virus-infected hepatocytes through Fas ligand (extrinsic pathway, caspase-8) and perforin/granzyme B; the executioner caspases (caspase-3) degrade the nucleus and cytoskeleton. 3) In apoptosis individual cells die, the cell shrinks, the chromatin condenses, the membrane is preserved, and apoptotic bodies are phagocytosed without inflammation. In necrosis, in contrast, groups of cells die, the cell swells, the membrane is disrupted, the nucleus undergoes pyknosis–karyorrhexis–karyolysis, and inflammation develops.

Quick review

Table 2.6. *Key facts on Topic 2*

Question	Answer
Signs of reversible injury	Cellular swelling, fatty change
Signs of irreversible injury	Severe mitochondrial dysfunction (amorphous densities), profound disruption of membranes
Tolerance to ischemia	Neuron 3–5 min, cardiomyocyte 20–30 min, skeletal muscle 2–3 h
Enzymes activated by Ca2+	Phospholipases, proteases, endonucleases, ATPases
The most reactive free radical	Hydroxyl radical (•OH)
Antioxidant enzymes	Catalase, superoxide dismutase, glutathione peroxidase
Nuclear changes	Pyknosis → karyorrhexis → karyolysis
Stages of necrosis	Paranecrosis → necrobiosis → death → autolysis
The most common necrosis	Coagulative (ischemia outside the brain)
Liquefactive necrosis	Brain (gray softening), abscess
Caseous necrosis	Tuberculosis — cheese-like, outlines not preserved
Fat necrosis	Acute pancreatitis, calcium soaps
Fibrinoid necrosis	Vasculitis, malignant hypertension
Color of gangrene	Iron sulfide — black
Sequestrum	Osteomyelitis
Outcomes of necrosis	Organization, encapsulation, petrification, ossification, cyst, purulent dissolution
Apoptosis — intrinsic pathway	BCL2/BAX-BAK → cytochrome c → apoptosome → caspase-9
Apoptosis — extrinsic pathway	Fas/FasL, TNFR1 → FADD → caspase-8
Executioner caspases	Caspase-3, caspase-6
“Eat me” signal	Phosphatidylserine
Clinical death	5–6 minutes
Rigor mortis	After 2–4 hours, from top to bottom

Glossary of terms

- **Gangrene** (Greek gangraina — fire) — necrosis of tissues in contact with the external environment that turns black as a result of the formation of iron sulfide.

- **Infarction** (Latin infarcire — to stuff, to fill in) — tissue necrosis resulting from impaired circulation (ischemia).
- **Karyopyknosis** — shrinkage of the nucleus, condensation of chromatin.
- **Karyorrhexis** — fragmentation of the nucleus.
- **Karyolysis** — dissolution of the nucleus.
- **Malacia** (Greek malakos — soft) — softening of dead tissue (for example, in the brain).
- **Marantic necrosis** — bedsores occurring in exhausted elderly people with signs of cachexia and marasmus.
- **Mummification** (Latin mumificatio — drying) — drying and hardening of dead tissue.
- **Necrobiosis** (Greek nekros — dead, bios — life) — irreversible degenerative changes preceding necrosis; a process of slow dying.
- **Necrosis** (Greek nekros — dead) — death in a living organism with cessation of the vital activity of cells and tissues.
- **Noma** (Greek nome — gnawing sore, “water cancer”) — wet gangrene of the facial soft tissues in debilitated children (often after measles and other severe infections).
- **Pathobiosis** — necrobiosis prolonged in time.
- **Plasmolysis** — hydrolytic breakdown and dissolution of the cytoplasm.
- **Plasmorrhexis** — fragmentation of the cytoplasm.
- **Sequestrum** (Latin sequestrum — separated) — a part of dead tissue that does not undergo autolysis, is not replaced by connective tissue and lies freely among living tissues.

TOPIC 3

Degenerations: types and pathogenetic mechanisms. Parenchymal and stromal-vascular degenerations

Learning objectives

The student should know:

- the concept of degeneration (dystrophy) and its relation to impaired trophism;
- the morphogenetic mechanisms of degenerations: infiltration, decomposition (phanerosis), perverted synthesis, transformation;
- the classification of degenerations (by location, type of disturbed metabolism, extent and origin);
- parenchymal protein, fatty and carbohydrate degenerations: causes, gross and microscopic appearance, outcomes;
- stromal-vascular protein degenerations: mucoid swelling, fibrinoid change, hyalinosis, amyloidosis (general concept);
- stromal-vascular fatty (obesity, atherosclerosis) and carbohydrate degenerations;
- the histochemical stains used to identify degenerations.

The student should be able to:

- describe gross specimens of fatty degeneration of the liver (“goose liver”), fatty degeneration of the myocardium (“tiger heart”), fatty infiltration of the heart and hyalinosis of the splenic capsule (“glazed spleen”);
- describe microscopic slides of hyaline droplet and hydropic degeneration of the renal tubular epithelium, fatty degeneration of the liver (Sudan III), arteriolar hyalinosis and mucoid swelling;
- determine the type of degeneration by histochemical reactions;
- distinguish reversible from irreversible degenerations.

3.1. The concept of degeneration (dystrophy)

Degeneration (dystrophy) (Greek dys — disorder, trophe — nourishment) is a complex pathological process in which structural changes appear in cells and intercellular substance as a result of **disturbed tissue (cellular) metabolism**. Degeneration is one of the types of **alteration** (damage).

Degeneration is based on a disturbance of **trophism**. **Trophism** is the set of mechanisms that ensure the metabolism and structural organization of a tissue. The mechanisms that maintain trophism are divided into two groups:

- **cellular mechanisms** — the cell's own structures and their regulation (cellular autoregulation);
- **extracellular mechanisms** — transport (blood and lymphatic vessels, microcirculation), the intercellular substance, and the integrative systems (neuroendocrine and neurohumoral regulation). Accordingly, the direct causes of degeneration are: disturbed cellular autoregulation (hyperfunction, toxins, radiation, hereditary enzyme deficiency — **enzymopathy**), disturbed transport systems (**hypoxia** — the most common cause of degeneration), and disturbed regulation by the integrative systems (diseases of the endocrine and nervous systems).

REMEMBER

In degeneration, substances that are **not normally present** accumulate in the cell or intercellular substance, or **normally present substances** accumulate in excess or in an unusual location; sometimes normal

substances disappear (for example, loss of glycogen in cardiomyocytes). Most degenerations are **reversible** processes, but profound changes can lead to necrosis (necrobiosis).

CAUTION

The **Robbins** textbook hardly uses the term “dystrophy”: these processes are discussed as reversible cell injury (cellular swelling, fatty change), **intracellular accumulations**, **hyaline change**, amyloidosis and pathologic calcification. Moreover, the English word dystrophy is used mainly for muscular dystrophies (Duchenne). In CIS and Uzbek textbooks (Strukov, Abdullakhodjaeva), however, the classification of degenerations (dystrophies) is fully preserved, and it is exactly this classification that is asked at the exam.

Morphogenetic mechanisms of degenerations

Table 3.1. *Morphogenetic mechanisms of degenerations*

Mechanism	Essence	Example
Infiltration	Excessive entry of substances from blood and lymph into the cell or intercellular substance, exceeding the capacity to metabolize them	Protein infiltration of the proximal tubular epithelium in nephrotic syndrome (hyaline droplet degeneration); cholesterol infiltration of the aortic intima in atherosclerosis
Decomposition (phanerosis)	Breakdown of complex substances of cells and tissues (lipoprotein membranes, glycoproteins), so that their components become “visible”	Fatty degeneration of cardiomyocytes in diphtheritic intoxication; breakdown of connective tissue in fibrinoid change
Perverted (abnormal) synthesis	Synthesis of abnormal substances that are not normally present in the cell or tissue	Synthesis of amyloid (amyloidosis); alcoholic hyaline in hepatocytes in alcoholism (Mallory bodies)
Transformation	Formation of a different type of substance from the initial products normally used to form a given substance	Conversion of carbohydrate and protein components into fat; polymerization of large amounts of glucose into glycogen

MEMORY AID

Remember the four mechanisms as **“IDAT” — “I Do Avoid Trash”**: **I**nfiltration (excess enters), **D**ecomposition (breaks down), **A**bnormal (perverted) synthesis (an abnormal substance), **T**ransformation (turns into another substance).

Classification of degenerations

Table 3.2. *Classification of degenerations (after A.I. Strukov)*

Feature	Types
By location (parenchyma or stroma)	1) Parenchymal (cellular); 2) Stromal-vascular (mesenchymal, extracellular); 3) Mixed
By type of disturbed metabolism	Protein (dysproteinoses), fatty (lipidoses), carbohydrate; in mixed degenerations — chromoproteins (pigments), nucleoproteins and mineral metabolism
By extent	Generalized (systemic) and local
By origin	Acquired and congenital (hereditary) — storage diseases (thesaurismoses)

3.2. Parenchymal protein degenerations (dysproteinoses)

In parenchymal protein degenerations, the metabolism of **proteins in the cell cytoplasm** is disturbed. They include **hyaline droplet, hydropic (vacuolar)** and **horny (keratinizing)** degenerations.

> **CAUTION**
> In older textbooks "**granular degeneration**" ("cloudy swelling") was given as a separate type. Electron microscopy showed that the cytoplasmic "granules" are swollen mitochondria and fragments of the ER, i.e., an expression of the cell's **functional strain (hyperplasia of organelles)** or early injury. Therefore A.I. Strukov and V.V. Serov do not include it among the degenerations. The gross appearance — the organ is enlarged, dull and looks "boiled" — corresponds to "**cellular swelling**" in Robbins.

Hyaline droplet degeneration

Large, homogeneous, eosinophilic, hyaline-like protein droplets appear in the cytoplasm; they merge with one another and fill the whole cell. In this degeneration the cell organelles are destroyed, and the process is **irreversible**, often leading to **coagulative necrosis**.

Location and causes:

- **Kidney** — in the epithelium of the proximal tubules. Cause: **nephrotic syndrome** (glomerulonephritis, amyloidosis, diabetic nephropathy, paraproteinemic nephropathy) — large amounts of protein pass through the glomerular filter, and the tubular epithelium takes it up by pinocytosis, but the lysosomes cannot digest it in time (**insufficiency of the vacuolar–lysosomal system**). The gross appearance is not specific — it is determined by the underlying kidney disease. Clinically — **proteinuria** and casts in the urine.
- **Liver** — **alcoholic hyaline — Mallory bodies** in hepatocytes (aggregates of cytokeratin intermediate filaments); they occur in alcoholic hepatitis, primary biliary cirrhosis and Wilson disease.
- **Plasma cells** — immunoglobulins accumulate in the cytoplasm to form **Russell bodies** (in chronic inflammation).
- **Myocardium** — very rare.

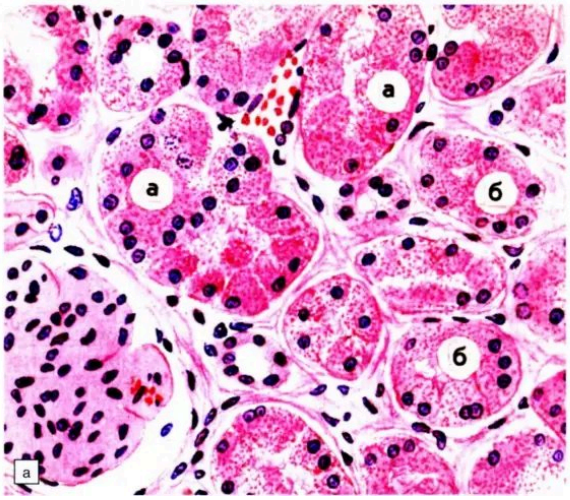


Figure 3.1. *Hyaline droplet degeneration of the renal tubular epithelium: large eosinophilic protein droplets in the cytoplasm (a, b) (Strukov, Serov, Fig. 2-4a)*

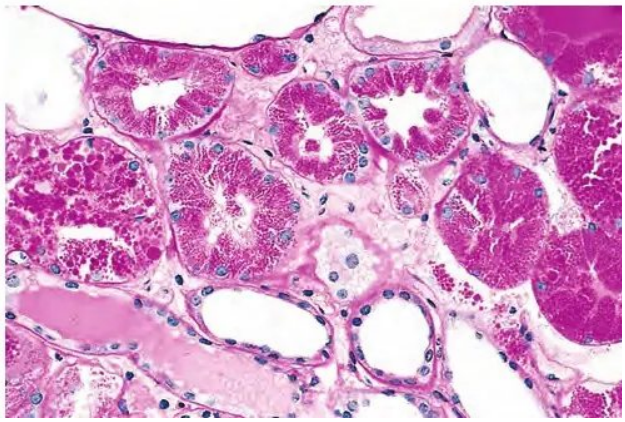


Figure 3.2. *Reabsorbed protein droplets in the epithelium of the proximal renal tubules (Robbins, 9th ed., Fig. 2-32)*

Hydropic (vacuolar) degeneration

Vacuoles filled with cytoplasmic fluid appear in the cell. The cause is increased permeability of cell membranes, failure of the Na+/K+ pump, and disturbed water–electrolyte and protein metabolism; **hydrostatic pressure** inside the cell increases. The cell enlarges, its cytoplasm fills with clear vacuoles, and the nucleus is displaced to the periphery or becomes vacuolated.

When the vacuoles merge and occupy the whole cell, the cell turns into a fluid-filled balloon — **ballooning degeneration**. This condition is regarded as **focal liquefactive necrosis** and is **irreversible**.

Location and causes:

- **Skin epithelium** — smallpox, chickenpox, burns, edema;
- **Renal tubules** — nephrotic syndrome, glycosuria (diabetes), potassium deficiency (hypokalemic nephropathy), toxic influences (ethylene glycol, some drugs) — **osmotic nephrosis**;
- **Liver** — viral hepatitis (especially hepatitis B) — **ballooning degeneration**;
- **Nerve cells, myocardium, adrenal cortex** — severe intoxications, hypoxia;
- **Tumor** cells (for example, chordoma, the clear cell type of renal cell carcinoma). The gross appearance hardly changes. Microscopically, the fluid in the vacuoles **does not stain** with fat stains (Sudan III) — this distinguishes it from fatty degeneration. Outcome: reversible when mild; in ballooning degeneration — necrosis.

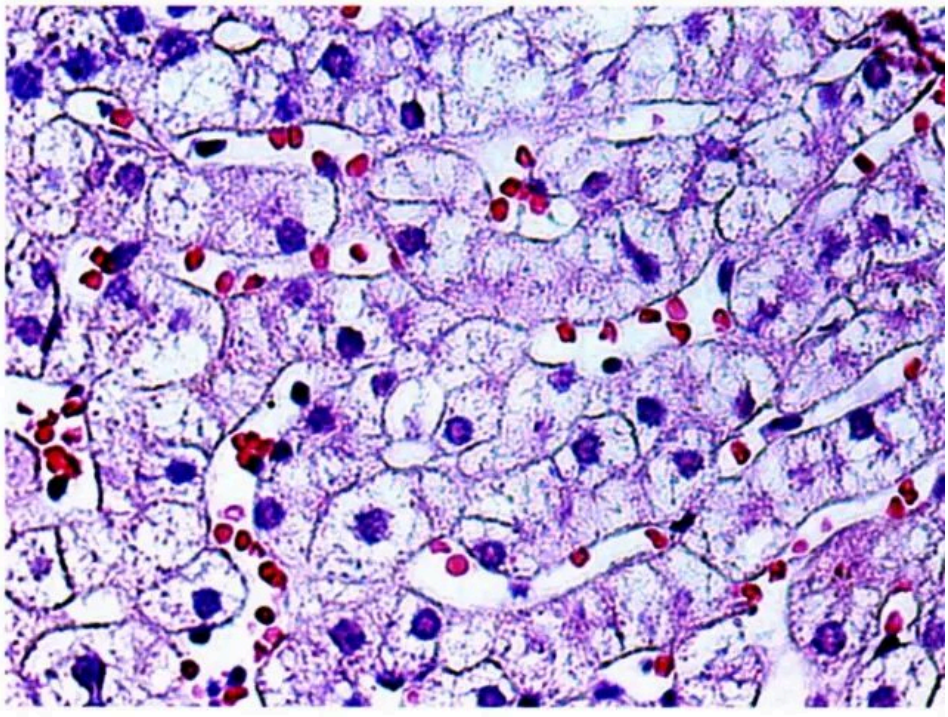


Figure 3.3. *Hydropic degeneration of the liver (biopsy): hepatocytes are enlarged, with vacuolated, clear cytoplasm (Strukov, Serov, Fig. 2-5)*

CLINICAL SIGNIFICANCE

In liver biopsy in viral hepatitis B, **ballooning degeneration** and **Councilman bodies** (Topic 2) are the main signs of hepatocyte injury. Their number and distribution are taken into account in grading the activity of inflammation.

Horny (keratinizing) degeneration

Formation of **horny substance (keratin)** in excess or in a site where it is not normally formed.

- **Hyperkeratosis** — excessive formation of horny substance in keratinizing stratified squamous epithelium: callus (calluses on the palm), ichthyosis (congenital hereditary hyperkeratosis — "fish skin"), warts.
- **Leukoplakia** (Greek leukos — white) — keratinization of mucous membranes lined by nonkeratinizing stratified squamous epithelium (oral cavity, tongue, esophagus, cervix), in the form of a white patch.
- In **keratinizing squamous cell carcinoma** — **"keratin pearls"** (concentric layers of horny substance).

> **CLINICAL SIGNIFICANCE**
> **Leukoplakia** is a clinical term ("white patch") behind which various histologic processes may lie: simple hyperkeratosis or dysplasia. Leukoplakia is considered a **precancerous** condition: some oral leukoplakias turn into squamous cell carcinoma. Therefore a biopsy is necessary in long-standing leukoplakia.

3.3. Parenchymal fatty degenerations (lipidoses)

Accumulation of **neutral fats (triglycerides)** in the cytoplasm of cells. In Robbins — **steatosis (fatty change)**. It occurs most often in cells actively involved in lipid metabolism — hepatocytes, cardiomyocytes and renal tubular epithelium.

Methods for detecting fats: on frozen sections (during paraffin processing, alcohol and xylene dissolve the fats — **empty vacuoles** remain in their place with hematoxylin and eosin):

- **Sudan III** — stains fats **orange-red**;
- **Scarlet red** — red;
- **Sudan IV, Nile blue, osmic acid** — black.

Fatty degeneration of the liver

Fatty degeneration of the liver is the most common fatty degeneration. **Causes:**

- **alcohol** — the most common cause (in developed countries);
- **obesity, diabetes mellitus, metabolic syndrome** — **nonalcoholic fatty liver disease**;
- **protein-energy malnutrition** (kwashiorkor) — synthesis of apoproteins decreases;
- **hypoxia** — anemia, chronic heart failure;
- **toxins and drugs** — CCl4, phosphorus, arsenic, glucocorticoids, tetracycline;
- **infections and intoxications** — diphtheria, tuberculosis, sepsis;
- acute fatty liver of pregnancy.

Mechanism. In the liver, fatty acids are handled as follows: they arrive from the blood → are converted to triglycerides → combine with apoproteins and are secreted into the blood as **lipoproteins**. If any link in this chain is disturbed, fat accumulates:

1. **Excessive delivery of fatty acids to the liver** — starvation, diabetes, alcohol (increased lipolysis in adipose tissue).
2. **Decreased oxidation of fatty acids** — hypoxia, mitochondrial damage (alcohol).
3. **Increased synthesis of triglycerides** — alcohol (excess NADH).
4. **Decreased synthesis of apoproteins** — CCl4, protein deficiency.

5. **Impaired secretion of lipoproteins** — alcohol, toxins. **Grossly:** the liver is enlarged and heavy (up to 3–6 kg), its surface is smooth, the edges are rounded, the consistency is soft and doughy, the color is **yellowish or clay-yellow**, and on cutting fat remains on the knife blade. Such a liver is called a "**goose liver**" (French foie gras — the fatty liver obtained by force-feeding geese).

Microscopically: small-droplet (initially) or **large-droplet** fat vacuoles in the cytoplasm of hepatocytes; a large vacuole pushes the nucleus to the periphery of the cell — the cell takes on a **"signet ring"** appearance. Fat droplets can rupture and form fatty cysts. By location: **centrilobular** (in the center of the lobule — hypoxia, alcohol, toxins), **periportal** (at the periphery of the lobule — phosphorus poisoning, protein deficiency), **diffuse**.

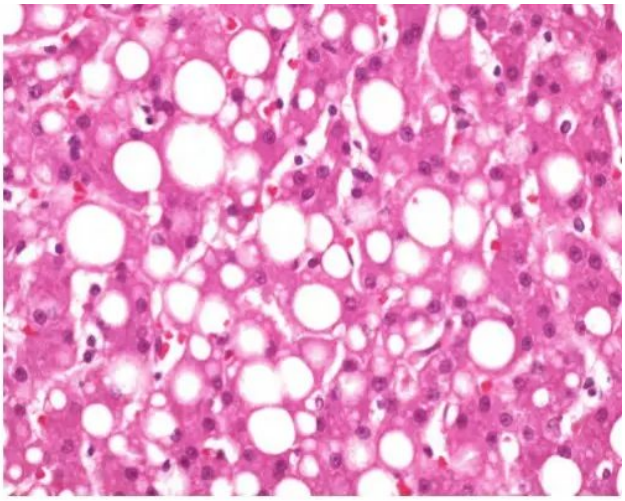


Figure 3.4. *Fatty degeneration of the liver (steatosis) — microscopic slide: large fat vacuoles in hepatocytes, nuclei pushed to the edge of the cytoplasm (hematoxylin and eosin) (Robbins, 9th ed., Fig. 2-30)*

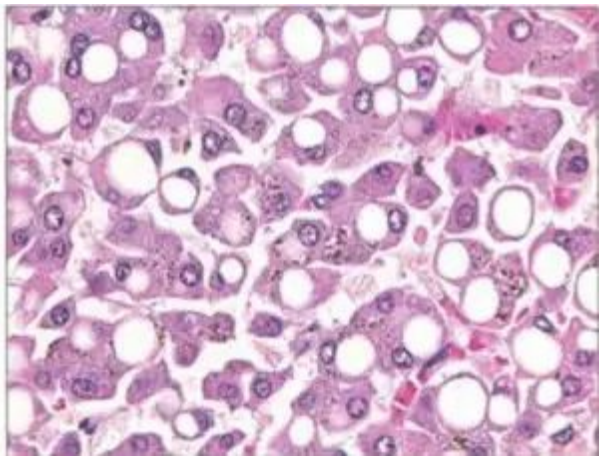


Figure 3.5. *Fatty degeneration of the liver: large and small clear vacuoles (Pathoma, Fig. 1.12)*

Outcomes: if the cause is removed, it is **reversible** (for example, 4–6 weeks after stopping alcohol). If it persists, fatty liver may progress to **steatohepatitis** (inflammation and hepatocyte necrosis), **fibrosis** and **liver cirrhosis**.

CLINICAL SIGNIFICANCE

Nonalcoholic fatty liver disease is today the most widespread chronic liver disease in the world and is closely associated with obesity and type 2 diabetes mellitus. It is often detected incidentally on ultrasound ("increased echogenicity of the liver"). When it progresses to steatohepatitis, transaminases rise; the basis of treatment is weight loss and correction of metabolic disorders.

Fatty degeneration of the myocardium

Causes: mainly **chronic hypoxia** (severe anemia, chronic cardiovascular insufficiency) and **intoxications** (diphtheria, alcohol, phosphorus, arsenic poisoning). Mechanism — damaged mitochondria reduce the oxidation of fatty acids, and fat droplets accumulate in cardiomyocytes; in diphtheria, breakdown of membrane structures (**decomposition**) is also important.

Grossly: the heart is slightly enlarged, its cavities are dilated, the myocardium is flabby and clay-yellow. From the endocardial side, especially in the papillary muscles and trabeculae of the left ventricle, **yellow-white transverse stripes** are seen — the "**tiger heart**". These stripes reflect a **focal** process: fatty degeneration develops in cardiomyocytes around the venules (which receive the least oxygen), while those around the arterioles are spared.

Microscopically: very fine droplets of fat in cardiomyocytes (**pulverized (dust-like) fatty change**), later small droplets; the cross-striation disappears.

Outcome: reversible when mild. When severe — necrosis of cardiomyocytes; myocardial contractility decreases — **heart failure**.

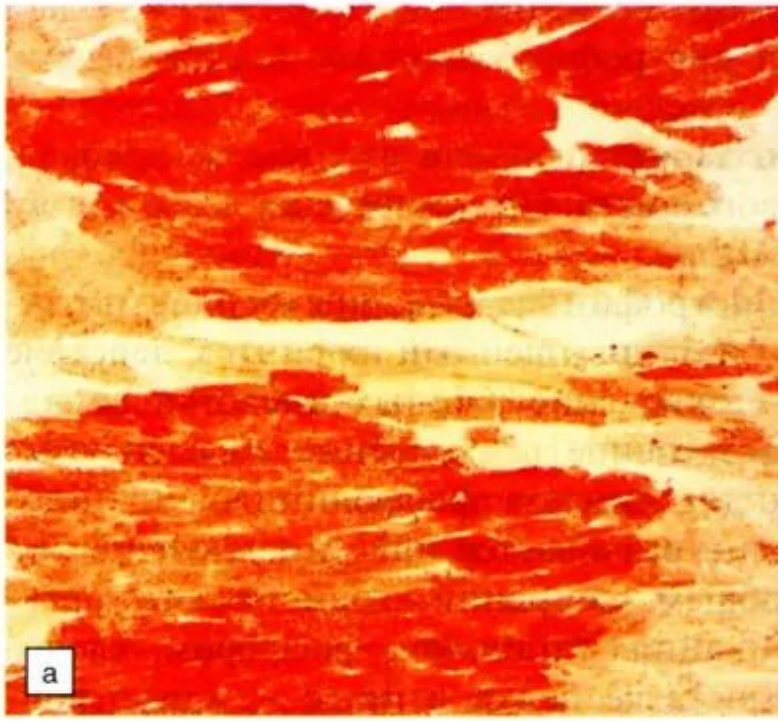


Figure 3.6. *Fatty degeneration of the myocardium (Sudan III): orange-red fat droplets in the cytoplasm of cardiomyocytes; the change is focal (Strukov, Serov, Fig. 2-6a)*

REMEMBER

"**Tiger heart**" is **parenchymal** fatty degeneration of the myocardium (fat inside cardiomyocytes). "**Fatty infiltration of the heart**" (adiposity of the heart) is **stromal-vascular** fatty degeneration (growth of adipose tissue beneath the epicardium and in the myocardial stroma). Confusing the two is one of the most common mistakes at the exam.

Fatty degeneration of the kidney

Neutral fats accumulate in the epithelium of the proximal and distal tubules. Mainly in **nephrotic syndrome** (hyperlipidemia and lipiduria — infiltration of fats), rarely in intoxications. The kidney is enlarged and soft; the cortex is swollen, gray, with yellow spots. It often occurs together with hyaline droplet and hydropic degeneration.

Hereditary lipidoses (storage diseases)

Because of hereditary deficiency of lysosomal enzymes, particular lipids accumulate (Topic 11):

- **Gaucher disease** — glucocerebrosides accumulate (glucocerebrosidase deficiency); **Gaucher cells** with cytoplasm resembling "**crumpled tissue paper**" in the spleen, liver and bone marrow;
- **Niemann–Pick disease** — sphingomyelin (sphingomyelinase deficiency); foamy macrophages;
- **Tay–Sachs disease** — GM2 ganglioside (hexosaminidase A deficiency); neurons are affected, with a "**cherry-red spot**" on the retina.

3.4. Parenchymal carbohydrate degenerations

Carbohydrates in the cell exist mainly as **glycogen** and **glycoproteins** (mucoids, mucins). Glycogen stains magenta with the **PAS reaction** (Schiff reagent) and red with **Best carmine**. Glycogen is water-soluble, so the tissue is fixed in absolute alcohol.

Disturbed glycogen metabolism

Diabetes mellitus is the main example of acquired disturbance of glycogen metabolism:

- in the **liver**, glycogen stores decrease (glucose does not enter the tissues), but glycogen accumulates in the **nuclei** of hepatocytes — the nucleus is clear and vacuolated ("**holed**" or "**empty nuclei**");
- in the **kidney** — because of glucosuria, glycogen accumulates in the epithelium of the loop of Henle and the narrow segment of the distal tubules — the epithelium is tall and its cytoplasm is clear and foamy (**Armanni–Ebstein change**);
- **diabetic microangiopathy** — glycoproteins accumulate in the glomeruli, and **nodular (Kimmelstiel–Wilson) glomerulosclerosis** develops. **Hereditary glycogenoses** — hereditary deficiency of enzymes that break down or synthesize glycogen: **von Gierke disease** (type I, glucose-6-phosphatase deficiency — glycogen in the liver and kidney), **Pompe disease** (type II, lysosomal acid maltase — heart, skeletal muscles, liver; cardiomegaly), **McArdle disease** (type V, muscle phosphorylase — skeletal muscles, pain and cramps on exertion).

Disturbed glycoprotein metabolism — mucous degeneration

In mucus-producing cells, **mucins and mucoids** accumulate in excess or their physicochemical properties change. Mucus gives **metachromasia** with **thionine and cresyl violet** (red-violet color with a blue dye) and is PAS-positive.

- **Catarrhal inflammation** — increased mucus secretion in mucous membranes (rhinitis, bronchitis, gastritis): goblet cells multiply, much mucus is produced, and cells desquamate. If mucus blocks the excretory ducts, **cysts** form.
- **Cystic fibrosis (mucoviscidosis)** — a hereditary disease: because of a mutation in the CFTR gene, chloride channels are defective and exocrine glands secrete **thick, viscous mucus**. The pancreatic ducts, bronchi, intestine and bile ducts become plugged, and cysts, chronic inflammation and sclerosis develop (Topic 11).
- **Colloid degeneration** — accumulation of colloid in colloid-forming glands (thyroid gland) — **colloid goiter**.
- In **tumors** — mucin-producing adenocarcinomas (**"signet ring cell" carcinoma** — mucus inside the cell, nucleus pushed to the periphery).

3.5. Stromal-vascular (mesenchymal) protein degenerations

Stromal-vascular degenerations develop in **connective tissue** and in the walls of vessels. They occur mainly within the **histion** — the microcirculatory bed and the connective tissue around it (ground substance, fibers, cells). Therefore they are often associated with impaired **permeability of the blood–tissue barrier**.

Stromal-vascular protein degenerations include **mucoid swelling, fibrinoid change (swelling), hyalinosis** and **amyloidosis**. The first three are often **successive stages of one process**: mucoid

swelling → fibrinoid swelling → hyalinosis. This sequence is called **disorganization** of connective tissue.

> **REMEMBER**
> Stages of connective tissue disorganization: **1) mucoid swelling** (superficial, **reversible**) → **2) fibrinoid swelling** (deep, **irreversible**) → **3) fibrinoid necrosis** → **4) hyalinosis** or sclerosis. This sequence is most evident in rheumatic diseases (rheumatic fever, systemic lupus erythematosus, rheumatoid arthritis) and in arterial hypertension.

Mucoid swelling

Mucoid swelling is a **superficial and reversible disorganization** of connective tissue. **Glycosaminoglycans** (mainly hyaluronic acid) accumulate and are redistributed in the ground substance. Because glycosaminoglycans are hydrophilic, tissue and vascular permeability increases, plasma proteins (mainly globulins) and glycoproteins enter the ground substance — the ground substance **swells**. The collagen fibers are preserved but swell and split into fibrils.

Microscopically: the ground substance is basophilic; when stained with **toluidine blue**, it turns **violet-red** instead of dark blue — **metachromasia** (a sign of accumulation of chromotropic substances). There are no gross changes.

Causes: infectious-allergic diseases, **rheumatic diseases** (rheumatic fever, especially in the heart valves and endocardium), atherosclerosis, endocrinopathies.

Outcome: if the cause is removed, the tissue is completely **restored**; otherwise it progresses to **fibrinoid swelling**.

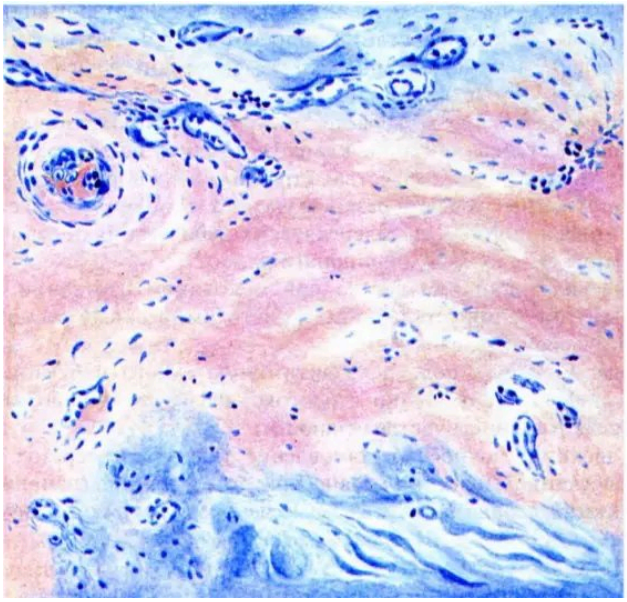


Figure 3.7. *Mucoid swelling: accumulation of glycosaminoglycans in connective tissue, violet-pink with toluidine blue — metachromasia (Strukov, Serov, Fig. 2-8)*



Figure 3.8. *Fibrinoid swelling and fibrinoid necrosis: in the capillaries of a renal glomerulus (arrows) — systemic lupus erythematosus (Strukov, Serov, Fig. 2-9a)*

Fibrinoid change (fibrinoid swelling and fibrinoid necrosis)

Fibrinoid swelling is a **deep and irreversible disorganization** of connective tissue: the ground substance and collagen fibers are destroyed, vascular permeability increases sharply, and plasma proteins, including **fibrinogen**, enter the tissue. As a result **fibrinoid** forms — a complex substance containing proteins (fibrin), polysaccharides, collagen breakdown products and often immune complexes.

Microscopically: the collagen fibers appear as a homogeneous, dense, **eosinophilic** structureless mass; special reactions for fibrin (Schueninov, Weigert) are positive; with the **van Gieson** method it stains **yellow** (not red like collagen); PAS-positive; metachromasia is weak or absent. Around fibrinoid swelling there is a macrophage reaction (for example, **Aschoff–Talalaev granulomas** in rheumatic fever).

The end of fibrinoid swelling is **fibrinoid necrosis** (Topic 2): complete destruction of connective tissue.

Types and causes:

- **Infectious-allergic** — in hyperergic reactions (tuberculosis, Arthus phenomenon);
- **Allergic (immunopathological)** — in **rheumatic diseases** (rheumatic fever, systemic lupus erythematosus, polyarteritis nodosa, scleroderma) and **vasculitis**;
- **Angioneurotic** — fibrinoid necrosis of the walls of arterioles and arteries in **malignant arterial hypertension**;
- **In chronic inflammation** — at the **floor of a chronic ulcer** of the stomach and duodenum, in appendicitis. **Outcome:** fibrinoid necrosis → sclerosis or **hyalinosis**. Organ function is impaired (for example, acute kidney injury in malignant hypertension, valvular defects in rheumatic fever).

> **MEMORY AID**
> Distinguishing by stains: **“Mucoid — Metachromasia, Fibrinoid — Fibrin reaction, Hyaline — Homogeneous”**. In mucoid swelling toluidine blue gives a violet-red color; in fibrinoid the reactions for fibrin are positive and it is yellow with van Gieson; hyaline is a dense, homogeneous, pale pink mass, yellow or red with van Gieson.

Hyalinosis

Hyalinosis (Greek hyalos — transparent, glassy) is the accumulation in connective tissue of a mass **resembling hyaline cartilage, homogeneous, translucent and dense — hyaline**. Hyaline is a **fibrillar protein** containing plasma proteins (fibrin, immunoglobulins, complement), lipids and glycoproteins. It is **eosinophilic**, PAS-positive, stains yellow or red with van Gieson, and is resistant to acids, alkalis and enzymes.

Hyalinosis is often the **end result** of plasmorrhagia, fibrinoid swelling, inflammation, necrosis and sclerosis. It is divided into **vascular hyalinosis** and **connective tissue hyalinosis**.

1. Vascular hyalinosis — mainly in **small arteries and arterioles**. Hyaline accumulates beneath the endothelium; the vessel wall thickens, and the lumen narrows or is completely occluded. Hyaline formation results from leakage of plasma into the vessel wall (**plasmorrhagia**) and accumulation of basement membrane material.

Three types of vascular hyaline:

- **simple hyaline** — from unchanged or slightly changed plasma components; in **benign hypertension** and atherosclerosis;
- **lipohyaline** — rich in lipids and beta-lipoproteins; in **diabetes mellitus**;
- **complex hyaline** — from immune complexes, fibrin and breakdown products of the vessel wall; in **immunopathological diseases** (rheumatic diseases). Vascular hyalinosis is **systemic** (in many organs) and is most pronounced in the vessels of the **kidney, brain, retina and pancreas** in hypertension and diabetes. Physiological hyalinosis occurs in the splenic arterioles (in adults and

the elderly, related to the blood-storing function of the spleen) and in the ovarian corpus luteum after the menstrual cycle.

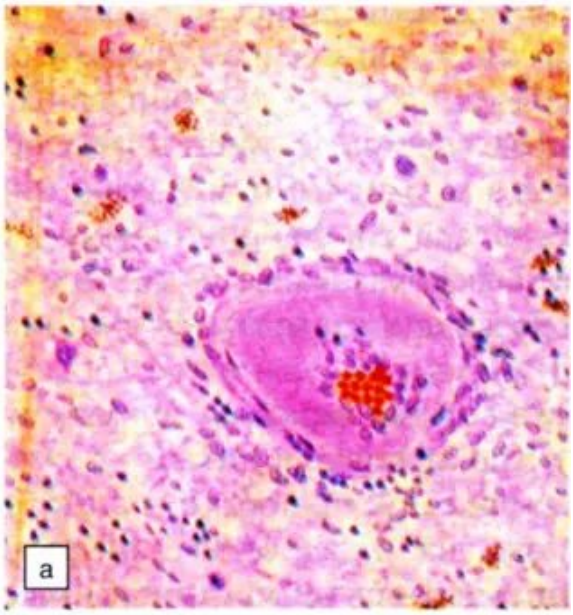


Figure 3.9. *Hyalinosis of the central artery of a splenic follicle: the wall is replaced by homogeneous pink hyaline masses (Strukov, Serov, Fig. 2-10a)*

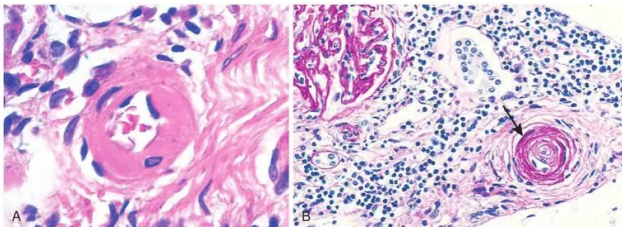


Figure 3.10. *Changes of arterioles in hypertension: A — hyaline arteriolosclerosis: the arteriolar wall is thickened by pink, homogeneous hyaline, and the lumen is markedly narrowed; B — hyperplastic arteriolosclerosis ("onion skin") (PAS reaction) (Robbins, 9th ed., Fig. 11-6)*

CLINICAL SIGNIFICANCE

In arterial hypertension and diabetes mellitus, hyalinosis of the renal arterioles leads to ischemia and sclerosis of the glomeruli (**arteriolosclerotic nephrosclerosis**): the kidney becomes small, its surface finely granular — the "**primary contracted kidney**", and chronic kidney disease develops as a result. Hyalinosis of the cerebral arterioles makes their walls brittle and increases the risk of **hemorrhagic stroke**.

2. Connective tissue hyalinosis — collagen fibers swell, fuse with one another and turn into a dense, homogeneous, cartilage-like mass; the cells are compressed and atrophy. It is usually the end result of fibrinoid change, inflammation and sclerosis. **Examples:**

- **scars** (keloid scar), fibrous adhesions of serous cavities;
- **heart valves** — after rheumatic endocarditis (the valves thicken and become deformed — a heart defect);
- **splenic capsule** — after perisplenitis: the capsule is thickened, dense, milky white and translucent — the "**glazed (sugar-coated) spleen**";
- **pleura** (outcome of pleurisy with hyalinosis), atherosclerotic plaques, organized thrombi and infarcts. **Outcome:** usually **irreversible**; sometimes hyaline may be resorbed in keloid scars. Its functional significance depends on location: arteriolar hyalinosis — atrophy and sclerosis of the organ; hyalinosis of the heart valves — a heart defect.

Amyloidosis (in brief)

Amyloidosis is a stromal-vascular protein degeneration characterized by deposition in tissues of **amyloid** — a fibrillar protein with a **beta-pleated sheet** structure formed from **misfolded proteins**. This degeneration develops by the mechanism of **perverted synthesis**. Amyloid stains red-orange with **Congo red** and shows **apple-green** birefringence in polarized light; it turns blue-violet with iodine and sulfuric acid (Virchow reaction) and gives **metachromasia** (red with methyl violet).

The causes, classification (AL, AA, ATTR and others), pathogenesis and organ changes of amyloidosis (sago and lardaceous spleen, “large white amyloid kidney”) are discussed in detail together with diseases of the immune system in **Topic 8**.

3.6. Stromal-vascular fatty degenerations

This group includes disorders of the metabolism of **neutral fats** and of **cholesterol and its esters**.

Disorders of neutral fat metabolism

An increase in adipose tissue stores can be **generalized** or **local**.

General obesity (adipositas) — excessive accumulation of neutral fats in the subcutaneous tissue, omentum, mesentery, mediastinum and epicardium. The **degree of obesity** is currently assessed by the **body mass index (BMI)**: BMI = weight (kg) / height^2 (m^2).

Table 3.3. *Degrees of obesity by body mass index (WHO)*

BMI (kg/m^2)	Assessment
Less than 18.5	Underweight
18,5–24,9	Normal weight
25,0–29,9	Overweight
30,0–34,9	Class I obesity
35,0–39,9	Class II obesity
40 or more	Class III (morbid) obesity

> **CAUTION**
> In older textbooks (Strukov), the degree of obesity was determined by the **percentage of excess weight over ideal weight**: grade I — 20–29%, II — 30–49%, III — 50–99%, IV — 100% or more. Current clinical practice uses the **WHO BMI classification**; it is useful to know both for the exam.

Causes and types of obesity: alimentary (overeating, physical inactivity), **cerebral** (damage to the hypothalamus — tumor, trauma, infection), **endocrine** (Cushing syndrome, hypothyroidism, hypogonadism, insulinoma), **hereditary** (mutations of the leptin or leptin receptor gene, Laurence–Moon–Biedl syndrome). Obesity is also distinguished by **enlargement** of fat cells (hypertrophic obesity — in adults, a relatively benign course) or **an increase in their number** (hyperplastic obesity — from childhood, difficult to treat).

Fatty infiltration of the heart (adiposity of the heart) — in general obesity the subepicardial adipose tissue increases, envelops the heart like a “sheath” and grows into the **myocardial stroma**, between the muscle fibers. The cardiomyocytes are compressed and undergo **atrophy**. The wall of the **right ventricle** is especially affected — it can be completely replaced by adipose tissue, which may lead to **cardiac rupture**. This is an example of false hypertrophy (Topic 1).



Figure 3.11. *Fatty infiltration of the heart: a thick yellow layer of fat beneath the epicardium (Strukov, Serov, Fig. 2-16b)*

Local accumulation of fat:

- **Dercum disease** (painful lipomatosis) — painful nodular deposits of fat in the subcutaneous tissue of the limbs and trunk;
- **vacant (ex vacuo) fatty replacement** — growth of adipose tissue in place of an atrophic organ (for example, in renal atrophy the perirenal adipose tissue grows and takes its place);
- **lipodystrophy** — loss or increase of adipose tissue at insulin injection sites.

CLINICAL SIGNIFICANCE

In obesity, adipose tissue is not simply a "store" but an active endocrine organ: it produces leptin, adiponectin, resistin and inflammatory cytokines (TNF, IL-6). Therefore obesity (especially the abdominal, visceral type) increases the risk of **insulin resistance, type 2 diabetes mellitus, arterial hypertension, atherosclerosis, nonalcoholic fatty liver disease**, obstructive sleep apnea, osteoarthritis and some cancers (endometrium, breast, colon).

Disorders of the metabolism of cholesterol and its esters

Disorders of the metabolism of cholesterol and its esters are most evident in **atherosclerosis**: cholesterol and lipoproteins accumulate in the intima of arteries (infiltration mechanism), forming **foam cells** (lipid-laden macrophages and smooth muscle cells) and **atheromatous plaques**. Atherosclerosis is studied in detail in systemic pathology.

Other examples:

- **xanthomas** — collections of cholesterol-laden macrophages in the skin and tendons (yellow nodules); in familial hypercholesterolemia;
- **xanthelasma** — yellow patches on the eyelids;
- **cholesterolosis** — cholesterol-laden macrophages in the gallbladder mucosa ("**strawberry gallbladder**").

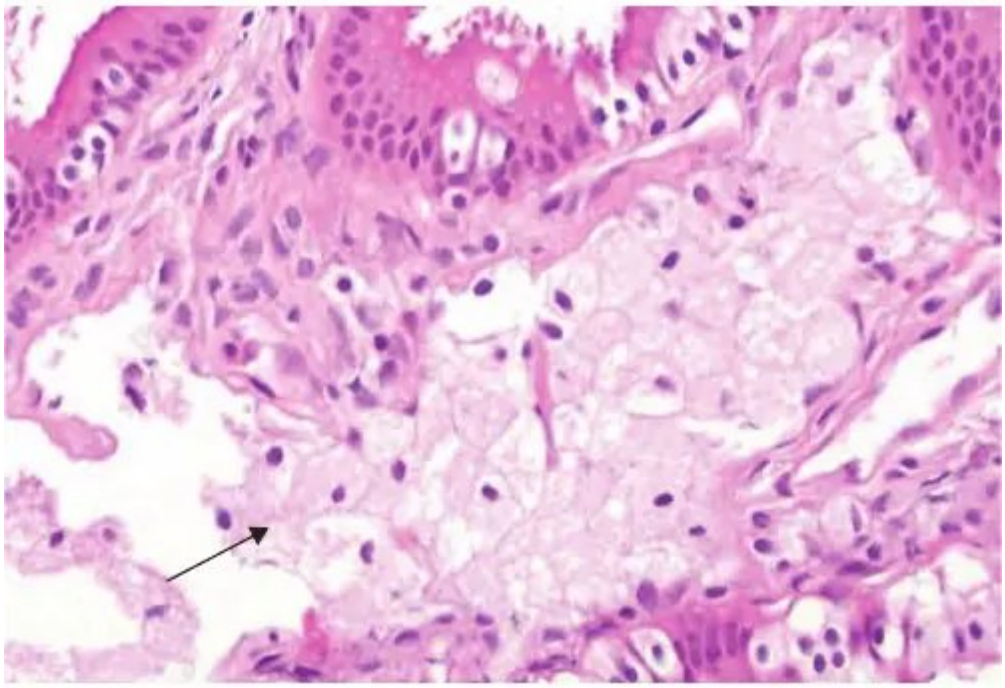


Figure 3.12. *Cholesterolosis of the gallbladder: cholesterol-laden foamy macrophages in the mucosal stroma (pointer) (Robbins, 9th ed., Fig. 2-31)*

3.7. Stromal-vascular carbohydrate degenerations

These degenerations are associated with disturbed metabolism of **glycoproteins and glycosaminoglycans**. A mucus-like substance accumulates in place of the collagen fibers of connective tissue — **mucinous change of tissues**. Connective, adipose, cartilaginous and bone tissues become translucent and mucoid, and their cells become stellate.

Causes: dysfunction of the endocrine glands — **myxedema** (hypofunction of the thyroid gland: mucus accumulates in the skin and in the stroma of the heart and other organs — “mucous edema”), cachexia (mucinous change of adipose tissue), chronic heart failure.

Outcome: the process can be reversible; if prolonged, liquefactive necrosis and cavities (**mucous cysts**) form.

3.8. Clinicopathological section

Biopsy and histochemistry — how do we identify the type of degeneration?

In a hematoxylin and eosin–stained slide, **clear vacuoles** in the cytoplasm of a cell can be one of three things: **fluid** (hydropic degeneration), **fat** (dissolved during paraffin processing) or **glycogen** (dissolved in water). Histochemical reactions are used to distinguish them.

Table 3.4. *Main histochemical stains for identifying degenerations*

Substance	Stain (reaction)	Color
Neutral fats	Sudan III / scarlet red / osmic acid, Sudan IV	Orange-red / red / black
Glycogen	PAS reaction (disappears after diastase treatment); Best carmine	Magenta; red
Mucus (mucin)	Thionine, cresyl violet (metachromasia); PAS; Alcian blue	Red-violet; red; blue

Substance	Stain (reaction)	Color
Glycosaminoglycans (mucoid swelling)	Toluidine blue	Violet-red (metachromasia)
Fibrinoid	Fibrin reactions (Weigert, Schueninov); van Gieson	Blue-violet; yellow
Collagen	Van Gieson	Red
Hyaline	Hematoxylin and eosin; van Gieson; PAS	Pink; yellow-red; red
Amyloid	Congo red (+ polarized light); methyl violet	Red (apple-green birefringence); red (metachromasia)

Clinical examples

- **Nephrotic syndrome.** A 6-year-old child has edema, 4 g of protein in the urine per day and hyperlipidemia. Kidney biopsy shows **hyaline droplet**, **hydropic** and **fatty degeneration** together in the epithelium of the proximal tubules — all of them result from **infiltration** of the tubules by proteins and lipids that have passed through the glomerular filter. When the underlying disease is treated and proteinuria decreases, the tubular changes largely regress.
- **Diphtheritic myocarditis.** Diphtheria exotoxin stops protein synthesis in cardiomyocytes and impairs the oxidation of fatty acids — **“tiger heart”**; myocardial contractility decreases, and there is a risk of sudden cardiac arrest in the 2nd week of the disease.
- **Malignant hypertension.** A patient with a blood pressure of 240/140 mm Hg develops acute kidney injury; the biopsy shows **fibrinoid necrosis** of the afferent arterioles and hyperplastic arteriolosclerosis with an “onion skin” appearance.

CLINICAL SIGNIFICANCE

In **rheumatic fever** all stages of connective tissue disorganization can be observed: in an acute attack — **mucoid swelling** of the heart valves (reversible; with proper treatment the valve recovers), then **fibrinoid change** and Aschoff–Talalaev granulomas, and then **sclerosis and hyalinosis** — the valve cusps thicken and fuse with one another, and a heart defect such as **mitral stenosis** forms. Therefore early diagnosis of rheumatic fever and prevention of its recurrence (benzathine benzylpenicillin (bicillin) prophylaxis) are very important.

Practical skills

Skill 1. Gross specimens

1. **“Fatty degeneration of the liver (‘goose liver’)”.** The liver is enlarged, with rounded edges, a smooth surface and a soft, doughy consistency, yellowish or clay-yellow; on cutting, fat remains on the knife blade. Conclusion: parenchymal fatty degeneration.
2. **“Fatty degeneration of the myocardium (‘tiger heart’)”.** The heart is slightly enlarged, its cavities are dilated, and the myocardium is flabby and clay-yellow; from the endocardial side, yellow-white transverse stripes are seen in the papillary muscles. Conclusion: parenchymal fatty degeneration.
3. **“Fatty infiltration of the heart”.** A thick layer of fat beneath the epicardium; the adipose tissue has grown into the thickness of the myocardium, especially in the right ventricle; the myocardium is thinned. Conclusion: stromal-vascular fatty degeneration (general obesity).
4. **“Hyalinosis of the splenic capsule (‘glazed spleen’)”.** The splenic capsule is unevenly thickened, dense, milky white and translucent, resembling glazed pottery. Conclusion:

connective tissue hyalinosis.

5. **“The aorta in atherosclerosis”.** Yellow spots and streaks (lipoidosis) and whitish-yellow fibrous plaques in the aortic intima. Conclusion: disturbed cholesterol metabolism (stromal-vascular fatty degeneration).
6. **“Colloid goiter”.** The thyroid gland is enlarged; on the cut surface there are follicles of various sizes filled with shiny, brownish-yellow colloid. Conclusion: colloid degeneration.

Skill 2. Microscopic slides

1. **“Hyaline droplet degeneration of the renal tubular epithelium”** (hematoxylin and eosin). Large, homogeneous, eosinophilic protein droplets in the cytoplasm of the proximal tubular epithelium; some cells lack a nucleus (necrosis); protein masses (casts) in the tubular lumen. Conclusion: hyaline droplet degeneration (nephrotic syndrome).
2. **“Hydropic degeneration of the renal tubular epithelium”** (hematoxylin and eosin). The tubular epithelium is enlarged, its cytoplasm is filled with clear vacuoles, and the nuclei are pushed to the edge or vacuolated; the vacuoles do not stain with Sudan III. Conclusion: hydropic (vacuolar) degeneration.
3. **“Fatty degeneration of the liver”** (stained with Sudan III). Orange-red fat droplets of various sizes in the cytoplasm of hepatocytes, especially in the center (or periphery) of the lobule; large droplets push the nucleus to the edge. With hematoxylin and eosin — “empty” vacuoles, “signet ring” cells. Conclusion: parenchymal fatty degeneration.
4. **“Fatty degeneration of the myocardium”** (Sudan III). Small orange fat droplets inside cardiomyocytes, focally distributed; the cross-striation is lost. Conclusion: parenchymal fatty degeneration.
5. **“Mucoid swelling”** (toluidine blue; for example, a heart valve in rheumatic fever). The ground substance is swollen, and the collagen fibers are swollen and split; the tissue stains violet-red (metachromasia). Conclusion: mucoid swelling.
6. **“Hyalinosis of the arterioles of the kidney, spleen or brain”** (hematoxylin and eosin). The arteriolar walls are thickened, with pale pink, homogeneous hyaline masses beneath the endothelium; the lumen is narrowed or occluded; the surrounding tissue is atrophic and sclerotic. Conclusion: vascular hyalinosis (hypertension or diabetes).

Review questions

1. What is degeneration (dystrophy)? State the concept of trophism and the mechanisms that maintain it.
2. Explain the morphogenetic mechanisms of degenerations with examples.
3. How are degenerations classified?
4. In which organs does hyaline droplet degeneration occur? What are its causes and outcomes?
5. What are hydropic (vacuolar) and ballooning degeneration?
6. State the types of horny degeneration and the clinical significance of leukoplakia.
7. Causes, mechanisms, gross and microscopic appearance of fatty degeneration of the liver.
8. How does the “tiger heart” develop? How can it be distinguished from “fatty infiltration of the heart”?
9. What stains are used to detect fats?
10. How do disorders of glycogen metabolism (diabetes mellitus, glycogenoses) manifest?

11. What is mucous degeneration? State the essence of cystic fibrosis.
12. List the stages of connective tissue disorganization.
13. What are the differences between mucoid swelling and fibrinoid swelling?
14. State the types of hyalinosis, the types of vascular hyaline and their causes.
15. Causes and degrees of general obesity and the significance of fatty infiltration of the heart.

Test questions

1. The main mechanism of accumulation of protein droplets in the renal tubular epithelium in nephrotic syndrome:
   A) Metaplasia
   B) Infiltration
   C) Transformation
   D) Perverted synthesis
   E) Decomposition
2. The morphogenetic mechanism of amyloid formation:
   A) Decomposition
   B) Atrophy
   C) Transformation
   D) Infiltration
   E) Perverted synthesis
3. An irreversible parenchymal protein degeneration:
   A) Hyperkeratosis
   B) Hyaline droplet
   C) Granular
   D) Mild hydropic
   E) Mucoid swelling
4. Alcoholic hyaline in hepatocytes is called:
   A) Aschoff bodies
   B) Councilman bodies
   C) Mallory bodies
   D) Lewy bodies
   E) Russell bodies
5. The disease in which ballooning degeneration is characteristic:
   A) Atherosclerosis
   B) Myxedema
   C) Viral hepatitis
   D) Rheumatic fever
   E) Essential hypertension
6. Keratinization of a nonkeratinizing mucous membrane:
   A) Leukoplakia
   B) Wart
   C) Callus
   D) Ichthyosis
   E) Hyperkeratosis

7. The stain that colors fats orange-red:
   A) PAS reaction
   B) Sudan III
   C) Toluidine blue
   D) Congo red
   E) Van Gieson

8. The “tiger heart” is:
   A) Brown atrophy of the heart
   B) Fatty infiltration of the heart
   C) Cardiac amyloidosis
   D) Myocardial hyalinosis
   E) Parenchymal fatty degeneration of the myocardium

9. The “goose liver” appearance is characteristic of:
   A) Hepatic amyloidosis
   B) Nutmeg liver
   C) Liver cirrhosis
   D) Fatty degeneration of the liver
   E) Brown atrophy of the liver

10. Accumulation of glycogen in the epithelium of the loop of Henle in diabetes mellitus:
   A) Armanni–Ebstein change
   B) Councilman bodies
   C) Kimmelstiel–Wilson change
   D) Gaucher cells
   E) Mallory change

11. Cystic fibrosis is an example of which degeneration?
   A) Parenchymal fatty
   B) Amyloidosis
   C) Stromal-vascular fatty
   D) Hyaline droplet
   E) Mucous (glycoprotein)

12. Superficial and reversible disorganization of connective tissue:
   A) Hyalinosis
   B) Amyloidosis
   C) Fibrinoid swelling
   D) Mucoid swelling
   E) Fibrinoid necrosis

13. A violet-red color on staining with toluidine blue (metachromasia) is characteristic of:
   A) Mucoid swelling
   B) Fibrinoid swelling
   C) Hydropic degeneration
   D) Fatty degeneration
   E) Hyalinosis

14. The type of hyaline formed in arterioles in diabetes mellitus:
    A) Simple hyaline
    B) Complex hyaline
    C) Alcoholic hyaline
    D) Lipohyaline
    E) Amyloid

15. The “glazed spleen” is:
    A) Splenic amyloidosis
    B) Splenic atrophy
    C) Hyalinosis of the splenic capsule
    D) Splenic hemosiderosis
    E) Splenic infarction

Clinical cases

Case 1

A 48-year-old man has consumed alcohol regularly for 20 years. On examination the liver extends 5 cm below the costal margin; it is smooth and painless. Ultrasound shows diffusely increased echogenicity of the liver. Liver biopsy: 60% of hepatocytes have large clear vacuoles in the cytoplasm, with nuclei pushed to the edge.

Questions: 1) Your histologic conclusion? 2) How can the content of the vacuoles be confirmed? 3) What is the mechanism of action of alcohol and the outcome of the process?

Answer: 1) Large-droplet fatty degeneration of the liver (alcoholic steatosis). 2) Staining a frozen section with Sudan III — the fat droplets stain orange-red (with hematoxylin and eosin, fat dissolves during paraffin processing and “empty” vacuoles remain). Vacuoles of hydropic degeneration do not stain with Sudan. 3) In alcohol metabolism excess NADH is formed, fatty acid oxidation decreases and triglyceride synthesis increases; delivery of fatty acids from adipose tissue increases; lipoprotein secretion is impaired. If alcohol is stopped, steatosis is reversible (within a few weeks). If drinking continues — alcoholic hepatitis (Mallory bodies, neutrophil infiltration), fibrosis and liver cirrhosis.

Case 2

A 5-year-old child was hospitalized with diphtheria. On the 12th day of the disease he died of sudden cardiac arrest. At autopsy the heart is slightly enlarged, its cavities are dilated, and the myocardium is flabby and clay-yellow; from the endocardial side, yellowish-white transverse stripes are seen in the papillary muscles.

Questions: 1) What is the change in the heart called? 2) Explain its mechanism. 3) What causes the stripes, and with what stain are they confirmed?

Answer: 1) Parenchymal fatty degeneration of the myocardium — “tiger heart” (in the setting of diphtheritic myocarditis). 2) Diphtheria exotoxin stops protein synthesis in cardiomyocytes and damages mitochondria: fatty acid oxidation decreases and the lipoprotein structures of membranes break down (decomposition) — small fat droplets appear in the cytoplasm. Contractility decreases and acute heart failure develops. 3) The fatty degeneration is focal: it develops mainly in cardiomyocytes around the venules, which are least supplied with oxygen, so the yellow stripes alternate with preserved red-brown myocardium. With Sudan III the fat inside cardiomyocytes stains orange.

Case 3

A 52-year-old woman has suffered from arterial hypertension and type 2 diabetes mellitus for 15 years. She died of chronic kidney disease. At autopsy both kidneys are small (80 g each), firm, with a finely granular surface; on the cut surface the cortex is thinned. Microscopically, the walls of the renal arterioles are thickened, with a pale pink homogeneous mass beneath the endothelium and a markedly narrowed lumen; many glomeruli are sclerosed.

Questions: 1) What is the change in the arterioles called? 2) What are its types, and which of them may be present in this patient? 3) State the mechanism and name of the changes in the kidneys.

Answer: 1) Arteriolar hyalinosis (hyaline arteriolosclerosis). 2) Types of vascular hyaline: simple hyaline (hypertension), lipohyaline (diabetes mellitus), complex hyaline (immunopathological diseases). In this patient simple hyaline and lipohyaline may be present. 3) Because of plasmorrhagia and accumulation of basement membrane material, the arterioles narrow — ischemia of the glomeruli, tubular atrophy and sclerosis develop. The kidney becomes small with a finely granular surface — arteriolosclerotic nephrosclerosis (“primary contracted kidney”); in diabetes, diabetic glomerulosclerosis is added. The result is chronic kidney disease.

Quick review

Table 3.5. *Key facts on Topic 3*

Question	Answer
Degeneration (dystrophy)	Changes in cells and intercellular substance resulting from disturbed trophism
Morphogenetic mechanisms	Infiltration, decomposition, perverted synthesis, transformation
Parenchymal protein degenerations	Hyaline droplet, hydropic (vacuolar), horny
Irreversible parenchymal degeneration	Hyaline droplet, ballooning
Mallory bodies	Alcoholic hyaline (cytokeratin) — in hepatocytes
Russell bodies	Immunoglobulins in plasma cells
Leukoplakia	Keratinization of a nonkeratinizing mucous membrane — a precancerous condition
Fat stain	Sudan III (orange-red), Sudan IV and osmium (black)
“Goose liver”	Fatty degeneration of the liver
“Tiger heart”	Parenchymal fatty degeneration of the myocardium (hypoxia, diphtheria)
Fatty infiltration of the heart	Stromal-vascular fatty degeneration (general obesity), risk of right ventricular rupture
Glycogen stain	PAS (magenta), Best carmine
Armanni–Ebstein	Glycogen in the loop of Henle in diabetes
Stages of disorganization	Mucoid swelling → fibrinoid swelling → fibrinoid necrosis → hyalinosis
Mucoid swelling	Reversible, metachromasia (toluidine blue)
Fibrinoid	Irreversible; fibrin reactions positive, yellow with van Gieson
Types of vascular hyaline	Simple (hypertension), lipohyaline (diabetes), complex (immune)
“Glazed spleen”	Hyalinosis of the splenic capsule

Question	Answer
Amyloid stain	Congo red + apple-green birefringence (Topic 8)
Obesity (WHO)	BMI of 30 or more

Glossary of terms

- **Degeneration (dystrophy)** (Greek dys — disorder, trophe — nourishment) — the appearance of structural changes in cells and intercellular substance as a result of disturbed tissue metabolism.
- **Trophism** — the set of mechanisms that ensure tissue metabolism and structural organization.
- **Infiltration** — excessive entry of substances from blood and lymph into the cell or intercellular substance.
- **Decomposition (phanerosis)** — breakdown of complex substances of cells and tissues, so that their components become “visible”.
- **Dysproteinosis** — a degeneration associated with disturbed protein metabolism.
- **Lipidosis** — a degeneration associated with disturbed fat (lipid) metabolism.
- **Thesaurismosis** (Greek thesaurismos — storing) — a storage disease of substances due to hereditary enzyme deficiency.
- **Hyaline** (Greek hyalos — glass) — a dense, homogeneous, translucent protein mass resembling hyaline cartilage.
- **Hyperkeratosis** — excessive formation of horny substance.
- **Leukoplakia** (Greek leukos — white, plax — plate) — keratinization of a nonkeratinizing mucous membrane (“white patch”).
- **Metachromasia** — staining of tissue in a color different from that of the dye (violet-red with toluidine blue).
- **Mucoid swelling** — a superficial, reversible disorganization of connective tissue with accumulation of glycosaminoglycans.
- **Fibrinoid** — a complex fibrin-like substance containing fibrin, plasma proteins and tissue breakdown products.
- **Plasmorrhagia** — leakage of plasma into and beyond the vessel wall.
- **Steatosis** — accumulation of neutral fats (triglycerides) in parenchymal cells.
- **Xanthoma** (Greek xanthos — yellow) — a yellow nodule composed of cholesterol-laden macrophages.
- **Myxedema** — accumulation of a mucus-like substance in tissues in hypofunction of the thyroid gland.

TOPIC 4

Mixed and mineral degenerations: types and pathogenetic mechanisms

Learning objectives

The student should know:

- the concept and classification of mixed degenerations;
- the classification of endogenous pigments: hemoglobinogenic, proteinogenic (tyrosinogenic) and lipidogenic pigments;
- the formation, methods of detection and significance of hemosiderin, ferritin, bilirubin, hematoidin and hematins;
- local and generalized hemosiderosis, hemochromatosis, types of jaundice (hemolytic, hepatocellular, obstructive);
- disorders of melanin metabolism (Addison disease, nevus, melanoma, albinism, vitiligo);
- lipofuscin and lipochrome; exogenous pigments (anthracosis);
- disorders of nucleoprotein metabolism: gout, uric acid stone disease, uric acid infarct;
- mineral degenerations: types of calcification (metastatic, dystrophic, metabolic), disorders of copper and potassium metabolism;
- stone formation: causes, mechanism, types and complications.

The student should be able to:

- describe gross specimens of brown induration of the lung, calcification of the aortic valve, gallstones and kidney stones, and gout;
- describe microscopic slides of hemosiderosis (Perls reaction), hemosiderin and hematoidin in a cerebral hemorrhage, calcification of the myocardium (calcinosis) and melanosis of the skin;
- distinguish hemosiderin, lipofuscin and melanin by histochemical reactions;
- distinguish the types of jaundice by clinical and laboratory signs.

4.1. The concept of mixed degenerations

Mixed degenerations are degenerations in which morphologic changes occur **both in the parenchyma and in the stroma and vessel walls**. They develop as a result of disturbed metabolism of **complex proteins** — chromoproteins and nucleoproteins — and of **minerals**.

Table 4.1. *Classification of mixed degenerations*

Group	Main types
Disorders of chromoprotein (endogenous pigment) metabolism	Hemoglobinogenic pigments (hemosiderin, ferritin, bilirubin, hematoidin, hematins, porphyrins); proteinogenic (tyrosinogenic) pigments (melanin, adrenochrome, pigment of enterochromaffin cells); lipidogenic pigments (lipofuscin, lipochrome, ceroid)
Disorders of nucleoprotein metabolism	Gout, uric acid stone disease, uric acid infarct
Disorders of mineral metabolism	Calcium (calcification, osteoporosis), copper (Wilson–Konovalov disease), potassium, iron; stone formation

Pigments (Latin pigmentum — paint) are substances that have their own color. They are **exogenous** (entering from outside — coal dust, tattoo ink) and **endogenous** (formed in the organism). Endogenous pigments are also present normally and perform particular functions: hemoglobin — oxygen transport, melanin — protection from ultraviolet rays, cytochromes — tissue respiration.

Disorders of pigment metabolism appear in three forms: an **increase** in pigment, a **decrease (loss)** of pigment, and its **appearance** in a site where it is not normally found.

4.2. Hemoglobinogenic pigments

Hemoglobinogenic pigments are formed as a result of **physiological or pathologic breakdown of hemoglobin**. Red blood cells are normally broken down (after living about 120 days) mainly in macrophages of the spleen, liver and bone marrow. When hemoglobin is broken down, **heme** and **globin** separate; **iron** is released from heme and reused, and **bilirubin** is formed from the iron-free porphyrin ring.

Table 4.2. *Hemoglobinogenic pigments*

Pigment	Iron	Conditions of formation	Detection
Ferritin	Present	Iron store (iron + the protein apoferritin); normally	Perls reaction (only when abundant), immunohistochemistry
Hemosiderin	Present	Ferritin polymer; inside cells (sideroblasts, siderophages) under aerobic conditions; 24 hours after hemorrhage	Perls reaction — blue (Prussian blue)
Hematoidin	Absent	Close to bilirubin; outside cells, under anaerobic conditions — in the center of a hemorrhage	Orange crystals, Perls-negative
Bilirubin	Absent	The main end product of hemoglobin breakdown; formed in macrophages and conjugated in the liver	Gmelin reaction (green, blue, violet color)
Hematins	Present (iron oxide, but Perls-negative)	Oxidized form of hemoglobin; dark brown-black crystals	Morphologically, in polarized light
Porphyrins	Absent	When heme synthesis is disturbed (porphyrias)	Red fluorescence in ultraviolet light

Hemosiderin and hemosiderosis

Hemosiderin is an iron-containing **golden-yellow or yellowish-brown granular** pigment. It is formed in **macrophages** (siderophages): the macrophage engulfs red blood cells (erythrophagocytosis), hemoglobin is broken down in lysosomes, and the released iron is converted to ferritin and then to hemosiderin. Cells that form hemosiderin are called **sideroblasts**, and macrophages that have accumulated it — **siderophages**. Hemosiderin stains **blue-green** with the **Perls reaction** (potassium ferrocyanide + hydrochloric acid) — **Prussian blue** is formed.

Local hemosiderosis — accumulation of hemosiderin in a limited area of tissue, in a focus of **hemorrhage**. Examples:

- **a bruise (hematoma)**: its color changes from red-bluish (hemoglobin) → green (biliverdin) → yellow (bilirubin) → golden-brown (hemosiderin);

- a **“rust-colored” cyst** in place of a **cerebral hemorrhage** — hemosiderin-laden macrophages in its wall and hematoidin crystals in the center;
- **brown induration of the lung** — chronic venous congestion of the lung in left ventricular failure, especially in **mitral stenosis**: red blood cells pass into the alveoli by diapedesis, and alveolar macrophages engulf them and accumulate hemosiderin — these cells are called **“heart failure cells”** (heart defect cells); the lung becomes firm (**induration** — sclerosis) and brown (Topic 7).

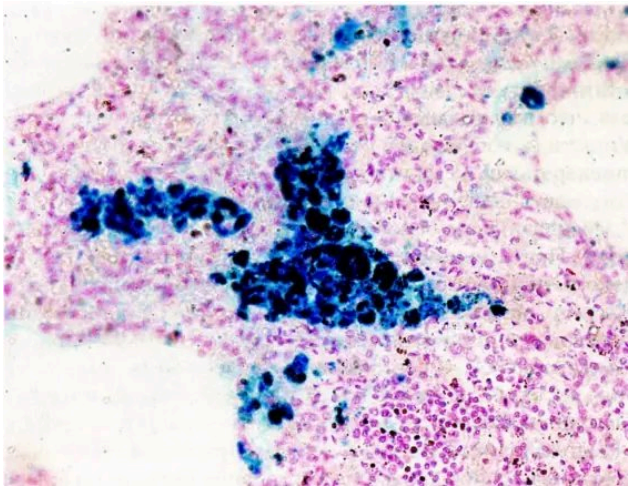


Figure 4.2. *Pulmonary hemosiderosis (Perls reaction): blue pigment granules in the cytoplasm of histiocytes and alveolar epithelium (sideroblasts and siderophages) (Strukov, Serov, Fig. 2-19)*



Figure 4.1. *Brown induration of the lung: the lung is enlarged, brown and firm — chronic venous congestion (TMA study guide, 2018)*

Generalized hemosiderosis — accumulation of hemosiderin in many organs as a result of **excessive intravascular breakdown of red blood cells** (**hemolysis**). Hemosiderin accumulates in macrophages of the spleen, liver, bone marrow and lymph nodes (the **reticuloendothelial system**), and also in the epithelium of the liver, kidney and lung. The organs become **rust-brown**. Causes:

- **hemolytic anemias** (hereditary: sickle cell anemia, thalassemia, spherocytosis; acquired: autoimmune);
- **infections** — malaria, sepsis, relapsing fever;
- poisoning with **hemolytic poisons** (Berthollet salt (potassium chlorate), phosphorus, mushrooms);
- incompatible blood **transfusion**, multiple transfusions;
- hemolytic disease of the newborn (Rh incompatibility).

Hemochromatosis

Hemochromatosis is **excessive accumulation of iron** in the organism, with deposition of this iron in **parenchymal cells** (hepatocytes, cardiomyocytes, pancreatic cells), which it **damages** (through free radicals — the Fenton reaction, Topic 2). Whereas the body normally contains 2–6 g of iron, in hemochromatosis it can exceed 50 g.

- **Primary (hereditary) hemochromatosis** is an autosomal recessive disease, most often due to mutation of the **HFE** gene (C282Y). As a result, **hepcidin** (the liver hormone that regulates iron absorption) decreases, and intestinal iron absorption increases continuously. It occurs more often and earlier in men (women lose iron through menstruation and pregnancy). • **Secondary hemochromatosis** — multiple transfusions (thalassemia), ineffective erythropoiesis, excessive iron intake, chronic liver diseases. **The clinicomorphologic triad:** 1) **pigment cirrhosis of the**

liver (the liver is brown, with micronodular cirrhosis); 2) **diabetes mellitus** (damage to the pancreatic islets); 3) **skin pigmentation** (bronze color — hemosiderin and melanin). Therefore it is also called "**bronze diabetes**". In addition, cardiomyopathy (arrhythmia, heart failure), hypogonadism and arthropathy are observed; the risk of hepatocellular carcinoma rises sharply.

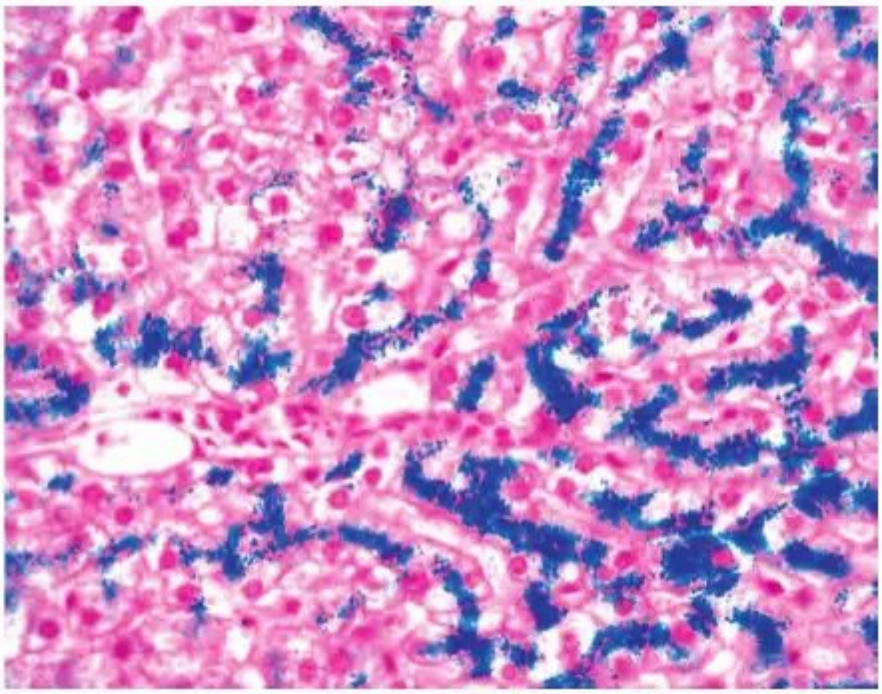


Figure 4.3. *Hereditary hemochromatosis (Perls reaction — Prussian blue): iron in hepatocytes in the form of blue granules; the parenchymal architecture is still normal (Robbins, 9th ed., Fig. 18-25)*

REMEMBER

Hemosiderosis — iron accumulates mainly in **macrophages** and usually **does not damage** the tissue. **Hemochromatosis** — iron accumulates in **parenchymal cells** and **damages** them: cirrhosis, diabetes, cardiomyopathy. The classic triad: **cirrhosis + diabetes + bronze skin**.

CLINICAL SIGNIFICANCE

If detected early, hemochromatosis is a disease that can be completely prevented by regular **bloodletting (phlebotomy)** — that is why serum ferritin and transferrin saturation are checked in patients' family members. In secondary hemochromatosis (thalassemia), iron-chelating drugs (**deferoxamine**) are used.

Bilirubin and jaundice

Bilirubin is the main end product of hemoglobin breakdown, a red-yellow pigment that **contains no iron**. The stages of its metabolism:

1. In macrophages, **biliverdin** and then **bilirubin** are formed from heme.
2. Water-insoluble **unconjugated (indirect)** bilirubin binds to **albumin** in the blood and is carried to the liver.
3. In hepatocytes it is **conjugated with glucuronic acid** by the enzyme **glucuronyltransferase** — water-soluble **conjugated (direct)** bilirubin is formed.
4. Conjugated bilirubin is excreted with bile into the intestine, where it is converted to **stercobilinogen** (stool color); part of it is absorbed and excreted in the urine as **urobilinogen**.

Jaundice (icterus) is the yellow coloring of the skin, sclera, mucous membranes, serous membranes and internal organs resulting from increased bilirubin in the blood (**hyperbilirubinemia**). Clinical jaundice becomes visible when blood bilirubin exceeds **34 µmol/L (2 mg/dL)** (normally up to 20 µmol/L).

Table 4.3. *Types of jaundice*

Type	Mechanism	Causes	Blood bilirubin	Urine and stool
Prehepatic (hemolytic)	Excessive breakdown of red blood cells — the liver cannot conjugate bilirubin in time	Hemolytic anemias, malaria, sepsis, incompatible blood transfusion, hemolytic disease of the newborn	Unconjugated increased	No bilirubin in the urine (bound to albumin), much urobilinogen; dark stool
Hepatic (hepatocellular)	Damage to hepatocytes — uptake, conjugation and excretion of bilirubin are impaired	Viral hepatitis, toxic hepatitis, cirrhosis; hereditary enzymopathies (Gilbert, Crigler–Najjar, Dubin–Johnson syndromes)	Both fractions increased	Dark urine (bilirubin present), paler stool
Posthepatic (obstructive, mechanical)	Obstruction of the bile ducts — bile does not pass into the intestine and returns to the blood	Gallstone disease, cancer of the head of the pancreas, tumors of the bile ducts, cicatricial strictures	Conjugated markedly increased	Dark urine (“beer-colored”), colorless (acholic) stool; itching of the skin (bile acids)

MEMORY AID

Determining the type of jaundice by laboratory tests: **“Hemolysis — indirect, Obstruction — direct, Liver — both”**. Colorless (pale) stools, beer-colored urine and itching skin mean that the bile duct is **obstructed** (obstructive jaundice).

CAUTION

Physiological jaundice in newborns (beginning on days 2–3 and resolving in 7–10 days) is due to the immaturity of the liver enzyme glucuronyltransferase. However, if unconjugated bilirubin is very high (hemolytic disease of the newborn), it crosses the blood–brain barrier, stains the subcortical nuclei of the brain yellow and damages them — **kernicterus (nuclear jaundice)**: severe neurological complications (Topic 11).

Hematoidin is an iron-free pigment chemically close to bilirubin, forming **bright orange crystals** (rhomboid plates or needles). It forms **outside cells**, in places that oxygen does not reach — in the center of old hematomas and infarcts. Hematoidin and hemosiderin often occur together: hematoidin in the center of the focus, hemosiderin at the periphery.

Hematins are an oxidized form of hemoglobin (heme), **dark brown or black** fine granules or crystals:

- **hemomelanin (malarial pigment)** — formed by the malaria plasmodium breaking down hemoglobin in the red blood cell; it accumulates in macrophages of the spleen, liver, bone marrow and brain and gives the organs a **gray-black (slate)** color;
- **hydrochloric acid hematin (hematin hydrochloride)** — formed by the action of hydrochloric acid on hemoglobin in hemorrhages and erosions of the gastric mucosa; it makes the floor of the

erosion black, and the vomitus looks like **“coffee grounds”**;

- **formalin pigment** — an artifact (dark brown needles) formed when blood-rich tissues are fixed in acidic formalin; it must be distinguished from pathologic pigments. **Porphyrins** are iron-free precursors of heme. In hereditary disorders of heme synthesis (**porphyrias**), porphyrins accumulate: the skin becomes extremely sensitive to light (blisters, ulcers, scars), the urine turns **red**, and the teeth and bones become reddish-brown; in acute intermittent porphyria, attacks of abdominal pain and neurological disturbances occur.

4.3. Proteinogenic (tyrosinogenic) pigments

These pigments include **melanin, adrenochrome** and the **pigment of enterochromaffin cells**.

Melanin

Melanin (Greek melas — black) is a **brown-black** pigment. It is synthesized in **melanosomes** of **melanocytes** (melaninocytes) of neuroectodermal origin from the amino acid **tyrosine** via dihydroxyphenylalanine (**DOPA**), with the participation of the enzyme **tyrosinase**. Melanocytes are located in the basal layer of the epidermis, hair follicles, the iris and retina, the pia mater and the substantia nigra of the brain. Melanin synthesis is increased by pituitary melanocyte-stimulating hormone (MSH) and adrenocorticotropic hormone (ACTH), and also by ultraviolet rays. Melanin is also often found in **melanophages** (macrophages that have engulfed melanin).

Melanin is detected with histochemical reactions: the **DOPA reaction** (melanocyte activity) and the **Fontana–Masson** method (black color with silver nitrate); it is **Perls-negative** and is **bleached** by oxidizing agents (hydrogen peroxide) — this distinguishes it from hemosiderin and lipofuscin.

Disorders of melanin metabolism:

Table 4.4. *Disorders of melanin metabolism*

Form	Type	Examples and mechanism
Hyperpigmentation — generalized (widespread), acquired	Addison disease	Bilateral damage to the adrenal cortex (tuberculosis, autoimmune adrenalitis, tumor metastases, amyloidosis): cortisol decreases → by feedback, ACTH and MSH increase → bronze pigmentation of the skin; hypotension, adynamia
	Other	Endocrinopathies (hypogonadism, hyperthyroidism), vitamin deficiencies (pellagra), cachexia
Hyperpigmentation — generalized, congenital	Xeroderma pigmentosum	Hereditary defect of DNA repair; extreme sensitivity of the skin to ultraviolet rays, patchy pigmentation; skin cancers and melanoma are frequent
Hyperpigmentation — local	Melanosis, nevus, melanoma	Melanosis of the colon (chronic constipation, laxatives); chloasma (melasma) from pregnancy and hormonal drugs; freckles (ephelides); pigmented nevus (birthmark — a benign tumor-like collection of melanocytes); melanoma — a highly malignant tumor arising from melanocytes (Topic 9)
Hypopigmentation — generalized, congenital	Albinism	Tyrosinase deficiency: melanocytes are present, but melanin is not synthesized; white skin and hair, reddish iris, nystagmus; risk of skin cancer
Hypopigmentation — local	Leukoderma, vitiligo	Vitiligo — sharply demarcated milky white patches on the skin resulting from autoimmune destruction of melanocytes; leukoderma — white patches after syphilis, leprosy, diabetes mellitus, burns and injuries

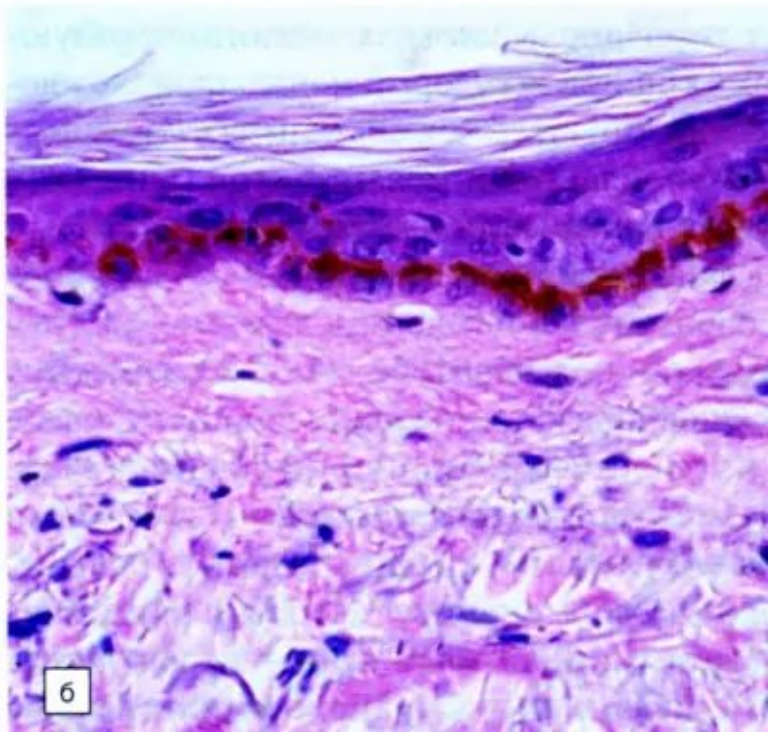


Figure 4.4. *Skin in Addison disease: accumulation of melanin in the basal layer of the epidermis, numerous melanophages in the dermis (Strukov, Serov, Fig. 2-20b)*

CLINICAL SIGNIFICANCE

Signs of transformation of a pigmented nevus into melanoma — the "**ABCDE**" rule: **A** (asymmetry) — asymmetry; **B** (border) — irregular border; **C** (color) — uneven color; **D** (diameter) — diameter greater than 6 mm; **E** (evolving) — change in size, shape or color. A pigmented spot with any of these signs should be removed and examined histologically.

Adrenochrome is an oxidation product of epinephrine (adrenaline) in the chromaffin cells of the adrenal medulla; it gives the **chromaffin reaction** (brown color with chromium salts). **Pheochromocytoma** (a catecholamine-producing tumor with attacks of arterial hypertension) develops from these cells.

Pigment of enterochromaffin cells — in the cells of the diffuse endocrine system of the digestive tract mucosa (associated with serotonin); **neuroendocrine tumors (carcinoid)** arise from these cells.

4.4. Lipidogenic pigments (lipopigments)

Lipidogenic pigments are fat-protein (lipid-protein) substances; they include **lipofuscin, lipochrome, ceroid** and the **vitamin D pigment**.

Lipofuscin (Latin fuscus — brown) is the **"aging pigment"**, the **"wear-and-tear pigment"**. It is formed during **autophagy**: undigested membrane lipids undergo peroxidation under the action of free radicals and, together with proteins, form insoluble residual bodies (**telolysosomes**). Lipofuscin is a **yellowish-brown, finely granular** pigment, often located **around the nucleus**

(perinuclear). It stains orange with Sudan III and blue-green with the Schmorl reaction; Perls-negative.

Lipofuscin **does not itself damage the cell**, but it is a sign of **free radical injury** and cellular **aging**. It accumulates in large amounts in the **heart, liver, neurons** and skeletal muscles in the elderly, in cachexia and in chronic diseases — the organ develops **brown atrophy** (Topic 1).

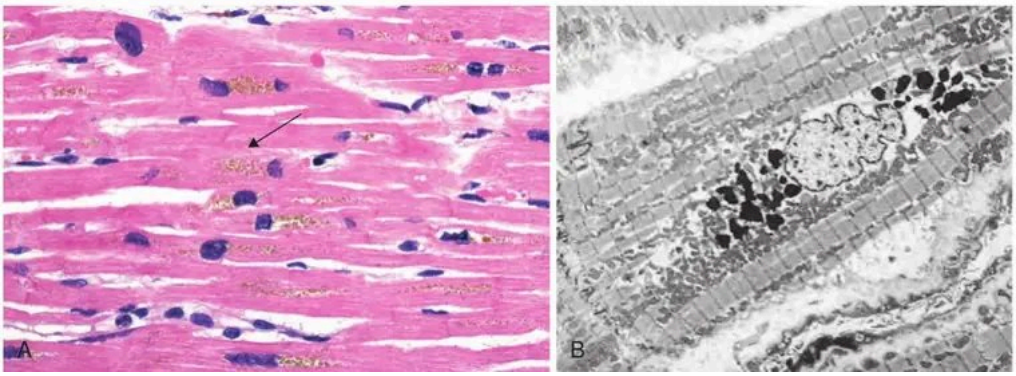


Figure 4.5. *Lipofuscin granules in a cardiomyocyte: A — light microscopy (arrow — yellowish-brown granules around the nucleus); B — electron microscopy (perinuclear, intralysosomal location) (Robbins, 9th ed., Fig. 2-33)*

Lipochrome — carotenoids (pigments taken in with food) that give a yellow color to adipose tissue, the adrenal cortex, blood serum and the corpus luteum of the ovary. In cachexia, when adipose tissue decreases, the concentration of lipochrome increases and the remaining fat becomes **orange**; the yellowish tint of the skin in diabetes mellitus (xanthosis) is also related to this pigment.

Ceroid — a pigment close to lipofuscin formed as a result of oxidation of fats in macrophages (for example, in vitamin E deficiency, in the intestinal mucosa, in endometriosis).

CAUTION

Distinguishing the three brown pigments is the task most often given in questions on microscopic slides:

- **Hemosiderin** — golden-brown, coarsely granular; **Perls-positive (blue)**; in hemorrhage and hemolysis.
- **Lipofuscin** — yellowish-brown, finely granular, **around the nucleus**; Perls-negative, Sudan-positive; in aging and atrophy.
- **Melanin** — brown-black; Perls-negative, Fontana–Masson-positive, bleached by hydrogen peroxide; in the skin, eye and melanocytic tumors.

4.5. Exogenous pigments

- **Anthracosis** (Greek anthrax — coal) — **coal dust** that enters through the respiratory tract, is engulfed by alveolar macrophages and accumulates in the lung tissue and hilar lymph nodes. In city dwellers and smokers the lungs have black spots beneath the pleura, and the lymph nodes are black. In coal miners large amounts of dust accumulate and fibrosis develops — **anthracotic pneumoconiosis**; if silicon is added to the dust — **anthracosilicosis**.
- **Tattoo** — ink introduced into the dermis of the skin is engulfed by dermal macrophages and persists for life.
- **Argyria** — gray-bluish discoloration of the skin with prolonged intake of silver compounds.

4.6. Disorders of nucleoprotein metabolism

Nucleoproteins consist of protein and nucleic acids. When their metabolism is disturbed, the end product of **purine bases** — **uric acid** — is produced in excess or excreted insufficiently (**hyperuricemia**). Uric acid and its salts (**urates**, mainly sodium urate) are deposited in tissues in the form of crystals.

Gout

Gout (Greek podagra: pous — foot, agra — trap, “foot trap”) is a disease characterized by deposition of **sodium urate crystals** in the joints and other tissues as a result of hyperuricemia.

- **Primary gout** — a hereditary disorder of purine metabolism (most often reduced renal excretion of uric acid; rarely enzyme defects, for example HGPRT deficiency in Lesch–Nyhan syndrome). Mainly in **men over 30 years of age**.
- **Secondary gout** — massive cell breakdown (leukemias, lymphomas, especially during **chemotherapy** of tumors — “tumor lysis syndrome”), chronic kidney disease, some drugs (thiazide diuretics). Attacks are triggered by **purine-rich food** (meat, offal), **alcohol** (especially beer), chilling, trauma and starvation.

Morphology:

1. **Acute gouty arthritis** — most often the **metatarsophalangeal joint of the big toe** (the first attack occurs here in **90%** of cases). In the synovial fluid, **needle-shaped urate crystals** are seen inside neutrophils; in polarized light they are **negatively birefringent** (yellow). Clinically — sudden, extremely severe pain beginning at night, with redness and swelling of the joint.
2. **Chronic gouty arthritis** — urate deposits in the articular cartilage, synovium, tendons and periarticular tissues; joint deformity, bone erosions (a “punched-out” sign on X-ray).
3. **Gouty nodules (tophi)** — the pathognomonic sign of gout: an **inflammatory reaction** around a collection of urate crystals — macrophages, lymphocytes, **foreign body giant cells** and fibroblasts (a granuloma). Tophi occur in the auricle, around joints, at the elbow and knee and in tendons; if the skin thins and ruptures, a chalky white mass is discharged.
4. **Gouty nephropathy** — urate crystals in the renal tubules and interstitium, uric acid stones, chronic interstitial nephritis; **kidney failure** is an important cause of death in patients with gout.



Figure 4.6. *Gout: white tophi in the joint of the big toe (A — amputated toe); B — gouty tophus: reactive fibroblasts, mononuclear cells and giant cells around a collection of dissolved urate crystals (Robbins, 9th ed., Fig. 26-47)*

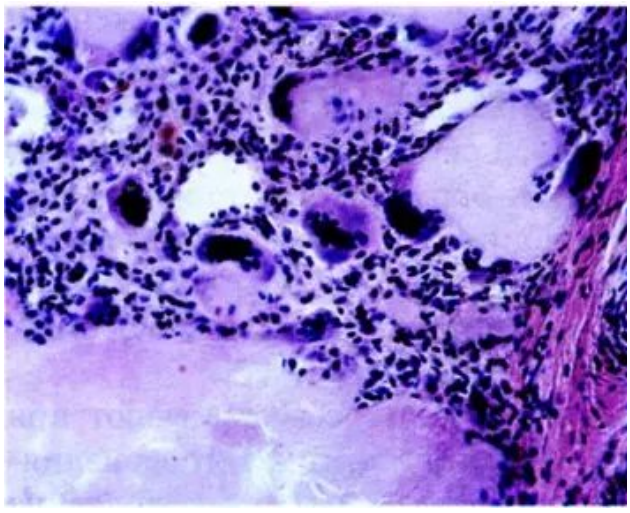


Figure 4.7. *Gout: giant cell inflammatory reaction around a deposit of uric acid salts (Strukov, Serov, Fig. 2-22)*

Uric acid stone disease and uric acid infarct

Uric acid stone disease — formation of **urate stones** in the kidneys and urinary tract; hyperuricemia and acidic urine (pH below 5.5) predispose to it. Urate stones are **radiolucent** (X-ray negative).

Uric acid infarct — deposits of **sodium and ammonium urate** in the kidneys (especially in the medullary papillae and collecting ducts) of newborns who **lived longer than the first 2 days** of life. On the cut surface, **yellowish-red streaks** directed toward the papillae are seen in the medulla. This is an **adaptive (physiological) state** related to the intense cell turnover of the newborn period and has no pathologic significance. The word “infarct” is used here in the sense of **stuffing** (Latin infarcire) — there is no necrosis.

> **CLINICAL SIGNIFICANCE**
> In forensic medical practice, finding a uric acid infarct in the kidney of a newborn shows that the infant **was born alive and lived for at least 2 days**.

4.7. Mineral degenerations

Minerals are part of cell structures, enzymes, hormones and pigments and maintain acid–base balance. To detect minerals in tissues, microincineration (spectrography), histochemical reactions and X-ray microspectral analysis are used.

Disorders of calcium metabolism — calcification (calcinosis)

Calcification (calcinosis, petrification) is the **precipitation** of calcium salts (mainly **calcium phosphate** — hydroxyapatite, less often calcium carbonate) from a dissolved state and their accumulation in tissues. With hematoxylin and eosin, calcium is **basophilic (blue-violet)**, **amorphous or granular**; it stains **black** with the **von Kossa** method (silver nitrate) and red with alizarin red. Grossly — whitish, stone-hard foci; the knife grates when cutting.

By mechanism, three types of calcification are distinguished:

1. Dystrophic calcification — deposition of calcium salts in **injured, necrotic or altered tissues** when **blood calcium is normal**. The cause is **local** physicochemical changes: alkalinization of the tissue, increased alkaline phosphatase activity, changes in collagen and elastic fibers, and phospholipids in the membranes of dead cells (a nucleation site for calcium crystallization). Calcium salts first accumulate in mitochondria and membrane vesicles. As a result **petrifications** form. Examples:

- foci of caseous necrosis (in tuberculosis — the **Ghon focus**, calcified lymph nodes);
- **atherosclerotic plaques** (atherocalcinosis);
- **heart valves** — rheumatic defects, **calcific aortic stenosis** in the elderly;
- infarcts, old thrombi (**phleboliths** — vein stones), dead parasites (echinococcus, trichinellae);
- a dead fetus — **lithopedion** (“stone child”);
- in some tumors — **psammoma bodies** (concentrically laminated calcium granules: papillary thyroid carcinoma, serous carcinoma of the ovary, meningioma).

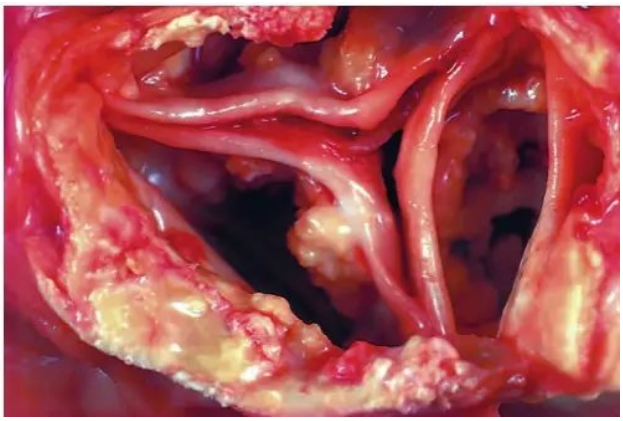


Figure 4.8. *Dystrophic calcification of the aortic valve: the valve cusps are thickened and fibrotic, with irregularly accumulated calcium masses behind them (calcific aortic stenosis) (Robbins, 9th ed., Fig. 2-34)*

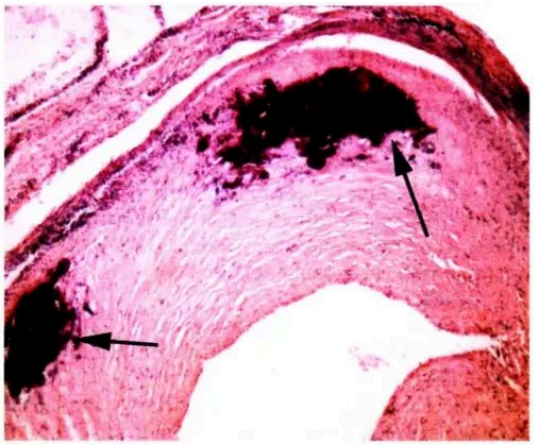


Figure 4.9. *Dystrophic calcification of an arterial wall: calcium deposits in the thickness of an atherosclerotic plaque (arrows) (Strukov, Serov, Fig. 2-23)*

2. Metastatic calcification (calcium metastases) — widespread deposition of calcium salts in **normal (healthy) tissues** as a result of **hypercalcemia**. Causes of hypercalcemia:

- **excess parathyroid hormone** — parathyroid adenoma (**hyperparathyroidism**), production of a parathyroid hormone–related protein by tumors;
- **bone destruction** — bone metastases, multiple myeloma, leukemias, prolonged immobilization, Paget disease;
- **vitamin D intoxication**, sarcoidosis (macrophages activate vitamin D);
- **chronic kidney disease** — phosphate retention develops, with secondary hyperparathyroidism. Calcium salts are deposited most often in organs that **excrete acid** and whose tissue is therefore **more alkaline**: the **lungs** (alveolar walls), the **gastric** mucosa (fundic glands), the **kidneys** (tubules — **nephrocalcinosis**), and also the **myocardium**, arterial walls and pulmonary veins. Calcium first accumulates inside cells — in mitochondria and phagolysosomes — and then in the intercellular substance.

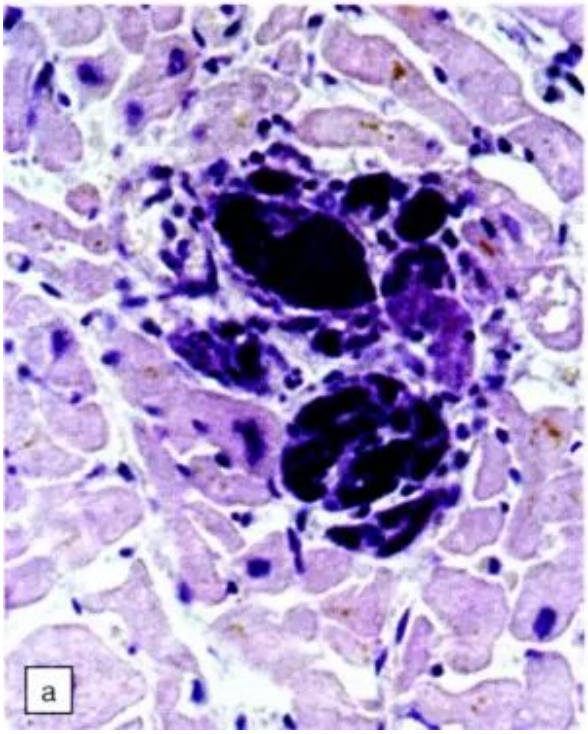


Figure 4.10. *Calcium metastases in the myocardium: A — calcified muscle fibers (black), microscopic view; B — calcium salts in mitochondrial cristae (electron micrograph) (Strukov, Serov, Fig. 2-23a)*

3. Metabolic (interstitial) calcification — deposition of calcium when blood calcium is normal and there is no evident local tissue damage. The cause is the inability of tissue buffer systems to hold calcium (disturbed calcium–phosphorus metabolism). There are two types: **systemic** (generalized) — in the skin, subcutaneous tissue, tendons, fasciae and muscles; **limited** — calcium plates in the skin and subcutaneous tissue, most often of the fingers and toes ("**calcium gout**"). It is caused by disorders of calcium–phosphorus metabolism and diseases such as scleroderma.

> **REMEMBER**
>
> The main criterion for distinguishing the types of calcification is **blood calcium** and the **state of the tissue**: **dystrophic** — normal calcium, damaged tissue (local); **metastatic** — hypercalcemia, healthy tissue (widespread; lung, stomach, kidney); **metabolic** — normal calcium, and the tissue is not damaged either.

Changes in the calcium content of bone: osteoporosis — a decrease in bone mass, the bones become brittle (old age, menopause, glucocorticoids, immobilization); **osteomalacia** and **rickets** — insufficient mineralization of the bone matrix in vitamin D deficiency.

Disorders of copper metabolism — Wilson–Konovalov disease

Hepatocerebral degeneration (Wilson–Konovalov disease) is an autosomal recessive disease: because of mutation of the **ATP7B** gene, excretion of copper into the bile and its incorporation into **ceruloplasmin** are impaired. Copper accumulates in the liver, then in the brain (especially the **subcortical nuclei** — the lentiform nucleus), the cornea and the kidneys.

- **Liver**: fatty degeneration, chronic hepatitis, **liver cirrhosis** (sometimes acute liver failure);
- **Brain**: death of neurons in the subcortical nuclei, cysts — tremor, movement disorders, mental changes;

- **Eye**: a green-brown ring at the edge of the cornea — the **Kayser–Fleischer ring**. Serum ceruloplasmin is decreased, and urinary copper is increased. Copper in tissue is detected with **rhodanine**. Treatment — copper-chelating drugs (penicillamine) and liver transplantation.

Disorders of potassium metabolism

- **Hyperkalemia** — kidney failure, Addison disease, extensive tissue necrosis (rhabdomyolysis), tumor lysis; risk of **cardiac arrest** (asystole).
- **Hypokalemia** — diarrhea, vomiting, diuretics, hyperaldosteronism; muscle weakness, arrhythmias, **hypokalemic (vacuolar) nephropathy** in the kidney; **paroxysmal myoplegia** — episodic muscle paralysis in a hereditary disorder of potassium metabolism.

4.8. Stone formation

Stones (concretions, calculi) are **dense (hard) structures** formed from substances contained in fluids in the cavities of hollow organs or in the excretory ducts of glands. Stones form most often in the **gallbladder and bile ducts** (**cholelithiasis** — gallstone disease) and the **urinary tract** (**urolithiasis** — urinary stone disease), and less often in the salivary glands, pancreatic ducts, bronchi (**broncholiths**), intestine (**coproliths**) and vessels (**phleboliths** — calcified thrombi).

Causes and mechanism of stone formation:

- **general factors** — metabolic disorders (of cholesterol, calcium, uric acid, oxalate, cystine), hereditary predisposition, dietary features, obesity;
- **local factors** — **stasis** of secretions (impaired outflow from a cavity), **inflammation** (desquamated mucosal cells, bacteria and proteins become the **organic matrix** of the stone and a crystallization center), and changes in the composition and pH of secretions. Stone formation consists of two processes: formation of an **organic matrix (nucleus)** and **crystallization of salts** (precipitation) on it.

Table 4.5. *Main types of gallstones and kidney stones*

Stone type	Composition and causes	Features
Cholesterol gallstones	Bile supersaturated with cholesterol; female sex, obesity, pregnancy, age over 40, estrogens	Whitish-yellow, multifaceted (faceted), radial crystalline structure on the cut surface; radiolucent
Pigment gallstones	Calcium salts of unconjugated bilirubin; black — chronic hemolysis (sickle cell anemia, spherocytosis), cirrhosis; brown — infection of the bile ducts	Small, black or brown, numerous
Calcium oxalate and phosphate kidney stones	The most common (about 80%); hypercalciuria, hyperoxaluria	Hard, irregular; radiopaque
Struvite (magnesium ammonium phosphate)	Urinary tract infection with urease-producing bacteria (Proteus), alkaline urine	Large, reproducing the shape of the pelvis and calyces — “staghorn calculi”
Uric acid (urate)	Hyperuricemia, gout, leukemias, acidic urine	Yellow-brown, smooth; radiolucent
Cystine stones	Hereditary cystinuria (impaired reabsorption of amino acids)	Rare, from childhood



Figure 4.11. *Cholesterol gallstones: the gallbladder wall is thickened and fibrotic due to chronic cholecystitis, and its cavity is filled with multifaceted yellowish stones (Robbins, 9th ed., Fig. 18-62)*



Figure 4.12. *Pigment gallstones: several faceted black stones in the gallbladder (chronic hemolysis) (Robbins, 9th ed., Fig. 18-63)*



Figure 4.13. *Nephrolithiasis: a large stone impacted in the renal pelvis (Robbins, 9th ed., Fig. 20-49)*

Complications of stones:

- **obstruction** of a hollow organ or duct: in the bile ducts — **obstructive jaundice**, in the kidney — **hydronephrosis** (pressure atrophy), colicky pain (**biliary and renal colic**);
- **pressure ulcer** of the wall (necrosis from pressure) and **perforation** — peritonitis, fistulas (for example, cholecystoenteric fistula, “gallstone ileus”);
- secondary **inflammation**: cholecystitis, cholangitis, pyelonephritis; if the pancreatic duct is obstructed — **acute pancreatitis**;
- risk of **gallbladder cancer** against the background of long-standing chronic inflammation.

4.9. Clinicopathological section

One hemorrhage — three different pigments

A patient with a cerebral hemorrhage died of pneumonia 3 months later. At autopsy, in place of the hemorrhage a **cyst** with **rust-brown** walls filled with yellow-brown fluid was found. Microscopically: macrophages laden with **hemosiderin** (blue with the Perls reaction) in the cyst wall; in the central part — **orange hematoidin crystals** lying outside cells; in the glia around the cyst — **foamy macrophages** that have engulfed lipids ("granular bodies"). This picture reflects the age of the hemorrhage and the stages of its resorption.

Jaundice — where is the "breakdown"?

- **A young woman** has anemia, an enlarged spleen, increased **unconjugated** bilirubin in the blood and no bilirubin in the urine — **hemolytic jaundice** (for example, hereditary spherocytosis). Prolonged hemolysis can lead to **pigment gallstones**.
- **A student** develops jaundice after fever and weakness, ALT is 1500 U/L, and both fractions are increased in the blood — **hepatocellular jaundice** (viral hepatitis).
- **A 65-year-old man** has gradually increasing painless jaundice, itching of the skin, colorless stools and an enlarged gallbladder (Courvoisier sign) — **obstructive jaundice**; the cause is most often **cancer of the head of the pancreas**.

> **CLINICAL SIGNIFICANCE**
> **Metastatic calcification** is a serious problem in patients with chronic kidney disease on dialysis: calcification of vessel walls (cardiovascular complications), calcification of skin vessels (**calciphylaxis** — skin necrosis and ulcers). Therefore serum calcium, phosphorus and parathyroid hormone are regularly monitored in such patients.

Practical skills

Skill 1. Gross specimens

1. **"Brown induration of the lung".** The lung is enlarged and firm; on the cut surface it is rust-brown with small brown spots. Conclusion: local hemosiderosis and sclerosis (chronic venous congestion — mitral valve defect).
2. **"Cyst in place of a cerebral hemorrhage ('rust-colored cyst')".** A smooth-walled cavity in the brain tissue, with a rust-brown wall. Conclusion: local hemosiderosis.
3. **"Calcification of the aortic valve".** The valve cusps are thickened, deformed and stone-hard, with whitish calcium masses; the valve orifice is narrowed. Conclusion: dystrophic calcification (calcific aortic stenosis).
4. **"Atherocalcinosis".** Hard, whitish, brittle calcified plaques in the aortic intima; they grate when cut. Conclusion: dystrophic calcification.
5. **"Gallbladder stones (cholelithiasis)".** The gallbladder wall is thickened; its cavity contains multifaceted yellowish (cholesterol) or small black (pigment) stones. Conclusion: gallstone disease, chronic cholecystitis.
6. **"Kidney stones, hydronephrosis".** A large stone (sometimes staghorn) in the renal pelvis; the pelvis and calyces are dilated, and the parenchyma is thinned. Conclusion: nephrolithiasis, hydronephrosis.

7. **“Gout”.** Nodules with a white, chalky mass (tophi) around a joint and in the auricle. Conclusion: gout.
8. **“Uric acid infarct in the kidney of a newborn”.** Yellowish-red streaks in the medulla directed toward the papillae. Conclusion: uric acid infarct (a physiological state).

Skill 2. Microscopic slides

1. **“Pulmonary hemosiderosis (brown induration of the lung)”** (hematoxylin and eosin and Perls reaction). The interalveolar septa are thickened (sclerosis), and the capillaries are dilated and filled with blood; macrophages with golden-brown granular pigment in the alveolar spaces and septa (“heart failure cells”); the pigment is blue with the Perls reaction. Conclusion: local hemosiderosis.
2. **“Cerebral hemorrhage”** (hematoxylin and eosin). Hemolyzed blood surrounded by hemosiderin-laden macrophages; orange hematoidin crystals in the center. Conclusion: hemorrhage, hemosiderosis and hematoidin.
3. **“Calcification of the myocardium (calcium metastases)”** (hematoxylin and eosin). Dark violet-blue, amorphous calcium deposits in some cardiomyocytes and in the stroma between them. Conclusion: metastatic calcification.
4. **“Gouty tophus”** (hematoxylin and eosin). Amorphous or needle-shaped crystalline masses (urate) surrounded by foreign body giant cells, macrophages, lymphocytes and fibrosis. Conclusion: gout.
5. **“Skin in Addison disease”** (hematoxylin and eosin). Numerous brown-black melanin granules in the cells of the basal layer of the epidermis, melanophages in the dermis. Conclusion: generalized acquired hyperpigmentation (melanosis).
6. **“Brown atrophy of the liver (lipofuscinosis)”** (hematoxylin and eosin). Fine yellowish-brown granules around the nuclei of hepatocytes (see Topic 1). Conclusion: accumulation of lipofuscin.

Review questions

1. What are mixed degenerations? State their classification.
2. How are endogenous pigments classified?
3. How is hemosiderin formed and how is it detected?
4. Causes and examples of local and generalized hemosiderosis.
5. What is hemochromatosis? The difference between primary and secondary hemochromatosis and their clinicomorphologic signs.
6. Explain the stages of bilirubin metabolism.
7. Compare the three types of jaundice by their causes, mechanisms and laboratory signs.
8. What are hematoidin, hematins and porphyrins?
9. How is melanin formed? State the types of disorders of melanin metabolism.
10. By what mechanism does skin pigmentation develop in Addison disease?
11. How can lipofuscin, lipochrome and melanin be distinguished from hemosiderin?
12. What is gout? Describe its causes and morphology.
13. What is the uric acid infarct of newborns?
14. State the types of calcification with their causes and examples.
15. The essence of Wilson–Konovalov disease.

16. Causes, mechanism, types and complications of stone formation.

Test questions

1. The pigment that stains blue with the Perls reaction:
   A) Melanin
   B) Hematoidin
   C) Lipofuscin
   D) Bilirubin
   E) Hemosiderin

2. An iron-free pigment formed outside cells under anaerobic conditions:
   A) Ferritin
   B) Hemomelanin
   C) Hematoidin
   D) Hydrochloric acid hematin
   E) Hemosiderin

3. The “heart failure cells” in the lung in mitral stenosis are:
   A) Giant cells
   B) Macrophages that have accumulated coal dust
   C) Cardiomyocytes that have accumulated lipofuscin
   D) Melanophages that have accumulated melanin
   E) Alveolar macrophages that have accumulated hemosiderin

4. Not part of the classic triad of hemochromatosis:
   A) Liver cirrhosis
   B) Diabetes mellitus
   C) Bronze color of the skin
   D) Itching of the skin and acholic stools
   E) All are included

5. Characteristic of obstructive (posthepatic) jaundice:
   A) Colorless stools and dark urine
   B) Absence of bilirubin in the urine
   C) Increase in unconjugated bilirubin in the blood
   D) Dark color of the stools
   E) Enlargement of the spleen

6. The pigment that gives a black color to the floor of gastric erosions:
   A) Hydrochloric acid hematin
   B) Melanin
   C) Lipochrome
   D) Bilirubin
   E) Hemosiderin

7. The pigment that gives the organs a gray-black color in malaria:
   A) Hemosiderin
   B) Melanin
   C) Hemomelanin
   D) Lipofuscin
   E) Porphyrin

8. Hyperpigmentation of the skin in Addison disease is associated with:
   A) Accumulation of lipofuscin
   B) Accumulation of hemosiderin
   C) Excessive synthesis of melanin
   D) Accumulation of carotene
   E) Increase in bilirubin

9. Absence of melanin synthesis due to tyrosinase deficiency:
   A) Chloasma
   B) Melanosis
   C) Vitiligo
   D) Albinism
   E) Leukoderma

10. Called the “aging pigment”:
   A) Lipofuscin
   B) Adrenochrome
   C) Melanin
   D) Hemosiderin
   E) Lipochrome

11. The pathognomonic sign of gout:
   A) Accumulation of hemosiderin
   B) Ghon focus
   C) Kayser–Fleischer ring
   D) Tophi (granulomatous reaction around urate crystals)
   E) Psammoma bodies

12. Deposition of calcium in healthy tissues against a background of hypercalcemia:
   A) Dystrophic calcification
   B) Petrification
   C) Metabolic calcification
   D) Osteoporosis
   E) Metastatic calcification

13. An example of dystrophic calcification:
   A) Calcification of the kidney in vitamin D intoxication
   B) Calcification of a caseous focus in tuberculosis (Ghon focus)
   C) Calcification of the stomach in multiple myeloma
   D) Calcification of the lung in hyperparathyroidism
   E) Calcification of vessels in chronic kidney disease

14. The element that accumulates in Wilson–Konovalov disease:
   A) Calcium
   B) Copper
   C) Potassium
   D) Iron
   E) Zinc

15. Large “staghorn” kidney stones formed in Proteus infection:
    A) Uric acid
    B) Struvite (magnesium ammonium phosphate)
    C) Cystine
    D) Calcium oxalate
    E) Cholesterol

Clinical cases

Case 1

A 50-year-old man has an enlarged liver, diabetes mellitus has been diagnosed, and his skin is bronze-gray; his libido is decreased. Serum ferritin is sharply elevated, and transferrin saturation with iron is 85%. Liver biopsy: golden-brown granular pigment in the cytoplasm of hepatocytes, fibrosis of the portal tracts.

Questions: 1) Your diagnosis? 2) How is the pigment detected, and in which cells has it accumulated? 3) State the mechanism and complications of the disease.

Answer: 1) Hereditary (primary) hemochromatosis — “bronze diabetes”. 2) The pigment is hemosiderin; it stains blue (Prussian blue) with the Perls reaction; it has accumulated in parenchymal cells (hepatocytes, pancreatic cells, cardiomyocytes) — this distinguishes hemochromatosis from hemosiderosis. 3) Mutation of the HFE gene leads to decreased hepcidin and a continuous increase in intestinal iron absorption. Iron generates free radicals, damages cells and causes fibrosis. Complications: pigment cirrhosis of the liver, hepatocellular carcinoma, diabetes mellitus, cardiomyopathy (arrhythmia, heart failure), hypogonadism, arthropathy. Treatment — regular phlebotomy.

Case 2

A 58-year-old man woke up at night with sudden, extremely severe pain in the joint of his right big toe; the joint is red and swollen. Yesterday at a wedding he ate a lot of meat and drank alcohol. Blood uric acid is high. He has had such attacks several times before, and there are small hard white nodules on his auricle.

Questions: 1) Your diagnosis? 2) What are the nodules on the auricle, and what is their microscopic structure? 3) To which type of degeneration does the disease belong, and what are its complications?

Answer: 1) Gout — an attack of acute gouty arthritis (triggered by purine-rich food and alcohol). 2) Tophi — collections of sodium urate crystals surrounded by foreign body giant cells, macrophages, lymphocytes and fibrosis (a granulomatous reaction). In the synovial fluid, needle-shaped crystals that are negatively birefringent in polarized light are found inside neutrophils. 3) A disorder of nucleoprotein metabolism (mixed degeneration). Complications: chronic gouty arthritis and joint deformity, gouty nephropathy, uric acid stones and kidney failure.

Case 3

A 60-year-old woman has been diagnosed with a parathyroid adenoma; blood calcium and parathyroid hormone are high. The patient died of kidney failure. At autopsy the skeletal bones are softened; there are numerous calcium deposits in the lungs, gastric mucosa and renal tubules, and stones in the kidneys.

Questions: 1) What is the change in the organs called? 2) Why are these particular organs affected? 3) How can it be distinguished from dystrophic calcification?

Answer: 1) Metastatic calcification (calcium metastases) — hypercalcemia due to hyperparathyroidism; nephrocalcinosis and nephrolithiasis in the kidney. 2) The lungs, stomach and kidneys excrete acid, so their tissue is more alkaline, which creates favorable conditions for the precipitation of calcium salts. 3) In dystrophic calcification, blood calcium is normal and calcium is deposited only in injured or necrotic tissue (atherosclerotic plaque, caseous focus, valve); in metastatic calcification there is hypercalcemia, and calcium is deposited widely in healthy tissues.

Quick review

Table 4.6. *Key facts on Topic 4*

Question	Answer
Mixed degenerations	Disorders of chromoprotein, nucleoprotein and mineral metabolism
Groups of endogenous pigments	Hemoglobinogenic, proteinogenic (tyrosinogenic), lipidogenic
Hemosiderin	Iron-containing, intracellular, aerobic; Perls — blue
Hematoidin	Iron-free, extracellular, anaerobic; orange crystals
“Heart failure cells”	Hemosiderin-laden alveolar macrophages (mitral stenosis)
Triad of hemochromatosis	Pigment cirrhosis + diabetes mellitus + bronze skin
Hemolytic jaundice	Unconjugated bilirubin ↑, no bilirubin in the urine
Obstructive jaundice	Conjugated bilirubin ↑, acholic stools, dark urine, itching
Hydrochloric acid hematin	Gastric erosions, “coffee grounds”
Hemomelanin	Malarial pigment, slate-colored organs
Melanin synthesis	Tyrosine → DOPA → melanin (tyrosinase), in melanocytes
Addison disease	Damage to the adrenal cortex → ACTH/MSH ↑ → bronze skin
Albinism / vitiligo	No tyrosinase / autoimmune destruction of melanocytes
Lipofuscin	“Aging pigment”, perinuclear, brown atrophy
Gout	Sodium urate; big toe; tophi; gouty nephropathy
Uric acid infarct	The newborn lived longer than 2 days; a physiological state
Dystrophic calcification	Ca normal, injured tissue (Ghon focus, plaque, valve)
Metastatic calcification	Hypercalcemia; lung, stomach, kidney (nephrocalcinosis)
Calcium stain	Von Kossa — black; H&E — basophilic
Psammoma bodies	Papillary thyroid carcinoma, serous carcinoma of the ovary, meningioma
Wilson–Konovalov	Copper; ATP7B; ceruloplasmin ↓; Kayser–Fleischer ring; cirrhosis
Staghorn calculus	Struvite (Proteus)

Glossary of terms

- **Pigments** (Latin pigmentum — paint) — substances that have their own color; they are exogenous and endogenous.
- **Hemosiderin** — an iron-containing golden-brown pigment formed in macrophages during the breakdown of hemoglobin.
- **Hemosiderosis** — excessive accumulation of hemosiderin in tissues (local or generalized).

- **Hemochromatosis** — a disease characterized by accumulation of iron in parenchymal cells, which it damages (“bronze diabetes”).
- **Hematoidin** — an iron-free orange crystalline pigment formed under anaerobic conditions in the center of a hemorrhage.
- **Icterus (jaundice)** — yellow coloring of tissues as a result of hyperbilirubinemia.
- **Kernicterus (nuclear jaundice)** — staining and damage of the subcortical nuclei of the brain by bilirubin in newborns.
- **Siderophage** — a macrophage that has accumulated hemosiderin.
- **Melanin** (Greek melas — black) — a brown-black pigment synthesized from tyrosine in melanocytes.
- **Melanosis** — excessive accumulation of melanin.
- **Albinism** (Latin albus — white) — congenital absence of melanin synthesis due to tyrosinase deficiency.
- **Vitiligo** — appearance of depigmented white patches on the skin as a result of destruction of melanocytes.
- **Lipofuscin** (Latin fuscus — brown) — the “aging pigment” formed from the residues of autophagy.
- **Anthracosis** (Greek anthrax — coal) — accumulation of coal dust in the lungs and lymph nodes.
- **Gout** (Greek podagra: pous — foot, agra — trap) — a disease characterized by hyperuricemia and deposition of urate crystals.
- **Tophus** — a granulomatous nodule formed around a collection of urate crystals in gout.
- **Calcinosis (calcification, petrification)** — deposition of calcium salts in tissues.
- **Petrification** (Greek petra — stone) — a calcified focus of necrosis.
- **Psammoma bodies** (Greek psammos — sand) — concentrically laminated calcium granules found in some tumors.
- **Lithopedion** — a dead and calcified fetus (“stone child”).
- **Concretion (stone, calculus)** — a hard structure formed by precipitation of substances in hollow organs and ducts.
- **Cholelithiasis** — gallstone disease.
- **Urolithiasis (nephrolithiasis)** — urinary (kidney) stone disease.
- **Phlebolith** — a calcified thrombus in a vein (“vein stone”).

TOPIC 5

Inflammation and repair. Definition and general manifestations of inflammation. Acute inflammation. Types

Learning objectives

The student should know:

- the definition, biological essence, causes and classification of inflammation;
- the local and general (systemic) signs of inflammation;
- the phases of inflammation: alteration, exudation, proliferation;
- the cellular and plasma mediators of inflammation and their effects;
- vascular and leukocyte reactions in acute inflammation (margination, adhesion, emigration, chemotaxis, phagocytosis);
- the differences between exudate and transudate;
- the types of exudative inflammation: serous, fibrinous (croupous, diphtheritic), purulent (abscess, phlegmon, empyema), putrefactive, hemorrhagic, catarrhal and mixed — their causes, morphology and outcomes;
- the outcomes of acute inflammation.

The student should be able to:

- describe gross specimens of fibrinous pericarditis (“hairy heart”), fibrinous pleurisy, diphtheritic colitis, catarrhal tracheitis, apostematous nephritis, purulent leptomeningitis, lung abscess and phlegmonous appendicitis;
- describe microscopic slides of serous, croupous and diphtheritic inflammation, purulent leptomeningitis, kidney abscess and hemorrhagic pneumonia;
- identify the type of exudate by its cellular composition;
- relate the type of inflammation to the clinical situation.

5.1. The concept of inflammation

Inflammation (Latin inflammatio, Greek phlogosis) is a **complex protective-adaptive vascular-mesenchymal reaction** of the organism that arises in response to an injurious agent and is aimed at eliminating or limiting it and restoring the damaged tissue. Inflammation occurs only in multicellular organisms with blood vessels; at its center are the **microcirculatory bed** and **leukocytes**.

> **REMEMBER**
> Inflammation is a **protective** reaction: without it, infections would spread uncontrollably and wounds would not heal. At the same time, inflammation can also be **harmful**: it underlies many diseases (anaphylactic reactions, rheumatoid arthritis, atherosclerosis, pulmonary fibrosis). Being able to distinguish the protective and damaging sides of the inflammatory process is the basis of clinical thinking.

The classic local signs of inflammation:

- **redness** (rubor), **swelling** (tumor), **local heat** (calor), **pain** (dolor) — the Roman scholar **A.C. Celsus** (1st century AD);
- **loss of function** (functio laesa) — added by R. Virchow (19th century).

These signs are based on vascular reactions: redness and heat — **arterial hyperemia**; swelling — **exudation**; pain — irritation and compression of nerve endings by mediators (bradykinin, prostaglandins) and edema.

From history. The protective role of **phagocytosis** in inflammation was discovered by the Russian scientist **I.I. Mechnikov** (1882; Nobel Prize, 1908). J. Cohnheim described the emigration of leukocytes through the vessel wall. T. Lewis established the effect of histamine on vascular permeability and laid the foundation for the doctrine of **mediators**.

Causes of inflammation

- **Infections** (bacteria, viruses, fungi, parasites and their toxins) — the most common cause;
- **tissue necrosis** — ischemia (infarction), trauma, physical and chemical agents (burns, frostbite, radiation, acids, alkalis);
- **foreign bodies** — dust particles, suture material, crystals (urates);
- **immune reactions** (hypersensitivity) — allergy, autoimmune diseases.

Classification of inflammation

Table 5.1. *Classification of inflammation*

Feature	Types
By course	Acute (from a few hours to 4–6 weeks), chronic (months, years), subacute
By the predominant phase (Strukov)	Alterative (alteration predominates), exudative (exudation predominates), productive (proliferative) (cell proliferation predominates)
By type of exudate	Serous, fibrinous, purulent, putrefactive, hemorrhagic, catarrhal, mixed
By etiology	Banal (nonspecific) — caused by various agents; specific — a particular causative agent forms characteristic granulomas (tuberculosis, syphilis, leprosy, scleroma)
By immune status	Normergic, hyperergic (immune inflammation), hypoergic
By location	The Latin or Greek suffix -itis is added to the name of the organ: gastritis, hepatitis, nephritis, myocarditis; some have special names — pneumonia, tonsillitis (angina), panaritium

CAUTION

Alterative inflammation is given as a separate type in the textbooks of Strukov and Abdullakhodjaeva: in it tissue damage (degeneration, necrosis) predominates, while exudation and proliferation are weak (for example, the myocardium in diphtheria, parenchymal organs in tuberculosis and viral hepatitis). The Robbins textbook does not distinguish such a type — these changes are considered as necrosis and cell injury. Many current sources also use the term **necrotic inflammation** instead of alterative inflammation.

5.2. Phases of inflammation

Inflammation consists of three closely interrelated phases:

1. **Alteration (damage)** — the initial phase: degeneration and necrosis of cells and intercellular substance; in this phase **mediators** are released and initiate inflammation.
2. **Exudation** — the reaction of the microcirculatory bed: increased vascular permeability, escape of the fluid part of the blood and of blood cells from the vessels (formation of **exudate**), phagocytosis.

3. **Proliferation (productive phase)** — proliferation of cells in the focus of inflammation (macrophages, fibroblasts, endothelium), formation of granulation tissue and restoration of the damaged tissue (repair).

5.3. Mediators of inflammation

Mediators of inflammation are biologically active substances formed or activated in the focus of inflammation that regulate its vascular and cellular reactions. They are divided into **cellular** (tissue) and **plasma** mediators.

Table 5.2. *The main mediators of inflammation*

Mediator	Source	Main effect
Histamine	Mast cells, basophils, platelets	Dilation of arterioles, rapid increase in venular permeability (the first mediator)
Serotonin	Platelets	Increased vascular permeability
Prostaglandins (PGE2, PGI2)	Many cells (from arachidonic acid, cyclooxygenase pathway)	Vasodilation, pain, fever
Thromboxane A_2	Platelets	Vasoconstriction, platelet aggregation
Leukotrienes (LTB4; LTC4, D4, E4)	Leukocytes, mast cells (lipoxygenase pathway)	LTB4 — potent chemotaxis; LTC4–E4 — bronchospasm, increased permeability
Platelet-activating factor	Leukocytes, endothelium	Platelet aggregation, permeability, leukocyte adhesion
Cytokines: TNF, IL-1	Macrophages, endothelium, mast cells	Endothelial activation (adhesion molecules), fever, acute phase proteins, cachexia, shock in sepsis
IL-6	Macrophages and others	Synthesis of acute phase proteins in the liver
Chemokines (IL-8 and others)	Macrophages, endothelium	Chemotaxis and activation of leukocytes
Nitric oxide (NO)	Endothelium, macrophages	Vasodilation, killing of microbes
Reactive oxygen species	Neutrophils, macrophages	Killing of microbes, tissue damage
Lysosomal enzymes	Neutrophils, macrophages	Breakdown of microbes and tissue
Complement: C3a, C5a	Plasma (synthesized in the liver)	Anaphylatoxins — release of histamine; C5a — potent chemotaxis
Complement: C3b	Plasma	Opsonization — enhancement of phagocytosis
Complement: C5b–C9	Plasma	Membrane attack complex — lysis of microbes
Bradykinin	Plasma (kinin system, when Hageman factor is activated)	Increased permeability, vasodilation, pain
Coagulation system (thrombin, fibrin)	Plasma	Formation of fibrin; thrombin activates endothelium and leukocytes

MEMORY AID

The main signs and their mediators: **fever — IL-1, TNF, PGE_2**; **pain — bradykinin, PGE_2**; **vasodilation — histamine, prostaglandins, NO**; **permeability — histamine, bradykinin, C3a/C5a, leukotrienes**; **chemotaxis — C5a, LTB_4, IL-8, bacterial peptides** (“**5-4-8**” — C5a, LTB**4**, IL-**8**).

CLINICAL SIGNIFICANCE

Aspirin and other nonsteroidal anti-inflammatory drugs (ibuprofen, diclofenac) block **cyclooxygenase** and reduce prostaglandin synthesis — they relieve pain, fever and inflammation. Glucocorticoids block the release of arachidonic acid (phospholipase A2) and the synthesis of cytokines — therefore their anti-inflammatory effect is much stronger, but they also suppress immunity.

5.4. Vascular reactions in acute inflammation

1. Changes in blood flow and vessel caliber:

- **brief arteriolar spasm** (a few seconds, neuroreflex);
- **arterial hyperemia** — arterioles dilate (histamine, NO), new capillaries open; **redness and heat**;
- **venous hyperemia and slowing of blood flow** — fluid escapes because of increased permeability and the blood thickens; red cells accumulate (**stasis**); leukocytes line up along the vessel wall (**margination**).

2. Increased vascular permeability is the most important feature of acute inflammation. Mechanisms:

- **contraction of endothelial cells** — gaps appear between endothelial cells in venules; under the action of histamine, bradykinin and leukotrienes it develops **rapidly** (in 15–30 minutes) and is short-lived — the **most common** mechanism;
- **direct injury to the endothelium** — burns, toxins: the endothelium undergoes necrosis and detaches; it begins immediately and lasts a long time;
- **leukocyte-mediated injury** — enzymes and ROS of activated leukocytes damage the endothelium;
- increased **transcytosis** (under the action of VEGF), the “leaky” wall of newly formed capillaries. **3. Reaction of lymphatic vessels** — lymph flow increases; secondary inflammation of lymphatic vessels (**lymphangitis**) and reactive inflammation of lymph nodes (**lymphadenitis**) may develop (for example, red streaks on the forearm and enlarged axillary lymph nodes after a hand wound).

Exudate and transudate

Table 5.3. *Differences between exudate and transudate*

Feature	Exudate	Transudate
Mechanism of formation	Increased vascular permeability (inflammation)	Increased hydrostatic pressure or decreased oncotic pressure (without inflammation)
Protein content	High (more than 30 g/L)	Low (less than 30 g/L)
Specific gravity	Above 1.018	Below 1.012
Cells	Many (leukocytes, red blood cells, macrophages)	Few (mainly desquamated mesothelium)
Clarity	Cloudy (turbid)	Clear
Rivalta test	Positive	Negative
Examples	Pleurisy, peritonitis, arthritis	Hydrothorax and ascites in heart failure; ascites in nephrotic syndrome and liver cirrhosis

5.5. Leukocyte reactions in acute inflammation

The main cells of acute inflammation are **neutrophils** (in the first 6–24 hours); later (from 24–48 hours) they are replaced by **monocytes-macrophages**. Neutrophils are numerous and respond quickly but live briefly (undergo apoptosis); macrophages live long and can proliferate. There are some exceptions: in viral infections lymphocytes arrive first, and in allergic reactions eosinophils predominate.

Escape of leukocytes from vessels (extravasation)

1. **Margination** — when blood flow slows, leukocytes move from the center of the vessel to the periphery (to the endothelial surface).
2. **Rolling** — leukocytes "roll" along the endothelium: endothelial **selectins** (E-selectin, P-selectin) bind **weakly** to carbohydrate ligands on the surface of leukocytes. Selectin expression is increased by histamine, TNF and IL-1.
3. **Firm adhesion** — **integrins** on the surface of leukocytes (LFA-1, VLA-4) are activated by chemokines and bind firmly to endothelial ligands (**ICAM-1, VCAM-1**).
4. **Transmigration (diapedesis)** — leukocytes pass mainly in **postcapillary venules** between endothelial cells (with the help of **PECAM-1 / CD31** molecules) and pierce the basement membrane with the help of collagenases.
5. **Chemotaxis** — movement of leukocytes toward increasing concentrations of **chemotactic factors** — to the focus of injury. Exogenous chemoattractants are bacterial peptides (N-formylmethionine); endogenous ones are **C5a, LTB_4, IL-8** and other chemokines.

MEMORY AID

Stages of leukocyte extravasation — **"Roll, Adhere, Transmigrate, Chase"**: **R**olling — **S**electins; **A**dhesion — **I**ntegrins (ICAM-1); **T**ransmigration — **P**ECAM-1 (CD31); **C**hemotaxis — following the signal (C5a, LTB4, IL-8). In short: **"S-I-P"**.

CLINICAL SIGNIFICANCE

In children with **hereditary defects of leukocyte adhesion** (LAD-1 — defect of the beta-2 chain of integrins; LAD-2 — defect of the selectin ligand), leukocytes cannot leave the vessels: there are very many leukocytes in the blood, but **no pus forms** in the focus of inflammation; recurrent severe bacterial infections and delayed separation of the umbilical cord stump are observed.

Phagocytosis

Phagocytosis is the engulfment and digestion of foreign particles, microbes and tissue debris by special cells (phagocytes). I.I. Mechnikov divided phagocytes into **microphages** (neutrophils) and **macrophages**. Stages of phagocytosis:

1. **Recognition and attachment** — phagocyte receptors recognize microbes directly (mannose receptors, "scavenger" receptors, Toll-like receptors) or through **opsonins**. The main opsonins are **IgG** (via the Fc receptor), the **C3b** fragment of complement and mannose-binding lectin.
2. **Engulfment** — the phagocyte surrounds the microbe with pseudopodia, and a **phagosome** forms; it fuses with a lysosome to become a **phagolysosome**.
3. **Killing and degradation:** – **the oxygen-dependent mechanism** (the most effective) — the "**respiratory burst**": NADPH oxidase (phagocyte oxidase) produces superoxide and H2O2; **myeloperoxidase**, in the presence of chloride ions, converts H2O2 into the potent bactericidal **hypochlorite ($HOCl^-$)**; – **nitric oxide** (in macrophages) and peroxynitrite; – **the oxygen-**

independent mechanism — lysosomal enzymes (lysozyme, elastase), defensins, cathelicidins, lactoferrin, major basic protein of eosinophils. 4. **Neutrophil extracellular traps (NETs)** — neutrophils form a net of chromatin and antimicrobial proteins that traps microbes (with neutrophil death). **Complete phagocytosis** — the engulfed microbe is killed and digested. **Incomplete phagocytosis (endocytobiosis)** — the microbe survives and even multiplies inside the phagocyte; this is characteristic of tubercle mycobacteria, gonococci, leishmaniae and the causative agent of leprosy. In such cases the phagocyte becomes a "vehicle" that spreads infection through the organism.

CLINICAL SIGNIFICANCE

Chronic granulomatous disease — a hereditary defect of NADPH oxidase (most often X-linked): phagocytes engulf microbes, but because there is no "respiratory burst" they cannot kill catalase-positive microbes (staphylococcus, Aspergillus) — recurrent severe infections and granulomas. **Chédiak–Higashi syndrome** — impaired fusion of phagosomes and lysosomes (giant granules), albinism and infections.

5.6. General (systemic) manifestations of inflammation

Local inflammation affects the whole organism — the **acute phase reaction** (systemic inflammatory response). It develops under the action of cytokines (**TNF, IL-1, IL-6**) and prostaglandins:

- **fever** — cytokines (endogenous pyrogens) increase PGE2 synthesis in the hypothalamus and raise the "set point" of the thermoregulatory center;
- **acute phase proteins** — the liver increases synthesis of **C-reactive protein (CRP)**, fibrinogen, serum amyloid A (SAA) and hepcidin; fibrinogen makes red cells stick together in "stacks" (rouleaux) — the **ESR increases**; in chronic inflammation SAA leads to **AA amyloidosis**, and hepcidin to **anemia**;
- **leukocytosis** — leukocytes increase in the blood (neutrophils in bacterial infections, a left shift — young forms); **lymphocytosis** in viral infections, **eosinophilia** in allergy and parasitic diseases; in some infections (typhoid fever, some viruses) — **leukopenia**;
- tachycardia, changes in blood pressure, chills, loss of appetite, drowsiness, weakness;
- in severe bacterial infection — **sepsis**: excessive release of cytokines leads to **septic shock**, DIC and multiple organ failure (Topic 7).

5.7. Types of exudative inflammation

In exudative inflammation, **exudate formation** predominates. By the composition of the exudate, the types are as follows:

Table 5.4. *Types of exudative inflammation*

Type	Composition of the exudate	Main causes	Examples
Serous	Protein 2–5%, few cells; clear or slightly cloudy	Burns, viruses, toxins, allergy, tuberculosis (in serous membranes)	Burn blister, serous pleurisy, serous meningitis, viral rhinitis, pneumonia with serous exudate
Fibrinous	Much fibrinogen — a fibrin membrane forms in the tissue	Diphtheria, dysentery, pneumococcus, tuberculosis, uremia (endogenous intoxication), mercuric chloride poisoning	Croupous (lobar) pneumonia, "hairy heart", diphtheria, dysentery (diphtheritic colitis)

Type	Composition of the exudate	Main causes	Examples
Purulent	Many neutrophils, their disintegrated remnants (pus corpuscles), debris, microbes	Pyogenic bacteria: staphylococcus, streptococcus, gonococcus, meningococcus, Pseudomonas aeruginosa	Abscess, phlegmon, empyema, purulent meningitis
Putrefactive (ichorous)	Anaerobic putrefactive bacteria added to a purulent or other exudate; foul odor	Anaerobic infection (clostridia)	War wounds, gangrenous lung, gangrenous appendicitis
Hemorrhagic	Many red blood cells	Highly virulent infections that damage the vessel wall	Influenza pneumonia, anthrax, plague, smallpox
Catarrhal	Mucus mixed with the exudate; only in mucous membranes	Infections (ARVI), allergy, thermal and chemical influences	Rhinitis, tracheitis, bronchitis, gastritis
Mixed	Two or more types combine	Addition of a new infection	Serous-purulent, serous-fibrinous, purulent-hemorrhagic, fibrinous-hemorrhagic

Serous inflammation

The exudate is a **serous fluid**: 2–5% protein (mainly albumins), a few neutrophils, desquamated epithelial or mesothelial cells. The exudate accumulates in **serous cavities** (serous pleurisy, pericarditis, peritonitis, arthritis), **mucous membranes** (serous catarrh), the **skin** (blisters in burns and viral infections — fluid accumulates beneath or within the epidermis) and the stroma of **parenchymal organs** (serous interstitial inflammation — liver, myocardium, kidney).

The outcome is usually **favorable**: the exudate is resorbed. But sometimes it is dangerous because of its location: cardiac tamponade in serous pericarditis; brain edema in serous inflammation of the meninges (serous meningitis); in the larynx — edema and asphyxia. If prolonged in serous cavities — adhesions; in parenchymal organs — diffuse sclerosis.

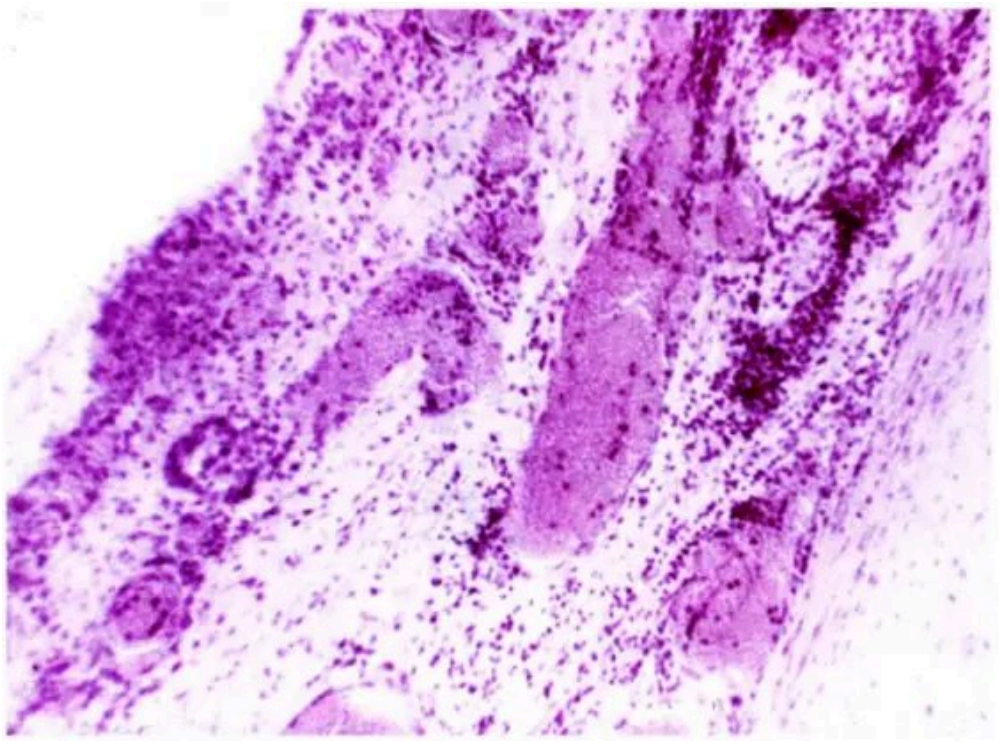


Figure 5.1. *Serous rhinitis: the nasal mucosa is edematous, with serous exudate and a few inflammatory cells in the stroma (Strukov, Serov, Fig. 5-7)*

Fibrinous inflammation

The exudate contains much **fibrinogen**, which is converted to **fibrin** in the damaged (necrotic) tissue. Tissue thromboplastin accelerates fibrin formation. Fibrinous inflammation develops mainly in **mucous and serous membranes**, less often in the thickness of an organ (in the lung). A **whitish-gray membrane (film)** forms on the surface of the mucous or serous membrane. By the depth of necrosis and the type of lining epithelium, there are two types:

1. Croupous inflammation (Scottish croup — membrane) — necrosis is **superficial**, the fibrin membrane is **thin** and **loosely** attached to the underlying tissue; it **detaches easily**, leaving a superficial defect. It occurs in mucous membranes lined by simple (columnar) epithelium or in serous membranes:

- **in the trachea and bronchi** — croupous tracheitis and bronchitis in diphtheria: the membranes can detach and block the lumen of the bronchi — **true croup** (risk of asphyxia);
- **in the lung — croupous (lobar) pneumonia** (pneumococcus; a whole lobe is affected);
- **in serous membranes** — fibrinous pleurisy, peritonitis, **pericarditis**: the pericardial surface is rough and covered with hairlike fibrin threads — the **“hairy heart”** (cor villosum); on auscultation — a **pericardial friction rub**; causes — uremia (chronic kidney disease), rheumatic fever, myocardial infarction, tuberculosis.



Figure 5.3. *Fibrinous epicarditis: the epicardium is dull and rough, with hairlike gray-yellowish fibrin deposits (TMA study guide, 2018)*

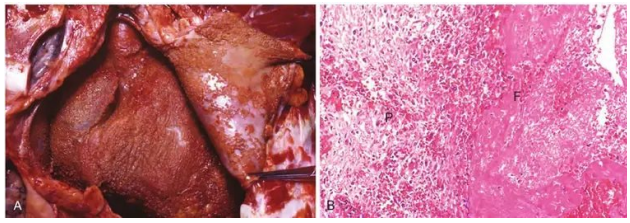


Figure 5.2. *Fibrinous pericarditis ("hairy heart"): A — fibrin deposits on the pericardial surface; B — a network of pink fibrinous exudate (F) covering the pericardial surface, P — pericardium (Robbins, 9th ed., Fig. 3-14)*

2. Diphtheritic inflammation (Greek diphthera — leather membrane) — necrosis is **deep**, the fibrin membrane is **thick**, penetrates deep into the necrotic tissue and is **firmly** attached to the underlying tissue; the membrane **detaches with difficulty**, leaving a deep **ulcer**. It occurs in mucous membranes lined by **stratified squamous epithelium** (oral cavity, pharynx, **tonsils**, esophagus, vagina, cervix) and in places where the subepithelial tissue is looser (**colon** — dysentery).

REMEMBER

In **diphtheria** the type of inflammation depends on location: in the **pharynx and tonsils** (stratified squamous epithelium) — **diphtheritic** inflammation: the membrane is firmly attached and does not block the airway, but much toxin is absorbed (myocarditis, neuritis); in the **larynx, trachea and bronchi** (columnar epithelium) — **croupous** inflammation: the membrane detaches easily and blocks the airway — **true croup**, asphyxia. This is why both types can occur in one disease.

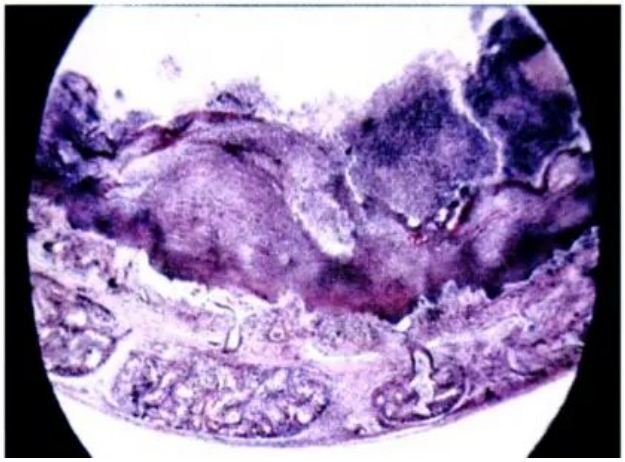


Figure 5.4. *Croupous tracheitis in diphtheria: the mucosa is superficially necrotic and covered with an easily detachable fibrin membrane (Strukov, Serov, Fig. 5-8)*

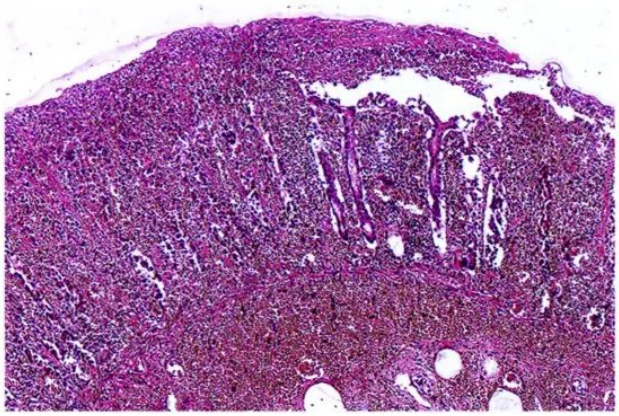


Figure 5.5. *Diphtheritic inflammation of the pharynx: deep necrosis; a thick fibrin-impregnated membrane firmly attached to the underlying tissue (Strukov, Serov, Fig. 5-9)*



Figure 5.6. *Diphtheritic colitis (dysentery): the colonic wall is markedly thickened, and the mucosa is covered with thick, gray-green, rough membranes (TMA study guide, 2018)*

Outcomes of fibrinous inflammation: in mucous membranes the membranes detach and **ulcers** form (superficial in croupous, deep in diphtheritic — healing with a scar). In serous membranes, fibrin can be resorbed or undergo **organization**: **adhesions** (scar bands) form between the serous layers; in the pericardium, complete obliteration of the cavity (**obliteration**) and calcification may occur — the "**armored heart**" (constrictive pericarditis). In the lung, organization of fibrin — **carnification** ("turning into flesh").

Purulent inflammation

Neutrophils predominate in the exudate. **Pus** is a thick, cloudy, yellow-green fluid containing living and dead neutrophils (**pus corpuscles**), lymphocytes, macrophages, tissue debris, microbes and protein-rich fluid. An important property of purulent exudate is its ability to **dissolve tissue** (**histolysis**): the proteolytic enzymes of neutrophils dissolve the tissue and form a cavity.

Purulent inflammation is caused by **pyogenic (pus-forming)** bacteria: staphylococcus, streptococcus, gonococcus, meningococcus, Fraenkel diplococcus (pneumococcus), Pseudomonas aeruginosa, Escherichia coli; less often aseptic suppuration (chemicals — turpentine).

Types of purulent inflammation:

1. Abscess — **localized** purulent inflammation of tissue in which the tissue dissolves and a **cavity filled with pus** forms. Around the abscess a **pyogenic capsule (pyogenic membrane)** forms — a layer of granulation tissue; it is rich in capillaries, through which leukocytes continuously enter the abscess cavity, and it absorbs pus. In an **acute abscess** the wall consists of one layer — the pyogenic membrane (granulation tissue); in a **chronic abscess** the wall has two layers: the inner **pyogenic** layer and the outer **fibrous** layer (mature connective tissue).

An abscess can occur in any organ (lung, liver, brain, kidney, subcutaneous tissue). Numerous small abscesses are called **apostematous** inflammation (for example, **apostematous nephritis** — many small abscesses in the kidney in sepsis). An abscess may rupture spontaneously onto the skin surface or into a cavity (forming a fistula).

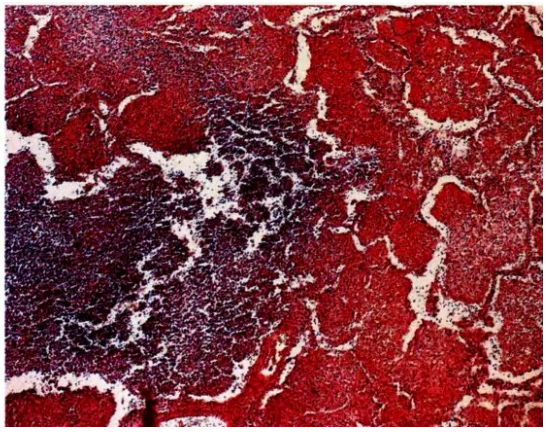


Figure 5.8. *Lung abscess (microscopic slide): the lung tissue has dissolved, forming a cavity filled with neutrophils (Strukov, Serov, Fig. 5-10)*

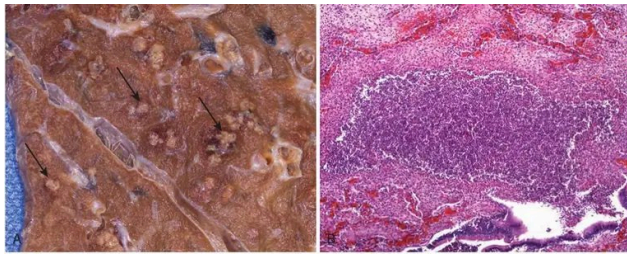


Figure 5.7. *Multiple bacterial abscesses in the lung (arrows) — a complication of bronchopneumonia: A — gross view; B — the abscess consists of neutrophils and cell debris, surrounded by congested vessels (Robbins, 9th ed., Fig. 3-15)*

2. Phlegmon — nonlocalized, **diffuse** purulent inflammation: the purulent exudate spreads between tissues, along fatty tissue, fasciae and neurovascular bundles, soaking and separating the tissues. Mainly in **soft tissues** — subcutaneous fatty tissue, between muscles, the mediastinum (**mediastinitis**), around the kidney (**paranephritis**), around the rectum (**paraproctitis**), and also in the wall of hollow organs (**phlegmonous appendicitis**, phlegmonous cholecystitis). **Soft phlegmon** — the necrotic foci liquefy; **hard phlegmon** — the tissue undergoes coagulative necrosis and sloughs off. Phlegmon often progresses to **sepsis**.

3. Empyema — accumulation of pus in a **preexisting natural cavity** or hollow organ: **pleural empyema** (pyothorax), empyema of the gallbladder, empyema of the appendix, empyema of the maxillary sinus, pericardial empyema (pyopericardium). Long-standing empyema may end in obliteration of the cavity.

General features and outcomes of purulent inflammation: usually an acute course; severe **intoxication**, fever, neutrophilic leukocytosis. If the pus is discharged (by rupture or surgery) — healing with a scar. Unfavorable outcomes: spread of pus to neighboring tissues (fistulas), erosion of vessel walls (**arrosive bleeding**), spread through lymph and blood — **sepsis**, **AA amyloidosis** (in chronic suppuration, Topic 8).



Figure 5.9. *Apostematous nephritis: numerous small round yellowish-gray abscesses on the surface and cut surface of the kidney (TMA study guide, 2018)*



Figure 5.10. *Purulent leptomeningitis: the leptomeninges of the brain are thickened, dull and soaked with yellow-green pus — a "purulent cap" (TMA study guide, 2018)*



Figure 5.11. *Phlegmonous-gangrenous appendicitis: the wall of the appendix is thickened and dirty gray, and its serosa is dull, with purulent-fibrinous deposits (TMA study guide, 2018)*

Putrefactive (ichorous) inflammation

It develops when **anaerobic putrefactive microorganisms** (clostridia and others) enter the focus of inflammation. The tissue putrefies, **foul-smelling** gases are formed, and the exudate becomes dirty green. It usually occurs in wounds with heavy contamination and destroyed tissue (war and road traffic injuries, military wounds); it is close to anaerobic (gas) gangrene. It runs a course with severe intoxication and has an unfavorable prognosis.

Hemorrhagic inflammation

The exudate contains many **red blood cells** and is red or brown. The cause is a **sharp increase** in the permeability of microcirculatory vessels and damage to the vessel wall. It is characteristic of highly dangerous infections: **plague, anthrax (Siberian ulcer)**, and also **severe forms of influenza**

(hemorrhagic pneumonia, hemorrhagic tracheobronchitis) and smallpox. The prognosis is usually serious. It often combines with other exudative types (serous-hemorrhagic, purulent-hemorrhagic).

Catarrhal inflammation

Catarrhal inflammation (catarrh) (Greek katarrheo — I flow down) develops **only in mucous membranes**: the exudate **flows down** from the mucosal surface and contains much **mucus** (hyperfunction of goblet cells and glands). By the type of exudate, **serous, mucous and purulent** catarrh are distinguished, and they can pass into one another (for example, in a common cold: first serous, then mucous, then mucopurulent discharge).

Causes: infections (**ARVI**, influenza), allergy (**allergic rhinitis**), thermal and chemical influences, endogenous intoxication (uremic catarrh).

Acute catarrh usually **resolves completely** in 2–3 weeks. **Chronic catarrh** is accompanied by **atrophy** (atrophic catarrh) or **hypertrophy** (hypertrophic catarrh, polyps) of the mucosa.

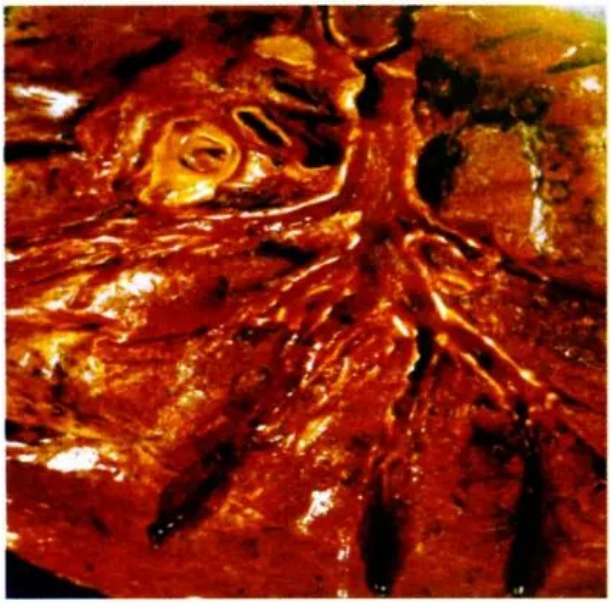


Figure 5.12. *Catarrhal bronchitis: mucous exudate in the bronchial lumen; the mucosa is red and swollen (Strukov, Serov, Fig. 5-11)*



Figure 5.13. *Catarrhal pharyngotracheitis in influenza: the swollen mucosa is covered with a large amount of cloudy mucous exudate (TMA study guide, 2018)*

5.8. Outcomes of acute inflammation

1. **Complete resolution** — if the injury was limited or brief: the mediators are neutralized, the exudate is resorbed through the lymphatics, neutrophils undergo apoptosis, macrophages clear the debris, and the parenchyma regenerates. For example, complete restoration of lung tissue in croupous (lobar) pneumonia.
2. **Scar formation (fibrosis)** — if much tissue was destroyed, nonregenerating tissue was damaged or much fibrin accumulated: the necrotic focus and exudate are replaced by granulation tissue, which turns into a scar (organization).
3. **Abscess formation** — in infection with pyogenic bacteria.
4. **Progression to chronic inflammation** — if the agent is not eliminated or the healing process is disturbed (Topic 6). **An ulcer (ulcus)** is a **local defect** of the surface of an organ or tissue formed by sloughing of necrotic and inflamed tissue. It occurs most often in the mucosa of the mouth, stomach, intestine and genitourinary tract and in the skin of the lower leg (in the elderly with impaired circulation and in people with diabetes). The most characteristic example is **gastric and duodenal ulcer**: in the acute phase there is neutrophilic inflammation at the ulcer edge; in the chronic phase layers of fibrinoid necrosis, granulation tissue and scar form at its floor. An

erosion, in contrast, is a defect only of the mucosal surface (not reaching the muscularis mucosae).

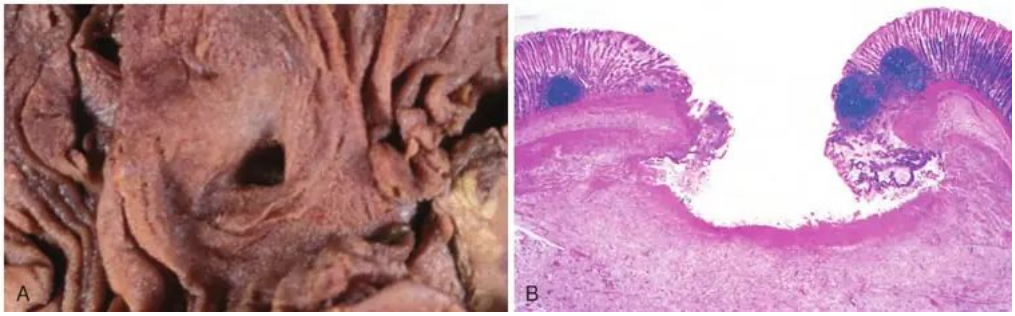


Figure 5.14. *Ulcer (chronic duodenal ulcer): A — gross view; B — the ulcer crater at low magnification, with acute inflammatory exudate at its floor (Robbins, 9th ed., Fig. 3-16)*

5.9. Clinicopathological section

Analysis of exudate — the key to diagnosis

Analysis of fluid (aspirate) obtained from the pleural or peritoneal cavity helps establish the diagnosis:

- **clear, protein-poor fluid — transudate** (heart failure, nephrotic syndrome, cirrhosis);
- **cloudy, protein-rich, lymphocytic exudate** — tuberculous pleurisy or tumor metastases (cytologic examination, search for tubercle mycobacteria);
- **purulent, neutrophilic exudate** — empyema (drainage is necessary);
- **hemorrhagic exudate** — tumor (mesothelioma, metastases), tuberculosis, pulmonary infarction, trauma.

Fibrinous pericarditis — a sign of uremia

In the final stage of chronic kidney disease (uremia), nitrogenous wastes accumulated in the blood are excreted through the serous membranes and damage them: **fibrinous pericarditis** (the pericardial friction rub is called the "death knell of uremia"), fibrinous pleurisy, fibrinous and catarrhal inflammation of the mucous membranes (uremic gastritis, colitis). Such a patient needs urgent **hemodialysis**.

> **CLINICAL SIGNIFICANCE**
> In **diphtheria** the main danger is not the local membranes but the **exotoxin**: it damages the myocardium (**toxic myocarditis**, fatty degeneration — "tiger heart", Topic 3), the peripheral nerves (**polyneuritis**, paralysis of the soft palate) and the adrenal glands. The cause of early death is **true croup (asphyxia)**; the cause of late death (in the 2nd week) is **cardiac arrest**. Therefore the basis of treatment is **diphtheria antiserum** (antitoxin), and of prevention — **vaccination** (DTP).

Principle of treating abscess and phlegmon

If the pus cavity is localized (abscess, empyema), antibiotics cannot penetrate adequately through the pyogenic capsule — the main treatment is **surgical evacuation of pus (drainage)**: "Ubi pus, ibi evacua" — "where there is pus, evacuate it". In phlegmon, wide incisions and antibiotics are needed, because the process spreads rapidly and can progress to sepsis.

Practical skills

Skill 1. Gross specimens

1. **“Fibrinous pericarditis (‘hairy heart’)”.** The epicardium is dull and rough and covered with hairlike gray-yellowish fibrin threads. Conclusion: croupous (fibrinous) inflammation.
2. **“Fibrinous pleurisy”.** A gray-whitish, easily detachable fibrin membrane on the surface of the visceral pleura; the pleura is dull. Conclusion: croupous inflammation.
3. **“Diphtheritic colitis (dysentery)”.** The colonic wall is thickened, and the mucosa is covered with thick, gray-green membranes that are difficult to detach, with ulcers in places. Conclusion: diphtheritic inflammation.
4. **“Catarrhal pharyngotracheitis (influenza)”.** The mucosa is swollen, hyperemic and covered with cloudy mucous exudate. Conclusion: catarrhal inflammation.
5. **“Apostematous nephritis”.** The kidney is enlarged; on its surface and cut surface there are numerous small (millet-sized) yellowish-gray abscesses with a red rim around them. Conclusion: purulent inflammation (abscesses), most often in sepsis.
6. **“Purulent leptomeningitis”.** The leptomeninges of the brain are thickened, dull and soaked with yellow-green pus, especially over the convex surface of the brain — a “purulent cap”. Conclusion: purulent inflammation (of the phlegmon type), most often meningococcus or pneumococcus.
7. **“Lung abscess”.** A round cavity filled with pus on the cut surface of the lung, its wall bounded by a pyogenic capsule. Conclusion: abscess.
8. **“Phlegmonous appendicitis”.** The appendix is enlarged and thickened; its serosa is dull and hyperemic, with fibrinous-purulent deposits; on the cut surface the wall is soaked with pus. Conclusion: phlegmon.



Figure 5.15. *Fibrinous pleurisy: a gray-whitish membrane on the surface of the visceral pleura; the pleura is dull (TMA study guide, 2018)*

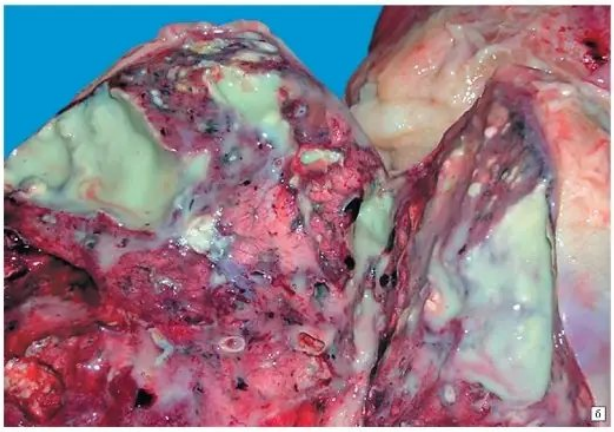


Figure 5.16. *Lung abscesses: round, gray-yellowish cavities filled with pus on the cut surface (TMA study guide, 2018)*

Skill 2. Microscopic slides

1. **“Fibrinous pericarditis”** (hematoxylin and eosin). Pink fibrillar and net-like fibrin masses on the epicardial surface, with neutrophils between them; the epicardium is hyperemic, edematous

and infiltrated by leukocytes. Conclusion: fibrinous (croupous) inflammation.

2. **“Croupous tracheitis in diphtheria”** (hematoxylin and eosin). The tracheal mucosa is superficially necrotic, without epithelium; on its surface is a membrane of fibrin and leukocytes that separates easily from the underlying tissue; the submucosa is edematous and hyperemic. Conclusion: croupous inflammation.
3. **“Diphtheritic tonsillitis (fauces)”** (hematoxylin and eosin). Deep necrosis (debris) of the stratified squamous epithelium and tonsillar tissue, impregnated with fibrin and infiltrated by neutrophils; the membrane is firmly attached to the deep layers. Conclusion: diphtheritic inflammation.
4. **“Purulent leptomeningitis”** (hematoxylin and eosin). The pia mater is thickened and diffusely infiltrated by neutrophils; the vessels are hyperemic. Conclusion: purulent (phlegmonous) inflammation.
5. **“Kidney abscess”** (hematoxylin and eosin). Localized foci of accumulated neutrophils in the renal cortex and medulla; in their center the tissue is dissolved, surrounded by granulation tissue (pyogenic capsule), with colonies of bacteria (basophilic). Conclusion: abscess.
6. **“Hemorrhagic pneumonia (influenza)”** (hematoxylin and eosin). The alveolar spaces are filled with red blood cells, a few neutrophils and fibrin; the interalveolar septa are thickened and the capillaries are hyperemic. Conclusion: hemorrhagic inflammation.

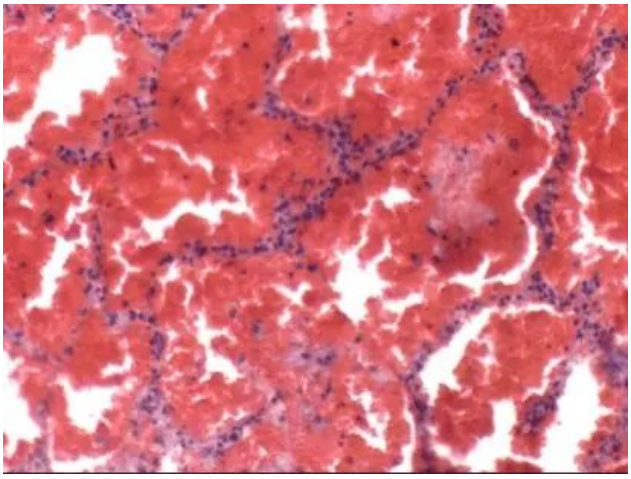


Figure 5.18. *Hemorrhagic pneumonia (microscopic slide): the alveoli are filled with red blood cells (TMA study guide, 2018)*

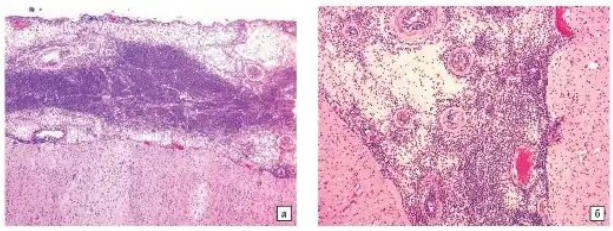


Figure 5.17. *Purulent meningitis (microscopic slide): the pia mater is thickened and diffusely infiltrated by neutrophils (a, b) (TMA study guide, 2018)*

Review questions

1. Define inflammation. What is its biological essence?
2. State the classic local signs of inflammation and their mechanisms.
3. How is inflammation classified?
4. Describe the phases of inflammation.
5. List the cellular and plasma mediators of inflammation. Which mediators cause fever, pain and chemotaxis?
6. Vascular reactions in acute inflammation and the mechanisms of increased vascular permeability.
7. State the differences between exudate and transudate.
8. Stages of leukocyte extravasation and the molecules involved.
9. Stages of phagocytosis. Complete and incomplete phagocytosis.

10. Systemic manifestations of inflammation (acute phase reaction).
11. Causes, locations and outcomes of serous inflammation.
12. Differences between croupous and diphtheritic inflammation. Why do both types occur in diphtheria?
13. Types of purulent inflammation: abscess, phlegmon, empyema. What is a pyogenic capsule?
14. Describe putrefactive, hemorrhagic and catarrhal inflammation.
15. State the outcomes of acute inflammation.

Test questions

1. The scientist who added “loss of function” to the classic signs of inflammation:
   A) Galen
   B) Cohnheim
   C) R. Virchow
   D) Celsus
   E) I.I. Mechnikov

2. The author of the theory of phagocytosis:
   A) I.I. Mechnikov
   B) J. Cohnheim
   C) A.I. Abrikosov
   D) T. Lewis
   E) R. Virchow

3. The first mediator that rapidly increases vascular permeability:
   A) Histamine
   B) Leukotriene B4
   C) TNF
   D) IL-1
   E) Bradykinin

4. The main mediators that cause fever:
   A) C3b and IgG
   B) Thromboxane A2
   C) Histamine and serotonin
   D) IL-1, TNF and prostaglandin E2
   E) Leukotrienes C4 and D4

5. The molecules that mediate “rolling” of leukocytes along the endothelium:
   A) ICAM-1
   B) Selectins
   C) Integrins
   D) Immunoglobulins
   E) PECAM-1

6. The main opsonins:
   A) IL-8 and TNF
   B) C5a and LTB4
   C) Histamine and serotonin
   D) Bradykinin and kallikrein
   E) IgG and C3b

7. Characteristic of an exudate:
   A) Specific gravity below 1.012
   B) High protein (more than 30 g/L), many cells, cloudy
   C) Low protein, clear
   D) Forms only in heart failure
   E) Negative Rivalta test

8. The “hairy heart” is:
   A) Fibrinous pericarditis
   B) Serous pericarditis
   C) Purulent pericarditis
   D) Fatty infiltration of the heart
   E) Hemorrhagic pericarditis

9. Characteristic of croupous inflammation:
   A) Deep necrosis, firmly attached membrane
   B) Exudate containing red blood cells
   C) Only in stratified squamous epithelium
   D) Mucous exudate
   E) Superficial necrosis, the membrane detaches easily

10. The inflammation that develops in the pharynx and tonsils in diphtheria:
    A) Croupous
    B) Serous
    C) Catarrhal
    D) Hemorrhagic
    E) Diphtheritic

11. Accumulation of pus in a preexisting natural cavity:
    A) Phlegmon
    B) Abscess
    C) Ichorosis
    D) Empyema
    E) Hydrocele

12. The layer of granulation tissue forming the wall of an abscess is called:
    A) Granuloma
    B) Sequestrum
    C) Demarcation line
    D) Pyogenic membrane
    E) Fibrous capsule

13. The inflammation characteristic of the severe form of influenza, anthrax and plague:
    A) Serous
    B) Hemorrhagic
    C) Catarrhal
    D) Putrefactive
    E) Fibrinous

14. Inflammation that develops only in mucous membranes, with much mucus in the exudate:
    A) Serous
    B) Phlegmon
    C) Catarrhal
    D) Hemorrhagic
    E) Croupous

15. Which system is defective in chronic granulomatous disease?
    A) Myeloperoxidase
    B) Complement C3
    C) NADPH oxidase
    D) Selectins
    E) Integrins

Clinical cases

Case 1

An unvaccinated 5-year-old child had fever and sore throat, followed by progressively worsening stridorous breathing and a “barking” cough, and died of asphyxia. At autopsy there are thick gray membranes on the tonsils, firmly attached to the underlying tissue and difficult to detach; in the larynx, trachea and bronchi there are thin, easily detachable gray membranes, some of which block the lumen of the bronchi.

Questions: 1) Your diagnosis? 2) What types of inflammation are present in the tonsils and trachea? Why do they differ? 3) The cause of death and other dangerous complications of the disease.

Answer: 1) Diphtheria (diphtheria of the pharynx and respiratory tract). 2) In the tonsils — diphtheritic inflammation (stratified squamous epithelium, deep necrosis, firmly attached membrane); in the larynx, trachea and bronchi — croupous inflammation (columnar epithelium, superficial necrosis, the membrane detaches easily). The difference depends on the type of epithelium and the structure of the submucosa. 3) The cause of death is true croup (asphyxia) as a result of detached membranes blocking the airway. Other dangerous complications are related to the action of the exotoxin: toxic myocarditis (a cause of late death), polyneuritis, hemorrhage into the adrenal glands.

Case 2

A 63-year-old man has had chronic glomerulonephritis for 15 years. In recent months he has had nausea, weakness and itching of the skin; blood creatinine and urea are sharply increased. On auscultation a friction rub is heard over the heart. The patient died.

Questions: 1) What change was present in the pericardium? 2) What are its cause and mechanism? 3) What outcome could this process have led to if the patient had survived?

Answer: 1) Fibrinous pericarditis (“hairy heart”) — the epicardium is dull, rough and covered with hairlike fibrin threads; microscopically — a pink fibrin network and leukocytes. 2) Uremia (the final

stage of chronic kidney disease): nitrogenous wastes accumulated in the blood are excreted through the serous membranes, damage them and increase vascular permeability — a fibrinogen-rich exudate escapes and fibrin forms. The friction rub is the rubbing together of the fibrin-covered pericardial layers. 3) The fibrin would undergo organization, adhesions would form between the pericardial layers, and the cavity could become obliterated and calcified — the “armored heart” (constrictive pericarditis).

Case 3

A 34-year-old woman received an injection into the thigh with a contaminated needle 5 days ago. The thigh is swollen, red and painful, with indistinct borders; temperature 39.5 °C, neutrophilic leukocytosis with a left shift in the blood. At surgery the subcutaneous fatty tissue and intermuscular spaces are diffusely soaked with yellow-green fluid; there is no localized cavity.

Questions: 1) What type and form of inflammation is this? 2) How does it differ from an abscess? 3) Possible complications and the principle of treatment.

Answer: 1) Purulent inflammation — phlegmon (subcutaneous and intermuscular phlegmon), caused by pyogenic bacteria (most often staphylococcus, streptococcus). 2) An abscess is a localized cavity with a pyogenic capsule around it; a phlegmon is nonlocalized and diffuse, and the pus spreads along tissue spaces, fasciae and neurovascular bundles. 3) Complications: spread of pus, erosion of the vessel wall (arrosive bleeding), thrombophlebitis, sepsis; if chronic — AA amyloidosis. Treatment — evacuation of pus through wide incisions (drainage), antibiotics and detoxification.

Quick review

Table 5.5. *Key facts on Topic 5*

Question	Answer
Inflammation	A protective-adaptive vascular-mesenchymal reaction
Classic signs	Rubor, tumor, calor, dolor (Celsus) + functio laesa (Virchow)
Discoverer of phagocytosis	I.I. Mechnikov
Phases	Alteration → exudation → proliferation
The first mediator	Histamine (mast cells)
Mediators of fever	IL-1, TNF, PGE2
Mediators of pain	Bradykinin, PGE2
Chemotaxis	C5a, LTB4, IL-8, bacterial peptides
Opsonins	IgG, C3b
Extravasation	Selectins (rolling) → integrins/ICAM-1 (adhesion) → PECAM-1 (transmigration) → chemotaxis
The main acute cell	Neutrophil (6–24 hours), then macrophage (24–48 hours)
The most potent bactericidal system	NADPH oxidase + myeloperoxidase → HOCl⁻
Exudate protein	More than 30 g/L
Croupous	Superficial necrosis, detaches easily; columnar epithelium, serous membranes
Diphtheritic	Deep necrosis, difficult to detach; stratified squamous epithelium
“Hairy heart”	Fibrinous pericarditis (uremia, rheumatic fever, infarction)

Question	Answer
Abscess	Localized, pyogenic membrane
Phlegmon	Diffuse purulent inflammation
Empyema	Pus in a natural cavity
Hemorrhagic	Influenza, anthrax, plague
Catarrhal	Only mucous membranes, mucous exudate
Outcomes of acute inflammation	Resolution, scar, abscess, chronic inflammation

Glossary of terms

- **Inflammation** (Latin inflammatio) — a vascular-mesenchymal reaction that develops in response to various injuries and is aimed at eliminating the pathogenic agent.
- **Alteration** (Latin alteratio — change) — damage to tissues in the form of degeneration and necrosis of various degrees.
- **Exudation** (Latin exudatio) — escape of the fluid part of blood plasma and formed elements from the vessels.
- **Proliferation** (Latin proliferatio — growth) — multiplication of capillary endothelium and connective tissue cells.
- **Pinocytosis** — uptake of tissue fluid by a cell (macrophage, endothelium).
- **Endocytobiosis** — incomplete phagocytosis: survival of a microbe inside the phagocyte.
- **Abscess** (Latin abscessus — boil, purulent sore) — localized purulent inflammation forming a cavity. An acute abscess is separated from the surrounding tissue by a pyogenic membrane; the wall of a chronic abscess has two layers — pyogenic (inner) and fibrous (outer).
- **Phlegmon** (Greek phlegmone — inflammation) — diffuse purulent inflammation.
- **Empyema** (Greek en — inside, pyon — pus) — accumulation of pus in natural cavities or hollow organs (pleura, appendix, gallbladder, maxillary sinus).
- **Catarrhal inflammation** (Greek katarrheo — I flow down) — inflammation of a mucous membrane in which the exudate flows down together with mucus.
- **Croupous inflammation** (Scottish croup — membrane) — fibrinous inflammation with superficial necrosis, impregnation of the necrotic masses with fibrin and formation of an easily detachable membrane.
- **Diphtheritic inflammation** (Greek diphthera — leather membrane) — fibrinous inflammation with deep necrosis in mucous membranes and formation of a thick membrane that is difficult to detach.
- **Angina (tonsillitis)** (Latin ango — I strangle, I squeeze) — inflammation of the pharynx (tonsils).
- **Sialadenitis** (Greek sialon — saliva, aden — gland) — inflammation of the salivary glands.
- **Ichorosis** (Greek ichor — putrid fluid) — putrefactive (foul-smelling) inflammation.

TOPIC 6

Chronic inflammation. Types. Cell repair

Learning objectives

The student should know:

- the definition, causes and morphologic signs of chronic inflammation;
- the cells of chronic inflammation: macrophages (pathways of activation), lymphocytes, plasma cells, eosinophils, giant cells;
- the types of productive (proliferative) inflammation: interstitial, granulomatous, and inflammation forming polyps and condylomas;
- the definition, morphogenesis, classification and outcomes of granulomas;
- specific granulomas: the structure of the granulomas of tuberculosis, syphilis, leprosy and scleroma;
- the concept of regeneration (repair), its types, mechanisms and the factors influencing it;
- the structure of granulation tissue, the types of wound healing, scar formation;
- pathologic regeneration, sclerosis and cirrhosis.

The student should be able to:

- describe gross specimens of miliary tuberculosis, hepatic echinococcosis, gastric polyp, the liver in visceral syphilis, syphilitic aortic aneurysm and keloid scar;
- describe microscopic slides of a tuberculous granuloma, syphilitic gumma, foreign body granuloma, rhinoscleroma, interstitial myocarditis, granulation tissue and organization of a thrombus;
- distinguish specific granulomas by their structure;
- relate the types of wound healing to clinical situations.

6.1. The concept of chronic inflammation

Chronic inflammation is inflammation of prolonged duration (weeks, months, years) in which **active inflammation, tissue damage** and **attempts at healing (repair)** proceed simultaneously. It may follow acute inflammation or begin as chronic from the start (insidious, without symptoms).

Causes:

1. **Persistent infections** — microorganisms that are difficult to eradicate: Mycobacterium tuberculosis, Treponema pallidum (syphilis), the leprosy bacillus, fungi, some viruses (hepatitis B and C), parasites. They often elicit a **delayed-type hypersensitivity** (type IV) reaction and lead to **granulomatous** inflammation.
2. **Immune-mediated inflammatory diseases** — autoimmune diseases (rheumatoid arthritis, systemic lupus erythematosus, multiple sclerosis), allergic diseases (bronchial asthma), inflammatory bowel diseases (Crohn disease, ulcerative colitis).
3. **Prolonged exposure to endogenous or exogenous toxic agents** — silicon dioxide (silicosis), asbestos, atherosclerosis (lipids).
4. It has been established that **latent chronic inflammation** also underlies some **chronic diseases** (atherosclerosis, type 2 diabetes mellitus, obesity, Alzheimer disease).

Morphologic features of chronic inflammation (Robbins):

1. **Infiltration by mononuclear cells** — macrophages, lymphocytes, plasma cells.
2. **Tissue destruction** — by the causative agent or inflammatory cells.
3. **Attempts at healing** — formation of new vessels (**angiogenesis**) and **fibrosis** (growth of connective tissue).

Table 6.1. *Differences between acute and chronic inflammation*

Feature	Acute inflammation	Chronic inflammation
Onset	Fast (minutes, hours)	Slow (days)
Duration	Several days (up to 4–6 weeks)	Months, years
Main cells	Neutrophils	Macrophages, lymphocytes, plasma cells
Predominant phase	Exudation	Proliferation (productive)
Tissue damage and fibrosis	Usually mild and self-limited; little fibrosis	Often severe and progressive; marked fibrosis
Local and systemic signs	Prominent	Less prominent

6.2. Cells of chronic inflammation

Macrophages

Macrophages are the **dominant cells** of chronic inflammation. They are derived from blood **monocytes** (which become macrophages after entering the tissue) or are tissue-resident macrophages (Kupffer cells in the liver, alveolar macrophages in the lung, microglia in the brain, osteoclasts in bone). All tissue macrophages together make up the **mononuclear phagocyte system**.

Macrophages are activated in two ways:

Table 6.2. *Pathways of macrophage activation*

Feature	Classical activation (M1)	Alternative activation (M2)
Activators	Microbial products (endotoxin), IFN-gamma (from T lymphocytes)	IL-4, IL-13 (Th2 lymphocytes)
Main function	Killing of microbes (ROS, NO, lysosomal enzymes), amplification of inflammation (IL-1, TNF, IL-12)	Suppression of inflammation, repair and fibrosis (growth factors: PDGF, TGF-beta), angiogenesis
Result	Defense, but tissue damage	Repair, scar

Functions of macrophages in chronic inflammation: **phagocytosis** of microbes and dead tissue; **presentation** of antigen to T lymphocytes; production of **cytokines** that regulate inflammation; control of **repair and fibrosis** by stimulating fibroblasts and angiogenesis. Macrophages and T lymphocytes activate each other and maintain the chronicity of inflammation (IFN-gamma ↔ IL-12).

Lymphocytes, plasma cells and other cells

- **T lymphocytes** — CD4+ helper T cells regulate inflammation through cytokines: **Th1** (IFN-gamma — classical activation of macrophages; tuberculosis, autoimmune diseases), **Th2** (IL-4, IL-5, IL-13 — eosinophils and alternatively activated macrophages; allergy, parasites), **Th17** (IL-17 — recruits neutrophils).
- **B lymphocytes and plasma cells** — produce antibodies. The plasma cell has an eccentrically placed nucleus with "**cartwheel**" chromatin and basophilic cytoplasm; immunoglobulins can

accumulate in its cytoplasm to form **Russell bodies**. In long-standing inflammation, lymphocytes, plasma cells and antigen-presenting cells form **lymphoid follicles** (**tertiary lymphoid organs** — for example, in the synovium in rheumatoid arthritis, in Hashimoto thyroiditis).

- **Eosinophils** — numerous in IgE-mediated allergic reactions and **parasitic infections**; the **major basic protein** in their granules is toxic to parasites but also damages the epithelium.
- **Mast cells** — release histamine and other mediators and participate in allergic and chronic inflammation.
- **Neutrophils** — may also be present in chronic inflammation (for example, chronic osteomyelitis, chronic lung damage) — "**acute-on-chronic**" inflammation.

Multinucleated giant cells

Multinucleated cells 40–50 µm or more in diameter formed by fusion of macrophages (epithelioid cells):

- **Pirogov–Langhans cells** — the nuclei (20 or more) are arranged at the periphery of the cell in a **horseshoe or ring** shape; characteristic of the tuberculous granuloma but also found in other granulomas;
- **foreign body giant cells** — the nuclei are scattered **haphazardly** in the cytoplasm (often in the center); the cytoplasm may contain foreign particles;
- **Touton cells** — a ring of nuclei in the center and foamy (lipid-laden) cytoplasm at the periphery; in xanthomas and fat necrosis.

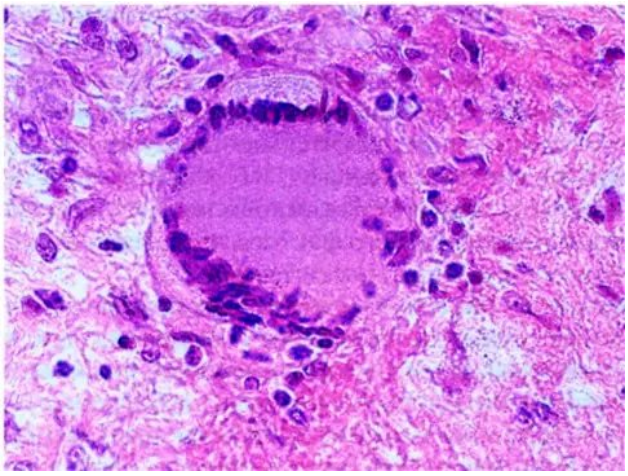


Figure 6.1. *Pirogov–Langhans giant cell: the nuclei are arranged in a horseshoe at the periphery of the cytoplasm (Strukov, Serov, Fig. 5-13)*

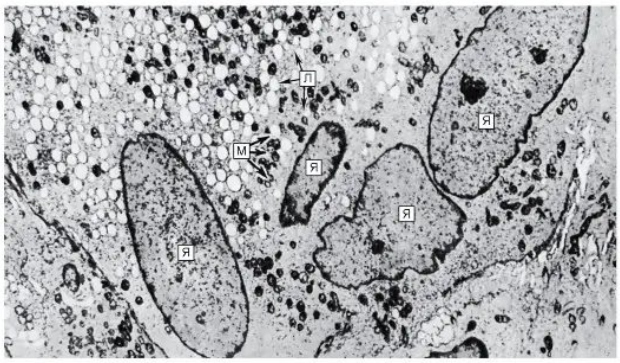


Figure 6.2. *Pirogov–Langhans giant cell (electron micrograph): many nuclei (N), mitochondria (M), lysosomes (L) (TMA study guide, 2018)*

6.3. Types of productive (proliferative) inflammation

In A.I. Strukov's classification, in **productive (proliferative) inflammation** cell proliferation predominates; it usually runs a chronic course. Four types are distinguished:

1. Interstitial (diffuse) inflammation — a **diffuse infiltrate** of lymphocytes, macrophages and plasma cells accumulates in the **stroma** of parenchymal organs (myocardium, liver, kidney, lung); degenerative and atrophic changes in the parenchyma. Examples: **interstitial myocarditis** (diphtheria, viral infections, rheumatic fever), interstitial nephritis, interstitial pneumonia, chronic hepatitis. Outcome — **diffuse sclerosis** of the organ (cardiosclerosis, nephrosclerosis, pneumosclerosis) and **cirrhosis**.

2. Granulomatous inflammation — accompanied by the formation of granulomas (in detail below).

3. Productive inflammation forming polyps and condylomata acuminata — in chronic inflammation of mucous membranes (or of surfaces lined by squamous epithelium), the stroma and epithelium grow and protrude outward:

- **inflammatory polyps** — in the mucosa of the nasal cavity, maxillary sinus, stomach, colon and cervix; polyps must be distinguished from tumors (adenomatous polyp);
- **condylomata acuminata** — in the squamous epithelium of the anus and external genitalia, most often associated with **papillomavirus** (types 6 and 11). **4. Productive inflammation around animal parasites** — around echinococcus, cysticercus and trichinella: macrophages, eosinophils, giant cells and a **fibrous capsule** form around the parasite (for example, the capsule around an echinococcal cyst).



Figure 6.4. *Hepatic echinococcosis: a white, round, fluid-filled cyst in the liver surrounded by a fibrous capsule (TMA study guide, 2018)*



Figure 6.3. *Gastric polyp: a soft, gray-reddish lesion with small papillae on a thin stalk on the greater curvature (TMA study guide, 2018)*

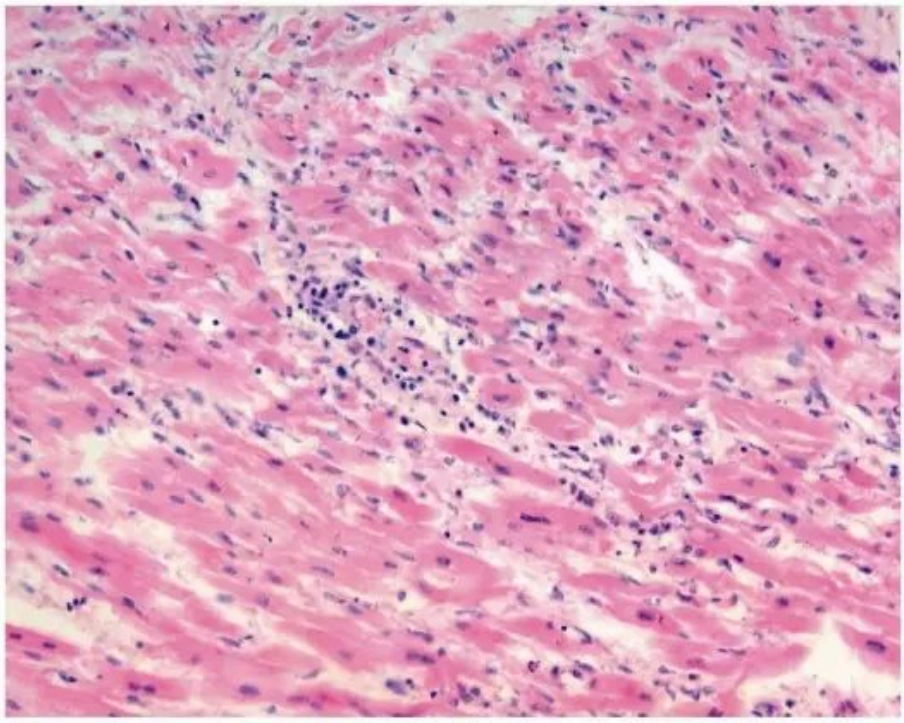


Figure 6.5. *Interstitial myocarditis: an infiltrate of lymphocytes, macrophages and plasma cells in the myocardial stroma; degenerative changes in cardiomyocytes (TMA study guide, 2018)*

6.4. Granulomatous inflammation

Granulomatous inflammation is a distinctive form of chronic inflammation whose main feature is the formation of **granulomas**. A **granuloma** (Latin granulum — grain, Greek -oma — tumor; i.e., a “nodule”) is a **localized focal** productive inflammation up to 1–2 mm in diameter, consisting mainly of a collection of **activated macrophages** (epithelioid cells), often surrounded by lymphocytes. Granuloma formation is the organism's attempt to **wall off** an agent that is difficult to eradicate.

Morphogenesis of the granuloma

1. Accumulation of **monocytes** in the focus of inflammation.
2. Their transformation into **macrophages** and formation of a macrophage granuloma.
3. Transformation of macrophages into **epithelioid cells** — large cells with pale pink, finely granular cytoplasm, indistinct borders and an elongated oval nucleus (resembling a “shoe sole”); they lie closely packed like epithelium. Epithelioid cells are poorly adapted to phagocytosis and better adapted to secretion (cytokines, enzymes).
4. Fusion of epithelioid cells (or macrophages) to form **multinucleated giant cells**.

By mechanism there are two types of granulomas:

- **Immune granulomas — type IV hypersensitivity** (T cell–mediated immune response): Th1 lymphocytes produce **IFN-gamma**, activate macrophages and transform them into epithelioid cells. In tuberculosis, syphilis, leprosy, sarcoidosis, fungal infections and Crohn disease.
- **Nonimmune granulomas (foreign body granulomas)** — without an immune response, around large inert foreign bodies (suture material, talc, coal particles, keratin, urate crystals). Epithelioid cells are few, **foreign body giant cells** are numerous; the foreign body is often visible in polarized light.

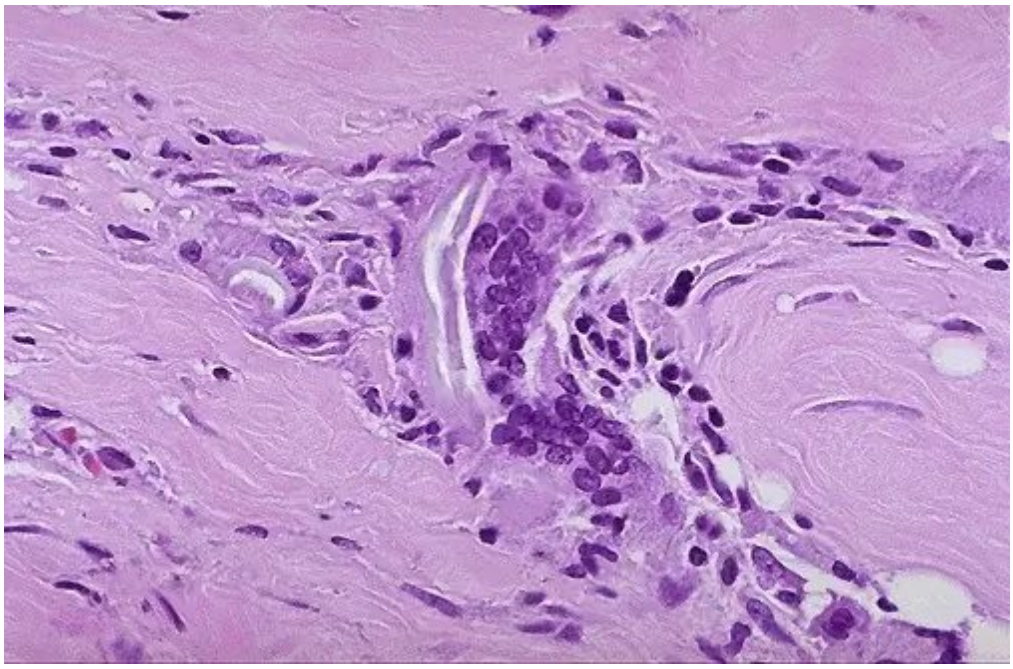


Figure 6.6. *Foreign body granuloma: in the center two giant cells surround foreign particles (TMA study guide, 2018)*

Classification of granulomas

Table 6.3. *Classification of granulomas (after V.V. Serov and V.V. Shekhter, simplified)*

Feature	Types
By etiology	Infectious (tuberculosis, syphilis, leprosy, scleroma, brucellosis, tularemia, fungi, parasites); noninfectious (foreign bodies, dust — silicosis, asbestosis; drugs); of unknown etiology (sarcoidosis, Crohn disease, granulomatosis with polyangiitis)
By pathogenesis	Immune (hypersensitivity) and nonimmune (foreign body) granulomas
By structure	Mature macrophage granuloma (simple phagocytoma); epithelioid cell granuloma (epithelioidocytoma); giant cell granuloma
By level of metabolism	Slow-turnover (inert foreign bodies) and fast-turnover (toxic agents — tuberculosis) granulomas
By specificity	Specific — the causative agent can be suspected from the structure (tuberculosis, syphilis, leprosy, scleroma); nonspecific — the others

Outcomes of granulomas: 1) **resorption** (the infiltrate disappears); 2) **fibrosis** — the granuloma turns into a scar (fibrous nodule); 3) **necrosis** — caseous necrosis in a tuberculous granuloma, later **calcification** (petrification); 4) **suppuration** (in fungal granulomas).

Specific granulomas

Specific inflammation is chronic inflammation caused by a particular agent (tuberculosis, syphilis, leprosy, scleroma) that forms granulomas of characteristic structure. Its general features: a specific causative agent; a **wavelike** chronic course (periods of exacerbation and remission); granulomas of characteristic structure; frequent **necrosis** (caseous or coagulative); a change in the tissue reaction depending on the immune status of the organism (with high resistance — a productive reaction, with low resistance — an alterative and exudative reaction).

> **CAUTION**
> The term **“specific inflammation”** is widely used in CIS textbooks (Strukov, Abdullakhodjaeva). In current international literature this concept is not singled out: these diseases are included in the group of **granulomatous inflammation**, and the final diagnosis is made on the basis of the granuloma structure together with **identification of the causative agent** (Ziehl–Neelsen staining, silver impregnation, PCR, culture). The histologic appearance of a granuloma only gives a direction.

1. The tuberculous granuloma (tubercle) — the causative agent is Mycobacterium tuberculosis (Koch bacillus). Structure (from the center to the periphery):

- in the center — a focus of **caseous necrosis** (a structureless, granular, pink mass);
- around it — a layer of **epithelioid cells** (arranged like a palisade);
- among the epithelioid cells — **Pirogov–Langhans giant cells**;
- at the periphery — a layer of **lymphocytes**, a small number of plasma cells and macrophages;
- there are **no blood vessels** inside the granuloma (silver impregnation shows a network of argyrophilic fibers);
- with Ziehl–Neelsen staining, **acid-fast red bacilli** are found in epithelioid and giant cells. A granuloma without caseous necrosis is an **epithelioid cell (proliferative) tubercle**. Granulomas merge to form large foci; when they are surrounded by a capsule — a **tuberculoma**. Scattered small (millet-sized) granulomas in the organs — **miliary tuberculosis**.

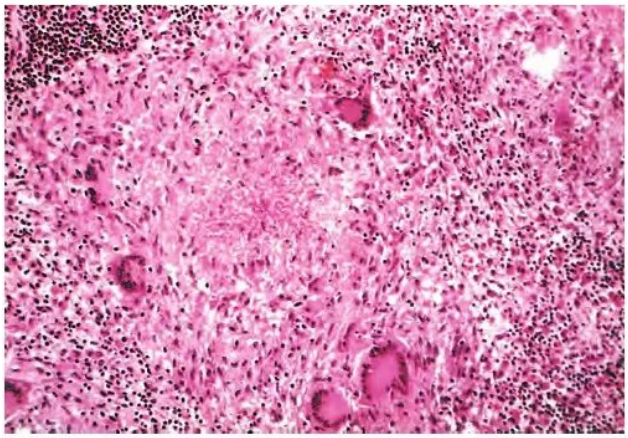


Figure 6.7. *Tuberculous granuloma: caseous necrosis in the center, surrounded by epithelioid cells, numerous Pirogov–Langhans giant cells and lymphocytes (Robbins, 9th ed., Fig. 3-23)*

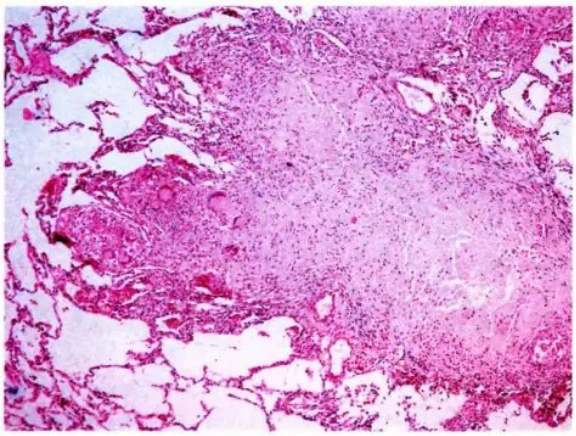


Figure 6.8. *Caseous necrosis in the center of a tuberculous granuloma (Strukov, Serov, Fig. 5-12)*

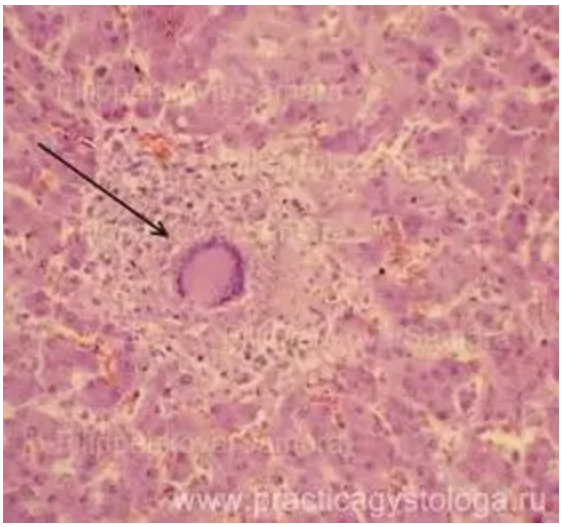


Figure 6.9. *Epithelioid cell granuloma in the liver (tuberculosis): epithelioid cells in the liver tissue, with a Pirogov–Langhans giant cell in the center (arrow) (TMA study guide, 2018)*

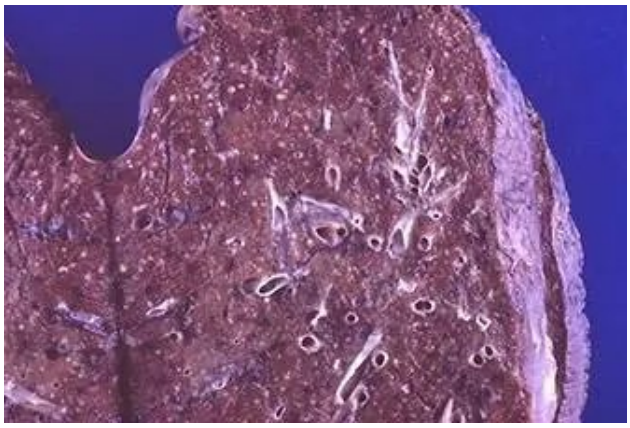


Figure 6.10. *Miliary tuberculosis of the lung: numerous hard, whitish-yellow, millet-sized nodules beneath the pleura and on the cut surface (TMA study guide, 2018)*

2. The syphilitic granuloma — gumma (Latin gummi — gum) — the causative agent is Treponema pallidum; characteristic of **tertiary syphilis**. Structure:

- in the center — an extensive focus of **coagulative (gummatous) necrosis**: a gummy, stretchy mass in which the outlines of vessels are sometimes preserved (unlike caseous necrosis);
- around it — **lymphocytes, plasma cells** (very numerous), macrophages, fibroblasts, few epithelioid and giant cells;
- **productive endovasculitis** — the walls of small vessels thicken and their lumen narrows (**obliterating endarteritis**), with plasma cells around them (**perivasculitis**);
- at the periphery — a vascular connective tissue capsule. Gummas occur in the liver, bones, skin, brain, heart and other organs; on healing they leave a **deep stellate scar** (in the liver — **“hepar lobatum” (lobulated liver)**). Besides gummas, tertiary syphilis also produces **gummatous infiltration** (diffuse) — most importantly **syphilitic mesaortitis**: a lymphoplasmacytic infiltrate around the vasa vasorum in the media of the ascending aorta and arch, destruction of elastic fibers and a “**shagreen skin**” (tree-bark) appearance of the intima; its complications — **aortic aneurysm** and aortic valve insufficiency.

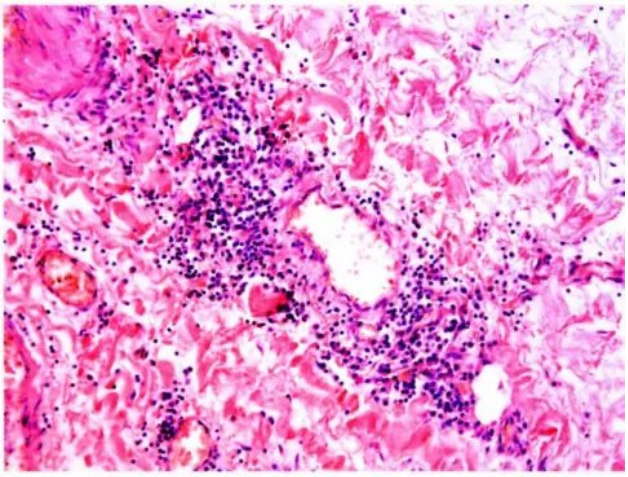


Figure 6.11. *Syphilitic granuloma (gumma): lymphocytes, plasma cells, fibroblasts and productive vasculitis in small vessels (Strukov, Serov, Fig. 5-14)*

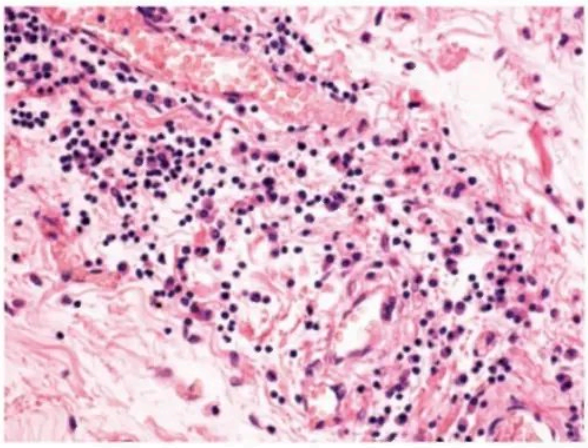


Figure 6.12. *Syphilitic mesaortitis: an infiltrate of lymphoid, plasma and epithelioid cells in the aortic wall around the vasa vasorum (TMA study guide, 2018)*



Figure 6.13. *The liver in visceral syphilis: the liver is enlarged, hard, coarsely bumpy and divided by scars into lobes ("hepar lobatum") (TMA study guide, 2018)*



Figure 6.14. *Syphilitic aneurysm of the aortic arch (a complication of syphilitic mesaortitis) (TMA study guide, 2018)*

3. The leprosy granuloma — leproma (Greek lepra — leprosy) — the causative agent is Mycobacterium leprae (Hansen bacillus). Structure: macrophages, epithelioid and giant cells, lymphocytes, plasma cells. Inside macrophages, numerous mycobacteria accumulate in the form of a "**cigarette pack**"; fat vacuoles appear in them, and large **Virchow (lepra) cells** form. Lepromas occur in the skin, mucous membranes, **peripheral nerves**, lymph nodes and internal organs.

Depending on the immune status, leprosy has two main forms:

- **lepromatous form** — low immunity (Th2 response): numerous Virchow cells **filled with mycobacteria**, skin nodules ("**leonine face**"), highly contagious;
- **tuberculoid form** — high immunity (Th1 response): tuberculosis-like epithelioid cell granulomas, **few** mycobacteria, mainly nerves are affected (loss of sensation, trophic ulcers).

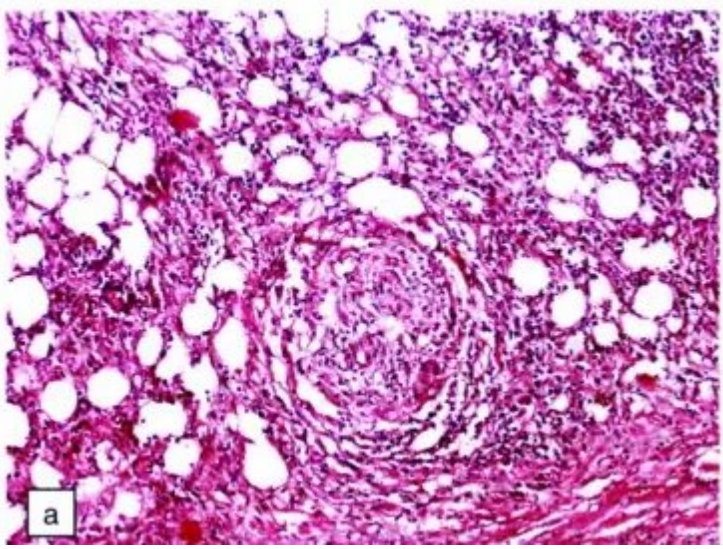


Figure 6.15. *Lepromatous leprosy — leproma: a nodule composed of macrophages, lymphocytes and plasma cells (Strukov, Serov, Fig. 5-15a)*

4. The scleroma granuloma — the causative agent is Klebsiella rhinoscleromatis (Volkovich–Frisch bacillus). Mainly in the upper respiratory tract — the nose (**rhinoscleroma**), larynx and trachea. Structure: plasma cells, epithelioid cells, lymphocytes, **hyaline (Russell) bodies**; most characteristic are **Mikulicz cells**: large macrophages with pale, foamy cytoplasm containing Volkovich–Frisch bacilli. Outcome — coarse sclerosis and scars, narrowing of the airways (stenosis, risk of asphyxia).

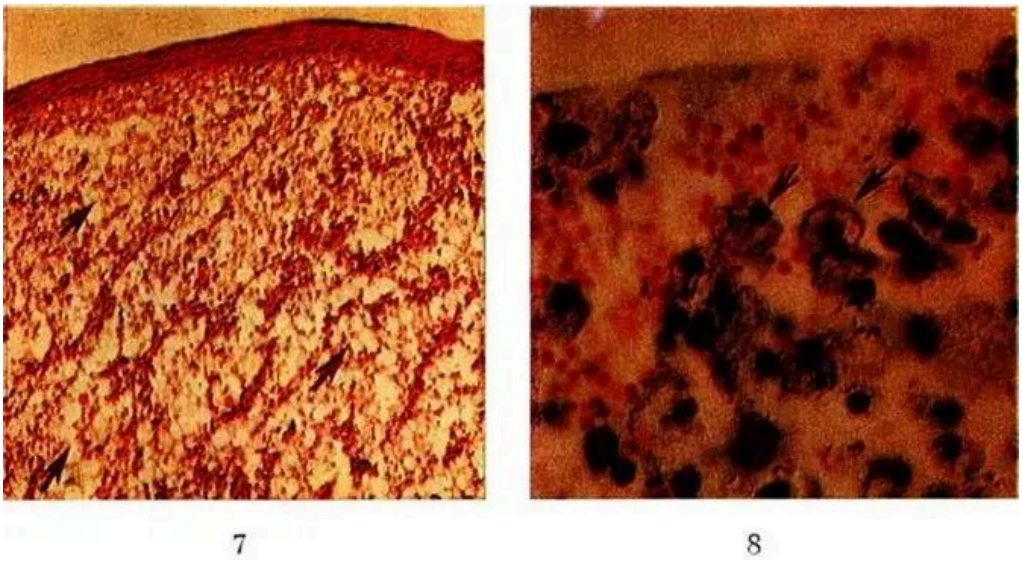


Figure 6.16. *Rhinoscleroma: a specific granuloma composed of various cells — plasma cells, Mikulicz cells, hyaline bodies (TMA study guide, 2018)*

Table 6.4. *Comparative characteristics of specific granulomas*

Disease	Causative agent	Granuloma	Necrosis	Characteristic cells
Tuberculosis	M. tuberculosis	Tubercle	Caseous (in the center)	Epithelioid cells, Pirogov–Langhans giant cells, lymphocytes; no vessels
Syphilis	T. pallidum	Gumma	Coagulative (gummatous), extensive	Many plasma cells, lymphocytes; productive endovasculitis
Leprosy	M. leprae	Leproma	Usually absent	Virchow cells (foamy macrophages filled with bacteria)

Disease	Causative agent	Granuloma	Necrosis	Characteristic cells
Scleroma	K. rhinoscleromatis	Scleroma granuloma	Absent	Mikulicz cells, plasma cells, Russell bodies
Sarcoidosis	Unknown	“Stamped-out” granulomas	Absent	Epithelioid cells, giant cells (Schaumann and asteroid bodies)

MEMORY AID

Remember the specific granulomas by their “**marker cell**”: **Tuberculosis — Pirogov–Langhans (horseshoe)**, **Syphilis — Plasma cell and vessel (endovasculitis)**, **Leprosy — Virchow (“cigarette pack”)**, **Scleroma — Mikulicz (foamy)**. “**Tuberculosis — Caseous, Syphilis — Gummatous, Sarcoidosis — Noncaseating**”.

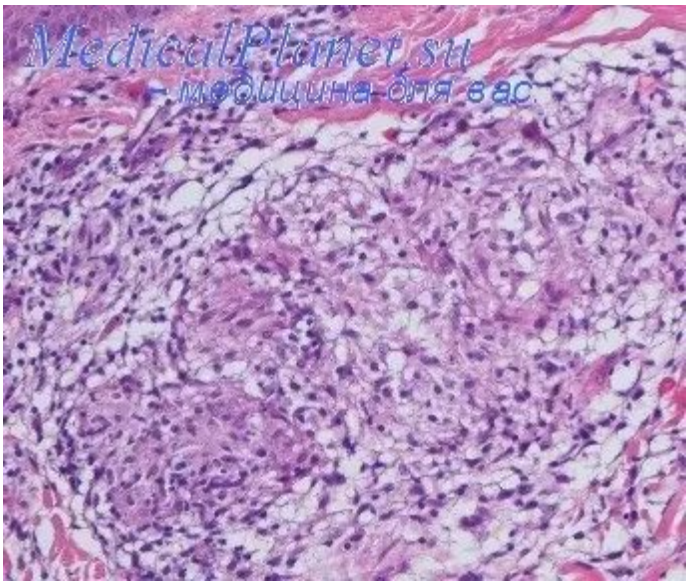


Figure 6.17. *Sarcoidosis: uniform noncaseating epithelioid cell granulomas, lymphoid cells and a few giant cells (TMA study guide, 2018)*

6.5. Repair (regeneration)

Regeneration (Latin regeneratio — renewal) is the **restoration (re-formation)** of the structural elements of cells and tissues. Restoration of damaged tissue is called **repair**, and it occurs in two ways: **regeneration** (by proliferation of cells) and **scar formation** (replacement by connective tissue).

Types of regeneration

1. **Physiological regeneration** — continuous renewal of worn-out cells throughout life: blood cells, epithelium of the skin and mucous membranes, endometrium (menstrual cycle).
2. **Reparative (restorative) regeneration** — restoration of tissue damaged by pathological processes. It is of two kinds:

– **restitution (complete regeneration)** — the defect is filled with **exactly the same** tissue; mainly in labile and stable tissues (epithelium, connective tissue, bone) and when the damage is small; – **substitution (incomplete regeneration)** — the defect is filled with **connective tissue**

(scar), while the remaining parenchymal cells enlarge (**regenerative hypertrophy**) and maintain the function of the organ. For example, in myocardial infarction the necrotic area is replaced by a scar, and the surrounding cardiomyocytes hypertrophy. 3. **Pathologic regeneration** — a disturbance of the

regeneration process: regeneration is **excessive** (hyperregeneration) or **insufficient** (hyporegeneration), or the tissue turns into **another type** (metaplasia). **Forms of regeneration** (D.S. Sarkisov): **cellular regeneration** — an increase in the number of cells (by mitosis; epithelium, connective tissue, bone); **intracellular regeneration** — an increase in the number and size of intracellular structures (organelles) (heart, skeletal muscles, neurons — nondividing cells). In most organs both forms occur together.

Mechanisms of regeneration

Regeneration is controlled by **growth factors** and the **extracellular matrix (ECM)**:

- **growth factors** — stimulate cell proliferation, migration and differentiation: epidermal growth factor (**EGF**) and transforming growth factor alpha (**TGF-alpha**) — epithelium; platelet-derived growth factor (**PDGF**) — migration of fibroblasts, smooth muscle cells and macrophages; vascular endothelial growth factor (**VEGF**) — **angiogenesis**; fibroblast growth factor (**FGF**) — angiogenesis and fibroblasts; **hepatocyte growth factor (HGF)** — liver regeneration; **transforming growth factor beta (TGF-beta)** — the main factor of **fibrosis** (increases collagen synthesis and reduces its breakdown) and it suppresses inflammation;
- **extracellular matrix** — collagen, elastin, fibronectin, laminin, proteoglycans; it serves as a scaffold for the tissue and stores growth factors. **If the basement membrane and scaffold are preserved**, complete regeneration is possible; if the scaffold is destroyed, a scar forms;
- **stem (cambial) cells** — tissue stem cells (for example, intestinal crypts, the basal layer of the skin, hepatic ductal cells).

Regeneration of various tissues

- **Blood** — physiological regeneration is very active; after blood loss the bone marrow becomes hyperplastic, the yellow marrow of the long bones turns into red marrow, and foci of hematopoiesis outside the marrow (**extramedullary hematopoiesis**) appear.
- **Blood vessels** — capillaries are restored by **budding** (endothelial proliferation — angiogenesis); large vessels regenerate only in their inner layer, and defects of the media and adventitia are replaced by a scar.
- **Connective tissue** — through **granulation tissue** (below).
- **Epithelium** — stratified squamous epithelium (skin) is completely restored from the edges of the defect and from the basal layer; the epithelium of mucous membranes (stomach, intestine) is rapidly restored from cambial cells in the crypts and glands.
- **Liver** — has a high capacity for regeneration (**restitution**): if 2/3 of the liver is removed, the remaining part reaches its original mass within a few weeks. But in chronic injury regeneration is disturbed: pseudolobules form and **cirrhosis** develops.
- **Bone** — a **bony callus** forms at the fracture site: first fibrin and granulation tissue (**preliminary connective tissue callus**), then an **osteoid-cartilaginous callus**, then a **coarse-fibered bony callus**, and finally lamellar bone. If the fragments are mobile, much cartilage forms and healing slows; if connective tissue remains between the fragments, a **false joint (pseudarthrosis)** forms.
- **Skeletal muscles** — if the sarcolemma is preserved, they are restored by **satellite cells**; large defects are replaced by a scar.
- **Cardiac muscle** — cardiomyocytes do not divide; the defect is replaced by a **scar**, with intracellular regeneration and **regenerative hypertrophy** in the remaining cardiomyocytes.

- **Nervous tissue** — in the central nervous system neurons are not restored (the defect is replaced by **gliosis** — a glial scar or a cyst). In peripheral nerves, if the neuron body is preserved, the axon is restored by growing from its central end (1–2 mm per day); if the path of growth is blocked, an **amputation neuroma** (a nodule of nerve fibers and connective tissue) forms.

Granulation tissue and scar formation

Granulation tissue is young, newly formed connective tissue **rich in capillaries and fibroblasts**. Its gross appearance: a **red-pink, soft, granular** (Latin granulum — grain; the granules are capillary loops) surface that bleeds easily. Granulation tissue begins to appear **3–5 days** after injury.

The **six layers** of granulation tissue (after N.N. Anichkov, from the surface to the depth):

1. **Superficial leukocytic-necrotic layer** — leukocytes, debris, fibrin.
2. **Superficial layer of vascular loops** — vertically arranged capillary loops with polyblasts (various cells) between them.
3. **Layer of vertical vessels** — perivascular fibroblasts.
4. **Maturing layer** — fibroblasts begin to lie horizontally, and collagen fibers form.
5. **Layer of horizontal fibroblasts**.
6. **Fibrous layer** — mature connective tissue.

Stages of scar formation:

1. **Angiogenesis** — new capillaries bud from existing vessels (under the action of VEGF); the new capillaries are “leaky” — which is why granulation tissue is edematous.
2. **Migration and proliferation of fibroblasts**, synthesis of collagen and other ECM components (under the action of TGF-beta, PDGF, FGF).
3. **Remodeling** — collagen matures and is reorganized, the number of vessels decreases, and granulation tissue turns into a **dense, whitish, poorly vascularized scar**. The balance between collagen synthesis and degradation (matrix metalloproteinases) determines the size of the scar.

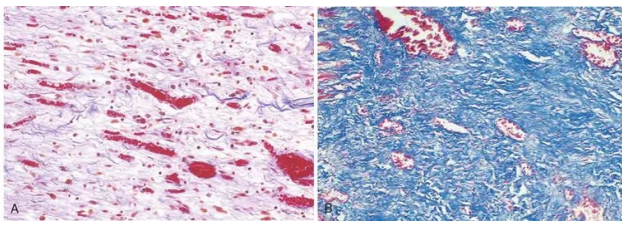


Figure 6.18. *Granulation tissue and scar: A — young granulation tissue with numerous capillaries, edema and inflammatory cells; B — mature scar: dense collagen fibers (trichrome stain — blue), few vessels (Robbins, 9th ed., Fig. 3-27)*

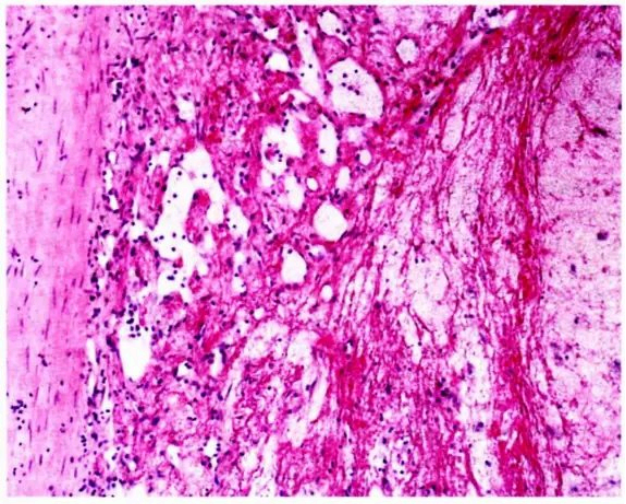


Figure 6.19. *Organization and canalization of a thrombus in a vein: granulation tissue has grown into the thrombotic mass (Strukov, Serov, Fig. 8-1)*

Wound healing

Healing of a skin wound is the classic example of tissue repair. Four types are distinguished:

1. **Direct closure of an epithelial defect** — a superficial defect involving only the epithelium is closed by ingrowth of epithelium (erosions of the cornea and mucous membranes).
2. **Healing under a scab** — a superficial defect is covered with a scab of dried blood, fibrin and lymph; epithelium grows beneath it, and the scab falls off in 3–5 days.
3. **Healing by first intention (primary union)** — wounds with closely apposed edges, **clean (uninfected)**, with little tissue loss (for example, a sutured surgical incision): the incisional gap fills with fibrin; neutrophils appear on day 1, macrophages and granulation tissue by day 3, collagen bridges by day 5, and by the 2nd week the wound is completely united by a scar; the epithelium covers the defect in 24–48 hours. The result is a **thin, almost invisible scar**. 4. **Healing by second intention (secondary union)** — wounds with a **large tissue defect**, widely separated edges, **infection** and much necrotic tissue: intense inflammation and abundant **granulation tissue** form; **wound contraction** occurs due to **myofibroblasts** (within 6 weeks the defect may shrink to 5–10% of its original size); the result is a **wide, coarse scar**.

REMEMBER

The strength of a wound scar: after the sutures are removed (at the end of the 1st week) it equals only **10%** of the strength of normal skin; it increases rapidly over the next 4 weeks and reaches **70–80%** by the 3rd month, but never reaches 100%.

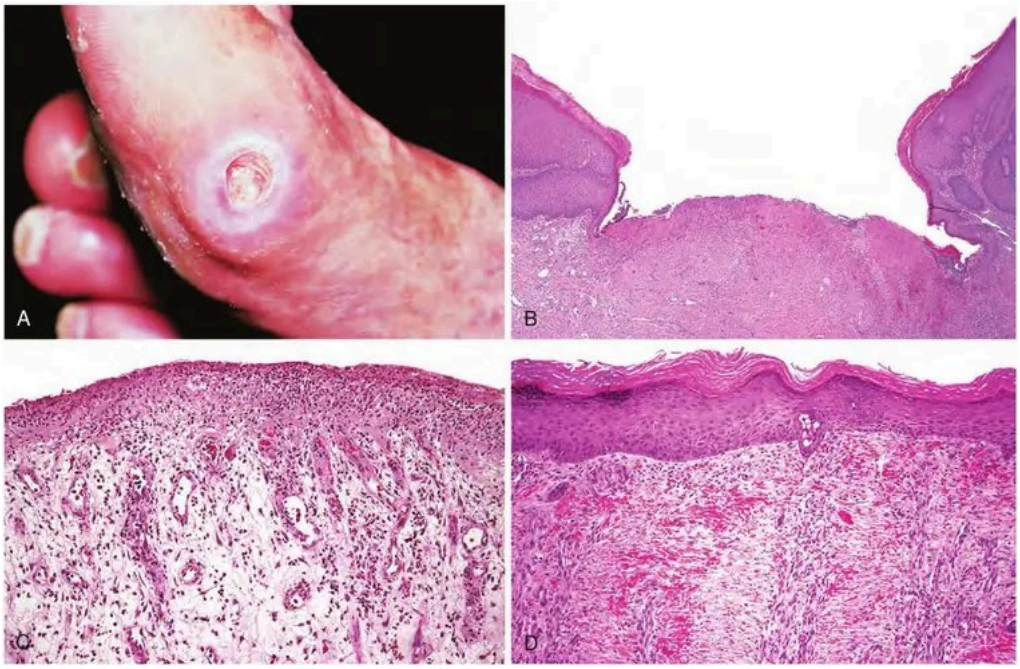


Figure 6.20. *A healing skin ulcer in a diabetic patient: A — a pressure ulcer (wide defect); B — thin epithelium between the ulcer edges and extensive granulation tissue in the dermis; C — continuing epithelialization; D — wound contraction (Robbins, 9th ed., Fig. 3-30)*

Factors influencing repair

- **Local factors: infection** (the most important factor — it prolongs inflammation), mechanical factors (motion of the wound edges), **inadequate blood supply** (atherosclerosis, varicose veins, diabetes), **foreign bodies** (sutures, bone fragments), the type, size and location of the wound.
- **Systemic factors: nutrition** (protein, **vitamin C** — necessary for collagen synthesis, zinc deficiency), **diabetes mellitus**, **glucocorticoids** (suppress inflammation and TGF-beta, reducing fibrosis), age (healing slows in the elderly), immune status.

Pathologic regeneration and disorders of repair

- **Keloid scar** (Greek chele — crab claw) — excessive collagen formation: the scar **extends beyond** the wound margins and bulges like a tumor, and it tends to recur. There is a hereditary predisposition (more common in people of African descent). Microscopically — broad, hyalinized collagen bundles.
- **Hypertrophic scar** — collagen is excessive, but the scar does not extend beyond the wound margins and may shrink with time.
- **"Proud flesh"** (exuberant granulation tissue) — granulation tissue protrudes above the skin level and interferes with epithelialization.
- **Contractures** — scars after burns contract and limit joint movement.
- **Insufficient repair** — wound opening (**wound dehiscence**, especially in the abdominal wall), chronic (trophic) ulcers.
- **Desmoid (aggressive fibromatosis)** — tumor-like proliferation of fibroblasts, a borderline tumor.
- **False joint** and **amputation neuroma** (above).

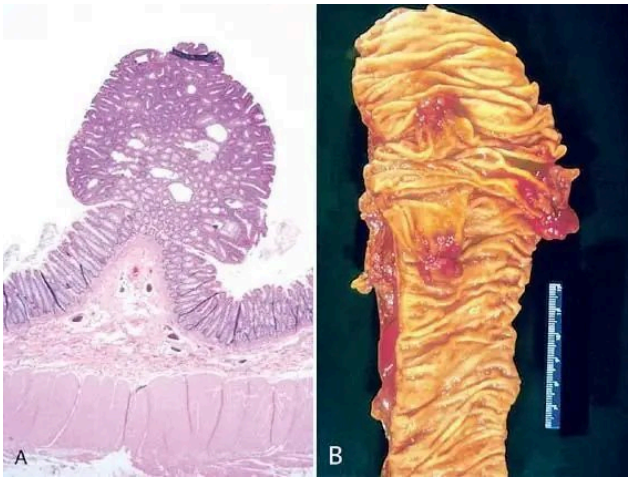


Figure 6.22. *Adenomatous polyp of the intestine: A — microscopic slide; B — gross specimen (TMA study guide, 2018)*



Figure 6.21. *Keloid: A — a bulging scar on the skin resulting from excessive accumulation of collagen; B — a thick mass of connective tissue in the dermis (Robbins, 9th ed., Fig. 3-32)*

Organization, encapsulation, sclerosis and cirrhosis

- **Organization** — replacement of necrotic foci, thrombi, fibrinous exudate (fibrinous inflammation) or a hematoma by connective tissue. For example, **organization and canalization of a thrombus** — granulation tissue grows into the thrombotic mass from the wall, new vascular channels appear in it, and blood flow is partially restored (Topic 7).
- **Encapsulation** — formation of a connective tissue capsule around a nonresorbable dead mass, foreign body or parasite.
- **Sclerosis** (Greek sklerosis — hardening) — growth of connective tissue in an organ, displacing the parenchyma and making the organ firm (cardiosclerosis, nephrosclerosis, pneumosclerosis). Its causes: chronic inflammation (interstitial inflammation), organization of necrotic foci, chronic hypoxia, activation of fibroblasts.
- **Cirrhosis** (Greek kirrhos — tawny, orange-yellow) — the final stage, in which sclerosis is accompanied by **restructuring** of the parenchyma (pseudolobules, regenerative nodules) and **deformation** of the organ (liver cirrhosis, pneumocirrhosis).

6.6. Clinicopathological section

A granuloma has been found — what is the next step?

If **epithelioid cell granulomas** are found in a lymph node biopsy, the pathologist seeks answers to the following questions:

- **Is there necrosis?** Granulomas with caseous necrosis — first of all **tuberculosis** (or a fungal infection); uniform, noncaseating, "stamped-out" granulomas — **sarcoidosis** is more likely.
- **Is the causative agent found? Ziehl–Neelsen** staining (acid-fast bacilli), PAS and Grocott silver stain (fungi), PCR (mycobacterial DNA).
- **Is there a foreign body?** It is seen in polarized light (talc, sutures). The answers to these questions completely change the treatment: in tuberculosis — antituberculosis drugs; in sarcoidosis — glucocorticoids (giving glucocorticoids in tuberculosis is dangerous — the infection spreads!).

> **CLINICAL SIGNIFICANCE**
> **Chronic inflammation and cancer** are interrelated: tumors develop more often against a background of long-standing chronic inflammation — Helicobacter pylori gastritis — gastric adenocarcinoma and MALT lymphoma; chronic viral hepatitis B and C — hepatocellular carcinoma; ulcerative colitis — colorectal cancer; Barrett esophagus — adenocarcinoma; sinus tracts of chronic osteomyelitis — squamous cell carcinoma. Mechanism: continuous cell proliferation, DNA damage by ROS, cytokines and growth factors.

Wound healing and diabetes mellitus

In diabetes mellitus, wounds heal slowly and become infected easily. The causes: **micro- and macroangiopathy** (reduced blood supply), **neuropathy** (because of loss of sensation the patient does not feel the injury — the "diabetic foot"), **impaired neutrophil function** in hyperglycemia (chemotaxis and phagocytosis are impaired), and impaired collagen synthesis. As a result, chronic trophic ulcers and gangrene develop (Topic 2).

Practical skills

Skill 1. Gross specimens

1. **"Miliary tuberculosis of the lung".** Numerous millet-sized (1–2 mm), whitish-gray or yellowish, hard nodules beneath the pleura and on the cut surface of the lung. Conclusion: granulomatous (specific) inflammation — hematogenously disseminated tuberculosis.
2. **"The liver in visceral syphilis".** The liver is enlarged, hard and coarsely bumpy, divided by deep stellate scars into irregular lobes ("hepar lobatum"). Conclusion: tertiary syphilis — gummas and scars.
3. **"Syphilitic aortic aneurysm (syphilitic mesaortitis)".** The ascending aorta and arch are saccularly dilated; the intima is uneven, with a "shagreen skin" appearance and deep scarred folds. Conclusion: a complication of productive inflammation (syphilitic mesaortitis).
4. **"Echinococcosis of the liver (or kidney)".** White, round, fluid-filled cyst(s) in the organ, separated from the surroundings by a thick fibrous capsule. Conclusion: productive inflammation around animal parasites (encapsulation).
5. **"Gastric (intestinal) polyp".** A soft, gray-reddish lesion on a thin stalk on the mucosal surface. Conclusion: productive inflammation forming polyps (or an adenomatous polyp — histologic

examination is necessary).

6. **“Keloid scar”.** A dense, smooth, raised scar on the skin extending beyond the wound margins. Conclusion: pathologic regeneration.

Skill 2. Microscopic slides

1. **“Tuberculous granuloma (lung or lymph node)”** (hematoxylin and eosin). Structureless pink caseous necrosis in the center, surrounded by a layer of epithelioid cells with Pirogov–Langhans giant cells among them (nuclei in a horseshoe), and lymphocytes at the periphery; no vessels inside the granuloma. Conclusion: tuberculous granuloma.
2. **“Syphilitic gumma”** or **“syphilitic mesaortitis”** (hematoxylin and eosin). Extensive coagulative necrosis surrounded by lymphocytes, numerous plasma cells and fibroblasts; productive endovasculitis in small vessels. In the aorta — a lymphoplasmacytic infiltrate around the vasa vasorum in the media, with destroyed elastic fibers. Conclusion: productive inflammation in syphilis. 3. **“Rhinoscleroma”** (hematoxylin and eosin). Plasma cells, lymphocytes, hyaline bodies and large foamy Mikulicz cells beneath the nasal mucosa. Conclusion: scleroma granuloma. 4. **“Foreign body granuloma”** (hematoxylin and eosin). Foreign body giant cells (haphazard nuclei), macrophages and fibroblasts around a foreign body (suture). Conclusion: nonimmune granuloma. 5. **“Interstitial myocarditis”** (hematoxylin and eosin). A diffuse lymphocytic-macrophage infiltrate and edema in the myocardial stroma; degeneration of cardiomyocytes. Conclusion: interstitial productive inflammation. 6. **“Granulation tissue”** (hematoxylin and eosin). Numerous vertically oriented thin-walled capillaries with fibroblasts, macrophages, lymphocytes, plasma cells and neutrophils between them; edematous intercellular substance. Conclusion: granulation tissue (repair). 7. **“Organization of a thrombus”** (hematoxylin and eosin). Granulation tissue consisting of fibroblasts and capillaries has grown into the thrombotic mass from the wall; new channels lined by endothelium. Conclusion: organization and canalization.

Review questions

1. Define chronic inflammation. What are its causes and morphologic features?
2. State the differences between acute and chronic inflammation.
3. How do the classical and alternative activation of macrophages differ?
4. The role of lymphocytes, plasma cells and eosinophils in chronic inflammation.
5. Describe the types of giant cells.
6. List the types of productive inflammation and give examples.
7. What is a granuloma? Its morphogenesis and outcomes.
8. The difference between immune and nonimmune granulomas.
9. General features of specific inflammation.
10. Compare the structure of the granulomas of tuberculosis, syphilis, leprosy and scleroma.
11. Types and forms of regeneration. What are restitution and substitution?
12. The growth factors that regulate regeneration and the role of the extracellular matrix.
13. The structure of granulation tissue and the stages of scar formation.
14. Describe the types of wound healing. The difference between healing by first and second intention.
15. Factors influencing repair and the types of pathologic regeneration.

Test questions

1. The main cell of chronic inflammation:
   A) Basophil
   B) Platelet
   C) Eosinophil
   D) Macrophage
   E) Neutrophil

2. The main cytokine that activates macrophages by the classical (M1) pathway:
   A) IL-10
   B) IFN-gamma
   C) IL-13
   D) IL-4
   E) TGF-beta

3. A giant cell with nuclei arranged in a horseshoe at the periphery:
   A) Virchow cell
   B) Mikulicz cell
   C) Touton cell
   D) Foreign body giant cell
   E) Pirogov–Langhans cell

4. The basis of immune granuloma formation:
   A) Complement activation
   B) Type III hypersensitivity
   C) Type IV (delayed) hypersensitivity
   D) Type II hypersensitivity
   E) Type I hypersensitivity

5. The type of necrosis in the center of a tuberculous granuloma:
   A) Liquefactive
   B) Fibrinoid
   C) Fat necrosis
   D) Coagulative (gummatous)
   E) Caseous

6. Characteristic of a syphilitic gumma:
   A) Caseous necrosis and absence of vessels
   B) Numerous plasma cells and productive endovasculitis
   C) Noncaseating granulomas
   D) Virchow cells
   E) Mikulicz cells

7. Foamy macrophages filled with bacteria in leprosy are called:
   A) Pirogov–Langhans cells
   B) Virchow cells
   C) Epithelioid cells
   D) Mikulicz cells
   E) Anichkov cells

8. The cell characteristic of the scleroma granuloma:
   A) Mikulicz cell
   B) Touton cell
   C) Reed–Sternberg cell
   D) Virchow cell
   E) Councilman body

9. Noncaseating epithelioid cell granulomas of uniform structure are characteristic of:
   A) Tuberculosis
   B) Echinococcosis
   C) Leprosy
   D) Syphilis
   E) Sarcoidosis

10. Complete restoration of a defect by the same tissue:
    A) Substitution
    B) Organization
    C) Restitution
    D) Metaplasia
    E) Encapsulation

11. The main composition of granulation tissue:
    A) Neutrophils only
    B) Epithelioid cells and giant cells
    C) Smooth muscle cells
    D) Numerous new capillaries and fibroblasts
    E) Mature collagen fibers and few vessels

12. The main growth factor that stimulates angiogenesis:
    A) TGF-alpha
    B) Erythropoietin
    C) VEGF
    D) EGF
    E) Insulin

13. The main cytokine (growth factor) of fibrosis:
    A) IL-8
    B) TNF
    C) Histamine
    D) TGF-beta
    E) IL-1

14. Characteristic of healing by second intention:
    A) Large defect, much granulation tissue, contraction due to myofibroblasts, wide scar
    B) Little granulation tissue
    C) The scar is almost invisible
    D) Only in mucous membranes
    E) A clean wound with closely apposed edges

15. A scar that grows beyond the wound margins:
    A) Keloid
    B) Hypertrophic scar
    C) Contracture
    D) “Proud flesh”
    E) Amputation neuroma

Clinical cases

Case 1

A 28-year-old man has prolonged low-grade fever, night sweats, weight loss and enlarged cervical lymph nodes. Lymph node biopsy: numerous granulomas with a structureless, granular, pink mass in the center, surrounded by epithelioid cells and giant cells with nuclei arranged in a horseshoe; lymphocytes at the periphery.

Questions: 1) Your histologic conclusion? 2) State the structure of the granuloma and the name of the giant cell. 3) How can the diagnosis be confirmed, and what is the mechanism of granuloma formation?

Answer: 1) Granulomatous epithelioid cell lymphadenitis with caseous necrosis — tuberculosis. 2) Caseous necrosis in the center, surrounded by epithelioid cells with Pirogov–Langhans giant cells among them and lymphocytes at the periphery; no vessels inside the granuloma. 3) Ziehl–Neelsen staining (acid-fast red bacilli), PCR, culture, clinical tests (Mantoux, QuantiFERON). Mechanism — type IV (delayed) hypersensitivity: Th1 lymphocytes produce IFN-gamma and activate macrophages, which turn into epithelioid and giant cells; mycobacteria survive inside macrophages because of incomplete phagocytosis.

Case 2

A 55-year-old man died suddenly of cardiac tamponade. At autopsy the ascending aorta is saccularly dilated (aneurysm) and ruptured; the intima is uneven and folded, with a “shagreen skin” appearance. Microscopically: an infiltrate of lymphocytes and numerous plasma cells around the vasa vasorum in the aortic media, destroyed elastic fibers, scar tissue; endarteritis in small vessels.

Questions: 1) Your diagnosis? 2) What period of the disease is this, and what is the process in the aorta called? 3) Specific changes of syphilis in other organs and the structure of a gumma.

Answer: 1) A complication of syphilitic mesaortitis — rupture of an aortic aneurysm, cardiac tamponade. 2) Tertiary (visceral) syphilis; the process in the aorta is gummatous infiltration (productive mesaortitis): the lymphoplasmacytic infiltrate around the vasa vasorum, destruction of the elastic framework and scar formation in its place weaken the aortic wall and lead to an aneurysm. 3) Gumma — extensive coagulative (gummatous) necrosis in the center surrounded by lymphocytes, plasma cells, fibroblasts, few epithelioid and giant cells, and productive endovasculitis. Gummas occur in the liver (“hepar lobatum”), bones, skin, brain and other organs.

Case 3

After abdominal surgery, a 40-year-old woman's surgical wound became infected; the sutures were removed and the wound was treated open. A month later the wound healed but left a wide, coarse scar. A year later a thick, hard, itchy raised lesion extending beyond the wound margins appeared in the skin around the scar.

Questions: 1) By which intention did the wound heal? 2) Its differences from healing by first intention and its mechanism. 3) What is the subsequent change in the scar called, and how does it differ?

Answer: 1) By second intention. 2) In healing by first intention the wound is clean with closely apposed edges, there is little granulation tissue, and a thin scar remains. In healing by second intention, infection and a large defect cause intense inflammation and abundant granulation tissue; the wound contracts due to myofibroblasts (contraction), and a wide, coarse scar remains. 3) Keloid — pathologic regeneration resulting from excessive collagen synthesis (TGF-beta and other factors, hereditary predisposition); it differs from a hypertrophic scar in that it grows beyond the wound margins, does not regress spontaneously and tends to recur after excision.

Quick review

Table 6.5. *Key facts on Topic 6*

Question	Answer
Features of chronic inflammation	Mononuclear infiltrate + tissue destruction + angiogenesis and fibrosis
Main cell	Macrophage
M1 / M2 macrophages	IFN-gamma — killing of microbes / IL-4, IL-13 — repair and fibrosis
Types of productive inflammation	Interstitial, granulomatous, polyps and condylomas, around parasites
Basis of the granuloma	Epithelioid cells (activated macrophages)
Mechanism of the immune granuloma	Type IV hypersensitivity (Th1 → IFN-gamma)
Pirogov–Langhans	Horseshoe nuclei; tuberculosis
Foreign body giant cell	Haphazard nuclei
Tuberculous granuloma	Caseous necrosis, epithelioid cells, P–L cells, lymphocytes, no vessels
Gumma	Coagulative necrosis, plasma cells, endovasculitis
Leproma	Virchow cells
Scleroma	Mikulicz cells, Russell bodies
Sarcoidosis	Noncaseating granulomas
Restitution / substitution	Complete restoration / replacement by a scar
Angiogenesis factor	VEGF
Fibrosis factor	TGF-beta
Granulation tissue	Capillaries + fibroblasts; from day 3–5; 6 layers
First intention	Clean wound with closely apposed edges, thin scar
Second intention	Large, infected wound; myofibroblasts — contraction; wide scar
Scar strength	10% at the end of the 1st week, 70–80% at 3 months
Keloid	A recurring scar that grows beyond the margins

Glossary of terms

- **Granuloma** (Latin granulum — grain, Greek -oma — tumor) — a localized focus of productive inflammation; mainly a collection of activated macrophages.
- **Gumma** (Latin gummi — gum) — the specific infectious granuloma in syphilis.

- **Leproma** (Greek lepra — leprosy, -oma — tumor) — the specific infectious granuloma in leprosy.
- **Tuberculoma** (Latin tuberculum — small knob) — a focus of caseous necrosis in the lung surrounded by a connective tissue capsule, with elements of calcification.
- **Infiltrate** (Latin in — into, filtratum — filtered) — entry of various cellular elements into a tissue, making it firm.
- **Condyloma** (Greek kondyloma — growth) — papillary growth of squamous epithelium and the underlying stroma.
- **Polyp** (Greek poly — many, pus — foot) — papillary (pedunculated) outgrowth of the mucosal epithelium and the underlying tissue.
- **Specific inflammation** (Latin species — kind) — chronic inflammation caused by a particular biological agent and having characteristic morphologic features (not singled out in current international classifications).
- **Sclerosis** (Greek sklerosis — hardening) — growth of connective tissue in an organ, displacing the parenchyma.
- **Cirrhosis** (Greek kirrhos — tawny) — growth of connective tissue and restructuring of the parenchyma with deformation of the organ.
- **Myofibrosis** (Greek mys — muscle, Latin fibrosus — fibrous) — small-focal cardiosclerosis.
- **Fibrosis** (Latin fibra — fiber) — excessive formation of connective tissue fibers.
- **Stenosis** (Greek stenos — narrow) — narrowing of a canal or orifice.
- **Aneurysm** (Greek aneurysma — dilatation) — a localized bulging dilatation of the wall of the heart or a vessel.
- **Regeneration** (Latin regeneratio — renewal) — restoration of the structural elements of cells and tissues.
- **Restitution** — complete restoration of a defect by the same tissue.
- **Substitution** — replacement of a defect by connective tissue (scar).
- **Organization** — replacement of necrosis, a thrombus or exudate by connective tissue.
- **Keloid** (Greek chele — crab claw) — a collagen-rich pathologic scar that grows beyond the wound margins.

TOPIC 7

Hemodynamic disorders. Thrombosis, embolism, infarction, shock

Learning objectives

The student should know:

- the classification of disorders of blood and lymph circulation;
- arterial and venous hyperemia: types, causes, morphology and outcomes; “nutmeg liver”, brown induration of the lung;
- anemia (ischemia), and types of bleeding and hemorrhage;
- edema, its mechanisms and types;
- hemostasis and thrombosis: Virchow's triad, types of thrombi, outcomes; DIC;
- types of embolism and their outcomes, pulmonary thromboembolism;
- infarction: definition, types (white, red, white with a red rim), morphology, outcomes;
- shock: types, stages and “shock organs”.

The student should be able to:

- describe gross specimens of “nutmeg liver”, brown induration of the lung, cerebral hemorrhage, a thrombus in the aortic wall, pulmonary thromboembolism, ischemic infarction of the spleen, hemorrhagic infarction of the lung and myocardial infarction;
- describe microscopic slides of nutmeg liver, venous congestion of the skin, stasis in cerebral capillaries, a mixed thrombus, pulmonary fat embolism, ischemic infarction of the kidney and hemorrhagic infarction of the lung;
- distinguish a thrombus from a postmortem blood clot;
- determine the type of infarct according to the organ and the type of blood supply.

7.1. Classification of circulatory disorders

Circulatory disorders are divided into two large groups:

1. **Disorders of blood supply (blood filling):** arterial hyperemia, venous hyperemia (congestion), anemia (ischemia).
2. **Disorders of the rheological properties of blood and of vessel wall permeability:** bleeding and hemorrhage, plasmorrhagia, stasis, thrombosis, embolism, DIC, shock. In addition, **disorders of lymph circulation** (lymphostasis, lymphedema) and **changes in the amount of tissue fluid** (edema, exsiccosis) are also considered in this section. Circulatory disorders are **generalized** (heart failure, shock) and **local** (thrombosis, infarction).

7.2. Arterial hyperemia

Arterial hyperemia (active hyperemia) is the filling of an organ or tissue with blood as a result of increased arterial inflow. It is an **active** process: the arterioles dilate (neurogenic and humoral influences). The tissue is **red** (bright), warm and slightly enlarged.

General arterial hyperemia (plethora) — an increase in circulating blood volume (erythremia — Vaquez disease, multiple transfusions).

Local arterial hyperemia is physiological (physical work, blushing with embarrassment) or pathologic. **Types of pathologic hyperemia:**

- **angioneurotic (neuroparalytic)** — loss of the influence of vasoconstrictor nerves or excitation of vasodilator nerves (reddening of the facial skin);
- **collateral** — increased blood flow through collateral pathways when a main artery is occluded;
- **vacant (ex vacuo)** — filling of tissue with blood when barometric pressure falls (for example, when medical cupping glasses are applied);
- **postanemic** — a sudden rush of blood when a compressed artery is released (for example, the abdominal vessels when ascitic fluid is removed rapidly); it can lead to cerebral anemia and fainting;
- **inflammatory hyperemia** — the most common type (Topic 5);
- hyperemia due to an **arteriovenous fistula**.

7.3. Venous hyperemia (congestion)

Venous hyperemia (passive hyperemia, congestion) is an increase of blood in an organ or tissue as a result of **impaired venous outflow**. It is a **passive** process. The veins and capillaries dilate and fill with deoxygenated blood — the tissue becomes **bluish-red** (**cyanosis**), and its temperature falls.

Generalized venous congestion

The cause is heart failure (myocarditis, cardiosclerosis, heart defects, myocardial infarction, hypertension). It is acute or chronic.

Acute generalized venous congestion — in acute heart failure (myocardial infarction, acute myocarditis): capillary hypoxia and permeability increase, with **plasmorrhagia, edema and diapedetic hemorrhages**, degeneration and necrosis in the parenchyma; in the lung — **pulmonary edema**.

Chronic generalized venous congestion — in chronic heart failure. Prolonged hypoxia leads to **parenchymal atrophy** and **stromal sclerosis** (activation of fibroblasts) — the organs become **firm** — **cyanotic (congestive) induration**. Cyanosis of the skin, edema (anasarca), transudate in the cavities (ascites, hydrothorax, hydropericardium).

Changes in the organs:

1. Liver — "nutmeg liver". In right ventricular failure, blood stagnates in the inferior vena cava; the hepatic veins, then **the central veins of the lobules and the sinusoids of the central zones** fill with blood. Hepatocytes of the central zone undergo **atrophy and necrosis** because of hypoxia (the lobule center is the least supplied with oxygen), whereas hepatocytes in the periphery undergo **fatty degeneration**. As a result the liver is enlarged and firm, and on the cut surface it is mottled **red-brown (center) and yellowish (periphery)** — resembling the cut surface of a **nutmeg**. If the process is prolonged, **nutmeg fibrosis** and then **cardiac (nutmeg) cirrhosis** develop.

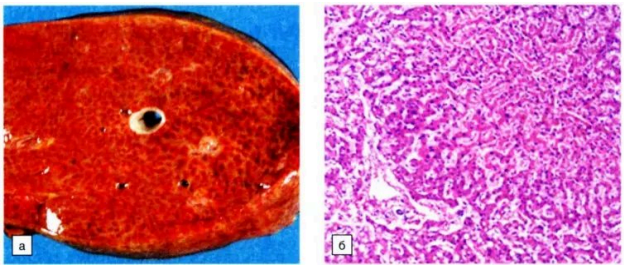


Figure 7.2. *Nutmeg liver: a — cut surface; b — in the center of the lobule the sinusoids are markedly dilated and filled with blood, and the hepatocytes are destroyed (Strukov, Serov, Fig. 4-2)*

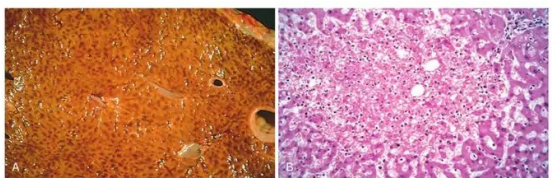


Figure 7.1. *Nutmeg liver: A — on the cut surface the central areas are red and slightly depressed, alternating with the surrounding yellowish parenchyma; B — centrilobular necrosis, hepatocyte degeneration and hemorrhage (Robbins, 9th ed., Fig. 4-3)*

2. Lung — brown induration. Left ventricular failure and especially **mitral stenosis** cause blood stagnation in the pulmonary circulation. The capillaries of the interalveolar septa dilate, and **diapedetic hemorrhages** occur; macrophages engulf red cells that have entered the alveoli and accumulate **hemosiderin** (**"heart failure cells"**, Topic 4); because of chronic hypoxia, **sclerosis** develops in the interalveolar septa. The lung becomes **brown (hemosiderosis)** and **firm (induration)**. Pulmonary hypertension leads to right ventricular hypertrophy.

3. Kidneys — cyanotic induration: the kidney is enlarged, firm and bluish, with a dark red medulla.

4. Spleen — cyanotic induration: enlarged, firm, dark cherry-colored.

5. Skin and limbs — cyanosis, edema, trophic ulcers.

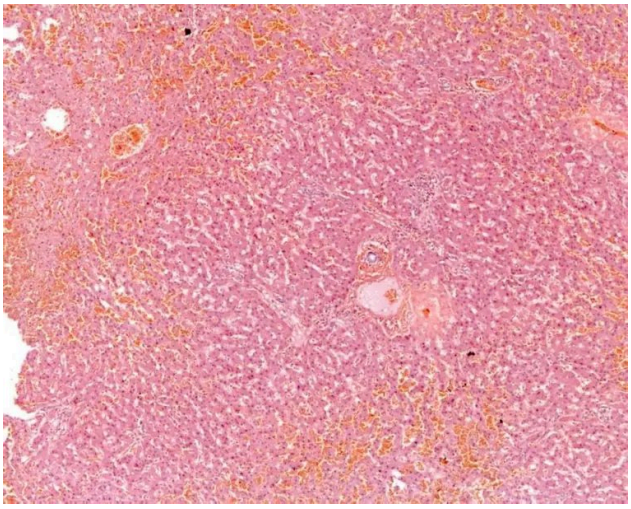


Figure 7.4. *Nutmeg liver (microscopic slide): in the center of the lobule the vein and sinusoids are dilated and filled with blood, and the central hepatocytes are atrophic (TMA study guide, 2018)*

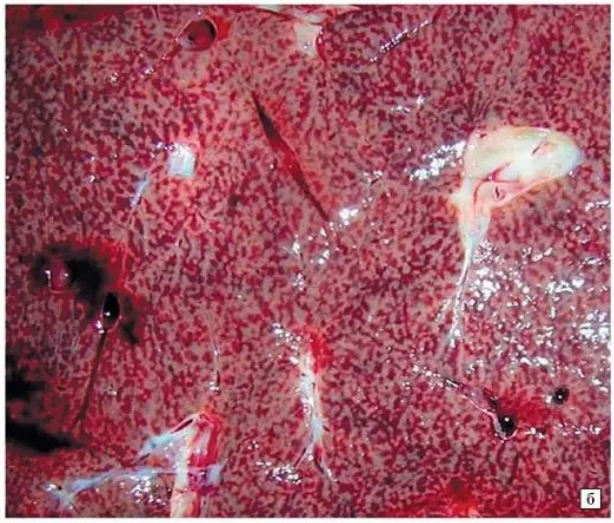


Figure 7.3. *Nutmeg liver (gross specimen): the liver is enlarged and firm, with alternating red-brown and yellowish spots on the cut surface (TMA study guide, 2018)*

Local venous congestion

It occurs when venous outflow from a particular area is impaired: when a vein is occluded by a thrombus or embolus, or compressed from outside (tumor, scar, cast, bandage). Examples: congestion in the **portal venous system in liver cirrhosis** (**portal hypertension**: ascites, splenomegaly, esophageal varices); **Budd–Chiari syndrome** (thrombosis of the hepatic veins); edema and cyanosis of the leg in thrombosis of the leg veins.

CLINICAL SIGNIFICANCE

In **portal hypertension**, blood passes from the portal vein to the inferior and superior venae cavae through **portocaval anastomoses**: the veins of the esophagus and gastric cardia (**varices** — sudden massive bleeding from them is one of the main causes of death in patients with cirrhosis), the rectal veins (**hemorrhoids**), and the veins of the anterior abdominal wall ("**caput medusae**").

7.4. Anemia (ischemia)

Local anemia (ischemia) is a **decrease or cessation of arterial blood flow** to an organ or tissue. Generalized anemia is **anemia** proper (blood diseases). Causes of local ischemia:

- **angiospastic** — arterial spasm (stress, cold, Raynaud syndrome, cocaine);
- **obturative** — occlusion of the arterial lumen by a thrombus, embolus or atherosclerotic plaque;
- **compressive** — compression of an artery from outside (tumor, tourniquet, ligature);

- **due to redistribution of blood** — ischemia of other areas during postanemic hyperemia. The outcomes depend on the speed and duration of ischemia and the sensitivity of the tissue: **acute ischemia** — degeneration and **infarction** (necrosis); **chronic ischemia** — **atrophy** of the parenchyma and **sclerosis** of the stroma (for example, atherosclerotic nephrosclerosis, chronic cerebral ischemia — dementia).

> **REMEMBER**
> The development of collateral circulation determines the outcome of ischemia: organs supplied by **functional end arteries** (heart, brain, kidney, spleen) are very sensitive to ischemia, and **infarction** develops easily in them; organs with a **dual** blood supply (lung, liver) and organs with many collaterals (arm, leg) are relatively resistant.

7.5. Bleeding and hemorrhage

Bleeding (hemorrhage) is the escape of blood **from a vessel or a cardiac chamber**: **external** (to the outside) and **internal** (into a body cavity or tissue). Accumulation of blood in tissue is called **hemorrhage (extravasation)**.

Types of bleeding by mechanism:

1. **Bleeding due to rupture** (per rhexin) — rupture of the wall of a vessel or the heart: trauma, rupture of an aneurysm, cardiac rupture in myocardial infarction, rupture of a cerebral vessel in essential hypertension.
2. **Bleeding due to erosion (arrosive)** (per diabrosin) — destruction of the vessel wall by inflammation, necrosis or tumor: at the floor of a gastric ulcer, in a tuberculous cavity, during tumor breakdown, in ectopic (tubal) pregnancy.
3. **Diapedetic bleeding** (per diapedesin) — escape of red cells from microvessels with **increased permeability** without disruption of the vessel wall: hypoxia, vasculitis, infections, intoxications, hemorrhagic diatheses (thrombocytopenia, hemophilia), DIC. Usually **small, pinpoint** hemorrhages.

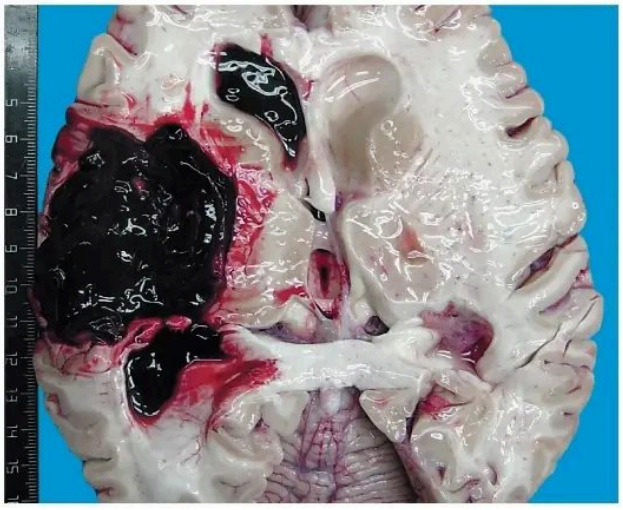


Figure 7.6. *Cerebral hemorrhage — hematoma: a blood clot in the region of the subcortical nuclei of the hemispheres, ruptured into the ventricles (TMA study guide, 2018)*

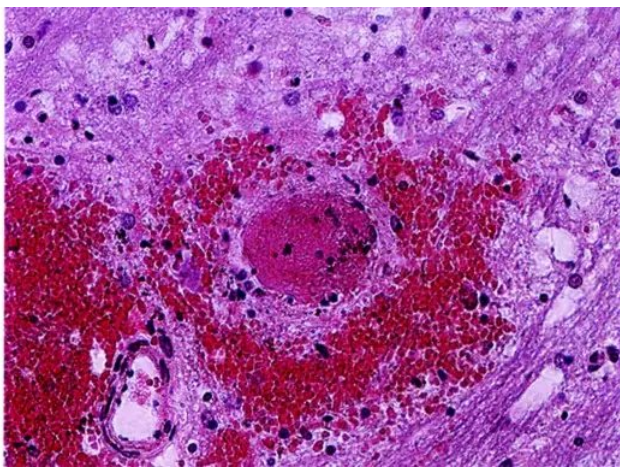


Figure 7.5. *Diapedetic hemorrhage in brain tissue: a collection of red cells around a capillary (Strukov, Serov, Fig. 4-7)*

Types of hemorrhage:

- **hematoma** — destruction of tissue with formation of a cavity filled with blood (brain, muscles, subcutaneous tissue);

- **hemorrhagic infiltration (imbibition)** — soaking of tissue with blood while its structure is preserved;
- **bruise (ecchymosis)** — a flat hemorrhage in the skin or mucosa larger than 1–2 cm;
- **petechiae** — pinpoint hemorrhages of 1–2 mm (in serous and mucous membranes; asphyxia, thrombocytopenia, sepsis);
- **purpura** — hemorrhages up to 3–5 mm (vasculitis, thrombocytopenia). **Accumulation of blood in cavities:** hemothorax (in the pleural cavity), **hemopericardium** (in the pericardium — **cardiac tamponade**), hemoperitoneum (in the peritoneal cavity), hemarthrosis (in a joint). Bleeding from organs: **nose** (epistaxis), **lung** (hemoptysis — coughing up blood), **stomach** (hematemesis — vomiting of blood, “coffee grounds”), **intestine** (melena — black, tarry stools), **urinary tract** (hematuria), **uterus** (metrorrhagia, menorrhagia).

Outcomes: resorption of blood (small hemorrhages); accumulation of hemosiderin with formation of a **“rust-colored” cyst** (in the brain); organization and encapsulation; suppuration. The significance depends on the volume and rate of blood loss (if more than 20% of the total blood volume is lost rapidly — **hemorrhagic shock**) and on the location (brainstem, pericardium).

7.6. Plasmorrhagia and stasis

Plasmorrhagia is the escape of blood plasma from the vascular lumen with **soaking** of the vessel wall and surrounding tissues (**plasmatic imbibition**). Mainly in microcirculatory vessels — essential hypertension, atherosclerosis, rheumatic diseases, diabetes mellitus. The vessel wall is thickened and homogeneous; the outcome is **hyalinosis** (arteriolohyalinosis) or **fibrinoid necrosis** (Topic 3).

Stasis (Greek stasis — standstill) is the **cessation of blood flow** in microcirculatory vessels (mainly capillaries). In the capillaries, red cells stick together to form homogeneous columns (**sludge phenomenon** — red cell aggregation), but there is no hemolysis or clotting (unlike a thrombus). Causes: infections, intoxications, shock, cold or heat. Stasis is reversible (blood flow is restored); if prolonged, it leads to necrosis.

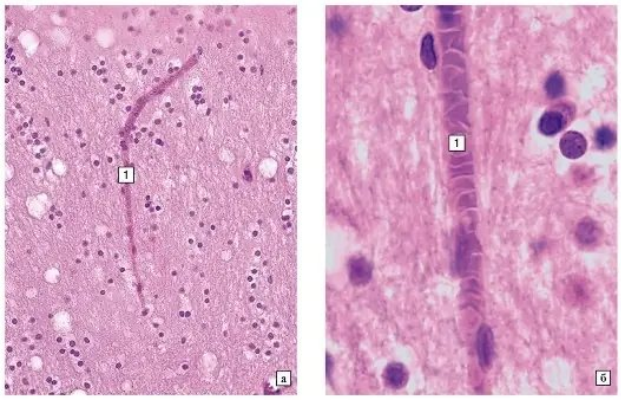


Figure 7.7. *Stasis in cerebral capillaries: the brain tissue is edematous, with columns of agglutinated red cells in the capillaries (TMA study guide, 2018)*



Figure 7.8. *Venous congestion of the skin: the veins and capillaries are markedly dilated and filled with blood (TMA study guide, 2018)*

7.7. Edema

Edema is an **increase in intercellular (interstitial) tissue fluid**. Accumulation of fluid in cavities is called **dropsy** (Greek hydrops): **hydrothorax** (pleura), **hydropericardium**, **ascites** (peritoneal cavity, hydroperitoneum), **hydrocele** (tunica vaginalis of the testis). Diffuse edema of the subcutaneous tissue — **anasarca**. Edema fluid is a **transudate** (low protein, clear; Topic 5).

Mechanisms of edema (Starling forces):

1. **Increased hydrostatic pressure in capillaries** — heart failure (generalized venous congestion), venous thrombosis, portal hypertension.
2. **Decreased plasma oncotic pressure (hypoproteinemia)** — **nephrotic syndrome** (albumin is lost in the urine), **liver cirrhosis** (albumin synthesis decreases), protein-energy malnutrition ("hunger edema").
3. **Increased vascular permeability** — inflammation, allergy (Quincke edema (angioedema)), toxins.
4. **Impaired lymph flow** (**lymphedema**) — involvement of lymph nodes by tumor or their removal (arm edema after breast cancer surgery), parasites (filariasis — **elephantiasis**), postirradiation sclerosis.
5. **Sodium and water retention** — kidney failure, secondary hyperaldosteronism (activation of the renin–angiotensin–aldosterone system in heart failure). **Types of edema: cardiac edema** (begins in the legs — gravitational, worse in the evening; anasarca); **renal edema** (begins in the face and eyelids, worse in the morning; widespread in nephrotic syndrome); **hepatic edema** (mainly ascites); **hunger edema**; **inflammatory and allergic edema**; **lymphedema**.

Morphology: edematous skin is pale and doughy, and a pit remains when pressed with a finger. Edematous organs are enlarged and heavy, and fluid flows from the cut surface. **Pulmonary edema** — the lung is heavy (2–3 times normal), and frothy pink fluid flows from the cut surface; the alveoli are filled with pale pink protein-containing fluid; in left ventricular failure (acute — myocardial infarction, hypertensive crisis) and toxic and allergic influences. **Cerebral edema** — the brain is enlarged, the gyri are flattened and the sulci narrowed; **displacement (herniation)** of the brain and **impaction into the foramen magnum** (herniation of the cerebellar tonsils into the foramen magnum) — the brainstem is compressed, and the respiratory and cardiac centers stop functioning — death.

> **CLINICAL SIGNIFICANCE**
>
> **The significance of edema** depends on its location: edema of the legs causes discomfort and predisposes to trophic ulcers; **laryngeal edema** (Quincke edema) — asphyxia; **pulmonary edema** — acute respiratory failure (emergency care: oxygen, diuretics); **cerebral edema** and brain herniation — a cause of death. With prolonged edema, sclerosis develops in the tissue.

Exsiccosis (dehydration) — a decrease in body water: severe diarrhea (cholera, food poisoning), vomiting, profuse sweating, diabetes mellitus and diabetes insipidus. The skin is dry and wrinkled, the eyes are sunken, and the blood thickens; in severe cases — hypovolemic shock.

7.8. Hemostasis and thrombosis

Normal hemostasis

Hemostasis is a complex system that stops bleeding when a vessel is injured and at the same time keeps the blood fluid in intact vessels. Its participants are the **endothelium, platelets** and **coagulation factors**. Stages of hemostasis:

1. **Brief arteriolar vasoconstriction** (under the action of endothelin).
2. **Primary hemostasis** — **platelet adhesion** to the exposed subendothelial matrix at the site of injury (via **von Willebrand factor**), their activation (release of ADP and thromboxane A2 from granules) and **aggregation** — a **platelet plug** forms.

3. **Secondary hemostasis** — **tissue factor** (factor III) triggers the coagulation cascade (via factor VII), and **thrombin** is formed; thrombin converts fibrinogen to **fibrin** and further activates platelets.
4. **Stabilization and contraction (retraction) of the clot**, followed by its dissolution by **fibrinolysis** (plasmin). Normal endothelium has **antithrombotic** properties: prostacyclin (PGI2) and NO (inhibit platelet aggregation), thrombomodulin (turns thrombin into an activator of protein C), heparin-like molecules (activate antithrombin III), tissue plasminogen activator (t-PA).

Thrombosis

Thrombosis is **clotting** of blood (formation of a clot) in the vascular lumen or cardiac chambers **during life**. The clot formed is a **thrombus**. Thrombosis is a pathologic activation of hemostasis.

Causes of thrombosis — Virchow's triad:

1. **Endothelial injury** — the most important factor (it alone can cause thrombosis): atherosclerotic plaques, endocardial injury in myocardial infarction, vasculitis, hypertension, trauma, bacterial toxins, cigarette smoke, hypercholesterolemia.
2. **Abnormal blood flow** — **slowing (stasis)** (in veins — heart failure, immobility, varicose veins; in the cardiac chambers — atrial fibrillation, dilated ventricles, aneurysms) or **turbulence** (in arteries — at plaques and bifurcations).
3. **Increased coagulability of blood (hypercoagulability, thrombophilia)**: – **primary (hereditary)** — **factor V mutation (Leiden)** — the most common; prothrombin gene mutation, deficiency of antithrombin III, protein C and protein S; – **secondary (acquired)** — prolonged bed rest, surgery and trauma, **malignant tumors**

(Trousseau syndrome — migratory thrombophlebitis), pregnancy and the postpartum period, **oral contraceptives**, smoking, obesity, **antiphospholipid syndrome**, heparin-induced thrombocytopenia, nephrotic syndrome.

> **MEMORY AID**
> Virchow's triad — "**Wall, Flow, Blood**": **W**all (endothelial injury), **F**low (stasis, turbulence), **B**lood (hypercoagulability). In arterial thrombosis the main factor is the **wall**; in venous thrombosis — the **flow**.

Stages of thrombus formation (morphogenesis): 1) **agglutination** of platelets (adhesion and aggregation); 2) **coagulation of fibrinogen**, formation of fibrin; 3) **agglutination** of red cells (trapping in the fibrin network); 4) **precipitation** of plasma proteins.

Types of thrombi

By structure and appearance:

- **White thrombus** — composed of platelets, fibrin and leukocytes; it forms **slowly**, in places with **rapid** blood flow — in **arteries** and on heart valves (in endocarditis).
- **Red thrombus** — composed of a fibrin network with entrapped red cells; it forms where clotting is **rapid** and blood flow is **slow** — in **veins**.
- **Mixed (laminated) thrombus** — the most common; it has a **head** (with the structure of a white thrombus, attached to the vessel wall), a **body** (mixed — alternating white and red layers: **lines of Zahn**) and a **tail** (red thrombus, free). It occurs in veins, the aorta and cardiac aneurysms.

- **Hyaline thrombi** — in microcirculatory vessels, composed of hemolyzed red cells, platelets and precipitated proteins; a homogeneous mass; in **DIC** and shock. **By relation to the vessel lumen: mural (parietal)** — partially occludes the lumen; **occlusive (obturating)** — completely occludes the lumen.

By location: arterial (coronary, cerebral, femoral and mesenteric arteries — usually over an atherosclerotic plaque, white or mixed, mural — leads to infarction), **venous (phlebothrombosis)** — most often in the **deep veins of the leg** (the source of pulmonary embolism), red, occlusive; in an inflamed vein — **thrombophlebitis**; **in the cardiac chambers** — **mural thrombi** (in myocardial infarction, aneurysm), thrombi in the auricles in atrial fibrillation, a free-lying **ball thrombus** in the atrium, **vegetations** on the valves (infective endocarditis) and a **propagating thrombus** (growing along the vessel toward the heart); capillary thrombosis.

Special types: dilatation thrombus — in aneurysms and varicose vessels; **marantic (stasis) thrombus** — in debilitated, cachectic elderly patients (because of slowed blood flow, most often in the dural venous sinuses and leg veins); **tumor thrombus** — when a tumor grows through the vessel wall and spreads along the vessel (in the renal vein in renal cell carcinoma); **septic thrombus** — a thrombus containing microbes (risk of purulent dissolution).

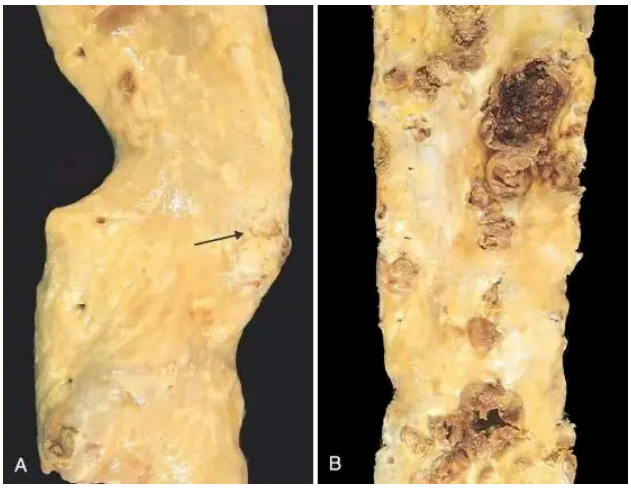


Figure 7.10. *Mural thrombus in the aorta: the intima is uneven, with thrombotic masses over atherosclerotic plaques (TMA study guide, 2018)*



Figure 7.9. *Mixed thrombus with a corrugated surface (a) and a ball thrombus in the left atrium (b) (Strukov, Serov, Fig. 4-11)*

CAUTION

Distinguishing a thrombus from a **postmortem blood clot** (Topic 2): a thrombus is **dull, dry, friable, with a rough (corrugated) surface, layered** and **attached** to the vessel wall; a postmortem clot is **smooth, shiny, elastic and moist**, follows the shape of the vessel, is not attached to the wall and is separated into "**chicken fat**" (yellow) and "**currant jelly**" (red) layers.

Outcomes of a thrombus

Favorable outcomes:

- **aseptic autolysis (dissolution)** — by the fibrinolytic system (plasmin), especially in small and fresh thrombi; **thrombolytic therapy** (alteplase, streptokinase) is based on this;
- **organization** — **granulation tissue** grows into the thrombotic mass from the wall and replaces it with connective tissue (beginning on days 5–6);
- **canalization and vascularization** — clefts and new endothelium-lined channels form in the organized thrombus, and blood flow is partially restored (**recanalization**);

- **calcification (petrification)** — in veins — **phleboliths** (vein stones).

Unfavorable outcomes:

- **occlusion** of the vessel — in an artery, **infarction** or **gangrene**; in a vein, **venous congestion and edema** in the distal area;
- **thromboembolism** — detachment of the thrombus or a fragment of it and its transport with the blood stream;
- **septic (purulent) dissolution** — when the thrombus is infected with microbes; **thromboembolic sepsis**, purulent metastases;
- **propagation** of the thrombus — its extension along the vessel.

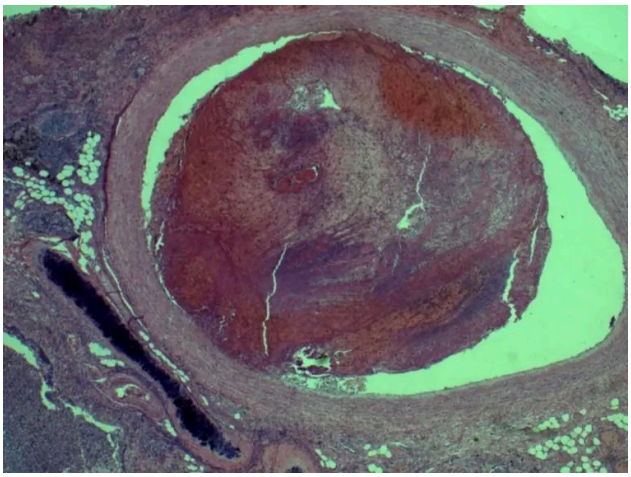


Figure 7.12. *Mixed thrombus in a vein (microscopic slide): the vessel lumen is filled with a mixed thrombus with layers of platelets, fibrin and red cells (TMA study guide, 2018)*

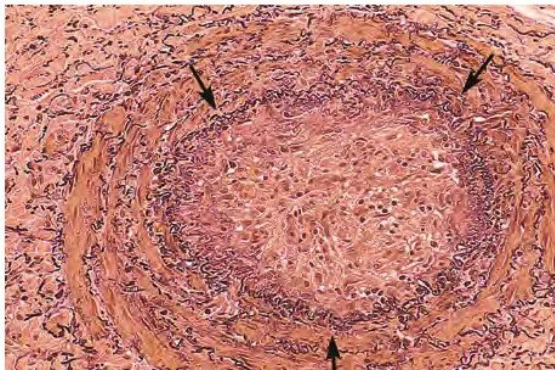


Figure 7.11. *Thrombosed artery (elastic stain): the original lumen is bounded by the internal elastic membrane (arrows); the thrombus is organized and perforated by several endothelium-lined recanalization channels (white spaces) (Robbins, 9th ed., Fig. 4-14)*

DIC

Disseminated intravascular coagulation (DIC) is not an independent disease but a **complication** of many severe conditions: numerous **microthrombi** (hyaline and fibrin thrombi) form in microcirculatory vessels, as a result of which platelets and coagulation factors are **consumed** (**consumptive coagulopathy**) and fibrinolysis is activated — **bleeding** develops. Therefore DIC is also called a "**thrombohemorrhagic syndrome**".

Causes: obstetric complications (premature placental separation, amniotic fluid embolism, a dead fetus), **sepsis** (especially gram-negative, meningococcemia), **shock**, extensive trauma and burns, malignant tumors (cancer of the lung, pancreas, prostate, acute promyelocytic leukemia), hemolytic transfusion reactions, snake venom.

Stages: 1) hypercoagulation; 2) consumptive coagulopathy; 3) **activation of fibrinolysis and hypocoagulation** — bleeding; 4) recovery (or death).

Morphology: microthrombi in the microvessels of the kidneys, lungs, brain, liver, heart and adrenal glands; numerous diapedetic **hemorrhages** (petechiae, ecchymoses), small foci of **necrosis** and **infarcts**; **symmetric necrosis** of the renal cortex, pituitary necrosis (**Sheehan syndrome** — postpartum), bilateral hemorrhage into the adrenal glands (**Waterhouse–Friderichsen syndrome** — in meningococcemia). The result is **multiple organ failure**.

7.9. Embolism

Embolism (Greek emballein — to throw in) is the circulation in blood or lymph of **particles (emboli)** not normally found there and the **occlusion** of vessels by them. Emboli are solid, liquid or

gaseous.

By the direction of movement of the embolus:

- **orthograde** — with the blood flow (the most common): from the systemic veins and the right heart — into the **pulmonary artery** system; from the left heart and aorta — into the **arteries of the systemic circulation** (brain, kidney, spleen, intestine, legs); from the intestinal veins — into the **portal vein** system (to the liver);
- **retrograde** — **against** the blood flow: a heavy embolus (for example, in the inferior vena cava system) descends under its own weight (into the hepatic and renal veins);
- **paradoxical** — an embolus from the systemic veins bypasses the lungs and passes directly into the **arteries** of the systemic circulation through a **septal defect of the heart** (patent foramen ovale, ventricular septal defect) (for example, a leg vein thrombus occludes a cerebral artery).

Types of embolism

1. Thromboembolism — the most common embolism (about 95–99% of emboli): the embolus is a detached thrombus or a fragment of it.

Pulmonary thromboembolism (PE) is one of the most important causes of death in hospitalized patients. Its source in **95% of cases is thrombosis of the deep veins of the leg** (popliteal, femoral, iliac veins). The consequences depend on the size of the embolus:

- **large emboli** — impact in the **trunk** of the pulmonary artery or at the bifurcation of its main branches (**“saddle” embolus**) — **sudden death** (acute right ventricular failure, “acute cor pulmonale”, cardiogenic shock); the **pulmocoronary reflex** (coronary artery spasm) is also important in the mechanism of death;
- **medium-sized emboli** — in the medium branches of the pulmonary artery; if the blood supply from the bronchial arteries is adequate, infarction does not develop, only hemorrhage; against a background of **left ventricular failure** (venous congestion) — **hemorrhagic infarction**;
- **multiple small emboli** — if recurrent, **pulmonary hypertension** and **chronic cor pulmonale**. **Systemic thromboembolism** — the source is mainly the **left heart** (mural thrombi in myocardial infarction — 2/3 of cases, left auricular thrombi in atrial fibrillation, left ventricular aneurysm, valvular vegetations — endocarditis), and also thrombi over atherosclerotic plaques in the aorta. The emboli impact in the arteries of the **brain** (ischemic stroke), **legs** (gangrene), **kidney, spleen and intestine** and cause **infarcts**.



Figure 7.14. *Pulmonary thromboembolism: dried, worm-like red-gray masses in the pulmonary trunk (TMA study guide, 2018)*



Figure 7.13. *Pulmonary thromboembolism: an embolus from deep vein thrombosis of the leg impacted at the bifurcation of the pulmonary artery (Robbins, 9th ed., Fig. 4-15)*



Figure 7.15. *Varicose veins of the lower leg — a source of venous thrombosis and thromboembolism (TMA study guide, 2018)*

2. Fat embolism — entry of fat droplets into vessels: **fractures of long bones** (yellow marrow fat), extensive crushing of subcutaneous fatty tissue, burns, erroneous intravascular injection of oily drugs. Fat droplets first occlude the **pulmonary capillaries**; if numerous, they pass through the lungs and reach the capillaries of the **brain and kidneys**. **Fat embolism syndrome** (1–3 days after injury): pulmonary insufficiency, neurological signs (numerous petechiae in the brain), anemia, thrombocytopenia, petechial skin rash. Fat is detected with **Sudan III** (on frozen sections).

Bone marrow embolism — fragments of bone marrow (hematopoietic cells and fat cells) in pulmonary vessels in bone fractures and cardiopulmonary resuscitation (rib fractures).

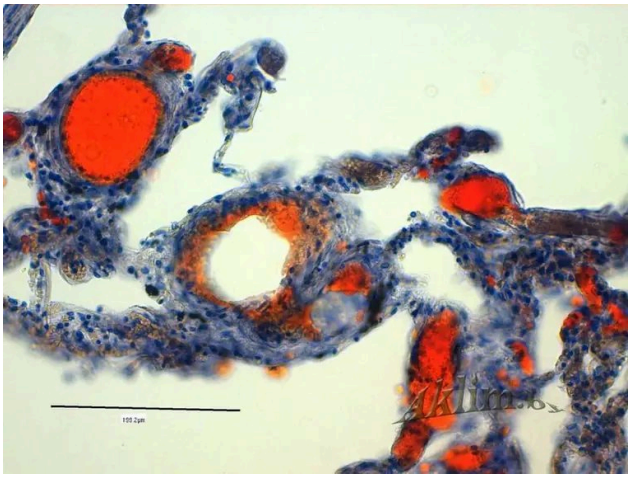


Figure 7.17. *Fat embolism of pulmonary capillaries (Sudan III): many capillaries are filled with orange-red fat droplets (TMA study guide, 2018)*

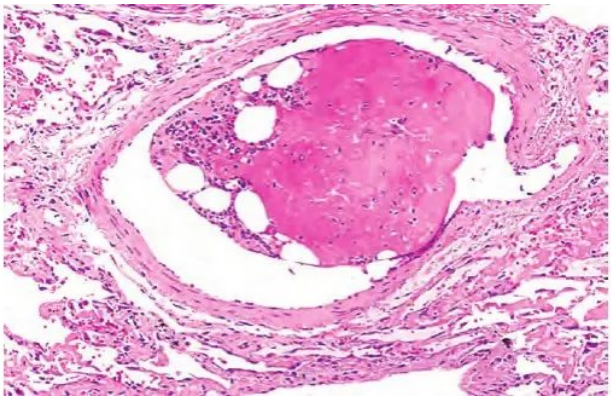


Figure 7.16. *Bone marrow embolus in a pulmonary vessel: hematopoietic cells (left) and fat vacuoles (Robbins, 9th ed., Fig. 4-16)*

3. Air embolism — entry of **atmospheric air** into the vessels: injury to the neck veins (negative pressure inside them), surgery (cardiac, pulmonary, neurosurgery — in the sitting position), breaches of technique during intravascular injections and infusions, artificial pneumothorax, childbirth. Air bubbles accumulate in the **right heart**, form frothy blood and occlude the pulmonary capillaries.

Usually **more than 100 mL** of air is needed to be clinically significant. At autopsy, to detect air embolism, the heart is opened **under water** (air bubbles escape).

4. Gas embolism — formation of bubbles from **dissolved gases** (mainly nitrogen) in the blood when atmospheric pressure falls rapidly — **caisson (decompression) disease**: in divers and caisson workers who surface rapidly and in pilots who climb rapidly to altitude in an unpressurized cabin. Gas bubbles occlude the vessels of joints and muscles (pain — "**the bends**"), the lungs (shortness of breath — "the chokes"), and the capillaries of the brain and spinal cord (paralysis); in the chronic form — **aseptic necrosis** of bones (femoral head). Gas also forms in tissue in gas gangrene.

5. Tissue (cellular) embolism — fragments of tissue or cells: **tumor cell embolism** — the basis of **metastases** (hematogenous and lymphogenous); fragments of tissue destroyed in trauma (brain, liver); **amniotic fluid embolism** — during childbirth, amniotic fluid (desquamated epithelium, lanugo hairs, meconium, fat) enters the maternal blood and occludes the pulmonary vessels — an extremely severe complication (mortality up to 80%): sudden dyspnea, cyanosis, shock, **DIC**.

6. Microbial (bacterial) embolism — clusters of microbes (bacterial emboli) or fragments of a purulently dissolved thrombus in the vascular lumen: in **sepsis**, **metastatic abscesses** in various organs (kidney, myocardium, brain) (for example, **embolic purulent nephritis**), septic infarcts.

7. Foreign body embolism — bullets, shell fragments, catheter fragments, calcium and cholesterol crystals (when an atherosclerotic plaque ruptures — cholesterol embolism).

8. Parasite embolism — echinococcus, ascarids.

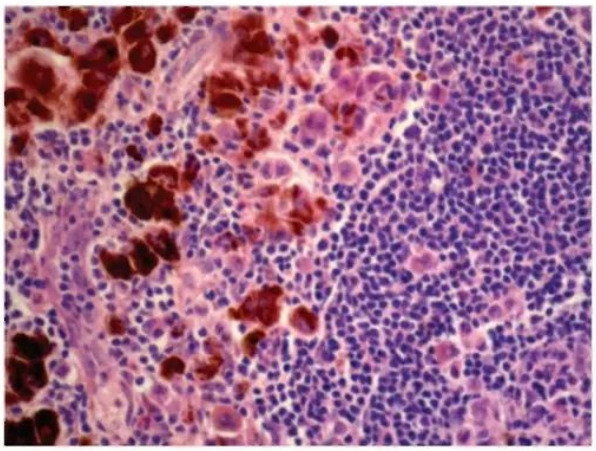


Figure 7.18. *Tumor cell embolism: melanoma cells in a lymph node — a cluster of cells with dark brown pigment (TMA study guide, 2018)*

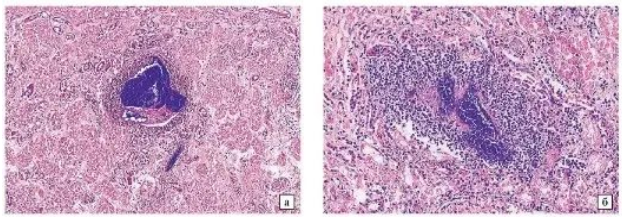


Figure 7.19. *Embolic purulent nephritis (microscopic slide): basophilic bacterial emboli in small vessels (a), surrounded by purulent inflammation (b) (TMA study guide, 2018)*

7.10. Infarction

Infarction (Latin infarcire — to stuff) is **vascular (ischemic) necrosis** resulting from sudden cessation of blood supply (**ischemia**). The most common causes of infarction are **arterial thrombosis** (usually over an atherosclerotic plaque) and **embolism**; less often — prolonged **spasm**, external compression of a vessel, complete cessation of venous outflow (for example, torsion of the testis or ovary), intestinal obstruction.

Factors influencing the development of infarction:

- **type of blood supply** — functional end arteries (heart, brain, kidney, spleen) — infarction develops easily; dual blood supply (lung — pulmonary and bronchial arteries; liver — hepatic artery and portal vein) — infarction is rare;

- **rate of occlusion** — if it develops slowly, collaterals have time to open;
- **sensitivity of the tissue to hypoxia** — neurons are irreversibly injured in 3–4 minutes, cardiomyocytes in 20–30 minutes; fibroblasts are much more resistant;
- oxygen saturation of the blood (anemia, heart failure).

Types of infarction

By shape: corresponding to the vascular branching of the organ — **wedge-shaped (triangular)** (kidney, spleen, lung — the base faces the organ capsule, the apex faces the hilum; fibrin on the capsule — reactive inflammation), **irregular** (heart, brain, intestine — organs without a trunk-type vascular branching).

By color and structure:

1. White (ischemic) infarct — whitish-yellow, firm, sharply demarcated; in **solid** organs with a **single** (end-arterial) blood supply: the **spleen, kidney**, less often the heart and brain.

2. White infarct with a red (hemorrhagic) rim — a zone of hemorrhage around a whitish-yellow focus (**demarcation zone**); this zone results from paralytic vasodilation after perifocal spasm and diapedetic hemorrhage. It occurs in the **heart** (myocardial infarction) and **kidney**.

3. Red (hemorrhagic) infarct — the necrotic focus is soaked with blood, dark red and sharply demarcated. Conditions for its development: a **dual blood supply** (lung) against a background of **venous congestion** (left ventricular failure), loose (spongy) tissues; **venous occlusion** (torsion of the testis or ovary); **restoration of blood flow** (reperfusion) in previously ischemic tissue. It occurs in the **lung and intestine**, less often in the brain (hemorrhagic transformation).

4. Septic (infected) infarct — when a fragment of vegetations (microbial embolus) in infective endocarditis or microbes enter the necrotic focus; the infarct turns into an **abscess**.



Figure 7.21. *White infarct of the kidney with a hemorrhagic rim (a) and an angiogram — no vessels in the infarct area (b) (Strukov, Serov, Fig. 3-4)*



Figure 7.20. *Red and white infarcts: A — a hemorrhagic, roughly wedge-shaped red infarct of the lung; B — a sharply demarcated white infarct of the spleen (Robbins, 9th ed., Fig. 4-18)*

Morphology and evolution of infarction

The main histologic appearance of an infarct is **ischemic coagulative necrosis** (in the brain — **liquefactive necrosis** — “gray softening”, Topic 2). Its evolution:

1. **First hours** — no gross changes; detected by histochemical methods (loss of glycogen and dehydrogenases). Histologic signs of necrosis appear **after 4–12 hours**.
2. **First days** — **demarcation inflammation** at the border of the focus: hyperemia, hemorrhage, neutrophil infiltration (days 1–3).
3. **Days 3–7** — macrophages phagocytose the dead tissue; **granulation tissue** begins to grow in from the periphery.
4. **Weeks to months** — the focus undergoes **organization** and is replaced by a **scar** (6–8 weeks in the myocardium). In the brain, necrotic foci are resorbed and a **cyst** forms (surrounded by a glial scar).

Infarcts in organs:

- **Myocardial infarction** — the most common and most important; mainly in the **left ventricle** and interventricular septum; the cause is atherosclerosis and thrombosis of the coronary arteries. **Subendocardial**, **intramural**, **subepicardial** and **transmural** (involving the full thickness of the wall) infarcts are distinguished. Complications: cardiogenic shock, arrhythmias, acute heart failure (pulmonary edema), **cardiac rupture** (hemopericardium, tamponade — on days 3–7), acute aneurysm, mural thrombus and thromboembolism, fibrinous pericarditis (Dressler syndrome). Outcome — **large-focal cardiosclerosis**, chronic aneurysm.
- **Cerebral infarction (ischemic stroke)** — atherosclerosis, thrombosis or embolism (from the heart); white (gray softening) or red (hemorrhagic transformation); outcome — a cyst.
- **Pulmonary infarction** — red (hemorrhagic), wedge-shaped, its base facing the pleura; fibrin deposits on the pleura (**fibrinous pleurisy**); cause — pulmonary thromboembolism + left ventricular failure; complications — **infarct pneumonia**, abscess, gangrene.
- **Renal infarction** — white with a red rim, wedge-shaped; most often embolic (from the left heart, aorta); outcome — a depressed scar.
- **Splenic infarction** — white, wedge-shaped; fibrin on its capsule (perisplenitis).
- **Intestinal infarction** — red (hemorrhagic); when a mesenteric artery (embolism, thrombosis) or vein (thrombosis) is occluded; it rapidly progresses to **gangrene** and **peritonitis**.

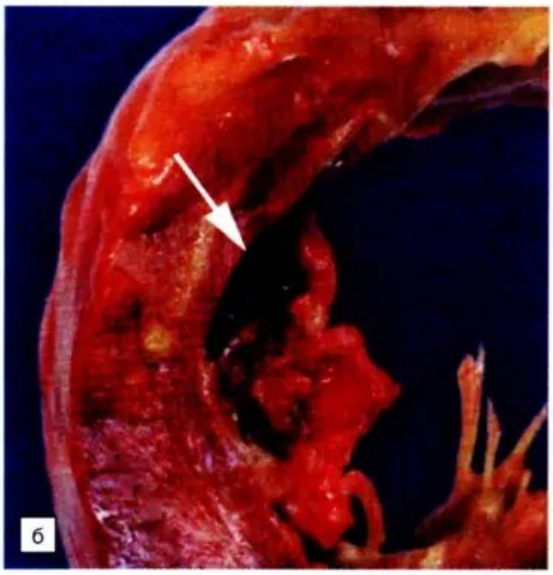


Figure 7.22. *Myocardial infarction: thrombosis of the left anterior descending coronary artery (b — arrow) (Strukov, Serov, Fig. 3-6b)*



Figure 7.23. *Ischemic myocardial infarction with a hemorrhagic rim: an irregular, yellowish focus with a red rim in the anterior wall of the left ventricle (TMA study guide, 2018)*



Figure 7.25. *Hemorrhagic infarction of the lung (gross specimen): a triangular, dark red, firm focus in the lung tissue with its base facing the pleura (TMA study guide, 2018)*

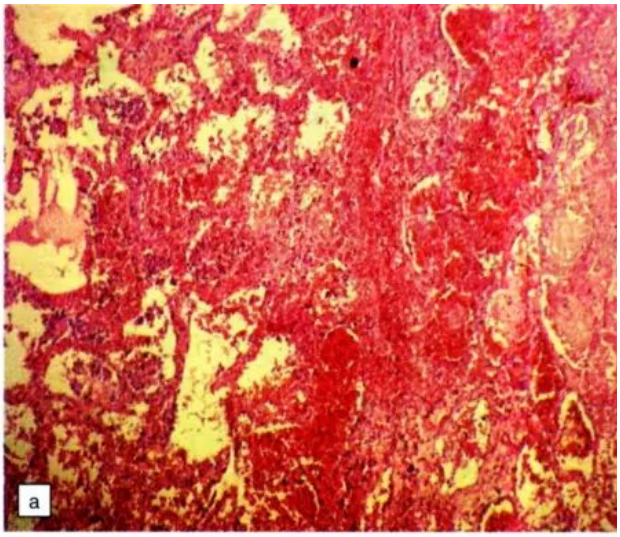


Figure 7.24. *Hemorrhagic infarction of the lung: the alveoli are filled with blood (Strukov, Serov, Fig. 3-5a)*

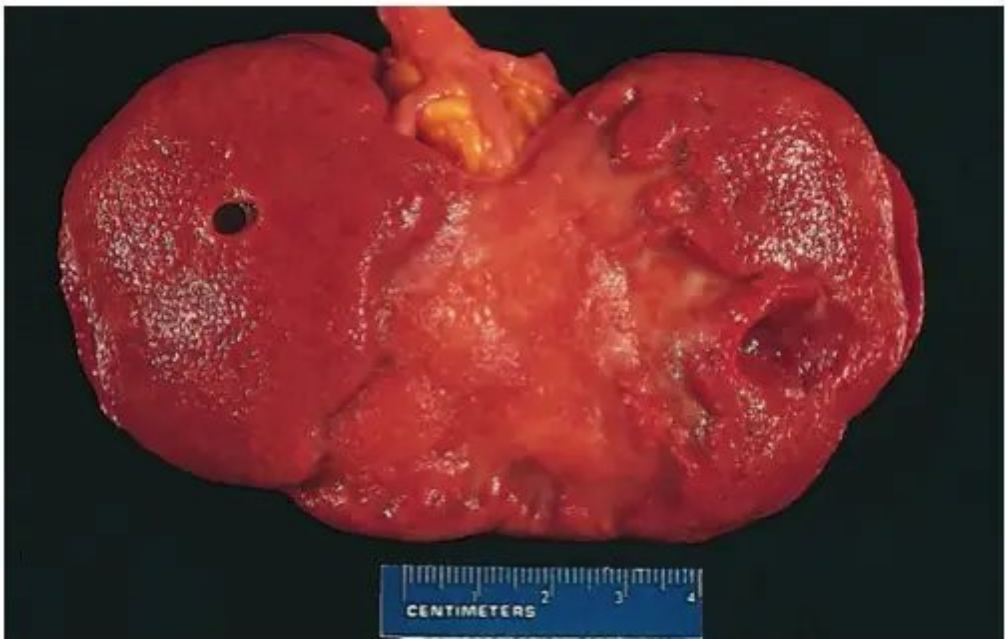


Figure 7.26. *An old infarct of the kidney replaced by a large fibrous scar (Robbins, 9th ed., Fig. 4-19)*

MEMORY AID

Remembering the color of infarcts by organ: "**Lung and bowel — red, spleen and kidney — white, heart — white with a red rim, brain — softens**". A red infarct needs "**Two routes and loose tissue**": a dual blood supply, venous congestion and loose tissue.

7.11. Shock

Shock is a state of **systemic hypoperfusion**: as a result of reduced cardiac output or an effective reduction in circulating blood volume, the blood supply and oxygen delivery to tissues are impaired, and cellular hypoxia and cell injury develop. Arterial blood pressure falls.

Table 7.1. *Types of shock*

Type	Mechanism	Causes
Cardiogenic	Failure of the pumping function of the heart	Myocardial infarction, arrhythmias, myocarditis, cardiac tamponade, pulmonary thromboembolism (obstructive)
Hypovolemic	Decreased blood or plasma volume	Blood loss (hemorrhagic), burns (plasma loss), severe diarrhea and vomiting (cholera)
Septic (distributive)	Systemic vasodilation, increased permeability, endothelial activation and DIC under the action of microbial products and cytokines	Bacterial (most often gram-negative, gram-positive) and fungal infections (sepsis)
Anaphylactic	IgE-mediated systemic vasodilation and increased permeability	Drugs (penicillin), insect stings, food allergy
Neurogenic	Loss of vascular tone	Spinal cord injury, complications of anesthesia
Traumatic	Combined factors of blood loss, pain and tissue crushing	Severe injuries

Stages of shock:

1. **Compensated (nonprogressive) stage** — reflex compensatory mechanisms (sympathetic system, catecholamines, renin–angiotensin, ADH) maintain perfusion of vital organs (brain, heart): tachycardia, peripheral vasoconstriction (cold, pale skin), renal water retention. In septic shock, however, the skin may be warm and flushed.
2. **Progressive stage** — tissue hypoxia, anaerobic glycolysis, **lactic acidosis**; the arterioles dilate, blood pools in the microcirculation, and cardiac output falls further; the endothelium is damaged and **DIC** develops.
3. **Irreversible stage** — cells and tissues are irreversibly injured; treatment is ineffective — death.

Morphologic changes in shock ("shock organs"): disturbed microcirculation (redistribution of blood, sludge, stasis, hyaline microthrombi — DIC), hemorrhages, degeneration and necrosis. The main "shock organs":

- **shock lung** — **diffuse alveolar damage** (acute respiratory distress syndrome): the lung is heavy, dark red and edematous, with **hyaline membranes** in the alveolar walls, atelectasis and hemorrhages; acute respiratory failure;
- **shock kidney** — **acute tubular necrosis** (necrotic nephrosis): the cortex is pale and swollen, the medulla dark red; necrosis of the proximal tubular epithelium; **acute kidney injury** (oliguria, anuria);
- **shock liver** — centrilobular necrosis and fatty degeneration;
- **heart** — subendocardial hemorrhages, small foci of necrosis (contraction band necrosis);
- **brain** — hypoxic encephalopathy, ischemic necrosis (in border zones);
- **digestive tract** — mucosal hemorrhages and **acute erosions and ulcers** ("stress ulcers"), hemorrhagic enteropathy;
- **adrenal glands** — lipid depletion of the cortex (consumption of cortisol), bilateral hemorrhage (Waterhouse–Friderichsen syndrome). As a result, **multiple organ dysfunction syndrome** develops.

7.12. Disorders of lymph circulation

Lymphatic insufficiency is mechanical (obstruction of lymphatic vessels by tumor, scar or parasites, or their congenital aplasia), **dynamic** (a sharp increase in tissue fluid formation) and **resorptive** (changes in tissue proteins and substances). As a result **lymphostasis**, dilatation of lymphatic vessels (**lymphangiectasia**) and **lymphedema** develop. In chronic lymphedema, **sclerosis** develops in the skin and subcutaneous tissue and the part becomes markedly enlarged — **elephantiasis** (Topic 1). When the thoracic duct is injured or obstructed, **chylous ascites** and **chylothorax** (accumulation of milky lymph — chyle) develop.

7.13. Clinicopathological section

The “silent killer” in hospital — deep vein thrombosis

A 68-year-old woman underwent surgery for a femoral neck fracture and was on bed rest for 10 days. When she got up for the first time to go to the toilet, she suddenly became short of breath and cyanotic and died. At autopsy there was a saddle thromboembolus in the pulmonary trunk and its main branches, and occlusive red thrombi in the deep veins of the left calf. The patient had all components of Virchow's triad: **slowing of blood flow** (immobility), **endothelial injury** (surgery, trauma) and **hypercoagulability** (postoperative state, old age). Therefore **prevention of thrombosis** after surgery (early mobilization, compression stockings, low-molecular-weight heparins) is mandatory.

> **CLINICAL SIGNIFICANCE**
> **Knowing the evolution of myocardial infarction has forensic and clinical significance**: the “age” of an infarct can be determined from its histologic features — neutrophils (1–3 days), macrophages and granulation tissue (3–10 days), scar (6–8 weeks). **Cardiac rupture** occurs most often on **days 3–7** — the period when the necrotic focus has been softened by macrophages but not yet replaced by a scar.

Practical skills

Skill 1. Gross specimens

1. **“Nutmeg liver”.** The liver is enlarged and firm with a smooth surface; on the cut surface red-brown (lobule centers) and yellowish (periphery) spots alternate — resembling a nutmeg. Conclusion: chronic generalized venous congestion.
2. **“Brown induration of the lung”.** The lung is enlarged and firm, brown-rust colored on the cut surface. Conclusion: chronic venous congestion in the pulmonary circulation (see Topic 4).
3. **“Cerebral hemorrhage (hematoma)”.** A cavity filled with a dark red blood clot in the region of the subcortical nuclei; the brain tissue is destroyed; the blood may have ruptured into the ventricles. Conclusion: bleeding due to rupture — hematoma (most often in essential hypertension).
4. **“Mural thrombus in the aorta”.** Atherosclerotic plaques in the aortic intima with dull, rough, layered gray-red masses firmly attached to the wall over them. Conclusion: mural mixed thrombus.
5. **“Pulmonary thromboembolism”.** Dry, dull, worm-like red-gray thromboemboli in the pulmonary trunk and its branches, not attached to the wall. Conclusion: thromboembolism.
6. **“Ischemic infarction of the spleen”.** A whitish-yellow, firm, triangular focus beneath the capsule with a red rim around it and fibrin on the capsule. Conclusion: white infarct.

7. **“Hemorrhagic infarction of the lung”.** A wedge-shaped, dark red, firm focus with its base facing the pleura; fibrin deposits on the pleura. Conclusion: red infarct.
8. **“Myocardial infarction”.** An irregular, yellowish-white, dull focus surrounded by a red rim in the wall of the left ventricle; there may be a mural thrombus on the endocardium. Conclusion: white infarct with a red rim.
9. **“Cancer metastases in the liver”.** Numerous sharply demarcated whitish-gray nodules of various sizes in the liver. Conclusion: a consequence of tumor cell embolism (hematogenous metastases).

Skill 2. Microscopic slides

1. **“Nutmeg liver”** (hematoxylin and eosin). In the center of the lobule the central vein and sinusoids are markedly dilated and filled with red cells; the central hepatocytes are atrophic or necrotic; fatty degeneration of peripheral hepatocytes. Conclusion: chronic venous congestion.
2. **“Venous congestion of the skin”** (hematoxylin and eosin). The veins and capillaries of the dermis are markedly dilated and filled with blood, with diapedetic hemorrhages and edema in places. Conclusion: venous congestion.
3. **“Stasis in cerebral capillaries”** (hematoxylin and eosin). The capillaries are dilated; red cells have stuck together to form homogeneous columns; perivascular edema. Conclusion: stasis.
4. **“Mixed thrombus in a vein”** (hematoxylin and eosin). A mass in the venous lumen attached to the wall, with alternating layers of platelets, fibrin (pink), leukocytes and red cells. Conclusion: mixed thrombus.
5. **“Fat embolism of pulmonary capillaries”** (Sudan III). The capillaries of the interalveolar septa are occluded by orange-red fat droplets. Conclusion: fat embolism.
6. **“Ischemic infarction of the kidney”** (hematoxylin and eosin). In the infarct area the outlines of tubules and glomeruli are preserved, but nuclei are absent (coagulative necrosis); hemorrhage and leukocyte infiltration at the border (demarcation zone). Conclusion: white infarct with a red rim.
7. **“Hemorrhagic infarction of the lung”** (hematoxylin and eosin). In the area of necrosis the alveoli are filled with red cells, and the nuclei of the interalveolar septa do not stain. Conclusion: red infarct.

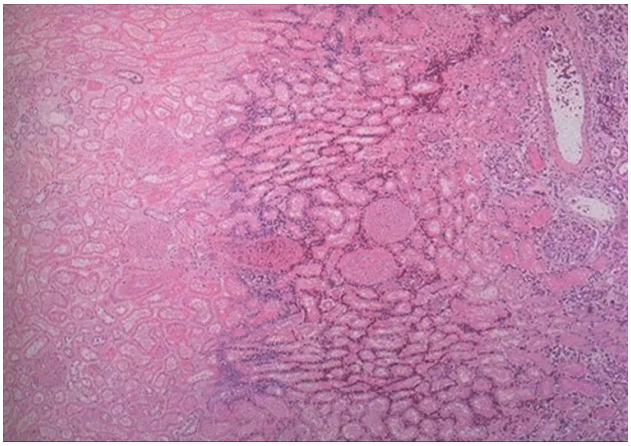


Figure 7.27. *Ischemic infarction of the kidney (microscopic slide): on the right — the infarct area (outlines preserved, no nuclei), on the left — normal tissue (TMA study guide, 2018)*

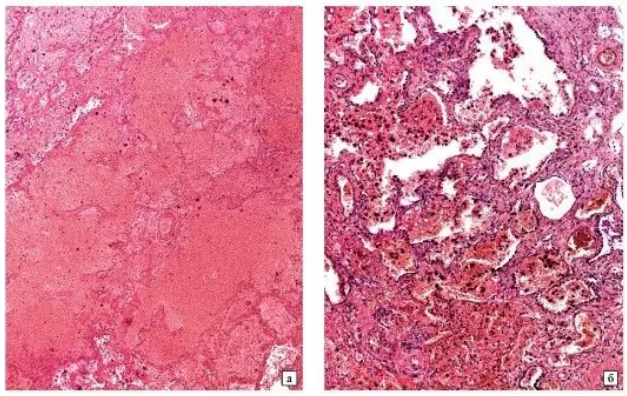


Figure 7.28. *Hemorrhagic infarction of the lung (microscopic slide): in the area of necrosis the alveoli are filled with red cells (TMA study guide, 2018)*

Review questions

1. How are circulatory disorders classified?
2. State the types of arterial hyperemia.
3. Describe the causes of generalized venous congestion and the changes in the organs. How does the “nutmeg liver” develop?
4. Local venous congestion and portal hypertension.
5. Causes and outcomes of ischemia.
6. Mechanisms of bleeding and types of hemorrhage.
7. What are plasmorrhagia and stasis?
8. Mechanisms and types of edema. Morphology of pulmonary and cerebral edema.
9. Stages of normal hemostasis and the antithrombotic properties of the endothelium.
10. Virchow's triad. Causes of hypercoagulability.
11. Types of thrombi and outcomes of thrombi. How can a thrombus be distinguished from a postmortem clot?
12. Causes, stages and morphology of DIC.
13. Types of embolism and directions of movement of emboli. What is paradoxical embolism?
14. The source and consequences of pulmonary thromboembolism.
15. What is infarction? Types of infarcts, conditions of their development, morphology and outcomes.
16. Types and stages of shock and the “shock organs”.

Test questions

1. The “nutmeg liver” is the result of which process?
   A) Fatty degeneration of the liver
   B) Hepatic amyloidosis
   C) Chronic generalized venous congestion
   D) Hepatic infarction
   E) Arterial hyperemia
2. Brown induration of the lung most often develops in:
   A) Pneumonia
   B) Anemia
   C) Right ventricular failure
   D) Mitral stenosis
   E) Portal hypertension
3. Escape of blood as a result of increased permeability while the integrity of the vessel wall is preserved:
   A) Arrosive bleeding
   B) Diapedetic bleeding
   C) Hemopericardium
   D) Hematoma
   E) Bleeding due to rupture

4. The main mechanism of edema in nephrotic syndrome:
    A) Decreased oncotic pressure (hypoalbuminemia)
    B) Increased hydrostatic pressure
    C) Impaired lymph flow
    D) Arterial hyperemia
    E) Increased vascular permeability

5. Not part of Virchow's triad:
    A) Endothelial injury
    B) Arterial hyperemia
    C) Hypercoagulability
    D) Slowing of blood flow
    E) All are included

6. The most common hereditary thrombophilia:
    A) Excess of protein C
    B) Hemophilia A
    C) Excess of antithrombin III
    D) Von Willebrand disease
    E) Factor V mutation (Leiden)

7. A thrombus that forms slowly in arteries with rapid blood flow:
    A) Red
    B) White
    C) Hyaline
    D) Dilatation
    E) Marantic

8. A feature characteristic of a thrombus:
    A) Not attached to the vessel wall
    B) “Chicken fat” and “currant jelly” layers
    C) Completely follows the shape of the vessel
    D) Dull, dry, friable, attached to the wall
    E) Smooth, shiny, elastic

9. The main source of pulmonary thromboembolism:
    A) Deep veins of the leg
    B) Dural venous sinuses
    C) Aorta
    D) Portal vein
    E) Mural thrombi of the left ventricle

10. Passage of a leg vein thrombus through a patent foramen ovale into a cerebral artery:
    A) Air embolism
    B) Fat embolism
    C) Retrograde embolism
    D) Orthograde embolism
    E) Paradoxical embolism

11. Caisson disease belongs to which type of embolism?
    A) Air
    B) Thromboembolism
    C) Gas
    D) Fat
    E) Tissue

12. The embolism that develops in fractures of long bones:
    A) Parasitic
    B) Gas
    C) Microbial
    D) Air
    E) Fat

13. The condition for the development of a red (hemorrhagic) infarct in the lung:
    A) Arterial hyperemia
    B) Solid tissue
    C) Dual blood supply and venous congestion
    D) A single blood supply only
    E) Good collateral flow

14. The infarct most often found in the spleen and kidney:
    A) White (ischemic)
    B) Venous
    C) Liquefactive
    D) Septic
    E) Red

15. The change that develops in the lung in shock:
    A) Caseous pneumonia
    B) Brown induration
    C) Anthracosis
    D) Diffuse alveolar damage (hyaline membranes)
    E) Emphysema

Clinical cases

Case 1

A 65-year-old woman had rheumatic fever 20 years ago, and mitral stenosis was diagnosed. In recent years she developed shortness of breath and blood-streaked sputum, then leg edema and heaviness under the right costal margin. She died of chronic heart failure.

Questions: 1) What changes are expected in the lungs? 2) What does the liver look like, and why? 3) What outcome can this process lead to in the liver?

Answer: 1) Brown induration of the lung — chronic venous congestion in the pulmonary circulation due to mitral stenosis: diapedetic hemorrhages, hemosiderin-laden macrophages (“heart failure cells”), sclerosis of the interalveolar septa. 2) “Nutmeg liver” — after right ventricular failure develops, congestion in the inferior vena cava: in the center of the lobule the vein and sinusoids are filled with blood, with atrophy and necrosis of the central hepatocytes (red-brown) and fatty

degeneration of the peripheral hepatocytes (yellowish). 3) Nutmeg fibrosis and cardiac (nutmeg) cirrhosis.

Case 2

A 52-year-old man died 2 hours after severe chest pain. At autopsy no gross changes were found, except for an occlusive thrombus over an atherosclerotic plaque in the anterior descending branch of the left coronary artery.

Questions: 1) What is the cause of death? 2) Why are there no gross changes in the myocardium, and how can the infarction be confirmed? 3) If the patient had survived, what complications and changes would have developed over the following days?

Answer: 1) Acute myocardial infarction (stage of acute ischemia), coronary artery thrombosis; death — probably from ventricular fibrillation or cardiogenic shock. 2) The gross signs of necrosis appear after 12–24 hours, and the histologic signs after 4–12 hours. Confirmation — histochemical methods: triphenyltetrazolium chloride (the area that has lost dehydrogenase activity does not stain), loss of glycogen with the PAS reaction; mitochondrial damage on electron microscopy. 3) On days 1–3 — neutrophilic demarcation inflammation; on days 3–7 — macrophages, risk of cardiac rupture (hemopericardium, tamponade), mural thrombus and thromboembolism, fibrinous pericarditis, acute aneurysm; at 6–8 weeks — scar (large-focal cardiosclerosis), chronic aneurysm.

Case 3

A 25-year-old man fractured his femur in a road traffic accident. On the second day he developed shortness of breath, cyanosis and confusion; pinpoint hemorrhages appeared on the skin of the chest and axillae; the blood platelet count decreased. The patient died.

Questions: 1) Your diagnosis? 2) Its mechanism and the changes in the lungs and brain. 3) By what method is it confirmed microscopically?

Answer: 1) Fat embolism (fat embolism syndrome). 2) Fat droplets from the yellow marrow of the fractured bone enter the veins and first occlude the pulmonary capillaries (pulmonary insufficiency); if abundant, they pass through the lungs and occlude the capillaries of the brain and other organs — numerous petechiae in the brain (“brain purpura”), neurological signs. In addition, fatty acids damage the endothelium, and platelet aggregation causes thrombocytopenia. 3) Staining frozen sections with Sudan III (or osmic acid) — orange-red (or black) fat droplets in the capillaries (with routine paraffin processing the fat dissolves).

Quick review

Table 7.2. *Key facts on Topic 7*

Question	Answer
Nutmeg liver	Chronic generalized venous congestion (right ventricular failure) → cirrhosis
Brown induration	Mitral stenosis, left ventricular failure; hemosiderosis + sclerosis
Signs of portal hypertension	Ascites, splenomegaly, esophageal varices, “caput medusae”
Mechanisms of bleeding	Rupture (rhexis), erosion (diabrosis), diapedesis
Mechanisms of edema	Hydrostatic ↑, oncotic ↓, permeability ↑, lymphatic obstruction, Na retention
Virchow's triad	Endothelial injury, abnormal blood flow, hypercoagulability

Question	Answer
The most common hereditary thrombophilia	Factor V (Leiden) mutation
White / red thrombus	Artery, rapid flow / vein, slow flow
Mixed thrombus	Head (white) + body (laminated) + tail (red)
Hyaline thrombus	Microvessels — DIC, shock
Outcomes of a thrombus	Dissolution, organization, canalization, calcification / embolism, septic dissolution
DIC	Microthrombi + consumptive coagulopathy → bleeding
Source of PE	Deep veins of the leg (95%)
Paradoxical embolism	From a vein to an artery through a septal defect of the heart
Fat embolism	Fracture of a long bone; Sudan III
Gas embolism	Caisson (decompression) disease — nitrogen
Air embolism	Injury to the neck veins; more than 100 mL
White infarct	Spleen, kidney (end arteries, solid tissue)
Red infarct	Lung, intestine (dual supply, venous congestion)
Myocardial infarction	White with a red rim; rupture on days 3–7; scar at 6–8 weeks
Cerebral infarction	Liquefactive necrosis → cyst
Types of shock	Cardiogenic, hypovolemic, septic, anaphylactic, neurogenic
Shock lung / kidney	Diffuse alveolar damage / acute tubular necrosis

Glossary of terms

- **Hyperemia** (Greek hyper — excess, haima — blood) — local filling with blood (plethora of a part).
- **Plethora** (Greek plethora — fullness) — an increase in the total blood volume of the organism.
- **Vacant hyperemia** (Latin vacuus — empty) — hyperemia associated with a fall in barometric pressure.
- **Erythema** (Greek erythros — red) — reddening of the skin as a result of active hyperemia.
- **Cyanosis** (Greek kyanos — dark blue) — bluish-red coloration of the skin and mucous membranes due to deoxygenated blood.
- **Induration** (Latin induratio — hardening) — sclerosis of tissues as a result of chronic venous congestion.
- **Ischemia** (Greek ischo — I hold back, haima — blood) — local anemia resulting from reduced blood inflow.
- **Hemorrhage** (Greek haimorrhagia) — bleeding.
- **Arrosive bleeding** (Latin arrodere — to gnaw away) — bleeding resulting from erosion of the vessel wall.
- **Diapedesis** (Greek dia — through, pedao — I leap) — passage of red and white blood cells through the vessel wall.
- **Hemopericardium** — accumulation of blood in the pericardial cavity.
- **Stasis** (Greek stasis — standstill) — cessation of blood flow in capillaries.

- **Sludge** (English sludge — mud) — blood in a state of red cell aggregation.
- **Transudate** (Latin trans — through, sudare — to seep) — edema fluid.
- **Anasarca** (Greek ana — over, sarx — flesh) — diffuse edema of the subcutaneous fatty tissue.
- **Ascites** (Greek askites — like a leather bag) — noninflammatory accumulation of fluid in the peritoneal cavity.
- **Hydrothorax / hydropericardium** — accumulation of edema fluid in the pleural / pericardial cavity.
- **Hydrocephalus** (Greek hydor — water, kephale — head) — accumulation of cerebrospinal fluid in the ventricles of the brain (internal) or the subarachnoid space (external).
- **Chylous ascites** (Greek chylos — juice) — accumulation of chyle (lymph) in the peritoneal cavity.
- **Exsiccosis** (Latin exsiccare — to dry out) — a decrease in interstitial fluid (dehydration).
- **Homeostasis** (Greek homoios — same, stasis — state) — the constancy of the internal environment of the organism.
- **Thrombosis** (Greek thrombosis — clotting) — clotting of blood in a vessel or cardiac chamber during life.
- **Coagulation** (Latin coagulatio — clotting) — clotting of blood.
- **Agglutination** (Latin agglutinatio — gluing together) — clumping and sticking together of cellular elements (platelets, red cells, bacteria).
- **Adhesion** (Latin adhaesio — sticking) — sticking to a surface.
- **Precipitation** (Latin praecipitatio — falling down) — deposition of a substance from a solution as a precipitate.
- **Retraction** (Latin retractio — contraction) — contraction (shrinkage) of a blood clot.
- **Occlusive (obturating) thrombus** (Latin obturare — to stop up) — a thrombus that completely occludes the vascular lumen.
- **Dilatation thrombus** (Latin dilatare — to widen) — a thrombus in aneurysms and dilated vessels.
- **Marantic thrombus** (Greek marasmos — wasting) — a (stasis) thrombus resulting from slowed blood flow in debilitated patients.
- **Embolism** (Greek emballein — to throw in) — circulation in blood or lymph of particles not normally found there and occlusion of vessels by them.
- **Retrograde embolism** — movement of an embolus against the blood flow.
- **Paradoxical embolism** — passage of an embolus from the systemic veins into the arteries, bypassing the lungs.
- **Metastasis** (Greek meta — elsewhere, stasis — placement) — transfer of elements capable of growth and multiplication (tumor cells, microbes) to another site with blood or lymph.
- **Infarction** (Latin infarcire — to stuff) — tissue necrosis resulting from impaired circulation (ischemia).
- **Shock** (English shock — blow) — a state of tissue hypoxia and cell injury resulting from systemic hypoperfusion.

TOPIC 8

Diseases of the immune system. Hypersensitivity reactions, autoimmune diseases, immunodeficiency. Amyloidosis

Learning objectives

The student should know:

- the structure of the immune system: central and peripheral organs, T and B lymphocytes, macrophages, natural killer cells, the major histocompatibility complex;
- morphologic changes in the immune organs in response to antigenic stimulation; accidental transformation of the thymus;
- the four types of hypersensitivity reactions: mechanisms, morphology and examples;
- the types of transplant rejection;
- autoimmune diseases: mechanisms of loss of tolerance, systemic lupus erythematosus, rheumatoid arthritis, systemic sclerosis (scleroderma), Hashimoto thyroiditis;
- primary and secondary immunodeficiency states; HIV infection and AIDS;
- amyloidosis: the structure and composition of amyloid, methods of detection, classification, pathogenesis, organ changes and clinical significance.

The student should be able to:

- describe gross specimens of the “sago” and “lardaceous” spleen, renal amyloidosis, Hashimoto goiter and lupus nephritis;
- describe microscopic slides of accidental transformation of the thymus, a lymph node in AIDS, amyloidosis of the spleen, kidney and adrenal glands (Congo red), and Hashimoto thyroiditis;
- identify the type of hypersensitivity from a clinical example;
- identify the type of amyloidosis from the clinical situation.

8.1. Structure of the immune system

The immune system protects the organism from foreign substances (antigens) and distinguishes **“self”** from **“nonself”**. It consists of **innate (nonspecific)** and **acquired (adaptive, specific)** immunity.

- **Innate immunity** — epithelial barriers, phagocytes (neutrophils, macrophages), dendritic cells, **natural killer (NK) cells**, the complement system; it recognizes microbes rapidly but nonspecifically (Toll-like receptors).
- **Acquired immunity** — **lymphocytes** and their products; specific and with memory. **Humoral immunity** — B lymphocytes and antibodies; **cell-mediated immunity** — T lymphocytes.

Organs of the immune system:

- **central (primary)** — the **thymus** (T lymphocytes mature) and **bone marrow** (B lymphocytes; in birds, the bursa of Fabricius — hence the name “bursa-dependent” or B lymphocytes);
- **peripheral (secondary)** — lymph nodes, spleen, mucosa-associated lymphoid tissue (tonsils, Peyer patches of the intestine — MALT).

Main cells:

- **T lymphocytes** (thymus-dependent): **$CD4^+$ helper T cells** (they produce cytokines and regulate the immune response: Th1, Th2, Th17), **$CD8^+$ cytotoxic T lymphocytes (killer T cells)** — kill virus-infected and tumor cells, **regulatory T lymphocytes**

(suppressor T cells) — suppress the immune response and maintain tolerance to self antigens. In the lymph node they lie in the paracortical zone, in the spleen around the arterioles. • **B lymphocytes** — under the action of antigen they turn into **plasma cells** and produce antibodies (immunoglobulins — IgG, IgM, IgA, IgE, IgD). They are located in the follicles of lymph nodes and in the follicles of the splenic white pulp. • **Macrophages and dendritic cells** — process antigen and **present** it to T lymphocytes (antigen-presenting cells). • **NK cells** — kill virus-infected and tumor cells without prior sensitization. **The major histocompatibility complex (MHC; in humans — HLA)** — proteins on the cell surface: **class I** (on all nucleated cells; presents antigen to CD8+ T lymphocytes) and **class II** (on antigen-presenting cells; to CD4+ T lymphocytes). HLA molecules determine tissue compatibility in transplantation; some HLA types are associated with predisposition to autoimmune diseases (for example, HLA-B27 — ankylosing spondylitis, HLA-DR4 — rheumatoid arthritis).

Changes in the immune organs in response to antigenic stimulation

Under the action of antigen the **lymph nodes and spleen** enlarge: in the **humoral** response — **follicular hyperplasia** (large germinal centers) and an increase in plasma cells (**plasmacytic transformation**), especially in the medullary cords of the lymph node; in the **cellular** response — **hyperplasia of the paracortical (T) zone**.

Accidental (stress) transformation (involution) of the thymus — rapid shrinkage of the thymus under the action of glucocorticoids in severe stress, infections, intoxication and tumors: lymphocytes in the cortex disappear by apoptosis (a **"starry sky"** picture — macrophages that have engulfed lymphocyte debris), then the border between the cortex and medulla disappears (inversion), and Hassall corpuscles become cystically dilated. **Thymomegaly** — congenital enlargement of the thymus; in older literature, the **"thymicolymphatic state"** (status thymicolymphaticus) was described as a condition with hyperplasia of the thymus, spleen and lymph nodes and hypoplasia of the adrenal glands, predisposing children to sudden death; it is now regarded as a form of congenital immunodeficiency.

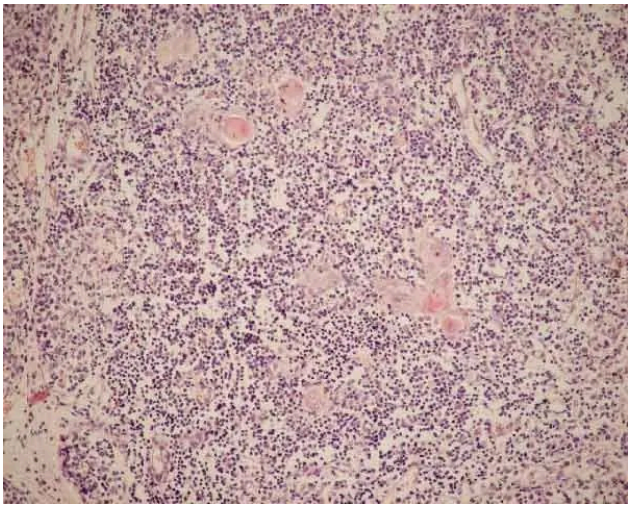


Figure 8.1. *Accidental transformation of the thymus: lymphocytes are depleted in the cortex, and the lobular borders are disrupted (TMA study guide, 2018)*

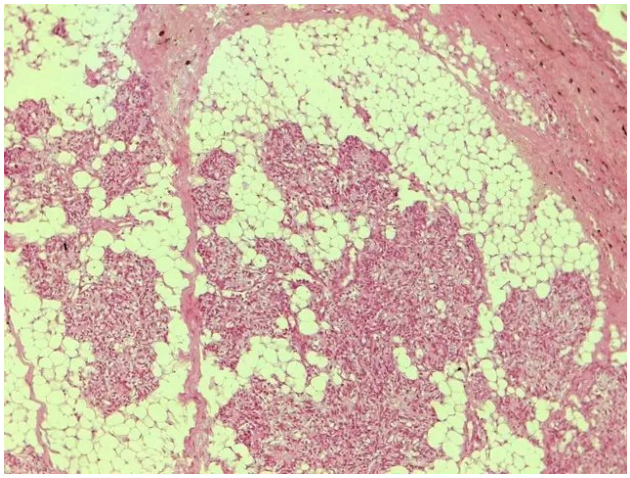


Figure 8.2. *Thymic dysplasia in Nezelof syndrome: the lobules are small, lymphocytes are few, and Hassall corpuscles are absent (TMA study guide, 2018)*

8.2. Hypersensitivity reactions

Hypersensitivity reactions are an **excessive or inappropriate** immune response of the organism to an antigen (exogenous or self), resulting in tissue damage. They are based on **allergy** (Greek allos — other, ergon — action) — altered reactivity of the organism. According to the Gell and Coombs classification, four types are distinguished:

Table 8.1. *Types of hypersensitivity reactions*

Type	Mechanism	Morphology	Examples
Type I — immediate (anaphylactic)	IgE bound to mast cells and basophils; when the antigen enters again, they degranulate (histamine, leukotrienes, prostaglandins)	Vasodilation, edema, smooth muscle contraction, excessive mucus secretion; eosinophil infiltration in the late phase	Anaphylactic shock, bronchial asthma, allergic rhinitis (pollinosis), urticaria, Quincke edema, food allergy
Type II — antibody-mediated (cytotoxic)	IgG or IgM binds to an antigen on the cell surface or in the tissue matrix → opsonization and phagocytosis, complement activation (lysis), inflammation, impaired receptor function	Cell destruction (hemolysis), inflammation, functional disturbance	Incompatible blood transfusion, hemolytic disease of the newborn, autoimmune hemolytic anemia, Goodpasture syndrome, rheumatic fever; Graves disease (antibodies stimulating the TSH receptor), myasthenia gravis (antibodies blocking the acetylcholine receptor)
Type III — immune complex	Antigen–antibody immune complexes circulate in the blood and are deposited in vessel walls → complement activation, neutrophils → inflammation	Vasculitis — fibrinoid necrosis, neutrophil infiltration; glomerulonephritis, arthritis	Systemic lupus erythematosus, postinfectious (poststreptococcal) glomerulonephritis, polyarteritis nodosa, serum sickness; local — Arthus phenomenon
Type IV — T cell–mediated (delayed)	Sensitized $CD4^+$ T lymphocytes release cytokines (IFN-gamma) and activate macrophages, or $CD8^+$ T lymphocytes kill cells directly; the reaction occurs in 24–72 hours	Lymphocytic-macrophage infiltrate, granulomas (Topic 6)	Tuberculin (Mantoux) test, tuberculosis, leprosy, contact dermatitis, type 1 diabetes mellitus, multiple sclerosis, rheumatoid arthritis, Crohn disease, transplant rejection

MEMORY AID

Remember the types of hypersensitivity with "**ACID**": **A** — Anaphylactic (I, IgE, immediate), **C** — Cytotoxic (II, IgG/IgM against cells), **I** — Immune complex (III), **D** — Delayed (IV, T lymphocytes). The first three are mediated by **antibodies**, the fourth by **cells**.

CLINICAL SIGNIFICANCE

Anaphylactic shock is the most severe form of type I reaction: within a few minutes of exposure to the antigen (penicillin, insect venom, nuts), systemic vasodilation, increased permeability, bronchospasm and laryngeal

edema develop — a risk of death. First aid — **epinephrine (adrenaline)** intramuscularly. At autopsy: laryngeal edema, pulmonary emphysema and edema, congestion and petechiae in the organs, eosinophils in the internal organs.

8.3. Transplant rejection

When an allogeneic organ (from another person) is transplanted, the recipient's immune system recognizes the donor **HLA** antigens as foreign. Types of rejection:

- **Hyperacute** — within minutes to hours: the recipient has **preexisting** antibodies against donor antigens (ABO incompatibility, previous transplantation, transfusion, pregnancy) — type III/II; thrombosis and fibrinoid necrosis in the vessels, **ischemic necrosis** of the organ. It is now rare thanks to pretransplant crossmatching.
- **Acute** — days, weeks, months: **cellular** (CD8+ cytotoxic T lymphocytes — interstitial lymphocytic infiltration, tubular damage — "tubulitis") and **humoral** (antibodies — vasculitis, endothelial damage, neutrophils in capillaries, C4d deposits) mechanisms; it responds to immunosuppressive therapy.
- **Chronic** — months, years: thickening of the arterial intima (**transplant arteriopathy**), interstitial fibrosis, parenchymal atrophy; gradual loss of organ function. **"Graft-versus-host" disease** — when bone marrow (immunocompetent cells) is transplanted, donor T lymphocytes attack the recipient's tissues: the skin (rash), liver (jaundice) and intestine (diarrhea) are affected.

8.4. Autoimmune diseases

In **autoimmune diseases**, tissues are damaged as a result of an immune response against the organism's **own antigens** (autoantibodies and self-reactive T lymphocytes). Normally **immunologic tolerance** to self antigens is maintained:

- **central tolerance** — lymphocytes that recognize self antigens are eliminated by apoptosis in the thymus and bone marrow (**clonal deletion**);
- **peripheral tolerance** — anergy, suppression by **regulatory T lymphocytes**, activation-induced death (Fas–FasL). **Causes of loss of tolerance: genetic predisposition** (HLA genes, PTPN22, AIRE and others); **infections** — **molecular mimicry** (a microbial antigen resembles a self antigen — streptococcal M protein and cardiac antigens in rheumatic fever), tissue damage and exposure of hidden antigens; hormonal factors (most autoimmune diseases are more common in **women**), drugs and chemicals, ultraviolet light (SLE).

Classification:

- **organ-specific** — Hashimoto thyroiditis, Graves disease, type 1 diabetes mellitus, autoimmune gastritis (pernicious anemia), myasthenia gravis, multiple sclerosis, Goodpasture syndrome;
- **systemic (non-organ-specific)** — **systemic lupus erythematosus**, **rheumatoid arthritis**, **systemic sclerosis (scleroderma)**, dermatomyositis, Sjögren syndrome, polyarteritis nodosa (most of them in the older classification are "**collagen diseases**" or **rheumatic diseases** — systemic involvement of connective tissue, the disorganization of Topic 3).

Systemic lupus erythematosus (SLE)

SLE is a systemic autoimmune disease affecting many organs and associated with **type III** hypersensitivity; mainly in **young women** (female:male ≈ 9:1). Its main feature is **antinuclear antibodies**

(ANA); specific ones are antibodies against **double-stranded DNA** and the **Smith (Sm) antigen**. The antibodies damage nucleated cells and form immune complexes.

Morphology:

- **vessels** — **fibrinoid necrosis** and acute necrotizing **vasculitis** in small arteries and arterioles; immune complex deposits;
- **kidneys — lupus nephritis** (in more than 50% of patients, one of the main causes of death): immune complexes are deposited in the glomeruli (subendothelially), and the capillary basement membrane thickens — "**wire loops**"; proliferative glomerulonephritis, **hyaline thrombi**, fibrinoid necrosis;
- **skin** — a **"butterfly" erythema** on the face (on the cheeks and bridge of the nose), photosensitivity; immunoglobulin and complement deposits at the dermoepidermal junction;
- **joints** — nonerosive synovitis (arthritis);
- **serous membranes** — fibrinous pericarditis, pleurisy;
- **heart** — **Libman–Sacks endocarditis** (small sterile vegetations on both surfaces of the valves), myocarditis;
- **spleen** — concentric periarterial fibrosis around the follicular arteries (**"onion skin"**);
- **blood** — hemolytic anemia, thrombocytopenia, leukopenia (type II); **antiphospholipid antibodies** — thromboses, miscarriage;
- **brain** — vasculitis, thrombosis, psychiatric disturbances. Histologic signs: **hematoxylin bodies** (LE bodies — nuclei altered by antibodies), **LE cells** in the blood (a neutrophil that has engulfed a hematoxylin body).

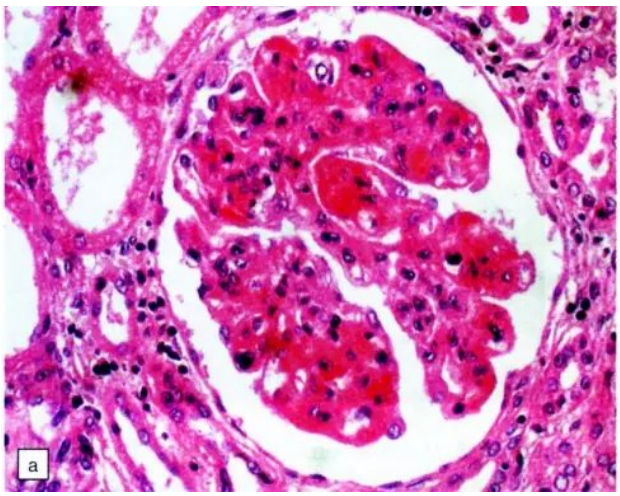


Figure 8.3. *Renal glomerulus in systemic lupus erythematosus: a — fibrinoid swelling and necrosis in the capillaries; b — fixation of IgG in fibrinoid areas (immunofluorescence) (Strukov, Serov, Fig. 6-6)*



Figure 8.4. *Lupus nephritis (electron micrograph): dense immune complex deposits in the thickened capillary basement membrane (TMA study guide, 2018)*

Rheumatoid arthritis

Rheumatoid arthritis is a chronic systemic autoimmune disease that mainly affects the **joints** (the small joints of the hands and feet — metacarpophalangeal and proximal interphalangeal joints, symmetrically). Mechanism — **type IV** (CD4+ T lymphocytes, TNF) and **type III** (immune complexes). In the blood — **rheumatoid factor** (IgM against the Fc portion of IgG) and **anti–cyclic citrullinated peptide antibodies (ACPA)**.

Morphology: chronic inflammation of the synovium — **hyperplasia of synovial cells**, lymphocytes, plasma cells, lymphoid follicles, new vessels; granulation tissue forming villous projections — the

pannus — grows into the cartilage and destroys it and the bone (**erosions**). Outcome — fibrous and bony **ankylosis**, joint deformities (ulnar deviation of the fingers, "**swan neck**"). Extra-articular — **rheumatoid nodules** (subcutaneous, at the elbow: fibrinoid necrosis in the center surrounded by palisading macrophages), vasculitis, pulmonary fibrosis, **AA amyloidosis**.

Systemic scleroderma (systemic sclerosis)

Systemic sclerosis is a disease characterized by activation of the immune system, damage to small vessels and **progressive fibrosis of the skin and internal organs**. Fibroblasts are activated by TGF-beta and other cytokines. The **skin** is thickened, dense and immobile; the fingers are thin and claw-like (**sclerodactyly**); the face is masklike; **Raynaud phenomenon**. **Internal organs:** esophagus (fibrosis, impaired motility), lungs (interstitial fibrosis, pulmonary hypertension), kidneys (thickening of the arterial intima — "**scleroderma renal crisis**"), heart. Antibodies: **Scl-70** (diffuse form), **anticentromere** (limited form — **CREST syndrome**: calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, telangiectasias).

Hashimoto thyroiditis

Hashimoto thyroiditis (autoimmune thyroiditis) is an organ-specific autoimmune disease and the most common cause of **hypothyroidism** (in iodine-sufficient areas). Antibodies against thyroid peroxidase and thyroglobulin; mainly a **type IV** (CD8+ T lymphocytes kill thyrocytes) and type II mechanism. **Morphology:** the thyroid gland is diffusely enlarged (goiter), firm and whitish-gray on the cut surface; a dense **lymphoplasmacytic infiltrate** in the parenchyma, **lymphoid follicles with germinal centers**, follicular atrophy; thyrocytes are large, with eosinophilic granular cytoplasm — **Hürthle (Askanazy) cells**; fibrosis. Complications — hypothyroidism (myxedema), risk of **B-cell lymphoma**.

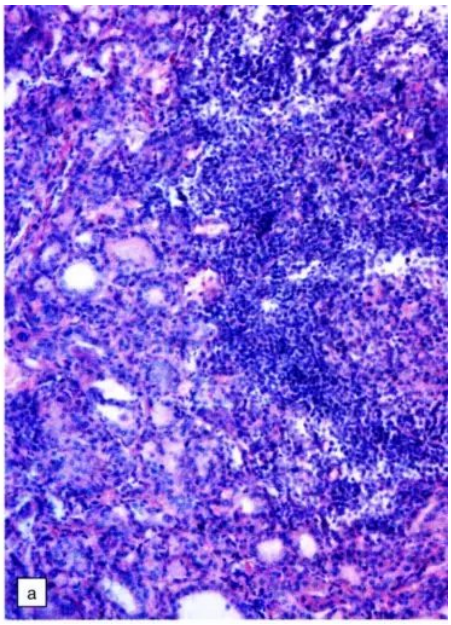


Figure 8.5. *Hashimoto thyroiditis: a — infiltration of thyroid tissue by lymphocytes, destruction of parenchymal elements; b — a lymphocyte passing between follicular cells (electron micrograph) (Strukov, Serov, Fig. 6-7)*

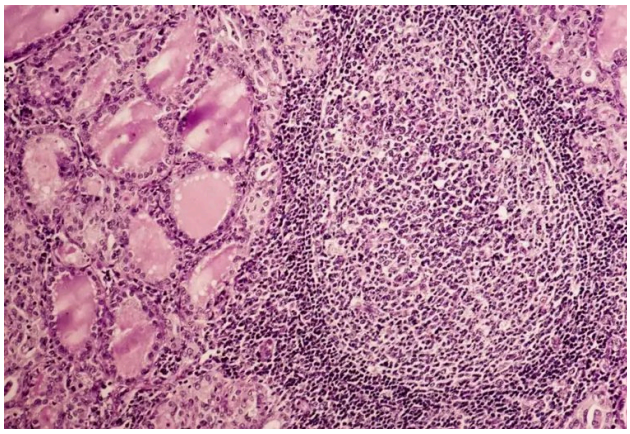


Figure 8.6. *Hashimoto goiter (microscopic slide): lymphoid follicles with germinal centers and a lymphocytic infiltrate in the parenchyma, follicular atrophy (TMA study guide, 2018)*



Figure 8.7. *Outcome of Hashimoto thyroiditis: the thyroid gland is diffusely enlarged and firm (TMA study guide, 2018)*

8.5. Immunodeficiency states

Immunodeficiency is a decrease in the organism's resistance to infections and tumors as a result of a deficiency of one or more components of the immune system. It is **primary (congenital, hereditary)** or **secondary (acquired)**. The common clinical feature is **recurrent, severe and opportunistic infections** (caused by opportunistic microbes) and a predisposition to tumors and autoimmune diseases.

Table 8.2. *Main primary immunodeficiency states*

Disease	Defect	Morphology and clinical features
Bruton disease (X-linked agammaglobulinemia)	Mutation of the BTK (Bruton tyrosine kinase) gene; B lymphocytes do not mature; only in boys	B lymphocytes and immunoglobulins are almost absent from the blood; no germinal centers or plasma cells in lymph nodes; recurrent bacterial infections (pneumonia, otitis) after 6 months of age (when maternal IgG is exhausted)
DiGeorge syndrome	22q11 deletion; failure of development of the 3rd and 4th pharyngeal pouches	Hypoplasia or aplasia of the thymus — T-cell deficiency; absence of the parathyroid glands — hypocalcemia, tetany; defects of the heart and great vessels, facial anomalies; viral and fungal infections
Severe combined immunodeficiency	Impaired function of T and B lymphocytes (IL-2 receptor gamma chain, adenosine deaminase deficiency)	The thymus is small and lacks lymphocytes; lymphoid tissue is almost absent; severe infections from the first months of life (candidiasis, pneumocystosis); treatment — bone marrow transplantation
Selective IgA deficiency	Impaired IgA production	The most common primary immunodeficiency; often asymptomatic; respiratory and intestinal infections, allergy, autoimmune diseases
Wiskott–Aldrich syndrome	X-linked (WAS gene)	Thrombocytopenia (bleeding), eczema, recurrent infections; risk of lymphoma

Disease	Defect	Morphology and clinical features
Complement defects	Deficiency of C2, C3, C5–C9	C3 — severe bacterial infections; C5–C9 — neisserial (meningococcal) infections; C1 inhibitor — hereditary angioedema
Phagocyte defects	NADPH oxidase, adhesion molecules, Chédiak–Higashi	Chronic granulomatous disease, leukocyte adhesion deficiency (Topic 5)

Secondary immunodeficiency is much more common than primary: **HIV infection (AIDS)**, protein-energy malnutrition, old age, malignant tumors (especially leukemias and lymphomas), **immunosuppressive therapy** (transplantation, chemotherapy, glucocorticoids), radiation, splenectomy (encapsulated bacteria — pneumococcal infections), diabetes mellitus, chronic kidney disease, nephrotic syndrome (loss of immunoglobulins).

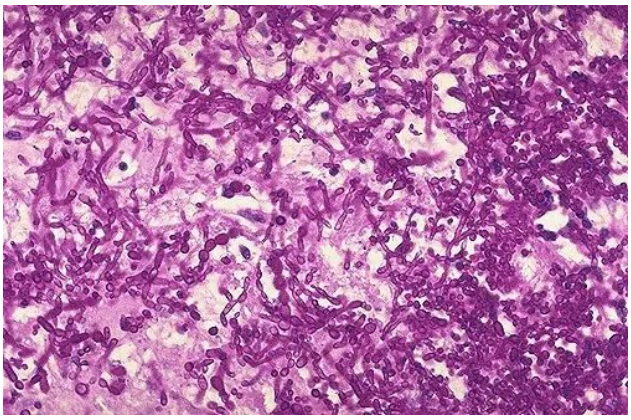


Figure 8.8. *Candida sepsis — fungal vegetations in the tissues of the esophagus (PAS reaction) — in immunodeficiency (TMA study guide, 2018)*

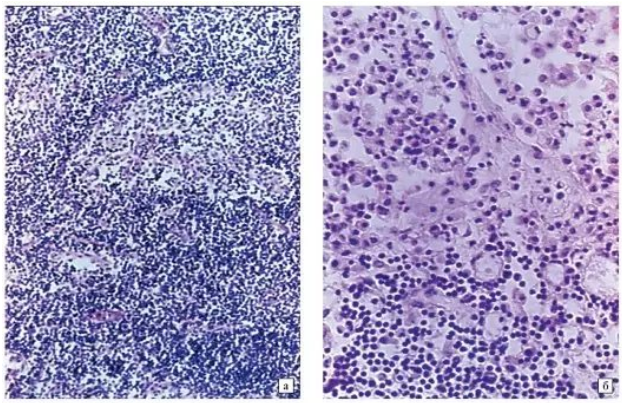


Figure 8.9. *A "depleted" lymph node in AIDS: lymphoid follicles and lymphocytes are markedly decreased (TMA study guide, 2018)*

HIV infection and AIDS

Acquired immunodeficiency syndrome (AIDS) is the final stage of infection caused by the **human immunodeficiency virus (HIV)**. HIV is a **retrovirus** (lentivirus); mainly HIV-1. **Routes of transmission:** sexual, parenteral (injecting drug use, contaminated instruments, blood transfusion), vertical (from mother to child — during pregnancy, childbirth and breastfeeding).

Pathogenesis: through its **gp120** protein the virus binds to the **CD4** molecule and coreceptors (**CCR5, CXCR4**) and infects **CD4$^+$ T lymphocytes**, macrophages and dendritic cells. The viral RNA is converted to DNA by **reverse transcriptase** and integrated into the cell genome. CD4+ T lymphocytes gradually die — cell-mediated immunity is profoundly impaired; macrophages become a **reservoir** of the virus and carry it to the brain.

Stages:

1. **Acute (primary) infection** — at 3–6 weeks: a mononucleosis-like syndrome (fever, lymphadenopathy, rash), high viremia, a transient decrease in CD4+ cells; seroconversion.
2. **Chronic (latent, clinically silent) phase** — on average 7–10 years: few symptoms, **persistent generalized lymphadenopathy**; the virus replicates actively in lymphoid organs, and CD4+ cells gradually decrease.
3. **AIDS** — CD4+ T lymphocytes **fewer than 200/μL**: **opportunistic infections**, **tumors**, cachexia, neurological disorders.

Morphology of AIDS:

- **lymph nodes** — first follicular hyperplasia, then **follicular involution** and a **sharp decrease** in lymphocytes (“depletion” of lymphoid tissue);
- **opportunistic infections** — **Pneumocystis jirovecii** pneumonia (the most common; foamy eosinophilic exudate in the alveoli), candidiasis (mouth, esophagus), cytomegalovirus infection (retina, intestine), atypical mycobacterioses and tuberculosis (the main cause of death in our region), cryptococcal meningitis, toxoplasmosis (brain abscesses), cryptosporidiosis (diarrhea), progressive multifocal leukoencephalopathy;
- **tumors** — **Kaposi sarcoma** (associated with HHV-8; red-violet patches and nodules on the skin), **B-cell lymphomas** (especially primary brain lymphoma), cervical and anal cancer (papillomavirus);
- **central nervous system** — HIV encephalopathy (microglial nodules, multinucleated giant cells) — **HIV-associated dementia**;
- **cachexia** (“wasting syndrome”).

CLINICAL SIGNIFICANCE

Modern **highly active antiretroviral therapy (ART)** stops HIV replication, restores CD4+ cell counts and prevents the development of AIDS — the life expectancy of people living with HIV can be almost normal. Therefore **early diagnosis** (HIV testing) and treatment are most important. ART is used during pregnancy to prevent mother-to-child transmission.

8.6. Amyloidosis

Amyloidosis is a **stromal-vascular protein degeneration** (Topic 3) characterized by **extracellular** deposition in tissues of **amyloid** — a fibrillar protein with a beta-pleated sheet structure formed from misfolded proteins. The name amyloid (Latin amylum — starch) was given by R. Virchow: he saw that this substance stains with iodine like starch.

Structure and composition of amyloid

Amyloid is a complex **glycoprotein**. On electron microscopy, amyloid consists of nonbranching, randomly arranged **fibrils** 7.5–10 nm in diameter. On X-ray crystallography the fibrils have a **cross-beta-pleated sheet** structure — this structure determines the characteristic staining of amyloid and its resistance to proteolysis. Composition of amyloid: the **fibrillar protein (F component)** — 95% (varying with the type of amyloidosis) and the **plasma component (P component)** — 5% (serum amyloid P component, glycosaminoglycans).

The main fibrillar proteins:

- **AL (amyloid light chain)** — immunoglobulin **light chains** (usually lambda); produced by **plasma cells**;
- **AA (amyloid-associated)** — derived from the acute phase protein **serum amyloid A (SAA)** synthesized in the liver; it increases in chronic inflammation under the action of IL-1 and IL-6;
- **ATTR (transthyretin)** — a thyroxine- and retinol-transporting protein; mutant (hereditary) or normal (senile);
- **A-beta-2-microglobulin** — in patients on hemodialysis;
- **A-beta (beta-amyloid)** — in brain plaques in Alzheimer disease;
- **AIAPP (amylin)** — in the pancreatic islets in type 2 diabetes mellitus;
- **AE (endocrine)** — procalcitonin (in medullary carcinoma of the thyroid).

Methods of detecting amyloid

- **Gross** — the **Virchow reaction**: when Lugol solution (iodine) and sulfuric acid are applied to the cut surface of an organ, amyloid turns **blue-violet** or green; with iodine alone — brown-orange.
- **Congo red** — the main method: amyloid stains **red-orange**; in **polarized light** — **apple-green birefringence** (dichroism) — pathognomonic for amyloid.
- **Metachromasia** — **red** with methyl (gentian) violet.
- **Thioflavin T** — yellow-green fluorescence on fluorescence microscopy.
- **Immunohistochemistry and mass spectrometry** — determination of the amyloid **type** (AL, AA, ATTR); necessary for choosing treatment.
- With hematoxylin and eosin, amyloid is an **extracellular**, **homogeneous, amorphous, eosinophilic (pale pink)** mass.

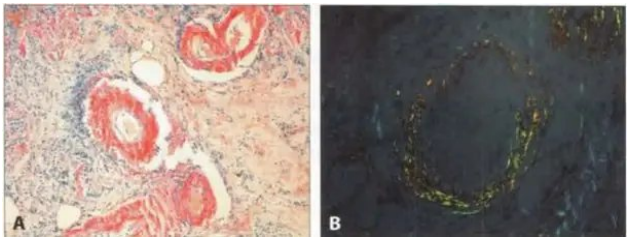


Figure 8.11. *Amyloid: A — a pink amorphous mass with hematoxylin and eosin (in vessel walls); B — apple-green birefringence with Congo red in polarized light (Pathoma, Fig. 1.13)*

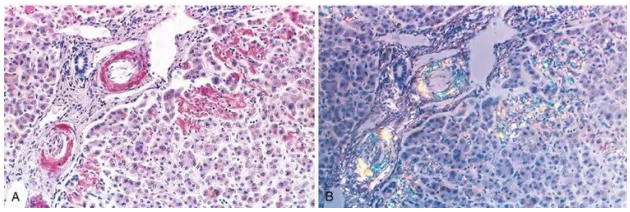


Figure 8.10. *Hepatic amyloidosis: A — pink-red amyloid deposits in vessel walls and along the sinusoids with Congo red staining; B — apple-green birefringence in polarized light (Robbins, 9th ed., Fig. 6-46)*

Classification of amyloidosis

Table 8.3. *The main clinicopathologic types of amyloidosis*

Type	Fibrillar protein	Causes	Main organs affected
Primary (associated with immunocyte dyscrasia) — systemic	AL	Monoclonal proliferation of plasma cells: multiple myeloma (5–15% of cases), B-cell lymphomas, Waldenström macroglobulinemia; often “idiopathic” without monoclonal gammopathy	Heart, kidneys, tongue (macroglossia), skin, peripheral nerves, digestive tract, vessels (mesenchymal type)
Secondary (reactive) — systemic	AA	Chronic inflammation: tuberculosis, bronchiectasis, chronic osteomyelitis, lung abscess, rheumatoid arthritis, ankylosing spondylitis, Crohn disease, ulcerative colitis; some tumors (renal cell carcinoma, lymphogranulomatosis); skin abscesses in drug users	Kidneys, spleen, liver, adrenal glands, intestine (parenchymal type)
Hereditary (familial)	AA — familial Mediterranean fever (periodic disease); ATTR (mutant) — familial amyloid polyneuropathies	Genetic	Kidneys (AA); peripheral and autonomic nerves, heart (ATTR)
Hemodialysis-associated	A-beta-2-microglobulin	Long-term hemodialysis	Joints, synovium, tendons (carpal tunnel syndrome)

Type	Fibrillar protein	Causes	Main organs affected
Senile systemic amyloidosis	ATTR (normal “wild” type)	Older than 70–80 years	Heart (cardiomyopathy)
Localized	AL, A-beta, AIAPP, AE	—	Nodules in the skin, eye, urinary bladder; the brain in Alzheimer disease (A-beta), islets in type 2 diabetes (AIAPP), endocrine tumors (AE)

CAUTION

In older textbooks (Strukov, Abdullakhodjaeva), amyloidosis is divided into **primary (idiopathic), hereditary, secondary (acquired) and senile** forms, and by the location of amyloid into **pericollagenous (mesenchymal)** — vascular adventitia, heart, muscles, nerves, skin (primary, hereditary amyloidosis) and **perireticular (parenchymal)** — vascular intima and basement membranes, spleen, kidney, liver, adrenal glands (secondary amyloidosis). The current classification is based on the **type of amyloid protein** (AL, AA, ATTR, etc.), because it is precisely the protein type that determines **treatment**.

Pathogenesis of amyloidosis

Amyloidosis is a **disease of protein misfolding**. Normally, misfolded proteins are corrected inside the cell by chaperones or degraded in proteasomes. In amyloidosis there are **two mechanisms**:

1. **A normal protein is produced in excess** and is prone to misfolding — **AA** (SAA increases in chronic inflammation; monocytes cannot degrade it adequately), **AL** (excess monoclonal light chains), A-beta-2-microglobulin (not removed by dialysis).
2. **A mutant protein** is prone to misfolding — **ATTR** (hereditary). The misfolded proteins form oligomers and fibrils and accumulate **outside cells**. Amyloid **mechanically compresses** the tissue and narrows the vessels, leading to parenchymal **atrophy**; the oligomers also have a direct toxic effect on cells. Amyloid is produced mainly with the participation of **macrophages** (amyloidoblasts), plasma cells, fibroblasts and cardiomyocytes.

Stages in the development of amyloidosis (V.V. Serov): the **preamyloid stage** — plasmacytic transformation of cells of the reticuloendothelial system, appearance of a clone of amyloidoblasts; **amyloid formation** — synthesis of the fibrillar protein and assembly of fibrils.

Changes in the organs

1. Kidneys — affected most often and most severely (especially in AA amyloidosis). Amyloid accumulates in the **glomerular mesangium and capillary basement membranes**, then in the tubular basement membranes and arteriolar walls; the glomeruli are gradually completely replaced by amyloid masses. **Stages:** latent → **proteinuric** → **nephrotic** (nephrotic syndrome: heavy proteinuria, hypoproteinemia, edema, hyperlipidemia) → **azotemic (uremic)**.

- In the nephrotic stage the kidneys are enlarged, firm and whitish, with a waxy luster on the cut surface — the **“large white (lardaceous) amyloid kidney”**; hyaline droplet, hydropic and fatty degeneration in the tubules.
- In the final stage the kidneys are small, firm and contracted — the **“amyloid contracted kidney”**; **chronic kidney disease** is the most common **cause of death** in amyloidosis.

2. Spleen — enlarged and firm. Two types:

- **“Sago spleen”** — amyloid accumulates in the **lymphoid follicles**; on the cut surface, translucent gray-white granules resembling **sago grains** (or boiled rice) are seen (early stage);
- **“Lardaceous (waxy) spleen”** — amyloid accumulates **diffusely** in the **red pulp**, along the sinusoids and reticular fibers; on the cut surface it is uniform, with a waxy luster, red-brown. **3. Liver** — enlarged, firm, shiny, whitish-yellow (“lardaceous”); amyloid is deposited in the **spaces of Disse** (around the sinusoids), in vessel walls and in the portal tracts; the hepatocytes are compressed and atrophy. Liver function is usually only slightly impaired (portal hypertension is rare).

4. Heart — the main organ affected in **AL** and **ATTR** amyloidosis: the heart is enlarged, with thickened, firm, waxy walls; amyloid is deposited between cardiomyocytes, in vessel walls and in the endocardium. **Restrictive cardiomyopathy**, arrhythmias (involvement of the conduction system) and heart failure are the main **cause of death** in AL amyloidosis. Low voltage on the ECG — inconsistent with the wall thickness.

5. Adrenal glands — amyloid in the cortex along vessels and reticular fibers; with bilateral involvement — **adrenal insufficiency** (Addison disease, Topic 4).

6. Other organs: tongue — **macroglossia** (in AL amyloidosis); **digestive tract** — diarrhea, malabsorption; **peripheral nerves** — polyneuropathy; **skin** — “**raccoon eyes**” around the eyelids (purpura, AL); **joints** and carpal tunnel syndrome; vessels (bleeding tendency).

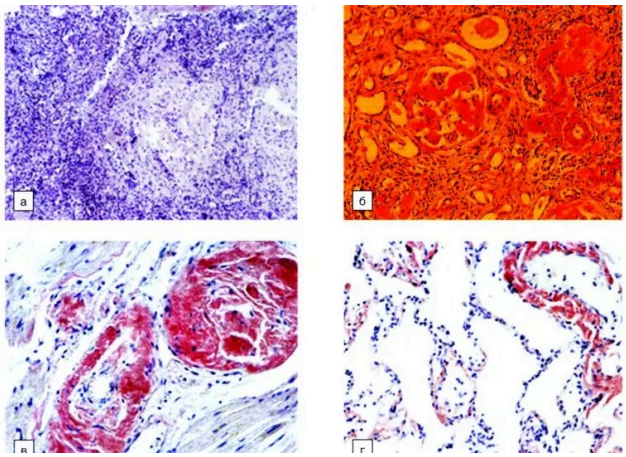


Figure 8.13. *Amyloidosis: a — amyloid in the splenic follicles (“sago” spleen); b — in the vessels of renal glomeruli; c — in the walls of cardiac vessels; d — in the walls of pulmonary vessels (Congo red) (Strukov, Serov, Fig. 2-15)*

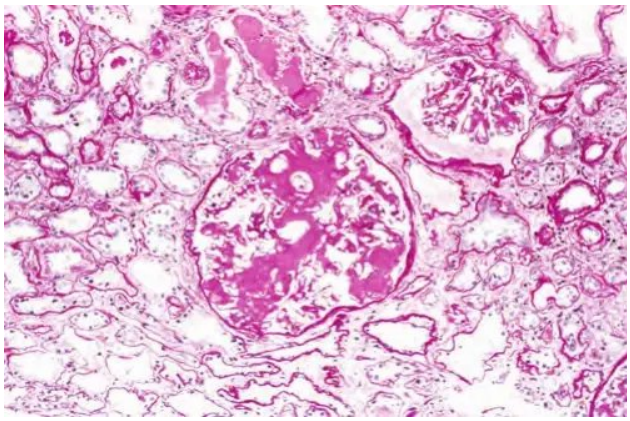


Figure 8.12. *Renal amyloidosis: the structure of the glomeruli is almost completely lost, with pink amorphous amyloid masses in the mesangium (Robbins, 9th ed., Fig. 6-47)*

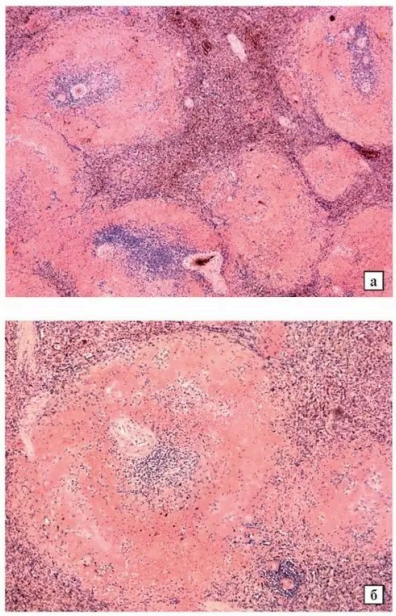


Figure 8.15. *Splenic amyloidosis (microscopic slide): amyloid masses in the lymphoid follicles and along the reticular fibers of the red pulp (a, b) (TMA study guide, 2018)*



Figure 8.14. *Splenic amyloidosis — "sago spleen": the spleen is enlarged and firm, with translucent gray granules on the cut surface (TMA study guide, 2018)*

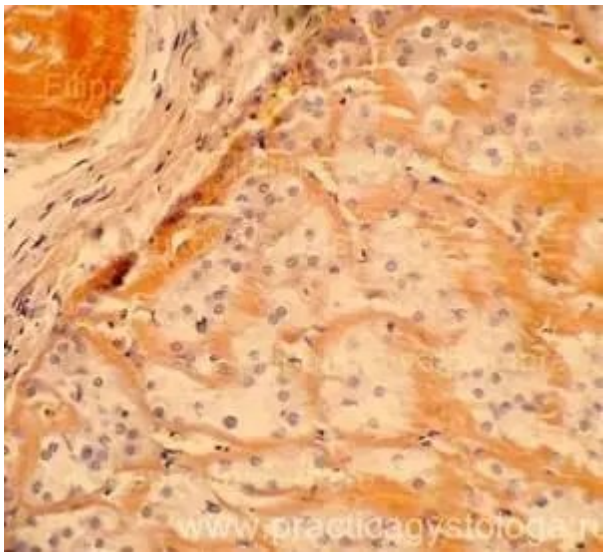


Figure 8.16. *Adrenal amyloidosis (Congo red): orange-red amyloid deposits along the arterial intima and reticular fibers (TMA study guide, 2018)*

Diagnosis of amyloidosis — biopsy when clinically suspected: the most convenient and safe is **abdominal fat pad aspiration biopsy**; rectal mucosa, gingival, kidney or heart biopsy; Congo red + polarized light, then determination of the amyloid type.

Outcome and treatment: amyloidosis is usually considered a **progressive** and irreversible process, but if the underlying cause is removed, the process may stop and even partially regress: **AA** — treatment of the chronic inflammation (treating tuberculosis or osteomyelitis, TNF inhibitors in rheumatoid arthritis), colchicine in familial Mediterranean fever; **AL** — elimination of the plasma cell clone (chemotherapy, bone marrow transplantation); **ATTR** — transthyretin-stabilizing drugs and liver transplantation.

CLINICAL SIGNIFICANCE

If **proteinuria appears in a patient with a chronic inflammatory disease** (for example, long-standing tuberculosis, osteomyelitis, bronchiectasis or rheumatoid arthritis), **AA amyloidosis** should be considered first. Formerly, tuberculosis and chronic suppurative diseases were the main causes of amyloidosis; today, in developed countries, they are rheumatoid arthritis and inflammatory bowel diseases.

Unexplained heart failure, nephrotic syndrome and macroglossia in an elderly patient — AL amyloidosis (multiple myeloma should be sought).

8.7. Clinicopathological section

Three diseases — not one mechanism

- **Poststreptococcal glomerulonephritis** (type III): 2–3 weeks after tonsillitis a child develops edema, hematuria ("meat-wash" colored urine) and arterial hypertension; immune complexes ("humps" — subepithelial deposits) and neutrophils in the glomeruli.
- **Goodpasture syndrome** (type II): a young man with hemoptysis and rapidly progressive kidney failure; antibodies against **type IV collagen in the basement membrane** of the pulmonary alveoli and renal glomeruli; on immunofluorescence — **linear** IgG deposits along the basement membrane.
- **Contact dermatitis** (type IV): an itchy vesicular rash under a nickel-containing watchstrap after 48 hours; an infiltrate of lymphocytes and macrophages in the dermis.

REMEMBER

On immunofluorescence microscopy, the **pattern** of immune deposits indicates the mechanism: **linear** (smooth along the basement membrane) — **type II** (anti–basement membrane antibodies, Goodpasture); **granular (lumpy-bumpy)** — **type III** (immune complexes: SLE, poststreptococcal glomerulonephritis).

Practical skills

Skill 1. Gross specimens

1. **"Splenic amyloidosis — sago spleen".** The spleen is enlarged and firm; on the cut surface there are translucent gray granules resembling sago grains (amyloid in the follicles). Conclusion: secondary (AA) amyloidosis.
2. **"Splenic amyloidosis — lardaceous spleen".** The spleen is enlarged and firm; on the cut surface it is uniform, shiny, red-brown and waxy. Conclusion: amyloidosis (diffuse deposition in the red pulp).
3. **"Renal amyloidosis — large white kidney".** The kidneys are enlarged, firm and whitish-gray, with a waxy luster on the cut surface; the cortex is widened. Conclusion: amyloid nephrosis (nephrotic stage).
4. **"Amyloid contracted kidney".** The kidneys are small and firm with an uneven surface. Conclusion: the azotemic stage of amyloidosis.
5. **"Hashimoto goiter".** The thyroid gland is diffusely enlarged, firm, whitish-gray and lobulated on the cut surface. Conclusion: autoimmune thyroiditis.
6. **"Croupous (lobar) pneumonia"** and **"caseous necrosis of a lymph node"** — studied together as sources of chronic inflammation leading to AA amyloidosis.



Figure 8.17. *Croupous (lobar) pneumonia (gray hepatization stage): the affected lobe is enlarged, firm and gray (TMA study guide, 2018)*

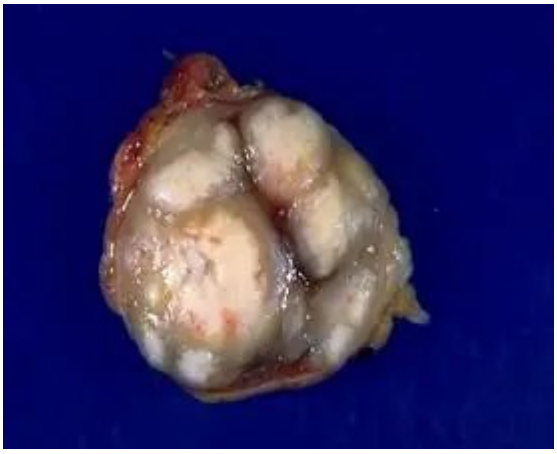


Figure 8.18. *Caseous necrosis of a lymph node (tuberculosis) — a possible source of secondary amyloidosis (TMA study guide, 2018)*

Skill 2. Microscopic slides

1. **“Accidental transformation of the thymus”** (hematoxylin and eosin). The thymic lobules are small; lymphocytes are depleted in the cortex (apoptosis, “starry sky”); the border between the cortex and medulla is indistinct; Hassall corpuscles are cystically dilated. Conclusion: stress involution.
2. **“A lymph node in AIDS”** (hematoxylin and eosin). The lymphoid follicles are involuted or absent, lymphocytes are markedly decreased, vessels are numerous and the stroma is exposed. Conclusion: lymphoid depletion (AIDS).
3. **“Splenic amyloidosis”** (Congo red or hematoxylin and eosin). Homogeneous amyloid masses, red-orange with Congo red, in the lymphoid follicles (sago) or along the reticular fibers of the red pulp. Conclusion: amyloidosis.
4. **“Renal amyloidosis”** (Congo red). Red amyloid in the glomerular mesangium and capillary walls, in arterioles and in tubular basement membranes; some glomeruli are completely replaced by amyloid. Conclusion: amyloid nephrosis.
5. **“Adrenal amyloidosis”** (Congo red). Red-orange amyloid along the capillaries and reticular fibers in the cortex; atrophy of cells. Conclusion: amyloidosis.
6. **“Hashimoto thyroiditis”** (hematoxylin and eosin). A dense lymphoplasmacytic infiltrate in the parenchyma, lymphoid follicles with germinal centers, follicular atrophy, Hürthle cells, fibrosis. Conclusion: autoimmune thyroiditis.

Review questions

1. The central and peripheral organs of the immune system, its main cells and their functions.
2. The major histocompatibility complex (HLA) and its significance.
3. Changes in lymph nodes and spleen on antigenic stimulation. Accidental transformation of the thymus.
4. The four types of hypersensitivity reactions, their mechanisms and examples.
5. The pathological anatomy of anaphylactic shock.

6. Types of transplant rejection.
7. Immunologic tolerance and the mechanisms of its breakdown.
8. The pathogenesis of systemic lupus erythematosus and the changes in the organs.
9. The morphology of rheumatoid arthritis. What is a pannus?
10. The morphology of systemic sclerosis and Hashimoto thyroiditis.
11. List the primary immunodeficiency states.
12. The pathogenesis and stages of HIV infection and the morphologic manifestations of AIDS.
13. What is amyloid? Its structure, composition and methods of detection.
14. Classification of amyloidosis (AL, AA, ATTR and others).
15. Changes in the kidney, spleen, liver and heart in amyloidosis.

Test questions

1. The main role in type I (anaphylactic) hypersensitivity is played by:
   A) IgM
   B) Immune complexes
   C) IgG
   D) IgE and mast cells
   E) CD8+ T lymphocytes

2. Systemic lupus erythematosus is mainly associated with which type of hypersensitivity?
   A) Type II
   B) Type III
   C) Type IV
   D) Type I
   E) None of them

3. The tuberculin (Mantoux) test is an example of which type of hypersensitivity?
   A) Type III
   B) Type II
   C) Anaphylactic
   D) Type IV
   E) Type I

4. Graves disease and myasthenia gravis belong to which type?
   A) Type III
   B) Transplant reaction
   C) Type II (antibodies against receptors)
   D) Type I
   E) Type IV

5. The most specific antibodies for SLE:
   A) Antibodies against thyroid peroxidase
   B) Antineutrophil cytoplasmic antibodies
   C) Rheumatoid factor
   D) Anticentromere antibodies
   E) Antibodies against double-stranded DNA and the Sm antigen

6. Thickening of the glomerular capillary basement membrane in lupus nephritis is called:
   A) “Humps”
   B) “Kimmelstiel–Wilson nodules”
   C) “Onion skin”
   D) “Crescents”
   E) “Wire loops”

7. The granulation tissue that destroys cartilage in rheumatoid arthritis:
   A) Tophus
   B) Granuloma
   C) Gumma
   D) Leproma
   E) Pannus

8. Hürthle (Askanazy) cells and lymphoid follicles are characteristic of:
   A) Hashimoto thyroiditis
   B) Myxedema
   C) Graves disease
   D) Colloid goiter
   E) Thyroid cancer

9. The organs that fail to develop in DiGeorge syndrome:
   A) Spleen and lymph nodes
   B) Adrenal glands
   C) Thymus and parathyroid glands
   D) Bone marrow
   E) Thyroid gland

10. Which cells does HIV mainly infect?
    A) CD4+ T lymphocytes and macrophages
    B) B lymphocytes
    C) CD8+ T lymphocytes
    D) Red blood cells
    E) Neutrophils

11. The causative agent of the most common opportunistic pneumonia in AIDS:
    A) Pneumococcus
    B) Pneumocystis jirovecii
    C) Legionella
    D) Mycoplasma
    E) Staphylococcus

12. The main method of detecting amyloid:
    A) Van Gieson
    B) Congo red and apple-green birefringence in polarized light
    C) Sudan III
    D) Perls reaction
    E) Ziehl–Neelsen

13. The type of amyloidosis that develops in chronic inflammation (tuberculosis, osteomyelitis, rheumatoid arthritis):
    A) AA
    B) A-beta
    C) AL
    D) A-beta-2-microglobulin
    E) ATTR

14. In multiple myeloma, amyloid is formed:
    A) From transthyretin
    B) From procalcitonin
    C) From immunoglobulin light chains (AL)
    D) From serum amyloid A
    E) From beta-2-microglobulin

15. In the “sago spleen”, amyloid accumulates mainly:
    A) Diffusely in the red pulp
    B) In the trabeculae
    C) In the capsule
    D) In the lymphoid follicles
    E) In the splenic vein

Clinical cases

Case 1

A 25-year-old woman has fever, joint pain, a red “butterfly” rash on the cheeks and bridge of the nose, and photosensitivity; protein and red cells in the urine. ANA and anti–double-stranded DNA antibodies are high in the blood, and complement is decreased. Kidney biopsy: the walls of the glomerular capillaries are thickened (“wire loops”); granular IgG and C3 deposits on immunofluorescence.

Questions: 1) Your diagnosis? 2) What is the mechanism of the disease? 3) What other organs may be affected, and what changes occur in them?

Answer: 1) Systemic lupus erythematosus, lupus nephritis. 2) As a result of loss of tolerance, autoantibodies against nuclear antigens are formed; immune complexes (type III) are deposited in the glomeruli and vessel walls, activate complement (which is why blood complement is decreased) and cause inflammation; antibodies against blood cells (type II) — anemia, thrombocytopenia. 3) Skin (erythema), joints (nonerosive arthritis), serous membranes (fibrinous pericarditis, pleurisy), heart (Libman–Sacks endocarditis), spleen (“onion skin” periarterial fibrosis), vessels (vasculitis with fibrinoid necrosis), brain (vasculitis, thrombosis).

Case 2

A 45-year-old man has had chronic osteomyelitis of the femur for 15 years, with constant pus discharge from a sinus tract. In the last year he developed edema of the legs and face and 6 g of protein in the urine per day. He died of chronic kidney disease. At autopsy the kidneys are enlarged, firm, whitish and waxy; the spleen is enlarged, with translucent granules on the cut surface.

Questions: 1) Your diagnosis and the type of amyloidosis? 2) How is amyloid detected? 3) The mechanism of the disease and the stages of the changes in the kidneys.

Answer: 1) Secondary (reactive) systemic amyloidosis — AA amyloidosis against the background of chronic osteomyelitis; “large white amyloid kidney” and “sago spleen”. 2) Grossly — Lugol solution and sulfuric acid (Virchow reaction — blue-violet); microscopically — Congo red (red-orange) and apple-green birefringence in polarized light; metachromasia with methyl violet; AA protein on immunohistochemistry. 3) In chronic inflammation, under the action of IL-1 and IL-6, the liver produces large amounts of serum amyloid A (SAA); its incompletely degraded fragments misfold, form AA fibrils and accumulate outside cells. Renal stages: latent → proteinuric → nephrotic (large white kidney) → azotemic (amyloid contracted kidney, uremia).

Case 3

A 32-year-old man has complained for 3 months of fever, marked weight loss, diarrhea, dry cough and shortness of breath. White plaques in the oral cavity (candidiasis), red-violet nodules on the skin. A history of injecting drug use. Blood CD4+ T lymphocytes 80/μL.

Questions: 1) Your diagnosis? 2) What could the skin nodules and the lung process be? 3) Briefly explain the pathogenesis of the disease.

Answer: 1) HIV infection, AIDS stage (CD4+ fewer than 200, opportunistic infections and tumors). 2) The skin nodules — Kaposi sarcoma (HHV-8); in the lung — Pneumocystis pneumonia (foamy eosinophilic exudate in the alveoli) or tuberculosis; in the mouth — candidiasis. 3) HIV binds via gp120 to CD4 and coreceptors (CCR5, CXCR4) and infects CD4+ T lymphocytes and macrophages; it integrates into the genome with the help of reverse transcriptase; CD4+ cells gradually die, and cell-mediated immunity is profoundly impaired; as a result, opportunistic infections and virus-associated tumors develop.

Quick review

Table 8.4. *Key facts on Topic 8*

Question	Answer
Central immune organs	Thymus (T), bone marrow (B)
Helper T / killer T / suppressor T	CD4+ / CD8+ / regulatory T lymphocytes
Accidental transformation	Apoptosis of thymic lymphocytes in stress (“starry sky”)
Type I	IgE, mast cells — anaphylaxis, asthma
Type II	IgG/IgM against cells — hemolysis, Goodpasture, Graves, myasthenia gravis
Type III	Immune complexes — SLE, poststreptococcal GN, Arthus
Type IV	T lymphocytes — Mantoux, tuberculosis, contact dermatitis, granulomas
Immunofluorescence	Linear — type II; granular — type III
SLE	ANA, ds-DNA, Sm; “wire loops”, Libman–Sacks, “onion skin”
Rheumatoid arthritis	Rheumatoid factor, ACPA; pannus; rheumatoid nodules
Hashimoto	Lymphoid follicles, Hürthle cells; hypothyroidism
Bruton	No B lymphocytes (BTK)
DiGeorge	No thymus + no parathyroid glands (22q11)
HIV	CD4, CCR5/CXCR4; AIDS — CD4 fewer than 200
AIDS-defining diseases	Pneumocystosis, candidiasis, tuberculosis, Kaposi sarcoma, lymphomas
Amyloid stain	Congo red + apple-green birefringence

Question	Answer
AL	Light chains — myeloma; heart, tongue
AA	SAA — chronic inflammation; kidney, spleen
ATTR	Transthyretin — hereditary (nerves), senile (heart)
Sago / lardaceous spleen	Amyloid in follicles / in the red pulp
Cause of death in amyloidosis	Kidney failure (AA), heart failure (AL)

Glossary of terms

- **Accidental** (Latin accidens — chance) — occurring by chance; accidental transformation of the thymus — its rapid involution under the influence of stress.
- **Allergy** (Greek allos — other, ergon — action) — altered (increased) reactivity of the organism to an antigen.
- **Amyloid** (Latin amylum — starch) — a glycoprotein consisting mainly of fibrillar protein.
- **Bursa-dependent (B) lymphocytes** (Latin bursa — pouch) — lymphocytes that mature in the bone marrow and carry out humoral immune reactions; so named because in birds they form in the bursa of Fabricius.
- **Thymus-dependent (T) lymphocytes** — lymphocytes that mature in the thymus and provide cell-mediated immunity.
- **Helper T cells** (English help) — helper lymphocytes that regulate the immune response (CD4+).
- **Killer T cells** (English killer) — cytotoxic lymphocytes that kill target cells (CD8+).
- **Suppressor T cells** (Latin suppressio — suppression) — (regulatory) lymphocytes that suppress the immune response.
- **Hypoplasia** — insufficient development of an organ or tissue.
- **Thymicolymphatic state** (status thymicolymphaticus) — the old name of a congenital immunodeficiency with hyperplasia of the thymus, spleen and lymph nodes and hypoplasia of the adrenal glands.
- **Autoimmunity** — an immune response of the organism against its own antigens.
- **Tolerance** (Latin tolerantia — endurance) — absence of a response of the immune system to a particular antigen (normally — to self antigens).
- **Immunodeficiency** — a deficiency of one or more components of the immune system.
- **Opportunistic infection** (Latin opportunus — convenient) — an infection caused by opportunistic microbes in a person with reduced immunity.
- **Pannus** (Latin pannus — piece of cloth) — synovial granulation tissue that destroys cartilage in rheumatoid arthritis.

TOPIC 9

Neoplasia. Biology, structure and growth of tumors. Types of tumors

Learning objectives

The student should know:

- the definition and main properties of a tumor (neoplasia) and precancerous processes;
- the structure of tumors: parenchyma and stroma, organoid and histioid structure;
- the types of atypia: tissue, cellular, biochemical, functional and antigenic atypia; differentiation and anaplasia;
- the types of tumor growth (expansive, infiltrative, appositional; exophytic, endophytic), the routes of metastasis and recurrence;
- the differences between benign and malignant tumors; the effects of tumors on the organism (local and systemic), paraneoplastic syndromes;
- carcinogenesis: etiologic factors (chemical, physical, viral), molecular basis (oncogenes, tumor suppressor genes, apoptosis and DNA repair genes), hallmarks of cancer;
- the classification and naming principles of tumors; the TNM system, grading and staging;
- epithelial (non-organ-specific) and mesenchymal tumors, tumors of melanin-forming tissue.

The student should be able to:

- describe gross specimens of uterine fibromyoma, breast fibroadenoma, ovarian cystadenoma, skin papilloma, intestinal polyp, gastric cancer, cancer metastases in the liver and osteosarcoma;
- describe microscopic slides of skin papilloma, fibroadenoma, keratinizing squamous cell carcinoma, adenocarcinoma, melanoma, leiomyoma, leiomyosarcoma and hemangioma;
- distinguish benign from malignant tumors by histologic features;
- correctly form the name of a tumor according to its tissue of origin and degree of malignancy.

9.1. The concept of a tumor

A **tumor** (tumor, neoplasma, blastoma) is a pathological process characterized by **uncontrolled, unlimited (autonomous) proliferation** of cells, in which the **growth, differentiation and morphology** of the cells are disturbed (**atypia**), and these properties are **hereditarily** transmitted to the progeny of tumor cells. According to the definition of the British pathologist R. Willis: “A tumor is an abnormal mass of tissue, the growth of which exceeds and is uncoordinated with that of the normal tissues and persists in the same excessive manner after cessation of the stimuli which evoked the change”.

The science that studies tumors is **oncology** (Greek onkos — tumor). Tumors are the second leading cause of death in the world after cardiovascular diseases.

REMEMBER

The **main properties** of a tumor: 1) **autonomous (unlimited) growth** — not obeying the regulatory mechanisms of the organism; 2) **atypia** — differences from the normal structure and function of the cell and tissue; 3) **monoclonality** — the tumor arises from a single mutated cell; 4) **hereditary transmission** of tumor properties to cell progeny; 5) **progression** — the tumor becomes increasingly malignant over time.

Tumor progression — as the tumor develops, new mutations accumulate and its properties (growth rate, invasion, metastasis, resistance to treatment) **become progressively more aggressive**; as a result, subclones with different properties appear within the tumor (**heterogeneity**).

Precancerous processes

A tumor usually develops from healthy tissue not suddenly but through **precancerous changes**:

1. **Background processes** — chronic inflammation, hyperplasia, metaplasia (Topic 1), disturbed regeneration.
2. **Focal hyperplasia and dysplasia** — **dysplasia** (mild, moderate, severe) — cellular atypia and disruption of tissue structure; severe dysplasia is a true precancerous condition.
3. **Noninvasive cancer — carcinoma in situ** (cancer in place) — the full thickness of the epithelium is replaced by atypical cells, but the **basement membrane is intact** (no invasion); it does not metastasize.
4. **Invasive cancer** — the tumor breaks through the basement membrane and grows into the stroma; the possibility of metastasis appears.

Table 9.1. *Important precancerous conditions*

Condition	Tumor that may develop
Squamous metaplasia and dysplasia of the bronchial epithelium (smoking)	Squamous cell carcinoma of the lung
Barrett esophagus	Esophageal adenocarcinoma
Chronic atrophic gastritis, intestinal metaplasia (H. pylori)	Gastric adenocarcinoma
Adenomatous polyps of the colon, familial adenomatous polyposis	Colonic adenocarcinoma
Cervical dysplasia (CIN; papillomavirus 16, 18)	Squamous cell carcinoma of the cervix
Atypical endometrial hyperplasia	Endometrial adenocarcinoma
Leukoplakia (with dysplasia)	Squamous cell carcinoma of the oral cavity
Liver cirrhosis (viral hepatitis B, C, alcohol)	Hepatocellular carcinoma
Ulcerative colitis, Crohn disease	Colorectal cancer
Dysplastic nevus, xeroderma pigmentosum	Melanoma

9.2. Structure of a tumor

Every tumor consists of two parts:

- **Parenchyma** — the **tumor cells** that make up the tumor itself; the biological properties of the tumor (malignant or benign) and its **name** depend on the parenchyma.
- **Stroma** — connective tissue, blood vessels, nerves and immune cells; it nourishes and supports the parenchyma. The stroma forms **under the influence of the tumor cells** (growth factors: VEGF, PDGF, FGF). The stroma often contains a **lymphocytic infiltrate** — a sign of an antitumor immune response (favorable for prognosis). By the ratio of stroma to parenchyma: **organoid tumors** — the parenchyma and stroma are clearly separated (most epithelial tumors); **histioid tumors** — the stroma is scant, only in the form of vessels (most mesenchymal tumors). If the stroma is abundant, the tumor is **hard** (**scirrhous** — carcinoma with a fibrous stroma, for example breast cancer — **desmoplasia**); if scant, it is **soft** (**medullary**, “brain-like” carcinoma).

Gross appearance: a nodule (encapsulated or unencapsulated), a polyp (pedunculated), mushroom-shaped, papillary (cauliflower-like), a plaque, a cyst (unilocular or multilocular), a diffuse infiltrate, an ulcer (with raised crater edges). Tumors range in size from millimeters to several tens of kilograms. On the cut surface, a **"fish-flesh"** appearance (sarcomas) and foci of necrosis and hemorrhage (malignant tumors) are seen.

9.3. Tumor atypia

Atypia — the properties of a tumor cell and tissue that differ from those of the normal cell and tissue.

1. **Morphologic atypia:** – **tissue atypia** — disturbance of the ratio of parenchyma to stroma, of the epithelial layers, of the shape and size of glands and of the orientation of fibers; the cells look normal. It is characteristic of **benign** tumors. – **cellular atypia** — characteristic of **malignant** tumors (in detail below). 2. **Biochemical atypia** — altered metabolism: increased glycolysis even in the presence of oxygen (**Warburg effect**), increased synthesis of proteins and nucleic acids. 3. **Functional atypia** — tumor cells lose their normal function or acquire **unusual functions** (for example, lung cancer produces hormones). 4. **Antigenic atypia** — new antigens appear on tumor cells (tumor-specific antigens, **embryonic antigens** — alpha-fetoprotein, carcinoembryonic antigen); they are used in diagnosis as **tumor markers**.

Differentiation and anaplasia

Differentiation is the **degree of resemblance** of tumor cells, in structure and function, to the normal tissue from which they arose. A **well-differentiated** tumor closely resembles its tissue of origin (for example, most benign tumors, well-differentiated adenocarcinoma); a **poorly differentiated** one shows little resemblance. **Anaplasia** (Greek "backward formation") is the loss of differentiation; an important sign of malignancy.

Signs of cellular atypia (anaplasia):

- **pleomorphism** — variation in the shape and size of cells and nuclei, **tumor giant cells**;
- **nuclear atypia** — nuclei are enlarged and **hyperchromatic** (dark-staining), with coarse chromatin and large nucleoli; **increased nuclear-to-cytoplasmic ratio** (normally 1:4–1:6, in cancer up to 1:1);
- **numerous mitoses** and **atypical (pathologic) mitoses** (tripolar, quadripolar);
- **loss of polarity** — the cells are arranged haphazardly;
- foci of **necrosis** (tumor vessels cannot keep pace with rapid growth).

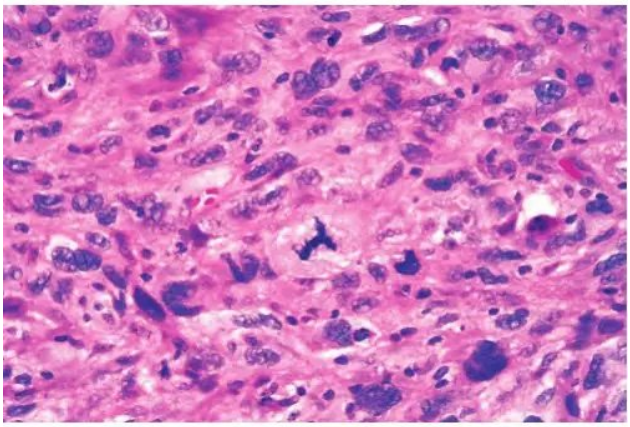


Figure 9.1. *Anaplastic tumor: the size and shape of cells and nuclei vary markedly; an atypical tripolar mitosis in the center (Robbins, 9th ed., Fig. 7-8)*

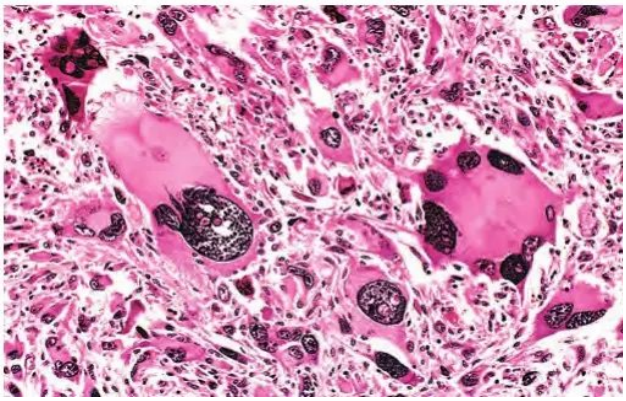


Figure 9.2. *Pleomorphic tumor of skeletal muscle (rhabdomyosarcoma): marked pleomorphism of cells and nuclei, hyperchromatic nuclei and tumor giant cells (Robbins, 9th ed., Fig. 7-9)*

9.4. Tumor growth and spread

Types of growth in relation to the surrounding tissue:

- **Expansive growth** — the tumor grows “by itself”, **compressing** and pushing aside the surrounding tissue; the atrophic tissue and condensed stroma around it form a **pseudocapsule**; the tumor border is sharp. It is characteristic of **benign** tumors.
- **Infiltrative (invasive, destructive) growth** — tumor cells **grow into the surrounding tissue, around vessels and nerves, and destroy it**; the border is indistinct. It is characteristic of **malignant** tumors; the name “**cancer**” (Latin cancer, Greek karkinos — crab) comes from the tumor spreading like the legs of a crab.
- **Appositional growth** — by transformation of surrounding normal cells into tumor cells (for example, multicentric growth). **In relation to a hollow organ: exophytic** — growing into the lumen of the organ (polyp, papillary tumor); **endophytic** — growing into the wall of the organ, in depth (often more malignant, ulcerates).

By the number of foci of growth: unicentric (from one focus) and **multicentric** (from several foci).



Figure 9.4. *Invasive ductal carcinoma of the breast (cut surface): the tumor is retracted and has grown infiltratively into the surrounding tissue; its border is indistinct, and it is stone-hard on palpation (Robbins, 9th ed., Fig. 7-13)*

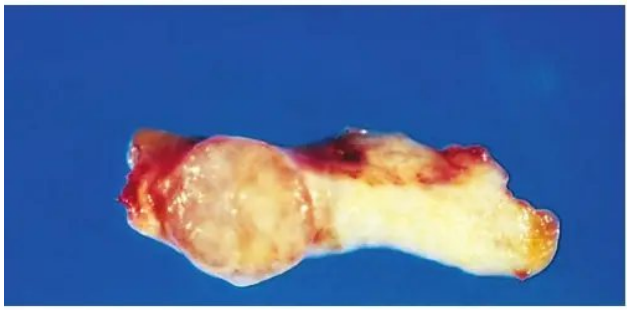


Figure 9.3. *Fibroadenoma of the breast: an encapsulated, yellowish tumor sharply demarcated from the surrounding whitish breast tissue (expansive growth) (Robbins, 9th ed., Fig. 7-11)*

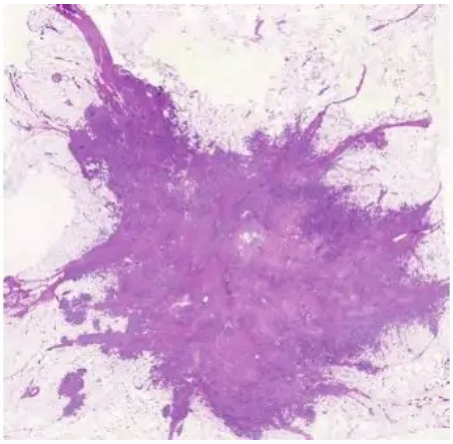


Figure 9.6. *Invasive breast cancer (microscopic slide): no capsule, irregular borders, infiltrative growth (Robbins, 9th ed., Fig. 7-14)*

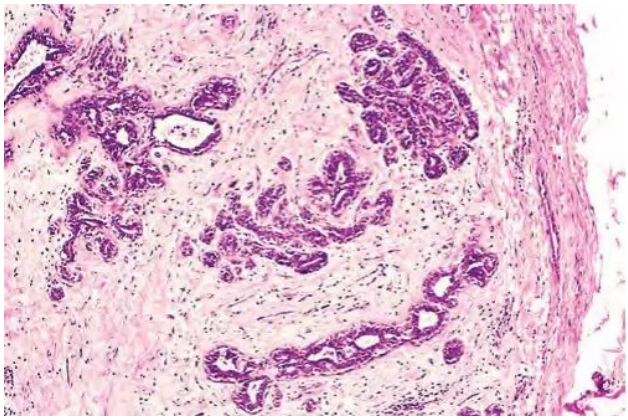


Figure 9.5. *Fibroadenoma of the breast (microscopic slide): a fibrous capsule (right) separates the tumor from the surrounding tissue (Robbins, 9th ed., Fig. 7-12)*

Invasion and metastasis

Metastasis (Greek metastasis — transfer) is the detachment of tumor cells from the primary focus, their transfer to another site and the formation there of a new tumor focus — a **secondary tumor (metastasis)**. **Metastasis is the most reliable sign of a malignant tumor**; benign tumors do not metastasize. About 30% of malignant tumors have already metastasized at the time of diagnosis.

Stages of the invasion–metastasis cascade: 1) detachment of tumor cells from one another (decreased E-cadherin); 2) attachment to the **basement membrane and extracellular matrix** and its destruction by **matrix metalloproteinases** (collagenase IV); 3) migration; 4) entry into a vessel (**intravasation**); 5) survival in the blood together with platelets (tumor emboli); 6) exit from the vessel (**extravasation**) and growth at the new site (angiogenesis).

Routes of metastasis:

1. **Lymphogenous** — the most characteristic early route for **carcinomas**: the regional lymph nodes are involved first. For example, in breast cancer — the axillary nodes; in gastric cancer — the left supraclavicular lymph node (**Virchow metastasis**), the ovaries (**Krukenberg metastasis**), the rectouterine pouch (**Schnitzler metastasis**). The **sentinel lymph node** is the first lymph node to receive lymph from the tumor; its biopsy is used in choosing treatment.
2. **Hematogenous** — characteristic of **sarcomas**, and of carcinomas in later stages. Through the portal vein — to the **liver** (tumors of the digestive tract); through the venae cavae — to the **lungs**. Common targets: liver, lungs, bones, brain, adrenal glands. Some tumors tend to metastasize to **bones**: cancers of the prostate, breast, lung, kidney and thyroid.
3. **Implantation (contact, across serous cavities)** — when the tumor reaches the serosa, cells fall into the peritoneal or pleural cavity and spread over the entire serous membrane (**carcinomatosis**): cancers of the ovary, stomach and colon; often with hemorrhagic exudate (ascites, pleurisy).
4. **Perineural** — along nerves (pancreatic and prostate cancer).
5. **Mixed. Recurrence (relapse)** — reappearance of a tumor **at the same site** after its removal or treatment; the cause is residual tumor cells or multicentric growth.

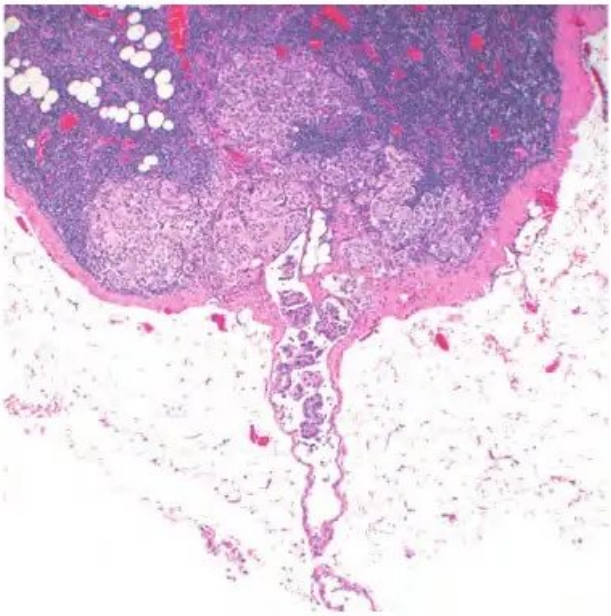


Figure 9.8. *Metastasis of breast cancer to an axillary lymph node: a cluster of tumor cells in the lymph node and a dilated lymphatic vessel (Robbins, 9th ed., Fig. 7-16)*

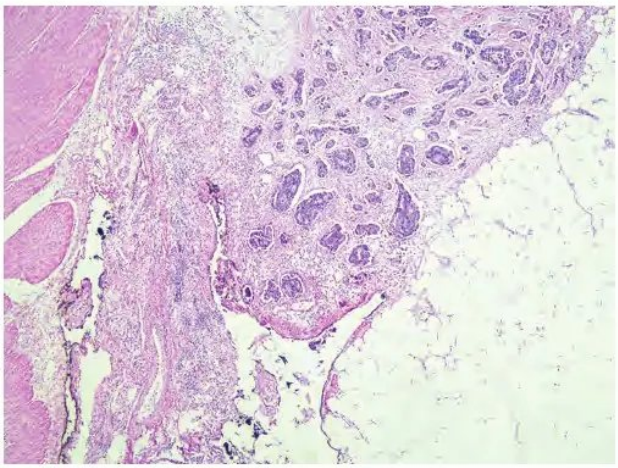


Figure 9.7. *Invasion of intestinal adenocarcinoma into the pericolic adipose tissue (Robbins, 9th ed., Fig. 7-15)*

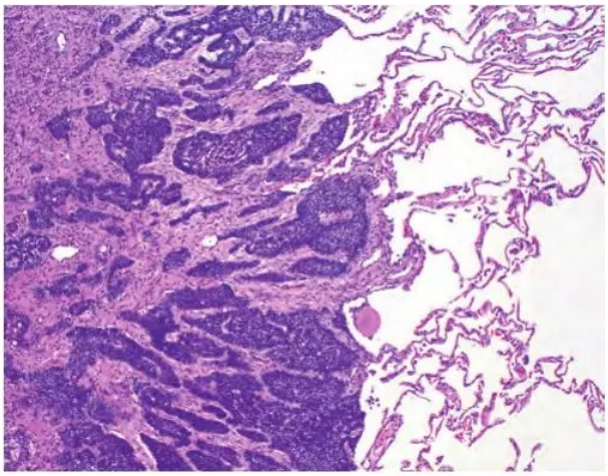


Figure 9.10. *Metastasis in the lung (microscopic slide): colonic adenocarcinoma has formed a metastatic nodule in the lung (Robbins, 9th ed., Fig. 7-18)*

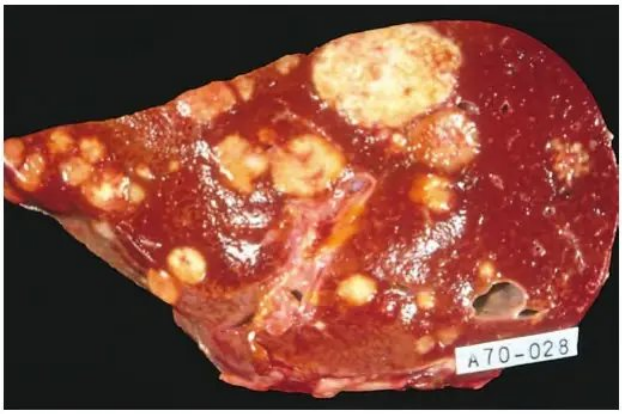


Figure 9.9. *Numerous metastatic cancer nodules in the liver (hematogenous metastases) (Robbins, 9th ed., Fig. 7-17)*

9.5. Benign and malignant tumors

Table 9.2. *Differences between benign and malignant tumors*

Feature	Benign tumor	Malignant tumor
Differentiation	High; resembles the tissue of origin	Variable; poorly differentiated or anaplastic
Atypia	Tissue atypia only	Tissue and cellular atypia
Mitoses	Few, typical	Numerous, atypical mitoses
Growth rate	Usually slow	Usually fast
Type of growth	Expansive, with a capsule (pseudocapsule)	Infiltrative (invasive), no capsule
Metastasis	Absent	Present
Recurrence	Rare (none if completely removed)	Frequent
Effect on the organism	Usually local (compression)	Local and systemic (cachexia, intoxication, paraneoplastic syndromes)
Necrosis, hemorrhage	Rare	Frequent

CAUTION

Tumors malignant by location: even a histologically benign tumor in a vital location (for example, the brain, next to the brainstem, in the spinal canal) can cause death by compression. There are also **borderline** tumors and **locally destructive** tumors (for example, basal cell carcinoma — destroys locally but almost never metastasizes). Hormone-producing benign tumors (insulinoma, pheochromocytoma) can be dangerous because of hormone excess.

Effects of tumors on the organism

- **Local effects: compression** of surrounding tissues and organs (brain, bile ducts — obstructive jaundice, intestine — obstruction), **obstruction** of hollow organs, destruction and **ulceration**, **bleeding** (tumor breakdown — anemia), secondary infection, **rupture** or **perforation** of an organ, **pathologic bone fracture**.

- **Systemic effects: cancer cachexia** (TNF and other cytokines, loss of appetite — Topic 1), anemia, fever, intoxication, **hormonal** effects (endocrine tumors).
- **Paraneoplastic syndromes** are signs not directly related to the location of the tumor or its metastases; they are often related to ectopic production of hormones or cytokines by the tumor. Examples: **Cushing syndrome** (small cell lung carcinoma — ACTH), **hypercalcemia** (squamous cell lung carcinoma, breast cancer — parathyroid hormone–related protein), the **syndrome of inappropriate antidiuretic hormone secretion** (small cell carcinoma — hyponatremia), **Trousseau syndrome** (migratory thrombophlebitis — adenocarcinoma of the pancreas and lung), **acanthosis nigricans**, **clubbing of the fingers** and hypertrophic osteoarthropathy, **myasthenia-like syndrome** (Lambert–Eaton), polycythemia (renal cell carcinoma — erythropoietin).

9.6. Carcinogenesis

Carcinogenesis is the process of transformation of a normal cell into a tumor cell. It is based on **genetic damage** (mutations) and **epigenetic changes**; cancer is a **genetic disease**.

Carcinogenic factors

1. Chemical carcinogens. The first observation was made by the English surgeon **P. Pott** (1775): scrotal skin cancer in chimney sweeps (exposure to soot). The Japanese scientists Yamagiwa and Ichikawa (1915) induced cancer experimentally by painting coal tar on a rabbit's ear.

- **Direct-acting agents** — react with DNA without modification: alkylating agents (some **chemotherapy drugs** — cyclophosphamide; they can cause secondary leukemias).
- **Indirect-acting agents (procarcinogens)** — act after activation in the body (cytochrome P-450): **polycyclic aromatic hydrocarbons** (**benzopyrene** — tobacco smoke, tar, smoked products; lung and bladder cancer), **aromatic amines and azo dyes** (aniline dyes — bladder cancer), **aflatoxin B1** (nuts and grain contaminated with the fungus Aspergillus flavus — liver cancer), **nitrosamines** (gastric cancer), vinyl chloride (hepatic angiosarcoma), **asbestos** (mesothelioma, lung cancer), arsenic, chromium, nickel, benzene (leukemia). **Stages of chemical carcinogenesis: initiation** — irreversible DNA damage (mutation) by the carcinogen; **promotion** — proliferation of the mutated cell under the action of **promoters** (hormones, phorbol esters, drugs) (a promoter is not itself carcinogenic and produces a tumor only if it acts after initiation); **progression**.

2. Physical carcinogens — radiation.

- **Ultraviolet rays** (sun, UVB) — form **pyrimidine dimers** in DNA: **basal cell** and **squamous cell carcinoma** of the skin, **melanoma**; the risk is high in fair-skinned people and in xeroderma pigmentosum (defect of DNA repair).
- **Ionizing radiation** (X-rays, gamma rays, radioactive substances) — double-strand breaks of DNA: **leukemias** (after Hiroshima and Nagasaki), **thyroid cancer** (in children after the Chernobyl accident — radioactive iodine), lung cancer (uranium miners, radon), bone sarcomas.

3. Biological (viral and microbial) carcinogens.

Table 9.3. *Viruses and microbes associated with tumors*

Causative agent	Tumors
Human papillomavirus (types 16, 18)	Cancer of the cervix, anus, pharynx and penis (types 6 and 11 — benign warts, condylomas)

Causative agent	Tumors
Epstein–Barr virus	Burkitt lymphoma, nasopharyngeal carcinoma, Hodgkin lymphoma, B-cell lymphomas in immunodeficiency
Hepatitis B and C viruses	Hepatocellular carcinoma
HHV-8 (Kaposi sarcoma herpesvirus)	Kaposi sarcoma (in AIDS), primary effusion lymphoma
HTLV-1 (human T-cell leukemia virus)	Adult T-cell leukemia/lymphoma
Helicobacter pylori	Gastric adenocarcinoma, gastric MALT lymphoma
Schistosomes (Schistosoma haematobium)	Squamous cell carcinoma of the urinary bladder
Liver flukes (opisthorchiasis, clonorchiasis)	Cholangiocarcinoma

4. Hereditary predisposition. Hereditary tumor syndromes: **retinoblastoma** (RB gene), **Li–Fraumeni syndrome** (TP53), **familial adenomatous polyposis** (APC), **hereditary breast and ovarian cancer** (BRCA1, BRCA2), **Lynch syndrome** (DNA repair genes — colon cancer), **neurofibromatosis** (NF1), **multiple endocrine neoplasia** (MEN1, RET); and also defects of DNA repair (xeroderma pigmentosum).

5. Hormonal factors — estrogen excess (endometrial and breast cancer), androgens (prostate cancer).

Molecular basis of carcinogenesis

Carcinogenesis is a **multistep** process: several mutations must accumulate in a single cell. Mutations occur in four groups of **target genes**:

1. **Proto-oncogenes → oncogenes** — genes that normally stimulate cell growth and division (growth factors, their receptors, signaling proteins, transcription factors). As a result of mutation, amplification or translocation, an **activated oncogene** gives a constant growth signal. Oncogenes act **dominantly** (mutation of one allele is enough). Examples: **RAS** (the most frequently mutated oncogene — cancers of the pancreas, colon and lung), **HER2/ERBB2** amplification (breast cancer — treated with trastuzumab), **MYC** (Burkitt lymphoma — t(8;14)), **BCR-ABL** (chronic myeloid leukemia — t(9;22), the “Philadelphia chromosome”; treated with imatinib), **BRAF** (melanoma), **EGFR** (lung adenocarcinoma). 2. **Tumor suppressor genes** — genes that normally **inhibit** cell growth; for a tumor to develop, **both alleles** must be inactivated (**Knudson “two-hit” hypothesis**). The main ones: **TP53** — the “guardian of the genome”: when DNA is damaged, it arrests the cell cycle and initiates repair, and if repair is impossible — apoptosis; it is mutated in more than 50% of human tumors; **RB** — controls the $G_1 \rightarrow S$ transition of the cell cycle; **APC** (colon), **BRCA1/2**, **NF1**, **VHL** (renal cell carcinoma), **PTEN**, **CDKN2A** (p16). 3. **Genes regulating apoptosis** — overexpression of **BCL2** (follicular lymphoma — t(14;18)), disturbed apoptotic mechanisms (Topic 2). 4. **DNA repair genes** — their defects accelerate the accumulation of mutations in other genes (**genomic instability**): mismatch repair genes (Lynch syndrome), BRCA1/2, nucleotide excision repair (xeroderma pigmentosum). In addition, **epigenetic changes** (hypermethylation of gene promoters — “silencing” of suppressor genes), **microRNAs** and activation of **telomerase** (limitless division) are important.

MEMORY AID

An oncogene is "**a car with a stuck gas pedal**" (one defect is enough — dominant); a tumor suppressor gene is "**a broken brake**" (both brakes — both alleles — must fail — recessive, "two hits"); TP53 is "**the driver**": when it sees a problem, it stops the car or destroys it (apoptosis).

Hallmarks of cancer (Hanahan and Weinberg)

The main properties of cancer cells ("hallmarks of cancer"):

1. **Self-sufficiency** in growth signals (oncogenes).
2. **Insensitivity** to growth-inhibitory signals (inactivation of suppressor genes).
3. **Evasion of apoptosis**.
4. **Limitless replicative potential** (activation of telomerase).
5. **Angiogenesis** (VEGF) — necessary for a tumor larger than 1–2 mm.
6. **Invasion and metastasis**.
7. **Altered cellular metabolism** (Warburg effect — aerobic glycolysis).
8. **Evasion of immune surveillance** (expression of PD-L1 — immune checkpoint inhibitors act against this). Enabling factors: **genomic instability** and **tumor-promoting inflammation**.

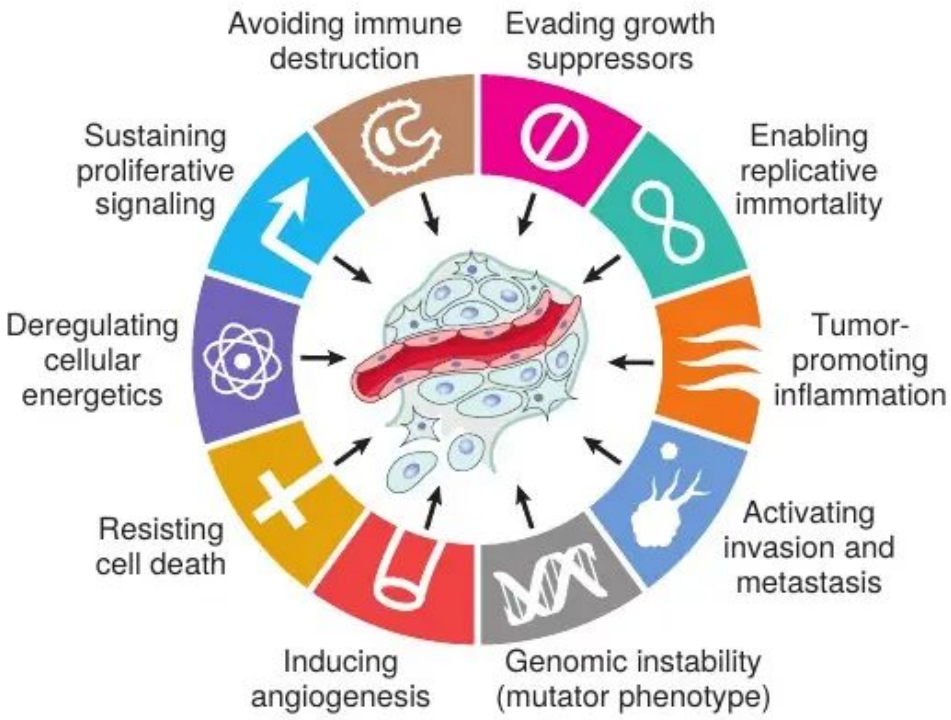


Figure 9.11. *Hallmarks of cancer (after Hanahan and Weinberg): self-sufficiency in growth signals, evasion of growth suppressors, evasion of apoptosis, limitless replication, angiogenesis, invasion and metastasis, reprogrammed energy metabolism, evasion of immune surveillance; enabling factors — genomic instability and tumor-promoting inflammation (Robbins, 9th ed., Fig. 7-24)*

9.7. Classification and naming of tumors

Tumors are classified mainly by the **histogenetic** principle (tissue of origin), by **degree of malignancy** and by **location** (WHO classification). **Naming rules:**

- **Benign tumors** — tissue name + the suffix **"-oma"**: fibroma, lipoma, chondroma, osteoma, leiomyoma, hemangioma. Benign epithelial tumors: **papilloma** (from squamous and transitional epithelium, papillary), **adenoma** (from glandular epithelium), cystadenoma (cystic adenoma).

- **Malignant epithelial tumors — cancer (carcinoma)**: squamous cell carcinoma, **adenocarcinoma** (glandular carcinoma), transitional cell carcinoma and others.
- **Malignant mesenchymal tumors — sarcoma** (Greek sarx — flesh — for its "fish-flesh" appearance on the cut surface): fibrosarcoma, liposarcoma, chondrosarcoma, osteosarcoma, leiomyosarcoma, rhabdomyosarcoma, angiosarcoma.
- **Tumors of hematopoietic and lymphoid tissue** — leukemias and lymphomas.
- **Exceptions** (ending in "-oma" although malignant): **melanoma, seminoma, lymphoma, mesothelioma, hepatoma**; named after an author or a distinctive feature: Wilms tumor (nephroblastoma), Ewing sarcoma, Hodgkin lymphoma.
- **Teratoma** — a tumor composed of tissues of several germ layers (ecto-, meso-, endoderm) (from totipotent germ cells of the ovary and testis): **mature teratoma (dermoid cyst)** — benign; **immature teratoma** — malignant.
- **Mixed tumors** — arising from one tissue but differentiating in different directions (pleomorphic adenoma of the salivary gland).
- **Hamartoma** — disorganized tumor-like growth of tissues native to the organ (a developmental anomaly); **choristoma** — normal tissue in an unusual location (heterotopia — pancreatic tissue in the gastric wall).



Figure 9.12. *Colonic polyp (adenoma): A — a pedunculated adenomatous (glandular) polyp; B — multiple polyps (Robbins, 9th ed., Fig. 7-1)*



Figure 9.13. *Cystic teratoma of the ovary (dermoid cyst): A — hair, sebaceous material and a tooth inside the cyst; B — skin, sebaceous glands, fat cells and neural tissue (arrow) (Robbins, 9th ed., Fig. 7-3)*

Grading and staging

Grading is the histologic assessment of a tumor by its **degree of differentiation** and the number of mitoses: **G1** (well differentiated), **G2** (moderately), **G3** (poorly differentiated), **G4** (undifferentiated, anaplastic).

Staging is the extent of spread of the tumor; according to the **TNM system**:

- **T** (tumor) — size and local spread of the primary tumor (Tis — in situ, T1–T4);
- **N** (nodus) — metastases to regional lymph nodes (N0 — none, N1–N3);
- **M** (metastasis) — distant metastases (M0 — none, M1 — present). Clinical stages I–IV are determined on the basis of TNM. **Staging is more important than grading for prognosis and the choice of treatment.**

9.8. Epithelial tumors (non-organ-specific)

Benign epithelial tumors

- **Papilloma** — from squamous or transitional epithelium; a papillary tumor (resembling a cauliflower or mulberry); each papilla has a vascular stromal core covered by epithelium; tissue

atypia (thickened epithelium, hyperkeratosis). Skin, oral cavity, pharynx, urinary bladder, urinary tract.

- **Adenoma** — from glandular epithelium; **acinar (alveolar)**, **tubular**, **trabecular**, **papillary**, **cystadenoma** (cystic — in the ovary), **fibroadenoma** (abundant stroma — breast). In mucous membranes — **adenomatous polyp**. Adenomatous polyps of the intestine are a precancerous condition.

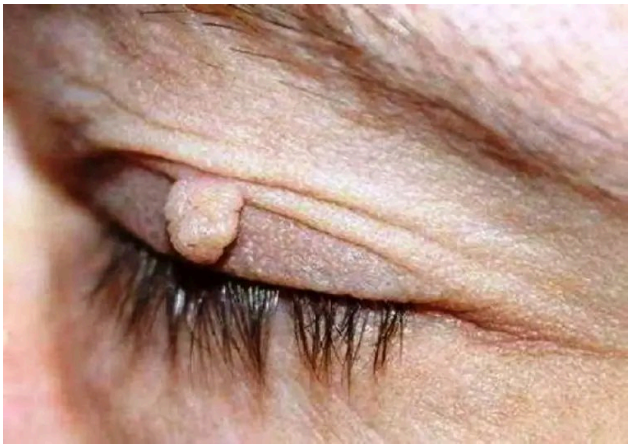


Figure 9.15. *Skin papilloma (gross specimen): a sharply demarcated, wart-like tumor on the surface of the upper eyelid (TMA study guide, 2018)*

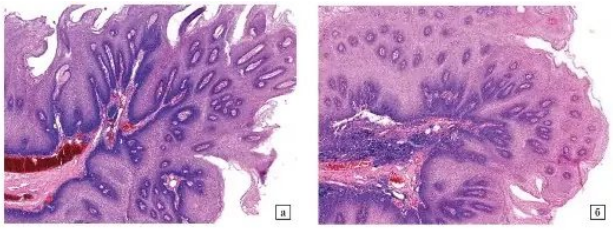


Figure 9.14. *Skin papilloma (microscopic slide): papillary projections with a vascular stromal core covered by thickened stratified squamous epithelium (TMA study guide, 2018)*

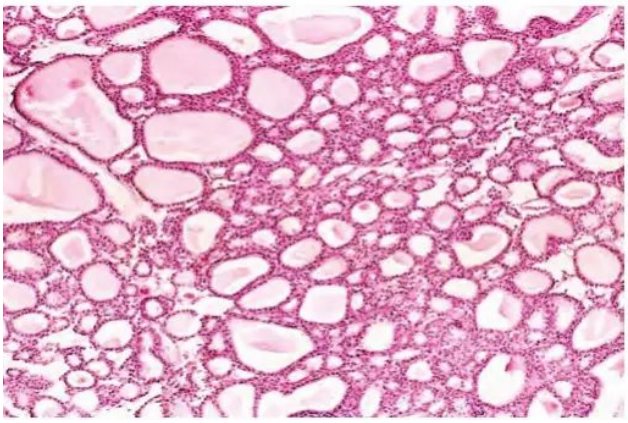


Figure 9.16. *Thyroid adenoma: normal-looking colloid-containing follicles (well differentiated) (Robbins, 9th ed., Fig. 7-5)*

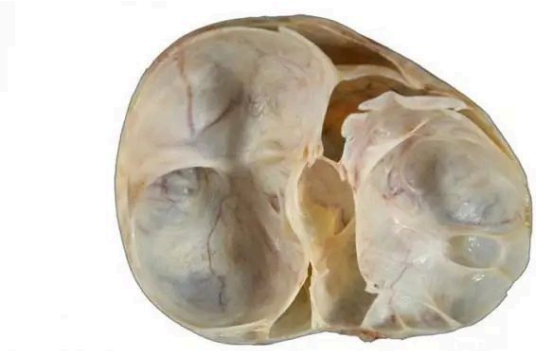


Figure 9.17. *Ovarian cystadenoma: a thin-walled cystic tumor much larger than the ovary, filled with colloid fluid (TMA study guide, 2018)*

Malignant epithelial tumors — carcinoma

Cancer (carcinoma) is the most common malignant tumor (80–90% of all malignant tumors). Mainly **after the age of 40**. It metastasizes first by the **lymphogenous** route.

The main histologic types:

1. **Carcinoma in situ** — cancer that has not broken through the basement membrane.
2. **Squamous cell (epidermoid) carcinoma** — in sites lined by stratified squamous epithelium (skin, mouth, esophagus, cervix) or at sites of metaplasia (bronchus). **Keratinizing** (well differentiated) — concentric layers of horny substance in the center of tumor cell cords — **"keratin pearls"** — and intercellular bridges; **nonkeratinizing** (moderately and poorly differentiated).
3. **Adenocarcinoma (glandular carcinoma)** — from glandular and columnar epithelium (stomach, intestine, breast, lung, endometrium, prostate); atypical glands of various shapes and sizes, often

producing mucus. Types: tubular, papillary, **mucinous (colloid, mucoid) carcinoma**, **signet ring cell carcinoma** (mucus inside the cell, nucleus pushed to the periphery — diffuse gastric cancer).

4. **Basal cell carcinoma (basalioma)** — from the basal layer of the skin; on the face, in the elderly; **locally destructive growth**, almost never metastasizes.
5. **Transitional cell (urothelial) carcinoma** — in the urinary tract.
6. **Solid carcinoma** — a poorly differentiated carcinoma in the form of trabeculae.
7. **Small cell carcinoma** — composed of small, lymphocyte-like cells (lung); highly malignant, neuroendocrine.
8. **Medullary (brain-like) carcinoma** — abundant parenchyma, scant stroma, soft; **scirrhous (fibrous) carcinoma** — abundant stroma, hard.

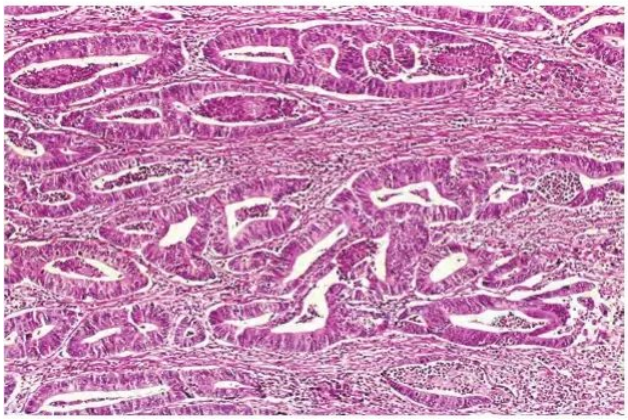


Figure 9.19. *Colonic adenocarcinoma: unlike normal glands, the cancerous glands are irregular in shape and size; invasion of the muscular layer (Robbins, 9th ed., Fig. 7-6)*

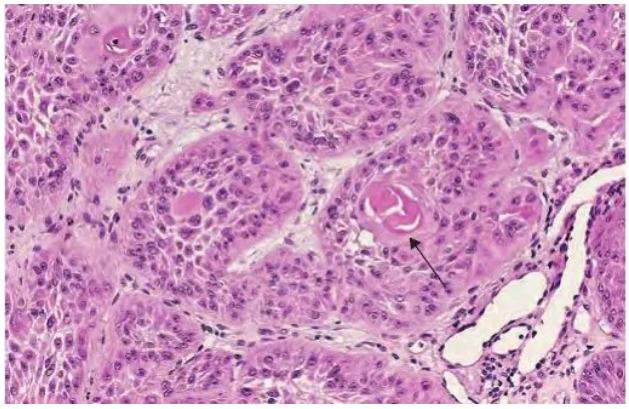


Figure 9.18. *Keratinizing squamous cell carcinoma of the skin: tumor cells resemble normal squamous epithelium, with intercellular bridges and keratin "pearls" (arrow) (Robbins, 9th ed., Fig. 7-7)*

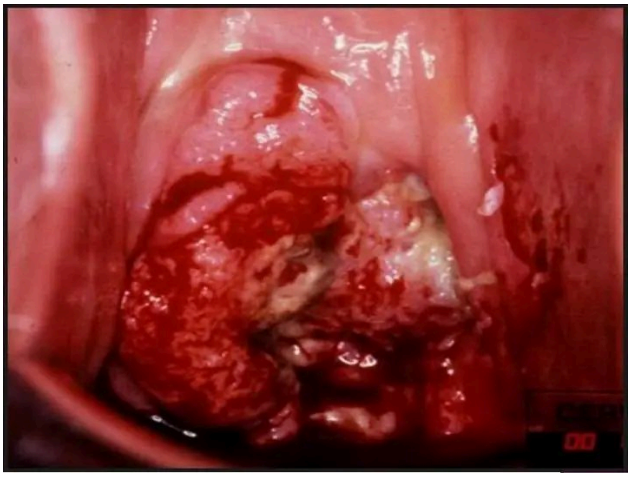


Figure 9.21. *Endophytic carcinoma of the cervix: the cervix is involved by disintegrating whitish tumor tissue (TMA study guide, 2018)*



Figure 9.20. *Diffuse gastric cancer (linitis plastica): the gastric wall is thickened and rigid, and the mucosal folds are flattened (TMA study guide, 2018)*

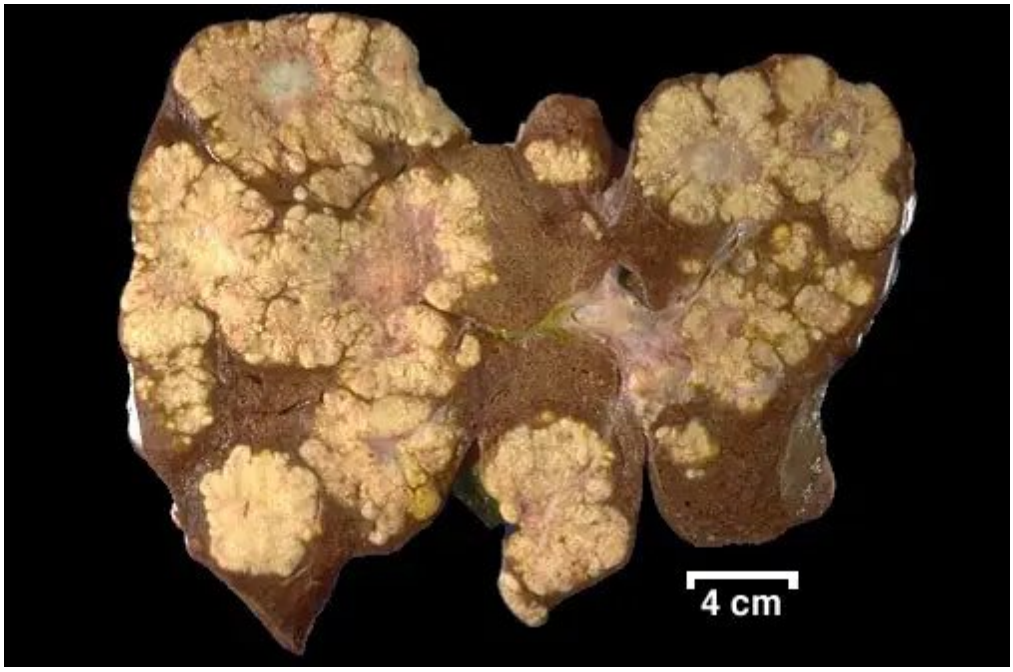


Figure 9.22. *Cancer metastases in the liver: the liver is markedly enlarged, its surface uneven, with numerous round gray-pink nodules (TMA study guide, 2018)*

9.9. Mesenchymal tumors

Mesenchymal tumors arise from connective, adipose, muscle, vascular, bone, cartilage and synovial tissues. Malignant mesenchymal tumors — **sarcomas** — are much less common than carcinomas, occur more often in young people and metastasize mainly by the **hematogenous** route.

Table 9.4. *Main mesenchymal tumors*

Tissue	Benign tumor	Malignant tumor
Fibrous connective tissue	Fibroma (hard and soft), dermatofibroma, desmoid (borderline)	Fibrosarcoma
Adipose tissue	Lipoma (the most common soft tissue tumor; encapsulated, lobulated)	Liposarcoma
Smooth muscle	Leiomyoma (most common in the uterus, “fibromyoma”)	Leiomyosarcoma
Striated muscle	Rhabdomyoma (in the heart, rare)	Rhabdomyosarcoma (the most common soft tissue sarcoma of children)
Blood vessels	Hemangioma (capillary, cavernous — liver, skin)	Angiosarcoma, Kaposi sarcoma
Lymphatic vessels	Lymphangioma	Lymphangiosarcoma
Cartilage	Chondroma, osteochondroma	Chondrosarcoma
Bone	Osteoma, osteoid osteoma	Osteosarcoma (in adolescents, around the knee); Ewing sarcoma
Synovial tissue	Giant cell tumor (of tendon sheath)	Synovial sarcoma
Mesothelium	Adenomatoid tumor	Mesothelioma (asbestos)

- **Leiomyoma (fibromyoma) of the uterus** is the most common tumor in women: sharply demarcated, whitish, firm nodules in the uterine wall (submucosal, intramural, subserosal) with a whorled fibrous pattern on the cut surface; histologically — bundles of smooth muscle cells

running in different directions (tissue atypia), without atypia or mitoses. Clinically — uterine bleeding, anemia, infertility.

- **Leiomyosarcoma** — atypical, pleomorphic spindle cells, numerous mitoses, necrosis.
- **Hemangioma** — the capillary type (in the skin, in children; often regresses spontaneously) and the cavernous type (in the liver — blood-filled spaces).
- **Osteosarcoma** is the most common primary malignant bone tumor; at 10–25 years of age, in the metaphysis of long bones (around the knee — the lower end of the femur and upper end of the tibia); the tumor cells form **osteoid**; it rapidly metastasizes to the lungs.



Figure 9.24. *Uterine fibromyoma (gross specimen): sharply demarcated, round, firm, encapsulated nodules in the uterine wall (TMA study guide, 2018)*

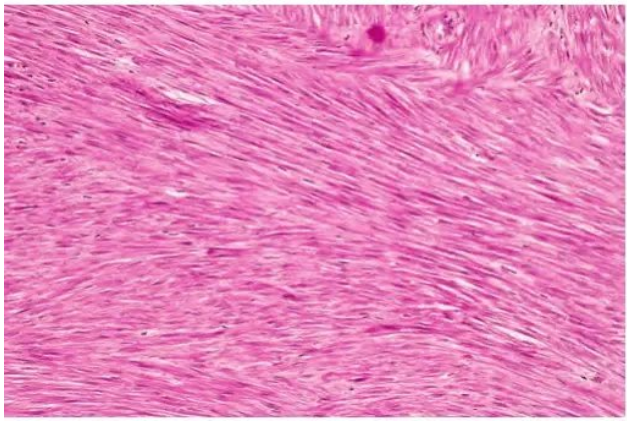


Figure 9.23. *Uterine leiomyoma (microscopic slide): interlacing bundles of smooth muscle cells, closely resembling normal myometrium (Robbins, 9th ed., Fig. 7-4)*

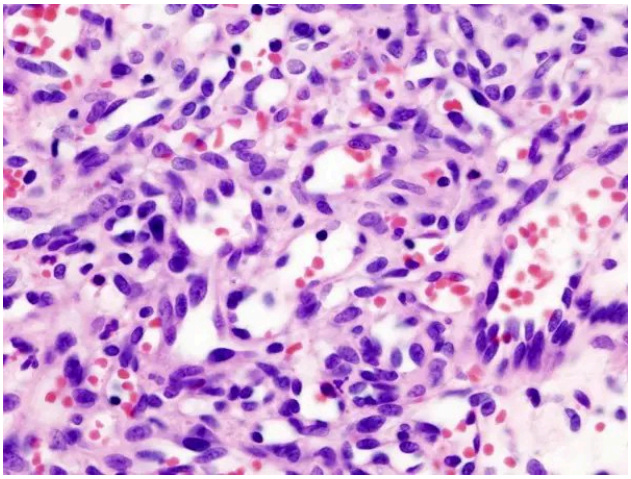


Figure 9.26. *Hemangioma (microscopic slide): numerous blood-filled vascular spaces lined by endothelium (TMA study guide, 2018)*

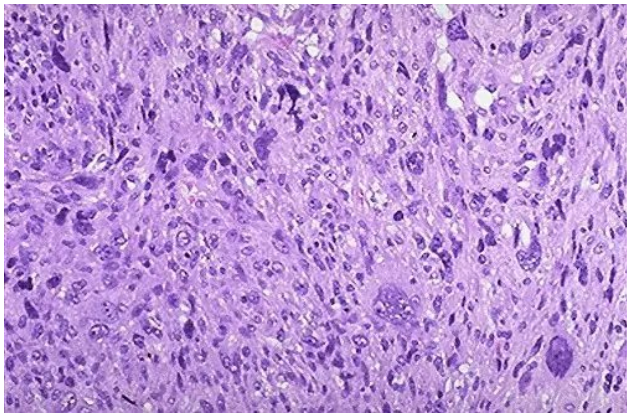


Figure 9.25. *Leiomyosarcoma (microscopic slide): atypical, pleomorphic spindle cells, numerous mitoses (TMA study guide, 2018)*

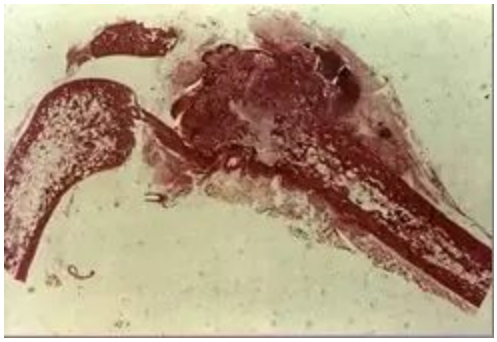


Figure 9.28. *Osteosarcoma of the knee region: a dense, poorly demarcated, brown-reddish tumor destroying the bone (TMA study guide, 2018)*



Figure 9.27. *Cavernous hemangioma of the liver: a sharply demarcated, spongy, blood-filled nodule in the liver (TMA study guide, 2018)*

9.10. Tumors of melanin-forming tissue

- **Pigmented nevus (birthmark)** — a benign tumor-like collection of melanocytes; by location: **junctional** (at the border of the epidermis and dermis), **dermal (intradermal)** and **compound (mixed)**. **Dysplastic nevus** — atypical melanocytes; high risk of melanoma.
- **Melanoma** is a **highly malignant** tumor arising from melanocytes; in the skin (often in a nevus or de novo), the eye (retina) and mucous membranes. Risk factors — ultraviolet rays (especially sunburns in childhood), fair skin, many nevi, family predisposition (CDKN2A), BRAF mutation. **Growth phases: radial** (horizontal, along the epidermis — low risk of metastasis) and **vertical** (deep into the dermis — high risk of metastasis). The main prognostic factor is the **thickness of the tumor (Breslow)**. Histologically: atypical melanocytes, numerous mitoses, melanin (there is also a nonpigmented — **amelanotic** — melanoma). It metastasizes very early by the lymphogenous and hematogenous routes (liver, lungs, brain, skin).

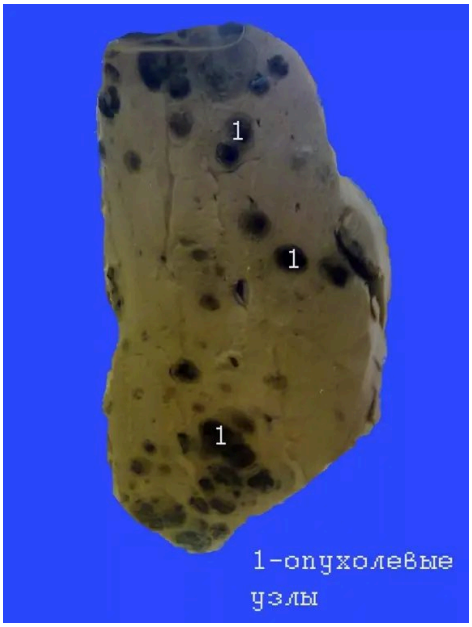


Figure 9.30. *Melanoma metastases in the liver: several round, dark nodules in the liver tissue (TMA study guide, 2018)*

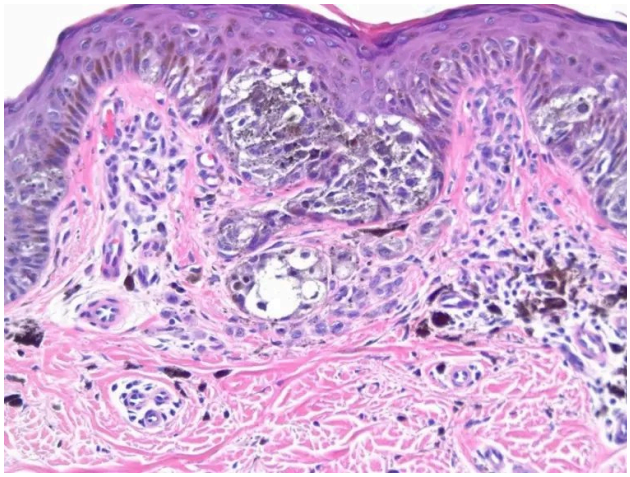


Figure 9.29. *Superficial spreading melanoma of the skin (microscopic slide): nests of atypical pigmented melanocytes in the epidermis and dermis (TMA study guide, 2018)*

CLINICAL SIGNIFICANCE

The “ABCDE” signs of melanoma (Topic 4) and any changing pigmented spot are indications for **excisional biopsy**. Early detection and removal of melanoma (Breslow thickness up to 1 mm) lead to almost complete cure; at a late stage the prognosis is unfavorable, but modern therapy (BRAF inhibitors, immune checkpoint inhibitors) considerably prolongs life.

9.11. Clinicopathological section

The pathologist and the oncologist: from diagnosis to treatment

The final diagnosis of a tumor is made only on the basis of **histologic** examination (biopsy, surgical specimen, cytology). The pathologist's report states: the histologic type of the tumor, its grade of differentiation (G), the depth of invasion, invasion of vessels and nerves, the resection margins (tumor present or absent), and the number of lymph nodes with metastases (pTNM). In modern diagnosis, **immunohistochemistry** (determining the origin of the tumor — cytokeratins for epithelium, vimentin for mesenchyme, CD45 for lymphoid tissue, S-100 and HMB-45 for melanoma) and **molecular tests** (for targeted therapy — estrogen and progesterone receptors and HER2 in breast cancer; EGFR and ALK in lung cancer; KRAS in colorectal cancer) are important.

CLINICAL SIGNIFICANCE

Tumor markers are substances produced by tumor cells and detected in the blood; they are used mainly to monitor the effect of treatment and recurrence: **alpha-fetoprotein** (hepatocellular carcinoma, yolk sac tumor), **carcinoembryonic antigen (CEA)** (colorectal cancer), **PSA** (prostate cancer), **CA-125** (ovarian cancer), **CA 19-9** (pancreatic cancer), **chorionic gonadotropin (beta-hCG)** (choriocarcinoma, testicular tumors). They are not sufficient for diagnosis on their own.

Screening — early detection of cancer

Screening programs are used to detect precancerous changes and early cancer: cervical smear (**Papanicolaou test**) and papillomavirus testing, **mammography** (breast cancer), **colonoscopy** and fecal occult blood testing (colorectal cancer), low-dose computed tomography (lung cancer in heavy smokers). Prevention: quitting smoking, **vaccination against papillomavirus and hepatitis B**, treatment of H. pylori, sun protection.

Practical skills

Skill 1. Gross specimens

1. **“Uterine fibromyoma”.** The uterus is enlarged, with sharply demarcated, round, firm, whitish nodules of various sizes in its wall; a whorled fibrous pattern on the cut surface. Conclusion: benign mesenchymal tumor (leiomyoma).
2. **“Fibroadenoma of the breast”.** A sharply demarcated, encapsulated, firm, whitish-gray, lobulated nodule. Conclusion: benign fibroepithelial tumor.
3. **“Ovarian cystadenoma”.** In place of the ovary — a thin-walled, smooth, unilocular or multilocular cyst filled with clear or mucinous fluid. Conclusion: benign epithelial tumor.
4. **“Skin papilloma”.** A sharply demarcated papillary (cauliflower-like) lesion protruding from the skin surface. Conclusion: benign epithelial tumor.
5. **“Colonic polyp”.** Pedunculated, gray-pink, lobulated lesion(s) on the mucosal surface. Conclusion: adenomatous polyp (precancerous).

6. **“Gastric cancer”.** Ulcerated tissue with raised edges (“saucer-shaped”) or diffusely thickened (linitis plastica), firm and whitish, in the gastric wall. Conclusion: malignant epithelial tumor.
7. **“Cancer metastases in the liver”.** The liver is enlarged, with sharply demarcated whitish-gray nodules of various sizes, with an “umbilicated” depression in the center. Conclusion: hematogenous metastases.
8. **“Osteosarcoma”.** A tumor in the bone metaphysis that destroys the bone and grows into the surrounding soft tissue, with indistinct borders; on the cut surface it is whitish-brown, with foci of hemorrhage and necrosis. Conclusion: malignant mesenchymal tumor.
9. **“Clear cell (hypernephroid) carcinoma of the kidney”.** A nodule at the pole of the kidney, variegated on the cut surface (yellow, red, brown) — foci of fat, hemorrhage and necrosis. Conclusion: malignant epithelial tumor.

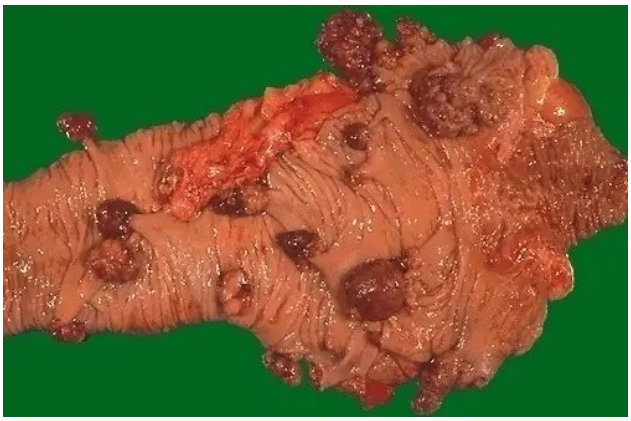


Figure 9.32. *Colonic polyps: numerous round, pedunculated lesions on the mucosal surface (TMA study guide, 2018)*

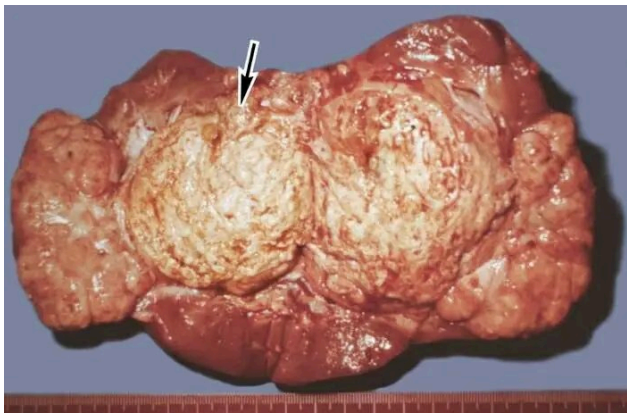


Figure 9.31. *Hypernephroid carcinoma of the kidney: the renal tissue is displaced by a tumor nodule that is variegated on the cut surface (TMA study guide, 2018)*

Skill 2. Microscopic slides

1. **“Skin papilloma”** (hematoxylin and eosin). Papillary projections with a core of connective tissue and vessels, covered by thickened stratified squamous epithelium with hyperkeratosis; the epithelial layers are preserved, and the basement membrane is intact. Conclusion: benign tumor (tissue atypia).
2. **“Fibroadenoma of the breast”** (hematoxylin and eosin). Surrounded by a capsule; elongated, compressed glands within a stroma-rich tumor. Conclusion: benign fibroepithelial tumor.
3. **“Keratinizing squamous cell carcinoma”** (hematoxylin and eosin). Cords and nests of atypical squamous epithelial cells have grown into the dermis; in the center of the nests — concentric layers of horny substance, “keratin pearls”; numerous mitoses. Conclusion: malignant epithelial tumor.
4. **“Adenocarcinoma”** (hematoxylin and eosin). Atypical glands of various shapes and sizes, multilayered epithelium, hyperchromatic nuclei, mitoses; invasion of the wall. Conclusion: glandular carcinoma.
5. **“Melanoma”** (hematoxylin and eosin). Nests of atypical, pleomorphic cells with brown-black melanin in the cytoplasm, numerous mitoses; invasion of the dermis. Conclusion: highly malignant tumor.
6. **“Leiomyoma (fibroleiomyoma)”** (van Gieson). Bundles of smooth muscle cells (yellow) and layers of connective tissue (red) running in different directions; no cellular atypia. Conclusion: benign tumor.

7. **“Leiomyosarcoma”** (hematoxylin and eosin). Atypical spindle cells, pleomorphism, numerous atypical mitoses, foci of necrosis. Conclusion: sarcoma.
8. **“Capillary hemangioma”** (hematoxylin and eosin). Numerous blood-filled capillary-type vessels lined by endothelium. Conclusion: benign vascular tumor.

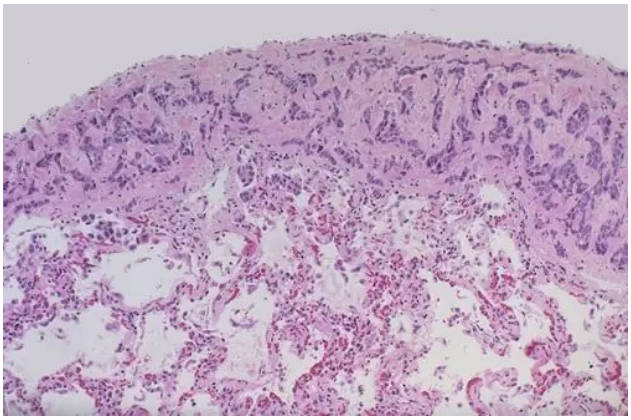


Figure 9.33. *Metastasis of breast carcinoma in the pleura (microscopic slide): tumor cells on the pleural surface (TMA study guide, 2018)*

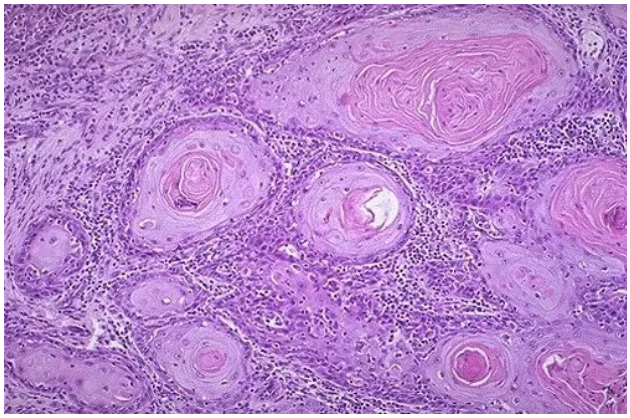


Figure 9.34. *Keratinizing squamous cell carcinoma of the skin (microscopic slide) (TMA study guide, 2018)*

Review questions

1. Define a tumor. What are the main properties of a tumor?
2. What are precancerous processes and carcinoma in situ?
3. The structure of a tumor: parenchyma and stroma. Organoid and histioid tumors.
4. Describe the types of atypia. Signs of cellular atypia.
5. The concepts of differentiation and anaplasia.
6. Types of tumor growth.
7. Routes of metastasis and the invasion–metastasis cascade.
8. Differences between benign and malignant tumors.
9. Local and systemic effects of tumors on the organism. Paraneoplastic syndromes.
10. Chemical, physical and viral carcinogenesis.
11. Oncogenes and tumor suppressor genes. The “two-hit” hypothesis. The role of TP53.
12. List the hallmarks of cancer.
13. Rules for naming tumors. Grading and TNM staging.
14. The main types of benign and malignant epithelial tumors.
15. Mesenchymal tumors and tumors of melanin-forming tissue.

Test questions

1. The most reliable sign of a malignant tumor:
    A) Tissue atypia
    B) Metastasis
    C) Rapid growth
    D) Presence of a capsule
    E) Large size

2. Characteristic of benign tumors:
   A) Expansive growth and tissue atypia only
   B) Cachexia
   C) Cellular atypia and atypical mitoses
   D) Hematogenous metastases
   E) Infiltrative growth

3. A noninvasive cancer that has not broken through the basement membrane:
   A) Medullary carcinoma
   B) Dysplasia
   C) Adenoma
   D) Scirrhous carcinoma
   E) Carcinoma in situ

4. Carcinomas most often metastasize first:
   A) By the hematogenous route
   B) Do not metastasize
   C) By the perineural route
   D) By the implantation route
   E) By the lymphogenous route

5. Metastasis of gastric cancer to the left supraclavicular lymph node:
   A) Recurrence
   B) Implantation metastasis
   C) Krukenberg metastasis
   D) Virchow metastasis
   E) Schnitzler metastasis

6. The tumor suppressor gene called the “guardian of the genome”:
   A) TP53
   B) HER2
   C) RAS
   D) BCL2
   E) MYC

7. The Knudson “two-hit” hypothesis was developed using the example of which gene?
   A) RAS
   B) RB (retinoblastoma)
   C) MYC
   D) EGFR
   E) BCR-ABL

8. The oncogene formed as a result of the “Philadelphia chromosome” in chronic myeloid leukemia:
   A) MYC
   B) APC
   C) BCL2
   D) BCR-ABL
   E) HER2

9. The virus associated with cervical cancer:
    A) Epstein–Barr virus
    B) HTLV-1
    C) HHV-8
    D) Hepatitis B virus
    E) Human papillomavirus (types 16, 18)

10. Aflatoxin B1 increases the risk of which tumor?
    A) Leukemia
    B) Lung cancer
    C) Hepatocellular carcinoma
    D) Bladder cancer
    E) Skin cancer

11. “Keratin pearls” are characteristic of:
    A) Adenocarcinoma
    B) Basal cell carcinoma
    C) Melanoma
    D) Keratinizing squamous cell carcinoma
    E) Sarcoma

12. Malignant mesenchymal tumors are called:
    A) Adenomas
    B) Sarcomas
    C) Teratomas
    D) Carcinomas
    E) Papillomas

13. A tumor ending in “-oma” although malignant:
    A) Lipoma
    B) Osteoma
    C) Melanoma
    D) Leiomyoma
    E) Fibroma

14. The letter “N” in the TNM system denotes:
    A) Metastases to regional lymph nodes
    B) Distant metastases
    C) Size of the primary tumor
    D) Necrosis
    E) Degree of differentiation

15. Develops as a result of ACTH production in small cell lung carcinoma:
    A) Polycythemia
    B) Trousseau syndrome
    C) Cushing syndrome
    D) Hypoglycemia
    E) Hypercalcemia

Clinical cases

Case 1

A 42-year-old woman has prolonged uterine bleeding and anemia. At surgery the uterus is enlarged, with numerous sharply demarcated, firm, whitish nodules from 0.5 to 6 cm in its wall, with a whorled fibrous pattern on the cut surface. Microscopically: bundles of smooth muscle cells running in different directions, no atypia or mitoses.

Questions: 1) Your diagnosis? 2) Is this tumor benign or malignant? Justify. 3) What type of atypia is observed?

Answer: 1) Uterine leiomyoma (fibromyoma) — multinodular. 2) Benign: sharply demarcated (expansive growth), no cellular atypia or atypical mitoses, resembles the tissue of origin (myometrium) — well differentiated. Clinical significance — bleeding and anemia (submucosal nodules). 3) Tissue atypia only — disordered arrangement of muscle bundles in different directions; the cells themselves resemble normal smooth muscle cells.

Case 2

A 62-year-old man has smoked for 40 years. Cough, hemoptysis, weight loss. Bronchoscopy: a tumor in the bronchial wall; biopsy: cords of atypical squamous epithelial cells with concentric layers of horny substance in their center, numerous mitoses. Blood calcium is increased. Later the cervical lymph nodes enlarged.

Questions: 1) Histologic diagnosis? 2) What precancerous changes and carcinogens caused this tumor? 3) Explain the hypercalcemia and the enlargement of the lymph nodes.

Answer: 1) Keratinizing (well-differentiated) squamous cell carcinoma of the lung — “keratin pearls”. 2) Under the action of tobacco smoke carcinogens (benzopyrene and other polycyclic aromatic hydrocarbons, nitrosamines): squamous metaplasia of the bronchial epithelium → dysplasia → carcinoma in situ → invasive cancer; mutations of TP53 and other genes. 3) Hypercalcemia — a paraneoplastic syndrome: the tumor produces a parathyroid hormone–related protein. Enlargement of the lymph nodes — lymphogenous metastases (characteristic of carcinomas).

Case 3

A 16-year-old adolescent complains of pain and swelling in the knee region. X-ray: a bone-destroying tumor in the lower metaphysis of the femur, with a “sunburst” sign. Biopsy: atypical, pleomorphic cells with osteoid material between them, numerous mitoses. A year later numerous nodules were found in the lungs.

Questions: 1) Your diagnosis? 2) To which group does this tumor belong, and what is the naming principle? 3) The lung nodules and the route by which they appeared.

Answer: 1) Osteosarcoma. 2) A malignant mesenchymal tumor — sarcoma; the name is the tissue of origin (bone — “osteo”) + “sarcoma”. It is common in young people, in the metaphysis of long bones (around the knee); the tumor cells form osteoid. 3) The lung nodules are hematogenous metastases: sarcomas metastasize mainly through the blood, and the lung is the first capillary filter.

Quick review

Table 9.5. *Key facts on Topic 9*

Question	Answer
Properties of a tumor	Autonomous growth, atypia, monoclonality, heritability, progression
Precancerous sequence	Hyperplasia/metaplasia → dysplasia → in situ → invasive cancer
Benign tumor	Tissue atypia, expansive growth, no metastasis
Malignant tumor	Cellular atypia, infiltrative growth, metastasis
Signs of anaplasia	Pleomorphism, hyperchromatic nuclei, nucleus/cytoplasm ↑, atypical mitoses, loss of polarity
Metastasis of carcinoma	First lymphogenous
Metastasis of sarcoma	Hematogenous (lung)
Virchow / Krukenberg / Schnitzler	Supraclavicular node / ovary / rectouterine pouch
Oncogenes	RAS, MYC, HER2, BCR-ABL, BRAF, EGFR — dominant
Suppressor genes	TP53, RB, APC, BRCA1/2, NF1, VHL — two hits
Chemical carcinogenesis	Initiation → promotion → progression
Viruses	Papillomavirus — cervix; EBV — Burkitt, nasopharynx; HBV/HCV — liver; HHV-8 — Kaposi
Naming	Benign — “-oma”; malignant epithelial — cancer (carcinoma); mesenchymal — sarcoma
Exceptions	Melanoma, seminoma, lymphoma, mesothelioma
Keratin pearls	Keratinizing squamous cell carcinoma
Signet ring cell	Diffuse gastric cancer
Grading / staging	Differentiation (G1–G4) / TNM
The most common tumor in women	Uterine leiomyoma
Prognosis of melanoma	Breslow thickness

Glossary of terms

- **Oncology** (Greek onkos — tumor) — the science that studies tumors.
- **Tumor** (blastoma — Greek blastano — I sprout; neoplasma — new formation; tumor) — a pathological process characterized by uncontrolled proliferation of cells.
- **Adenoma** (Greek aden — gland, -oma — tumor) — a benign tumor arising from glandular epithelium.
- **Adenocarcinoma** (aden — gland, carcinoma — a malignant tumor of epithelium) — glandular cancer arising from glandular (prismatic) epithelium.
- **Basal cell carcinoma (basalioma)** — a locally destructive tumor arising from the basal cells of the epidermis.
- **Paget disease of the breast** — a malignant tumor of the nipple and areola of the breast.
- **Hidradenoma** — a benign tumor arising from the secretory epithelium of skin glands.

- **Carcinoid** — a neuroendocrine tumor arising from the enterochromaffin (Kulchitsky) cells of the gastric and intestinal mucosa.
- **Melanoma** — a malignant tumor of melanin-forming tissue.
- **Neurinoma (schwannoma)** (Greek neuron — nerve, -oma) — a benign tumor arising from the Schwann cells (lemmocytes) of the nerve fiber sheath.
- **Nevus** — a benign tumor-like lesion of melanin-forming tissue.
- **Papilloma** (Latin papilla — nipple) — a benign papillary tumor arising from stratified squamous or transitional epithelium.
- **Cancer (carcinoma)** — a malignant tumor of epithelium.
- **Carcinoma in situ** — a noninvasive cancer (not breaking through the basement membrane) but with marked cellular atypia.
- **Clear cell (hypernephroid) carcinoma** — the most common malignant tumor of the kidney.
- **Seminoma (dysgerminoma)** — a malignant germ cell tumor (seminoma in the testis, dysgerminoma in the ovary).
- **Syringoadenoma** — a benign tumor of the sweat glands of the skin.
- **Thecoma** — a benign tumor arising from the sex cord stroma of the ovary.
- **Trichoepithelioma** — a benign tumor arising from hair follicles.
- **Choriocarcinoma (chorionepithelioma)** — a malignant tumor arising from the trophoblast after abortion, tubal pregnancy or hydatidiform mole.
- **Exophytic growth** (Greek exo — outside, phyton — growth) — growth of a tumor into the lumen of an organ.
- **Endophytic growth** (Greek endon — inside) — growth of a tumor into the wall of an organ.
- **Expansive growth** (Latin expansus — expanded) — growth of a tumor by itself, compressing the surrounding tissues.
- **Infiltrative (invasive) growth** — growth of tumor cells into the surrounding tissues with their destruction.
- **Metastasis** (Greek metastasis — transfer) — transfer of tumor cells to another site with formation of a new focus.
- **Recurrence** (Latin recidivus — returning) — reappearance of a tumor at the site of its removal.
- **Anaplasia** (Greek ana — backward, plasis — formation) — loss of differentiation.

TOPIC 10

General pathology of infectious diseases. Viral and bacterial infections. Tuberculosis. Syphilis. Sepsis

Learning objectives

The student should know:

- the general features of infectious diseases: causative agent, portal of entry, primary infectious complex, routes of spread, local and systemic changes;
- the mechanisms by which microorganisms damage tissue and the main types of tissue reactions to infection;
- the morphology of viral infections (influenza, measles, herpes, chickenpox, cytomegalovirus infection), viral inclusions;
- bacterial infections: staphylococcal and streptococcal infections, scarlet fever, diphtheria, meningococcal infection;
- tuberculosis: etiology, pathogenesis, primary, hematogenous and secondary forms of tuberculosis, tissue reactions in the tuberculous focus;
- syphilis: stages, morphology, visceral and congenital syphilis;
- sepsis: classification, pathogenesis and clinicomorphologic forms;
- the basic concepts of fungal and protozoal infections (candidiasis, aspergillosis, malaria, leishmaniasis).

The student should be able to:

- describe gross specimens of the primary tuberculous complex of the lung, miliary tuberculosis, cavitary tuberculosis, staphylococcal destruction of the lung, purulent leptomeningitis, the spleen in malaria and the liver in visceral syphilis;
- describe microscopic slides of measles pneumonia (giant cells), cytomegaly, tuberculous granuloma, myocarditis in diphtheria, esophageal candidiasis and the liver in malaria;
- formulate the pathological diagnosis of an infectious disease.

10.1. General characteristics of infectious diseases

Infectious (communicable) diseases are diseases caused by pathogenic microorganisms (viruses, bacteria, fungi, protozoa, helminths, prions) and often transmitted from person to person. Their development depends on the interaction of the **causative agent** (virulence, dose, tropism), the **host** (immune status, age, nutrition, comorbidities) and **environmental** conditions.

General features of an infectious disease:

1. **Portal of entry** — the site where the agent enters the organism: skin (wound, insect bite), respiratory tract (airborne droplets), digestive tract (fecal–oral), genitourinary tract (sexual), parenteral route (blood, injection), transplacental (vertical).
2. **Primary infectious complex** — changes at the portal of entry (**primary affect**) + **lymphangitis** as the agent spreads through the lymph + involvement of the regional lymph nodes (**lymphadenitis**). For example, the primary tuberculous complex, the hard chancre and regional lymphadenitis in syphilis.

3. **Routes of spread** — contiguous (to neighboring tissues), canalicular (along the bronchi, urinary and bile ducts), lymphogenous, hematogenous, perineural.
4. **Local and systemic changes** — local: inflammation; systemic: hyperplasia of lymphoid organs, degeneration in parenchymal organs, rash, intoxication.
5. **Cyclic course** — incubation period → prodromal period → period of main manifestations → recovery or complications.

Mechanisms by which microorganisms damage tissue

- **Direct cell damage** — viruses multiply inside the cell and destroy it (cytopathic effect), alter cell metabolism and induce apoptosis;
- **Toxins** — **exotoxins** (proteins secreted by bacteria: diphtheria, tetanus, botulinum and cholera toxins) and **endotoxin** (the lipopolysaccharide of the gram-negative bacterial wall — fever, septic shock, DIC);
- **Enzymes** — proteases, hyaluronidase (spread of streptococci), coagulase (staphylococci);
- **Immune-mediated damage** — hypersensitivity reactions (type IV in tuberculosis; type III in poststreptococcal glomerulonephritis; molecular mimicry in rheumatic fever).

The main types of tissue reaction to infection

Table 10.1. *Tissue reactions to infection (after Robbins)*

Reaction	Morphology	Examples
Purulent (suppurative)	Neutrophil infiltrate, tissue liquefaction, abscess	Staphylococcus, streptococcus, pneumococcus, meningococcus, gram-negative bacteria
Mononuclear and granulomatous	Lymphocytes, plasma cells, macrophages; granulomas	Tuberculosis, leprosy, syphilis, fungi, viruses (interstitial inflammation)
Cytopathic-cytoproliferative	Viral inclusions in cells, multinucleated giant cells, epithelial proliferation, blisters	Herpes, chickenpox, measles, cytomegalovirus, papillomavirus
Necrotizing	Little inflammation; the tissue rapidly undergoes necrosis	C. perfringens (gas gangrene), amebae, fulminant viral hepatitis of the liver
Chronic inflammation and scarring	Fibrosis, sclerosis	Chronic viral hepatitis — cirrhosis, schistosomiasis

10.2. Viral infections

Viruses are obligate intracellular parasites. The common features of most viral infections: **mononuclear (lymphocytic) infiltration**, **interstitial inflammation**, **degeneration and necrosis** of epithelial cells, **viral inclusions** (intranuclear or cytoplasmic), **multinucleated giant cells**, and addition of a secondary **bacterial infection**.

Influenza

Influenza is an acute respiratory infection caused by RNA-containing **influenza viruses (A, B, C)** and transmitted by airborne droplets. Because of the variability of the **hemagglutinin (H)** and **neuraminidase (N)** antigens of virus A, epidemics and **pandemics** occur (the 1918 “Spanish flu” — 50 million deaths). The virus damages the epithelium of the respiratory tract (**epitheliotropic**) and has a **toxic** effect on the vessels (increased permeability, hemorrhages).

Forms and morphology:

- **Mild form** — **serous-catarrhal** inflammation of the upper respiratory tract; degeneration and desquamation of the epithelium; basophilic and fuchsinophilic inclusions in the epithelium.
- **Moderate form** — catarrhal inflammation spreads to the trachea, bronchi and bronchioles; in the lung — **focal influenza pneumonia** — serous-hemorrhagic exudate, macrophages and desquamated epithelium in the alveoli; sometimes **hyaline membranes**.
- **Severe form:** a) **toxic influenza** — severe intoxication: **serous-hemorrhagic** inflammation and necrosis in the respiratory tract, **hemorrhagic pneumonia** and edema in the lung, edema and small hemorrhages in the brain; b) **influenza with pulmonary complications** — a secondary bacterial (staphylococcal, pneumococcal) infection is added, producing the **“large variegated influenza lung”**: focal hemorrhagic pneumonia, abscesses, bronchiectasis, pleurisy. Complications: pneumonia, lung abscess, otitis, sinusitis, meningitis, encephalitis, myocarditis, vascular thromboses.

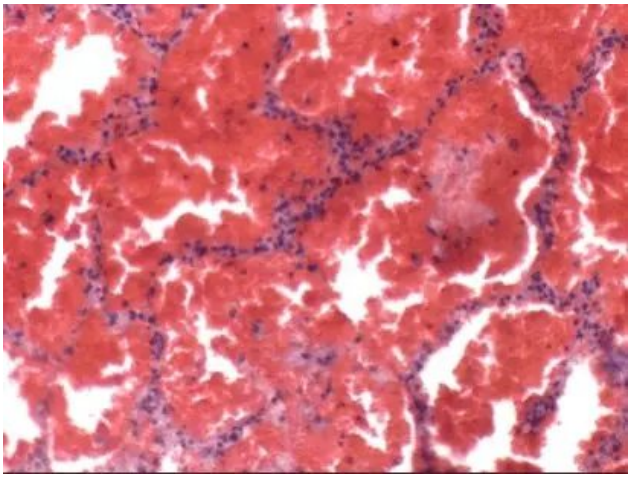


Figure 10.1. *Influenza pneumonia (hemorrhagic): the alveoli are filled with red cells (TMA study guide, 2018)*



Figure 10.2. *Catarrhal pharyngotracheitis in influenza (TMA study guide, 2018)*

Measles

Measles is a highly contagious acute infection, occurring mainly in children, caused by an RNA virus of the paramyxovirus family. The virus enters through the respiratory tract, multiplies in lymphoid tissue and spreads with viremia.

Morphology:

- **in the oral mucosa** (on the buccal mucosa opposite the lower molars) — whitish spots with a red halo — **Belsky–Filatov–Koplik spots** (the pathognomonic early sign of the disease, before the rash);
- **skin rash (exanthem)** — a maculopapular rash beginning on the face and behind the ears and spreading downward; perivascular lymphocytic infiltrate in the dermis; later **branny desquamation** and pigmentation;
- **respiratory tract** — catarrhal-purulent inflammation; in severe cases **necrotizing laryngitis and tracheitis** — “**false croup**”;
- **lung** — **giant cell interstitial pneumonia**: multinucleated (Warthin–Finkeldey) giant cells in the alveolar walls and spaces, with eosinophilic inclusions in their nuclei and cytoplasm; secondary bacterial pneumonia;
- **lymphoid organs** — follicular hyperplasia, **Warthin–Finkeldey giant cells**.

Complications: pneumonia (the main cause of death in children), **noma** (wet gangrene of the facial soft tissues in debilitated children), encephalitis, later **subacute sclerosing panencephalitis**; immunosuppression (the virus damages lymphocytes — reactivation of tuberculosis). **Prevention — vaccination**.

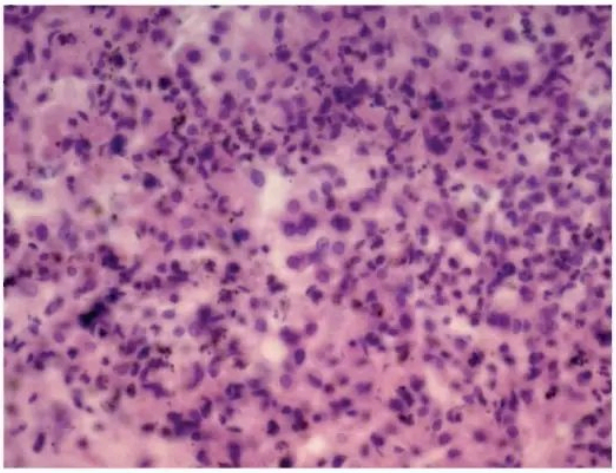


Figure 10.4. *Giant cell measles pneumonia (microscopic slide) (TMA systemic pathology study guide, 2018)*

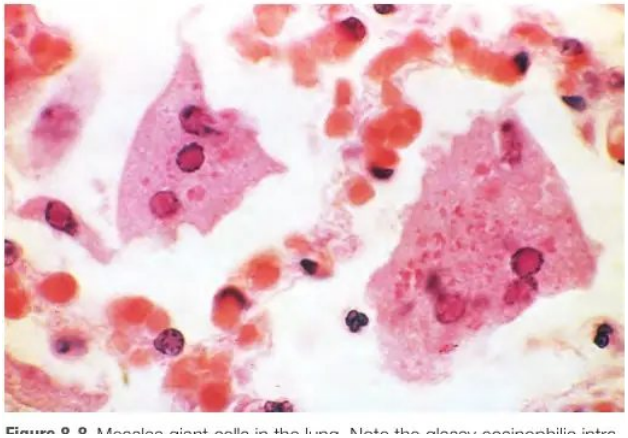


Figure 10.3. *Giant cells in the lung in measles: glassy eosinophilic intranuclear inclusions in multinucleated cells (Robbins, 9th ed., Fig. 8-8)*

Herpesvirus infections

- **Herpes simplex (HSV-1, HSV-2)** — **vesicles** on the skin and mucous membranes (lips, genitals): epithelial cells swell (ballooning degeneration) and rupture, forming an intraepidermal vesicle; **glassy eosinophilic intranuclear inclusions (Cowdry type A bodies)** and multinucleated giant cells in epithelial cells. The virus persists **latently in sensory ganglia** and recurs. Severe forms: **herpetic encephalitis** (hemorrhagic-necrotizing inflammation of the temporal lobes), generalized infection in newborns, esophagitis and pneumonia in immunodeficiency.
- **Chickenpox and herpes zoster (VZV)** — chickenpox in children: a vesicular skin rash (morphology similar to herpes); the virus persists latently in the dorsal root ganglia and later recurs as **herpes zoster (shingles)** (painful vesicles along a nerve).
- **Cytomegalovirus infection (CMV)** — infected cells are **markedly enlarged (cytomegaly)**, with a large basophilic intranuclear inclusion surrounded by a clear halo — the "**owl's eye**". The salivary glands, lungs, kidneys, liver and brain are most often affected. **Congenital** (intrauterine) infection — microcephaly, periventricular calcification, jaundice, hepatosplenomegaly, deafness; **in immunodeficiency** (AIDS, transplantation) — pneumonia, retinitis, colitis.
- **Epstein–Barr virus** — **infectious mononucleosis** (fever, tonsillitis, lymphadenopathy, splenomegaly, atypical lymphocytes in the blood); association with tumors (Topic 9).

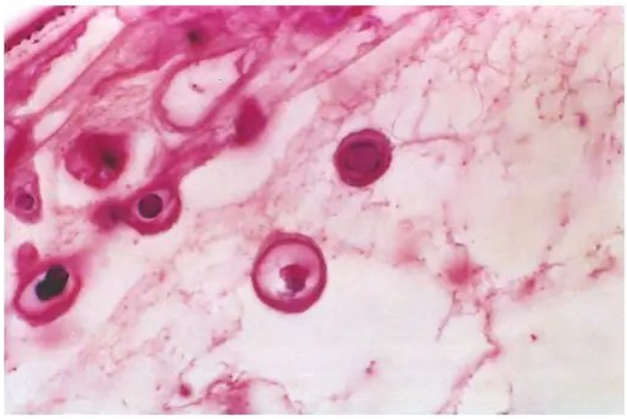


Figure 10.5. *Herpes vesicle: glassy intranuclear inclusions in epithelial cells (Robbins, 9th ed., Fig. 8-9)*

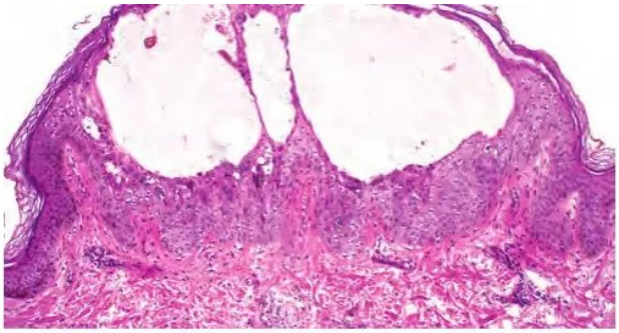


Figure 10.6. *Skin in chickenpox: an intraepidermal vesicle (Robbins, 9th ed., Fig. 8-10)*

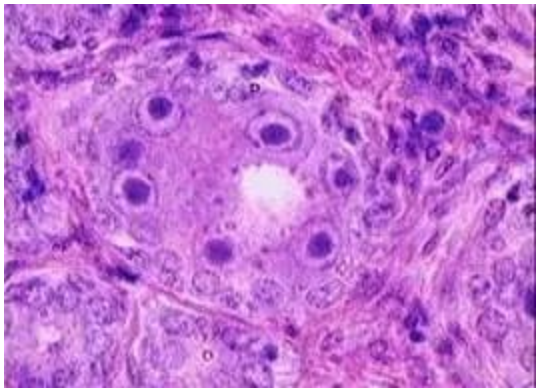


Figure 10.8. *Cytomegaly of a salivary gland: giant cells with an "owl's eye" appearance in the ductal epithelium (TMA systemic pathology study guide, 2018)*

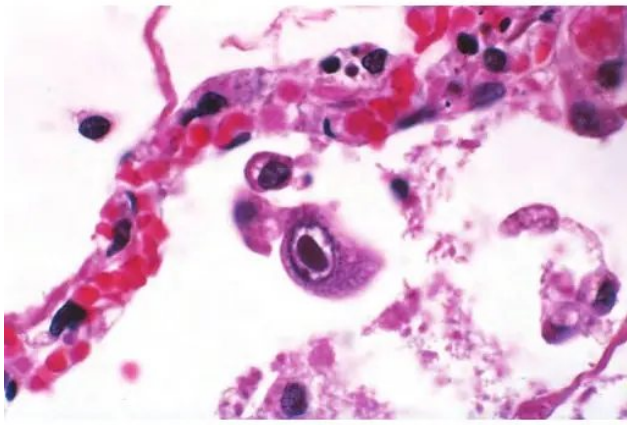


Figure 10.7. *Cytomegalovirus: a large cell in the lung with a distinct intranuclear inclusion and indistinct cytoplasmic inclusions (Robbins, 9th ed., Fig. 8-12)*

MEMORY AID

Viral inclusions: **"Owl's eye — CMV"**, **"Ground-glass nucleus — herpes"**, **"Giant cell + inclusion — measles (Warthin–Finkeldey)"**, **"Negri bodies — rabies"** (in the cytoplasm of neurons).

10.3. Bacterial infections

Staphylococcal and streptococcal infections

Staphylococci (Staphylococcus aureus) — gram-positive cocci arranged like bunches of grapes; they produce **coagulase**, hemolysins, leukocidin and **enterotoxins**. They cause mainly **purulent-necrotizing** inflammation that **tends to be localized** (abscess):

- skin — **furuncle** (purulent-necrotizing inflammation of a hair follicle and sebaceous gland), **carbuncle** (fusion of several furuncles), panaritium, impetigo;
- **lung** — staphylococcal pneumonia, especially in children and after influenza: abscesses and **staphylococcal destruction of the lung** (air spaces — bullae, pyopneumothorax);
- bone — **osteomyelitis**; heart — **infective endocarditis**; sepsis;
- toxin-mediated diseases — **food poisoning** (enterotoxin), **toxic shock syndrome**, **scalded skin syndrome** in newborns (exfoliative toxin). **Streptococci** (Streptococcus pyogenes — group A beta-hemolytic) — chain-forming cocci; they produce hyaluronidase and streptokinase and cause **diffuse** inflammation of tissue (**phlegmon**): **tonsillitis (angina)**, pharyngitis, **erysipelas** (diffuse

edematous inflammation of the dermis, sharply demarcated redness), impetigo, necrotizing fasciitis, sepsis. **Postinfectious immune complications: rheumatic fever** (molecular mimicry) and **poststreptococcal glomerulonephritis** (type III).

Scarlet fever is a childhood disease caused by group A streptococci producing a pyrogenic (erythrogenic) toxin: **necrotic tonsillitis** (necrosis of the tonsils), regional **lymphadenitis**, a fine punctate rash (pale nasolabial triangle — Filatov sign), **"raspberry tongue"**, followed by desquamation of the skin in large sheets. Severe **toxic** and **septic** forms; late complications (in weeks 2–3) — glomerulonephritis, rheumatic fever (the allergic period).



Figure 10.10. *Staphylococcal destruction of the lung: an irregular acute abscess and air spaces (TMA systemic pathology study guide, 2018)*

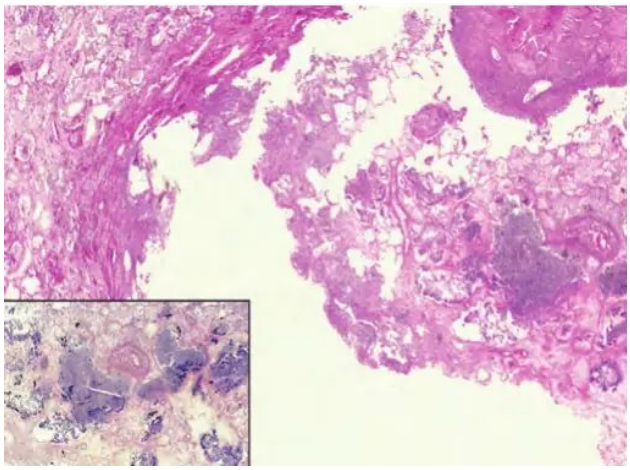


Figure 10.9. *Staphylococcal abscess of the lung: neutrophilic infiltrate and destruction of alveoli; clusters of bacteria on Gram staining (Robbins, 9th ed., Fig. 8-16)*

Diphtheria

Diphtheria is an acute infection caused by Corynebacterium diphtheriae; the main damaging factor is the **exotoxin** (it stops protein synthesis in cells). **Fibrinous inflammation** develops at the portal of entry (Topic 5):

- **in the pharynx and tonsils** (stratified squamous epithelium) — **diphtheritic** inflammation: a thick gray **membrane** firmly attached to the underlying tissue; much toxin is absorbed, and the soft tissues of the neck swell;
- **in the larynx, trachea and bronchi** (columnar epithelium) — **croupous** inflammation: an easily detachable membrane — **true croup**, asphyxia. **Systemic action of the toxin: toxic myocarditis** — fatty degeneration of cardiomyocytes ("tiger heart"), foci of myolysis, interstitial inflammation — **sudden cardiac arrest** in the 2nd week; **parenchymal neuritis** (paralysis of the soft palate, eye muscles and diaphragmatic nerves — in weeks 3–6); hemorrhage and necrosis in the adrenal glands; nephrosis. **Causes of death:** early — asphyxia (true croup) and toxic shock; late — cardiac paralysis (myocarditis), diaphragmatic paralysis.

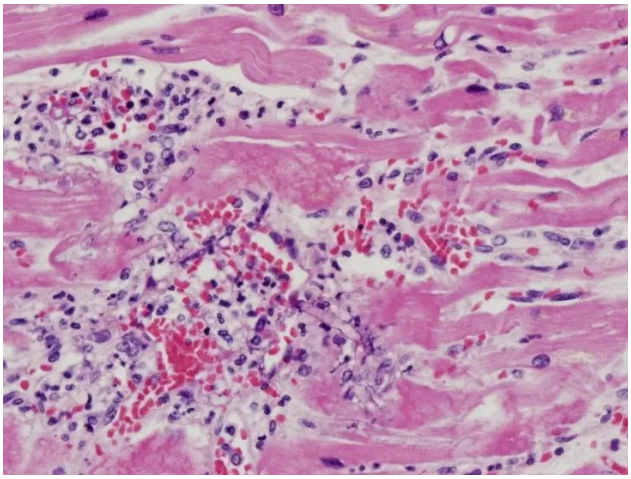


Figure 10.12. *Myocarditis in diphtheria: fatty degeneration of cardiomyocytes, foci of myolysis, edema and infiltrate in the stroma (TMA systemic pathology study guide, 2018)*

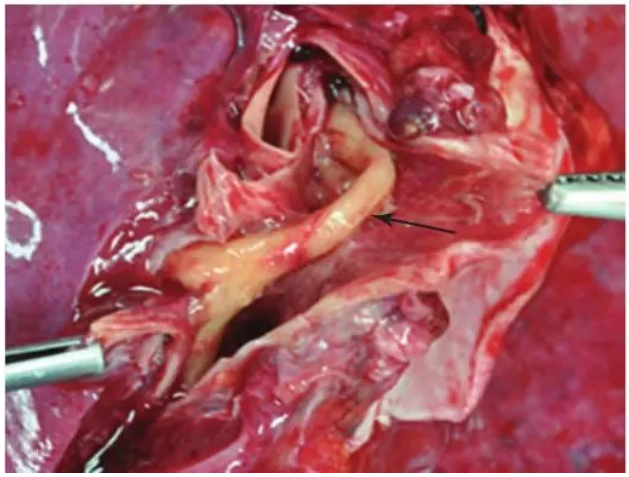


Figure 10.11. *A diphtheritic membrane in the bronchial lumen (arrow) (Robbins, 9th ed., Fig. 8-18)*

Meningococcal infection

Meningococcus (Neisseria meningitidis) is a gram-negative diplococcus; it is transmitted by airborne droplets; mainly in children. Forms:

- **meningococcal nasopharyngitis** — catarrhal inflammation;
- **purulent meningitis (leptomeningitis)** — the leptomeninges are thickened, dull and soaked with yellow-green pus — a **"purulent cap"** (mainly over the convex surface and base of the brain); microscopically, a diffuse neutrophilic infiltrate in the meninges; complications — **hydrocephalus** (obliteration of the CSF pathways), cerebral edema and herniation, encephalitis, abscess;
- **meningococcemia (meningococcal sepsis)** — severe **septic shock**, **DIC**, a **hemorrhagic rash** on the skin ("stellate" necrotic rash), bilateral hemorrhage into the adrenal glands — **Waterhouse–Friderichsen syndrome** (acute adrenal insufficiency) — leads rapidly to death.

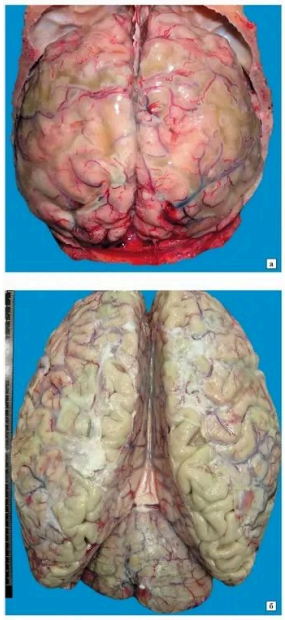


Figure 10.13. *Purulent leptomeningitis: the leptomeninges of the brain are thickened and soaked with yellow-green pus (TMA systemic pathology study guide, 2018)*

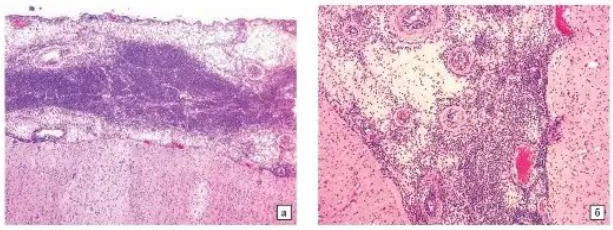


Figure 10.14. *Purulent meningitis (microscopic slide): the pia mater is diffusely infiltrated by neutrophils (TMA study guide, 2018)*

CAUTION

Within the curriculum, intestinal infections (typhoid fever, salmonellosis, dysentery, cholera) and highly dangerous infections (plague, anthrax) are studied in detail in systemic pathology. This chapter gives their examples important for general pathology: **diphtheritic colitis** in dysentery (Topic 5), **waxy necrosis** in typhoid fever (Topic 2), **hemorrhagic inflammation** in anthrax and plague (Topic 5).

10.4. Tuberculosis

Tuberculosis is a chronic infectious disease caused by Mycobacterium tuberculosis (Koch bacillus, 1882); it affects mainly the lungs but can involve any organ. Mycobacteria are **acid-fast** (red bacilli with the Ziehl–Neelsen method) and resistant to phagocytosis because of their waxy cell wall. Transmission is mainly by **airborne droplets** (from a patient with open tuberculosis), less often through the intestine (cow's milk — M. bovis). According to WHO, tuberculosis is one of the leading causes of death from a single infectious agent in the world; it runs a severe course together with **HIV infection**.

Pathogenesis

Mycobacteria are engulfed by alveolar macrophages, but they block the fusion of the phagosome with the lysosome and multiply inside the macrophage (**incomplete phagocytosis**). After about 3 weeks **cell-mediated immunity** develops: **Th1 lymphocytes** produce IFN-gamma and activate macrophages — these turn into **epithelioid cells** and **giant cells** and form a **granuloma** (Topic 6); at the same time **type IV hypersensitivity** develops — **caseous necrosis** and a positive tuberculin (Mantoux) test. Thus in tuberculosis immunity and hypersensitivity appear simultaneously.

Tissue reactions in the tuberculous focus

- **Alterative reaction** — caseous necrosis predominates (when immunity is low);
- **Exudative reaction** — serous-fibrinous or purulent exudate that rapidly turns into caseous necrosis (when hypersensitivity is high);
- **Productive reaction** — tuberculous granulomas (tubercles) (when immunity is good). Outcomes of the tuberculous focus: **favorable** — encapsulation, calcification (petrification), ossification, scar; **unfavorable** — liquefaction of the caseous mass and formation of a cavity (**cavern**), enlargement of the focus and dissemination.

Clinicomorphologic forms

Three main types of tuberculosis are distinguished:

1. Primary tuberculosis — on the first infection (in a nonsensitized organism), most often in children. Its morphologic expression is the **primary tuberculous complex**:

- **primary affect (Ghon focus)** — in the lung parenchyma, usually **in the lower part of the upper lobe or the upper part of the lower lobe**, beneath the pleura, a focus of caseous pneumonia 1–1.5 cm in size (with perifocal inflammation around it);
- **lymphangitis** — tuberculous tubercles along the lymphatic vessels;
- **regional lymphadenitis** — caseous necrosis of the hilar and peribronchial lymph nodes (larger than the primary affect). **Outcomes:** a) **healing** (in most people) — the foci are encapsulated and calcified: the calcified primary affect (**Ghon focus**) and the calcified lymph nodes together form the **Ranke complex** (visible on X-ray); mycobacteria may survive inside for years; b) **progression**: growth of the primary affect — **primary lung cavity**, caseous pneumonia;

lymphogenous spread (bronchoadenitis, extensive lymph node involvement); **hematogenous** spread (miliary tuberculosis, meningitis) — especially in young children and in immunodeficiency; c) **chronic course**.

2. Hematogenous tuberculosis — spread through the blood from foci that persisted after the primary infection (most often from lymph nodes). Forms:

- **generalized hematogenous tuberculosis** — **miliary tuberculosis**: millet-sized (1–2 mm) whitish-yellow tubercles in many organs (lungs, liver, spleen, kidneys, meninges); in severe cases — **tuberculous sepsis** (necrotic foci without reaction);
- **hematogenous tuberculosis with predominant lung involvement** — bilateral symmetric miliary or large-focal dissemination in the lungs;
- **extrapulmonary hematogenous tuberculosis** — individual organs: **bones and joints** (spine — **Pott disease**, cold abscesses, gibbus), **kidney** (cavities, "pyonephrosis"), genital organs (epididymitis, salpingitis — infertility), adrenal glands (**Addison disease**), **tuberculous meningitis** (gelatinous exudate at the base of the brain), skin, lymph nodes (scrofula). **3. Secondary (reinfection) tuberculosis** — in **adults** who previously had primary tuberculosis (endogenous reactivation or exogenous reinfection); mainly in the **lung apices** (segments I and II); the main route is **canalicular (bronchogenic)** spread. **8 forms** (phases) of secondary tuberculosis are distinguished:

Table 10.2. *Forms of secondary tuberculosis (classification of A.I. Strukov)*

Form	Morphology
Acute focal (Abrikosov focus)	Specific endobronchitis of a segmental bronchus in the lung apex, then a focus of acinar or lobular caseous bronchopneumonia; on healing — a calcified Aschoff–Puhl focus
Fibrous-focal	Sclerosis around healed and calcified foci, reactivation of new foci
Infiltrative	Extensive perifocal exudative inflammation around the focus (on X-ray — the Assmann–Redeker infiltrate)
Tuberculoma	An encapsulated focus of caseous necrosis 2–5 cm in diameter (resembles a tumor on X-ray)
Caseous pneumonia	Caseous necrosis involves a whole lobe or more; the lung is yellow and firm ("cheese-like"); severe, leads rapidly to death
Acute cavernous	The caseous mass is discharged through a bronchus and a cavity forms; its wall is lined internally by caseous masses
Fibrocavernous (chronic pulmonary tuberculosis)	The cavity wall has three layers: inner — caseous (pyogenic), middle — tuberculous granulation tissue, outer — fibrous; the process spreads from the apex downward (apicocaudally) and bronchogenically; the most common form
Cirrhotic	Deformation of the lung, extensive scar tissue, bronchiectasis, a scar in place of the cavity

Complications and causes of death: pulmonary **hemorrhage** (erosion of vessels in the cavity wall), **pneumothorax**, pleural empyema, **cardiopulmonary failure** (chronic cor pulmonale), **secondary AA amyloidosis** (Topic 8), additional tuberculosis of the bronchi, larynx and intestine (from swallowed sputum), tuberculous meningitis.

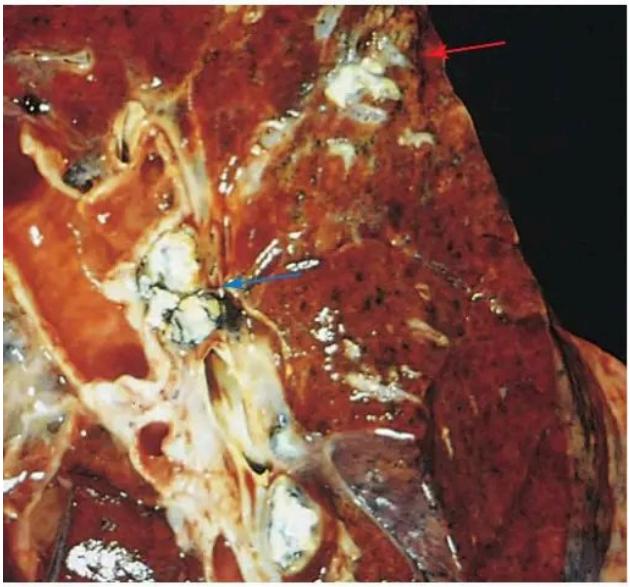


Figure 10.15. *Primary pulmonary tuberculosis — the Ghon complex: a gray-white parenchymal focus beneath the pleura in the lower part of the upper lobe (red arrow), caseous necrosis in the hilar lymph nodes (blue arrow) (Robbins, 9th ed., Fig. 8-26)*

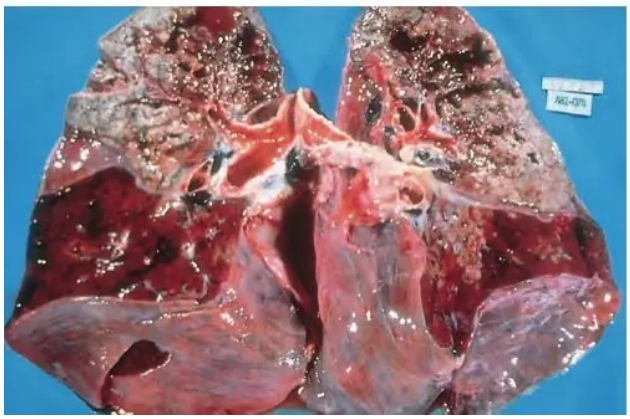


Figure 10.16. *Secondary pulmonary tuberculosis: the upper parts of both lungs are filled with gray-white areas of caseation and numerous cavities (Robbins, 9th ed., Fig. 8-28)*

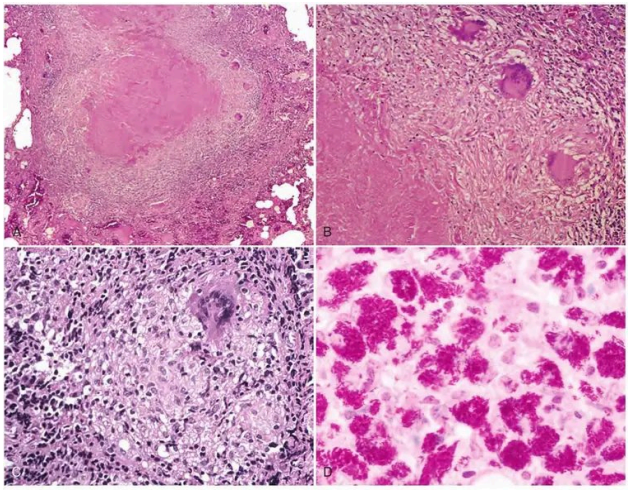


Figure 10.17. *The morphologic spectrum of tuberculosis: A — a characteristic tubercle with central granular caseous necrosis surrounded by epithelioid and multinucleated giant cells; B — high magnification; inset — red bacilli with acid-fast staining; C — noncaseating granulomas sometimes seen in immunocompetent people; D — in immunodeficiency — sheets of foamy macrophages filled with mycobacteria (Robbins, 9th ed., Fig. 8-27)*

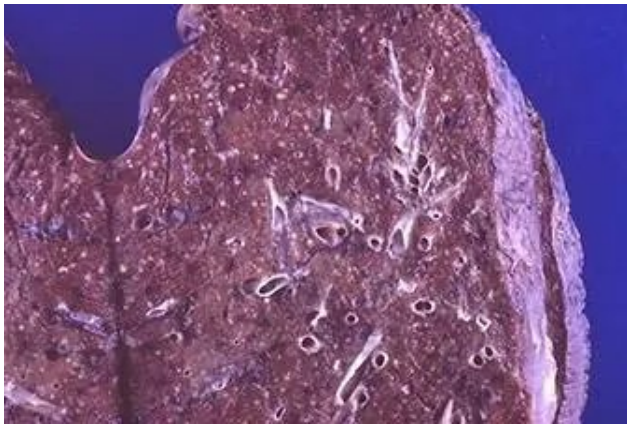


Figure 10.19. *Miliary tuberculosis of the lung (TMA study guide, 2018)*

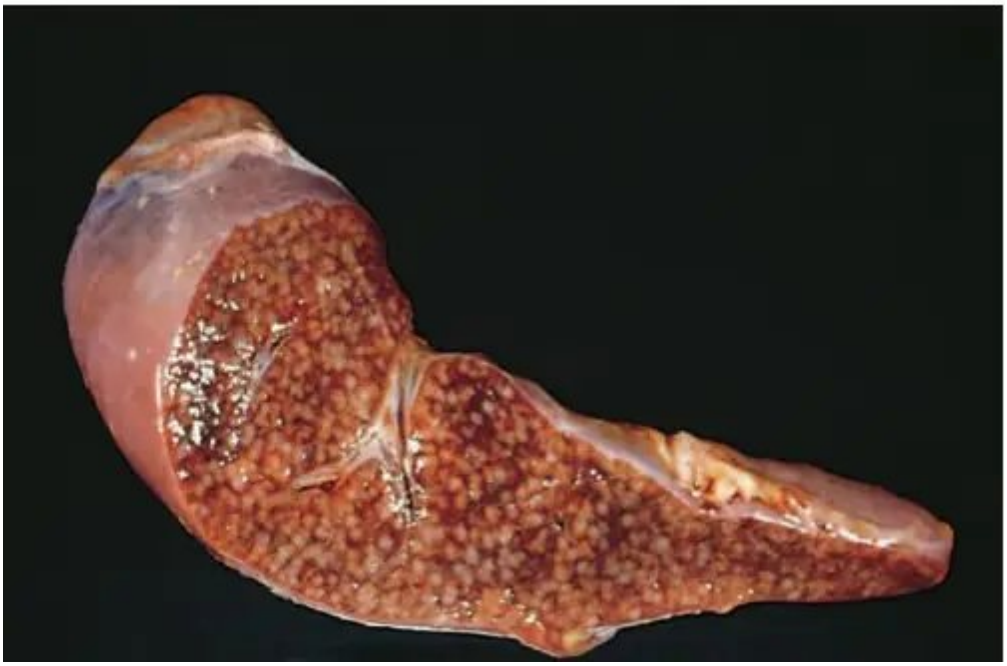


Figure 10.18. *Miliary tuberculosis of the spleen: numerous gray-white tubercles on the cut surface (Robbins, 9th ed., Fig. 8-29)*

CLINICAL SIGNIFICANCE

Tuberculosis is a relevant disease for Uzbekistan. Diagnosis: detection of acid-fast bacilli in sputum (microscopy), molecular testing (**GeneXpert** — mycobacterial DNA and rifampin resistance), culture, X-ray; latent infection — the Mantoux test and IFN-gamma release assays. **Drug-resistant tuberculosis** (MDR-TB) is a serious problem. To prevent severe forms in children (miliary tuberculosis, tuberculous meningitis), **BCG** vaccination is given in the maternity hospital.

10.5. Syphilis

Syphilis is a chronic infection with a wavelike course caused by Treponema pallidum; it is transmitted mainly **sexually**, and also vertically (transplacentally) and parenterally. Treponemes cannot be seen with ordinary stains — **silver impregnation**, dark-field microscopy, immunofluorescence are used; diagnosis is mainly by **serologic** tests (nontreponemal — RW (Wassermann reaction), RPR; treponemal — RIF, TPHA).

In every stage of syphilis the main morphologic feature is **damage to small vessels**: **obliterating endarteritis** (endothelial proliferation, thickening of the vessel wall) and a **perivascular lymphoplasmacytic infiltrate** (abundance of **plasma cells** is characteristic).

Table 10.3. *Stages of acquired syphilis*

Stage	Time	Morphology
Primary	3–4 weeks after infection	A hard chancre at the portal of entry — a painless ulcer with a firm base and smooth floor (on the genitals, lip, anus); lymphoplasmacytic infiltrate and endarteritis in the dermis; regional lymphadenitis (firm, painless) — the primary syphilitic complex; the chancre heals spontaneously in 4–6 weeks
Secondary	After 2–3 months	Hematogenous spread of treponemes — syphilides: macular (roseola), papular and pustular rashes on the skin (including the palms and soles) and mucous membranes; condylomata lata (around the anus); generalized lymphadenopathy; many treponemes in the lesions — highly contagious

Stage	Time	Morphology
Tertiary	After many years (3–5 years or more)	Gummas and gummatous infiltration (Topic 6): cardiovascular syphilis — syphilitic mesaortitis (“shagreen skin” intima), aortic aneurysm, aortic valve insufficiency, narrowing of the coronary ostia; neurosyphilis — general paresis (atrophy of the cerebral cortex, dementia), tabes dorsalis (demyelination of the posterior columns); visceral syphilis — gummas and scars in the liver (“hepar lobatum”), bones (destruction of the nasal septum — “saddle nose”), testes

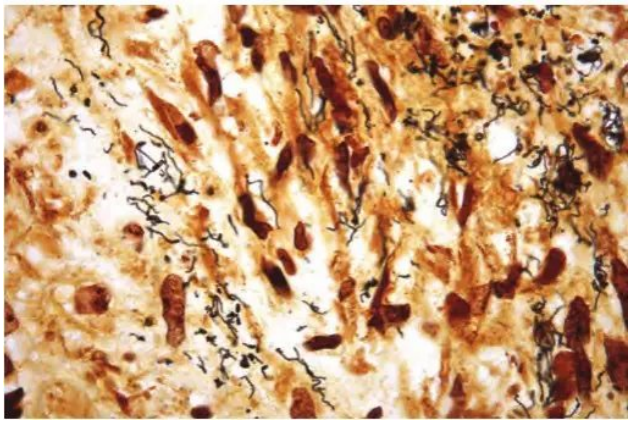


Figure 10.21. *Treponema pallidum (Steiner silver stain): spiral organisms (Robbins, 9th ed., Fig. 8-33)*

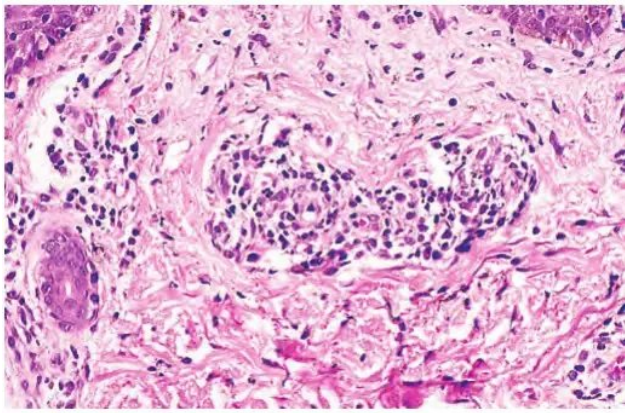


Figure 10.20. *Secondary syphilis: perivascular lymphoplasmacytic infiltrate and endothelial proliferation in the dermis (Robbins, 9th ed., Fig. 8-5)*



Figure 10.23. *The liver in visceral syphilis: the liver is enlarged, hard and divided by scars into lobes (TMA study guide, 2018)*

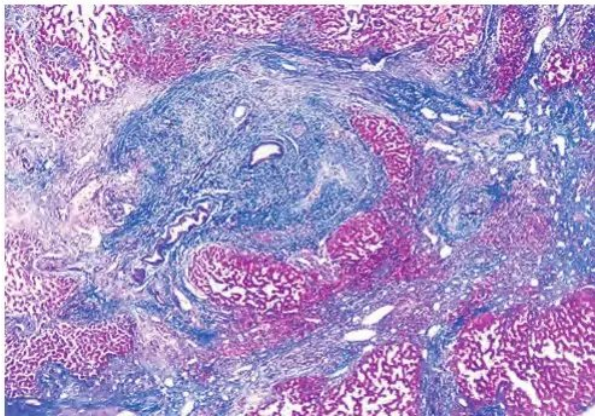


Figure 10.22. *Gumma in syphilis (trichrome stain): blue-stained scar — “hepar lobatum” (Robbins, 9th ed., Fig. 8-36)*

Congenital syphilis — intrauterine infection (after the 4th month of pregnancy, transplacentally):

- **stillbirth** or miscarriage (maceration);
- **early congenital syphilis** (in the newborn and early childhood): **liver** — diffuse interstitial hepatitis, fibrosis (“**flint liver**”), **lung** — “**white pneumonia**” (interstitial fibrosis, unexpanded alveoli), **bones** — **syphilitic osteochondritis** (a yellow band at the epiphyseal–diaphyseal border) and periostitis, skin — a vesicular rash (on the palms and soles), nasal obstruction (rhinitis, “snuffles”);

- **late congenital syphilis** (after 2 years of age): **Hutchinson triad** — notched, crescent-shaped deformity of the incisors (**Hutchinson teeth**), **interstitial keratitis**, inner ear damage (**deafness**); “saddle nose”, **saber shins**.

CLINICAL SIGNIFICANCE

Syphilis responds well to penicillin, but the spontaneous disappearance of symptoms in the early stages deceives the patient. Therefore serologic testing of **pregnant women** and blood donors is mandatory: timely treatment during pregnancy prevents congenital syphilis.

10.6. Sepsis

Sepsis is life-threatening **organ dysfunction** caused by a **dysregulated host response** to infection (Sepsis-3, 2016). In traditional pathological anatomy, sepsis is a **nonspecific** (similar with different agents), **noncyclic** generalized infectious disease characterized by constant or periodic entry of agents and toxins into the blood from a local infectious focus.

CAUTION

Sepsis terminology has changed. In the textbooks of Strukov and Abdullakhodjaeva, sepsis is divided into **septicemia** (without metastatic purulent foci) and **septicopyemia** (with metastatic abscesses), and “bacteremia” and “septic endocarditis” are considered separately. In the international **Sepsis-3** consensus (2016), “sepsis” is infection + organ dysfunction (an increase of 2 or more points on the SOFA score), and “**septic shock**” is hypotension requiring vasopressors with elevated lactate; the terms “septicemia” and “sepsis with SIRS” have been removed from practice. The exam requires knowledge of the classic classification, while the new definitions are used in the clinic.

Pathological concepts of sepsis:

- **Septic focus (primary focus)** — a local focus of purulent inflammation from which the agent enters the blood; it often coincides with the **portal of entry**; the septic focus often shows **purulent lymphangitis** and **thrombophlebitis** (purulently dissolved thrombi — a source of bacterial embolism).
- **Types of sepsis by portal of entry: therapeutic (parasepsis)** — from infection of internal organs; **tonsillogenic**; **surgical** (postoperative, wound); **uterine (obstetric)** — after childbirth and abortion; **otogenic**; **odontogenic**; **urogenic**; **umbilical sepsis** (in newborns); **cryptogenic** — when no focus is found.

Clinicomorphologic forms:

1. **Septicemia** — severe **intoxication** without metastatic purulent foci: hemorrhages in the skin and mucous membranes, **septic spleen** (enlarged, soft, bright red on the cut surface, yielding abundant pulp on scraping), hyperplasia of lymph nodes, degeneration and interstitial inflammation in parenchymal organs, hemolysis (jaundice), DIC; **septic shock**.
2. **Septicopyemia** — the main feature is **metastatic abscesses** in various organs (lungs, kidneys, myocardium, brain, liver) (the result of bacterial embolism, for example, **embolic purulent nephritis** — Topic 7); most often caused by staphylococci and Pseudomonas aeruginosa.
3. **Septic (bacterial) endocarditis** — **polypous-ulcerative endocarditis** of the heart valves (most often aortic and mitral; tricuspid in drug users): large thrombotic vegetations on the valves, destruction (perforation), microbial colonies; **thromboembolic syndrome** (infarcts of the kidney, spleen, brain), **immune complex syndrome** (glomerulonephritis, vasculitis, Roth spots, Osler nodes). Acute (staphylococcus, on a healthy valve) and subacute (viridans streptococcus — Streptococcus viridans, on a previously damaged valve) forms.

4. **Chronic sepsis** — prolonged suppuration (osteomyelitis, carious teeth), cachexia, amyloidosis.

> **CLINICAL SIGNIFICANCE**
> **Septic shock** is the main mechanism of death from sepsis: systemic vasodilation, endothelial damage, DIC and "shock organs" under the action of endotoxin and cytokines (TNF, IL-1) (Topic 7). The most important principle of treatment is giving antibiotics **within the first hour** and **eliminating the septic focus** (opening an abscess, removing an infected catheter or tissue).

10.7. Fungal and protozoal infections (in brief)

Candidiasis — Candida albicans (an opportunistic yeast-like fungus); it develops with antibiotics, glucocorticoids, diabetes mellitus, immunodeficiency (AIDS) and in newborns. White plaques in the oral cavity ("**oral thrush**"), the esophagus (candida esophagitis), the vagina; in severe cases **candida sepsis** (microabscesses in the kidney, brain, heart). Microscopically — yeast cells and **pseudohyphae** (PAS-positive).

Aspergillosis — Aspergillus fungi: **aspergilloma** (a "fungus ball" in old cavities), **allergic bronchopulmonary aspergillosis**, **invasive aspergillosis** (in neutropenia — invades vessels, causing hemorrhagic infarcts). The hyphae are thin, septate and branch **at acute angles**.

Malaria — caused by Plasmodium (falciparum, vivax, malariae, ovale); transmitted by female Anopheles mosquitoes. Plasmodia multiply in red cells and destroy them — **hemolysis**, attacks of fever. **Hemomelanin (malarial pigment)** accumulates in macrophages (Topic 4): the **spleen** is enlarged and gray-black ("**slate spleen**"), and the liver and bone marrow are dark. **Tropical malaria** (P. falciparum) — **cerebral malaria**: infected red cells stick to cerebral capillaries and occlude them — hemorrhages, necrosis (Dürck granulomas), coma; acute kidney injury, severe anemia.

Leishmaniasis — Leishmania (protozoa), transmitted by sand flies (Phlebotomus). **Cutaneous leishmaniasis** ("**pendinka**", Borovsky disease — found in Central Asia) — ulcerating nodules in the skin; **visceral leishmaniasis (kala-azar)** — leishmaniae multiply in macrophages: marked **splenomegaly**, hepatomegaly, bone marrow involvement (pancytopenia), cachexia; **Leishman–Donovan bodies** in macrophages.

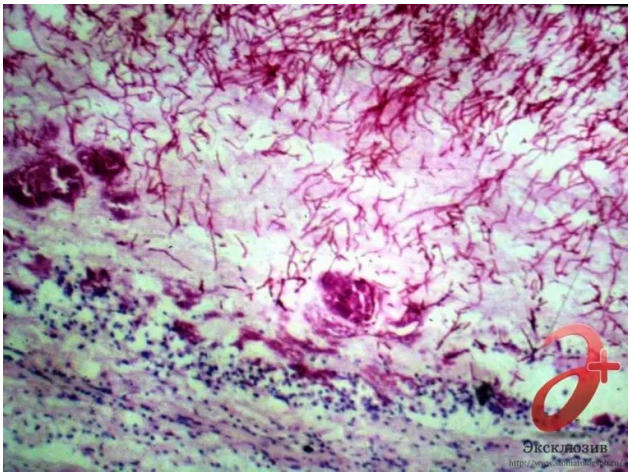


Figure 10.25. *Esophageal candidiasis (microscopic slide): necrosis of the mucosa, fungal filaments (TMA systemic pathology study guide, 2018)*

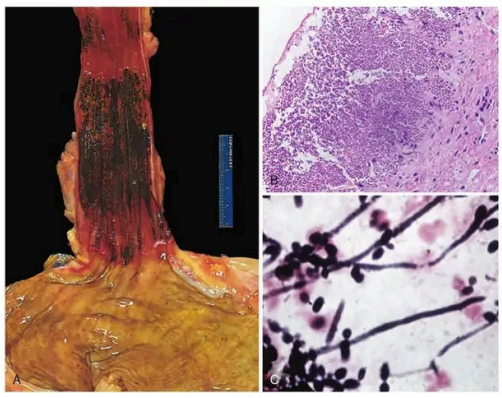


Figure 10.24. *Candidiasis: A — severe esophageal candidiasis; B — a dense fungal network in esophageal candidiasis; C — pseudohyphae and budding yeast cells (Robbins, 9th ed., Fig. 8-42)*



Figure 10.26. *Spleen and capillaries in malaria: the spleen is gray-black ("slate"), with malarial pigment — hemomelanin — in the capillaries (TMA systemic pathology study guide, 2018)*

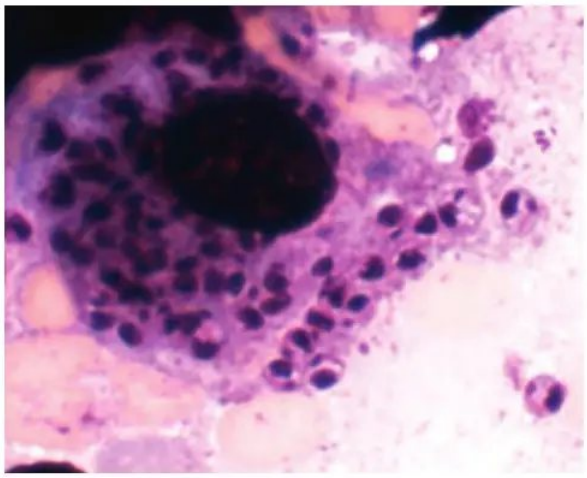


Figure 10.27. *Leishmania donovani parasites in a tissue macrophage (Giemsa stain) (Robbins, 9th ed., Fig. 8-49)*

10.8. Clinicopathological section

The pathological diagnosis of an infectious disease

The diagnosis of a patient who died of an infectious disease states: the **underlying disease** (nosological form, clinicomorphologic variant, laboratory-confirmed agent), **complications**, **concomitant diseases**. When an infectious disease is suspected, material is taken at autopsy for **bacteriologic, virologic and histobacterioscopic** examination; a special biosafety regime is used for highly dangerous infections; when an infectious disease is identified, the sanitary-epidemiological service is notified.

Example: "Underlying disease: fibrocavernous pulmonary tuberculosis (bacteriologically confirmed), a cavity in the upper lobe of the right lung. Complications: erosion of a vessel in the cavity wall, pulmonary hemorrhage, aspiration of blood. Concomitant disease: secondary amyloidosis of the kidneys and spleen."

CLINICAL SIGNIFICANCE

Prevention of infections — vaccination. The national immunization schedule against measles, diphtheria, pertussis, tetanus, poliomyelitis, hepatitis B, tuberculosis (BCG), papillomavirus infection and others prevents severe infectious diseases and deaths in children. In areas where vaccination coverage has declined, epidemics of measles and diphtheria are reappearing.

Practical skills

Skill 1. Gross specimens

1. **"Primary tuberculous complex".** A yellowish caseous focus 1–1.5 cm in size beneath the pulmonary pleura (primary affect); the hilar lymph nodes are enlarged and cheese-like on the cut surface. Conclusion: primary tuberculosis.
2. **"Miliary tuberculosis of the lung".** Numerous millet-sized whitish-yellow tubercles in the lung. Conclusion: generalized hematogenous tuberculosis.
3. **"Fibrocavernous tuberculosis".** A cavity with a thick, dense wall in the lung apex, its inner surface lined by caseous mass; sclerosis and foci of different ages around it; the pleura is thickened. Conclusion: secondary pulmonary tuberculosis.

4. **“Staphylococcal destruction of the lung”.** Irregular abscesses and air spaces in the lung tissue. Conclusion: staphylococcal pneumonia.
5. **“Purulent leptomeningitis”.** The leptomeninges are thick, dull and soaked with yellow-green pus. Conclusion: purulent meningitis (meningococcal).
6. **“The spleen in malaria”.** The spleen is enlarged, firm and gray-black (slate-colored) on the cut surface. Conclusion: malaria (hemomelanosis).
7. **“The liver in visceral syphilis”** and **“syphilitic aortic aneurysm”** — see Topic 6.
8. **“Septic spleen”.** The spleen is enlarged and soft, bright red on the cut surface, yielding abundant pulp on scraping. Conclusion: sepsis (septicemia).



Figure 10.29. *The liver in malaria (microscopic slide): black hemomelanin in Kupffer cells (1), lipofuscin in hepatocytes (2) (TMA systemic pathology study guide, 2018)*

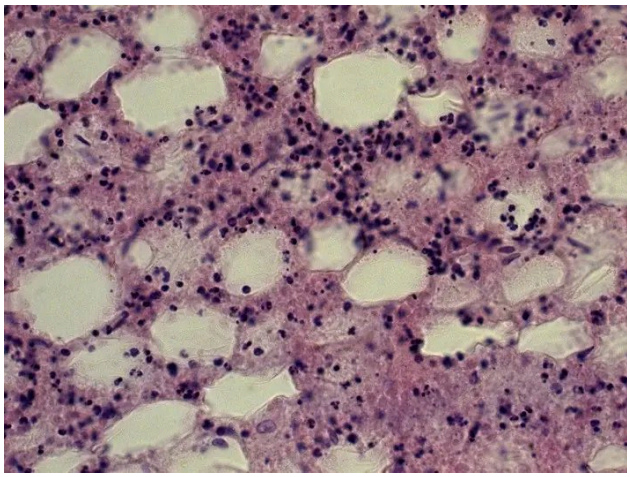


Figure 10.28. *Chronic brain abscess: a cavity with a smooth, dense wall in the frontal region (TMA systemic pathology study guide, 2018)*

Skill 2. Microscopic slides

1. **“Giant cell pneumonia in measles”** (hematoxylin and eosin). Serous exudate, macrophages and multinucleated giant cells (eosinophilic inclusions in the nucleus and cytoplasm) in the alveoli; lymphocytic infiltrate in the interalveolar septa. Conclusion: viral (measles) pneumonia.
2. **“Cytomegaly of a salivary gland”** (hematoxylin and eosin). Large cells in the ductal epithelium with a large basophilic intranuclear inclusion surrounded by a clear halo (“owl's eye”); lymphocytes in the interstitium. Conclusion: cytomegalovirus infection.
3. **“Tuberculous granuloma”** — see Topic 6.
4. **“Myocarditis in diphtheria”** (hematoxylin and eosin). Fatty degeneration of cardiomyocytes, foci of myolysis; edema, lymphocytes and macrophages in the stroma. Conclusion: toxic myocarditis.
5. **“Esophageal candidiasis”** (hematoxylin and eosin or PAS). Necrosis of the mucosa, with fungal pseudohyphae and yeast cells among the necrotic masses. Conclusion: candida esophagitis.
6. **“The liver in malaria”** (hematoxylin and eosin). Kupffer cells are enlarged, with black-brown granular pigment (hemomelanin) in the cytoplasm, Perls-negative. Conclusion: malaria.

Review questions

1. General features of infectious diseases. What are the portal of entry and the primary infectious complex?
2. Mechanisms by which microorganisms damage tissue.
3. The main types of tissue reaction to infection.

4. Forms and morphology of influenza.
5. Morphology and complications of measles.
6. Morphologic signs of herpes, chickenpox and cytomegalovirus infections.
7. Differences between staphylococcal and streptococcal infections. Scarlet fever.
8. Local and systemic changes in diphtheria, causes of death.
9. Forms of meningococcal infection. Waterhouse–Friderichsen syndrome.
10. The pathogenesis of tuberculosis and the tissue reactions in the tuberculous focus.
11. The primary tuberculous complex and its outcomes.
12. Forms of hematogenous tuberculosis.
13. Forms of secondary tuberculosis; morphology of fibrocavernous tuberculosis; complications.
14. Stages of syphilis and their morphology. Congenital syphilis.
15. Definition, classification and clinicomorphologic forms of sepsis.
16. The main morphologic signs of candidiasis, aspergillosis, malaria and leishmaniasis.

Test questions

1. The complex consisting of the change at the portal of entry, lymphangitis and regional lymphadenitis:
   A) Granuloma
   B) Septic focus
   C) Primary infectious complex
   D) Ranke complex
   E) Metastatic abscess

2. The pathognomonic early sign of measles:
   A) Warthin–Finkeldey cells
   B) Hutchinson teeth
   C) “Raspberry tongue”
   D) Cowdry bodies
   E) Belsky–Filatov–Koplik spots

3. Intranuclear inclusions with an “owl's eye” appearance are characteristic of:
   A) Cytomegalovirus infection
   B) Rabies
   C) Influenza
   D) Hepatitis B
   E) Measles

4. The inflammation characteristic of staphylococcal infection:
   A) Localized purulent-necrotizing (abscess, furuncle)
   B) Interstitial lymphocytic
   C) Catarrhal
   D) Diffuse phlegmon (erysipelas)
   E) Granulomatous

5. The cause of late death (in the 2nd week) in diphtheria:
   A) Meningitis
   B) Pneumonia
   C) Toxic myocarditis (cardiac paralysis)
   D) Sepsis
   E) True croup

6. Bilateral hemorrhage into the adrenal glands in meningococcemia:
   A) Addison disease
   B) Conn syndrome
   C) Waterhouse–Friderichsen syndrome
   D) Cushing syndrome
   E) Sheehan syndrome

7. Not part of the primary tuberculous complex:
   A) Lymphangitis
   B) Primary affect (Ghon focus)
   C) Regional lymphadenitis
   D) Cavity
   E) All are included

8. The calcified primary affect and calcified lymph nodes together:
   A) Aschoff–Puhl focus
   B) Ranke complex
   C) Tuberculoma
   D) Abrikosov focus
   E) Ghon focus

9. The main route of spread in secondary tuberculosis:
   A) Perineural
   B) Contiguous
   C) Hematogenous
   D) Canalicular (bronchogenic)
   E) Lymphogenous

10. The inner layer of the cavity wall in fibrocavernous tuberculosis:
    A) Granulation tissue
    B) Caseous (pyogenic)
    C) Epithelium
    D) Fibrous
    E) Muscular

11. The main morphologic feature in all stages of syphilis:
    A) Caseous necrosis
    B) Obliterating endarteritis and plasma cell infiltrate
    C) Neutrophilic infiltrate
    D) Viral inclusions
    E) Epithelioid granulomas

12. The Hutchinson triad is characteristic of:
    A) Neurosyphilis
    B) Early congenital syphilis
    C) Secondary syphilis
    D) Late congenital syphilis
    E) Primary syphilis

13. The form of sepsis with metastatic abscesses:
    A) Bacteremia
    B) Chronic sepsis
    C) Septicemia
    D) Septicopyemia
    E) Cryptogenic sepsis

14. The pigment that gives the spleen a gray-black color in malaria:
    A) Bilirubin
    B) Melanin
    C) Lipofuscin
    D) Hemosiderin
    E) Hemomelanin

15. The bodies found in macrophages in visceral leishmaniasis:
    A) Leishman–Donovan bodies
    B) Mallory bodies
    C) Negri bodies
    D) Councilman bodies
    E) Russell bodies

Clinical cases

Case 1

A 6-year-old child complains of fever, cough and weakness. X-ray: a small subpleural focus in the lower part of the upper lobe of the right lung and enlargement of the hilar lymph nodes. The Mantoux test is positive (it was recently negative). In the family, the grandmother has been coughing for a long time.

Questions: 1) Your diagnosis? 2) State its morphologic components. 3) Possible outcomes.

Answer: 1) Primary tuberculosis — the primary tuberculous complex (the grandmother is the probable source of infection, open tuberculosis). 2) The primary affect (Ghon focus) — a subpleural focus of caseous pneumonia; lymphangitis; regional (bronchopulmonary) caseous lymphadenitis. The conversion of the Mantoux test to positive indicates the development of type IV hypersensitivity. 3) Most often healing: encapsulation and calcification — the Ranke complex; in unfavorable cases progression — a primary cavity, caseous pneumonia, lymphogenous and hematogenous spread (miliary tuberculosis, tuberculous meningitis — especially dangerous in young children); later, secondary tuberculosis with reactivation.

Case 2

A 45-year-old man has had pulmonary tuberculosis for 10 years and was treated irregularly. Suddenly a large amount of blood came from his mouth and he died. At autopsy there is a thick-walled cavity 5

cm in size in the apex of the right lung, its inner surface lined by caseous mass, with an eroded vessel in its wall; small foci in the lower parts; blood in the bronchi. The kidneys are enlarged, firm, whitish and waxy.

Questions: 1) Which form of tuberculosis? 2) The structure of the cavity wall. 3) The cause of death and the changes in the kidneys.

Answer: 1) The fibrocavernous form of secondary tuberculosis (chronic pulmonary tuberculosis); the foci in the lower parts — bronchogenic spread. 2) Three layers: inner — caseous (pyogenic), middle — tuberculous granulation tissue (epithelioid and giant cells), outer — fibrous layer. 3) The cause of death is erosion of a vessel in the cavity wall, pulmonary hemorrhage and aspiration of blood (asphyxia). In the kidneys — secondary (AA) amyloidosis ("large white amyloid kidney"), a complication of the chronic purulent-necrotic process.

Case 3

A 32-year-old woman develops high fever, chills, jaundice and hypotension 5 days after childbirth. The uterus is tender, with a foul-smelling vaginal discharge. She died of multiple organ failure. At autopsy: purulent-necrotizing endometritis in the uterine mucosa, purulently dissolved thrombi in the uterine veins; numerous small abscesses in the lungs and kidneys; the spleen is enlarged, soft and bright red.

Questions: 1) Your diagnosis? 2) Where are the septic focus and the portal of entry? 3) How did the abscesses in the lungs and kidneys arise?

Answer: 1) Uterine (postpartum, obstetric) sepsis — septicopyemia, septic shock. 2) The septic focus is the purulent-necrotizing endometritis and purulent thrombophlebitis (uterine veins) in the uterus; the portal of entry is the placental site (the inner surface of the uterus). 3) Fragments of purulently dissolved thrombi and microbes were carried through the blood and occluded small vessels (bacterial embolism) — metastatic abscesses (lungs, kidneys — embolic purulent nephritis). Septic spleen — hyperplasia and congestion; jaundice — hemolysis and liver damage.

Quick review

Table 10.4. *Key facts on Topic 10*

Question	Answer
Primary infectious complex	Primary affect + lymphangitis + regional lymphadenitis
Endotoxin	Gram-negative LPS — fever, septic shock, DIC
Severe form of influenza	Hemorrhagic pneumonia; "large variegated influenza lung"
Measles	Koplik spots, Warthin–Finkeldey giant cells, giant cell pneumonia
CMV	Cytomegaly, "owl's eye"
Herpes	Vesicle, glassy intranuclear inclusions, latency in nerve ganglia
Staphylococcus	Localized purulent-necrotizing — furuncle, abscess, destruction
Streptococcus	Diffuse — erysipelas, phlegmon; rheumatic fever, glomerulonephritis
Diphtheria	Diphtheritic in the pharynx, croupous in the trachea; toxic myocarditis
Meningococcus	Purulent meningitis, meningococcemia, Waterhouse–Friderichsen
Primary tuberculosis	Ghon focus + lymphangitis + lymphadenitis → Ranke complex
Hematogenous tuberculosis	Miliary tuberculosis, extrapulmonary (bone, kidney), meningitis

Question	Answer
Secondary tuberculosis	Adults, apex, bronchogenic; 8 forms; fibrocavernous the most common
Cavity wall	Caseous + granulation + fibrous
Hallmark of syphilis	Endarteritis + plasma cells
Stages of syphilis	Hard chancre → syphilides → gumma, mesaortitis, neurosyphilis
Late congenital syphilis	Hutchinson triad
Forms of sepsis	Septicemia, septicopyemia, septic endocarditis, chronic
Malaria	Hemomelanin; slate spleen; cerebral malaria (P. falciparum)
Leishmaniasis	Leishman–Donovan bodies; splenomegaly

Glossary of terms

- **Cytomegaly** (Greek kytos — cell, megalos — large) — a disease characterized by the appearance of large cells (cytomegalic cells) in the epithelium.
- **Exanthem** (Greek exanthein — to blossom) — skin rash.
- **Enanthem** (Greek en — inside, anthein — to blossom) — rash on mucous membranes.
- **Staphylococci** (Greek staphyle — bunch of grapes, kokkos — grain) — gram-positive cocci arranged like bunches of grapes (family Staphylococcaceae).
- **Streptococci** (Greek streptos — chain) — chain-forming, spherical or oval, non-spore-forming, gram-positive, facultatively anaerobic bacteria (family Streptococcaceae).
- **Diphtheria** (Greek diphthera — leather membrane) — an acute infectious disease distinguished by the formation of a membrane as a result of fibrinous inflammation.
- **Meningococcus** (Neisseria meningitidis) — gram-negative diplococci of the genus Neisseria.
- **Candidiasis** (oral thrush) — a fungal infection caused mainly by Candida albicans; the fungi are opportunistic and cause disease when immunity is reduced.
- **Aspergillosis** — the general name of all diseases caused by fungi of the genus Aspergillus.
- **Malaria** (Italian mala aria — “bad air”) — a group of vector-borne infectious diseases caused by Plasmodium and transmitted by the bite of female Anopheles mosquitoes.
- **Leishmaniases** (named after W. Leishman) — a group of mainly zoonotic, vector-borne parasitic diseases with natural foci, caused by protozoa of the genus Leishmania and found in tropical and subtropical regions.
- **Primary affect** — the initial change that develops at the portal of entry of an infection.
- **Cavern (cavity)** (Latin caverna — cave) — a cavity formed as a result of discharge of necrotic tissue.
- **Tubercle** (Latin tuberculum — small knob) — the tuberculous granuloma.
- **Chancre** (French chancre — sore) — the primary affect of syphilis, a painless ulcer with a firm base.
- **Syphilides** — skin and mucosal rashes in secondary syphilis.
- **Sepsis** (Greek sepsis — putrefaction) — a generalized infectious disease with organ dysfunction resulting from a dysregulated response to infection.
- **Septicopyemia** (Greek pyon — pus) — sepsis with metastatic purulent foci.

TOPIC 11

Diseases of newborns and children. Prenatal and perinatal pathology. Hereditary diseases. Childhood tumors

Learning objectives

The student should know:

- the periods of prenatal pathology: gametopathies, blastopathies, embryopathies, fetopathies;
- congenital malformations: classification, causes (genetic factors, teratogens), main types;
- hereditary diseases: chromosomal diseases and monogenic diseases (modes of inheritance), their morphology;
- prematurity and intrauterine growth restriction;
- birth injury, respiratory distress syndrome (hyaline membrane disease), necrotizing enterocolitis;
- perinatal infections (the TORCH complex) and neonatal sepsis;
- hemolytic disease of the newborn and hydrops fetalis, kernicterus;
- sudden infant death syndrome;
- benign and malignant tumors of children (hemangioma, teratoma, neuroblastoma, Wilms tumor, retinoblastoma).

The student should be able to:

- describe gross specimens of polydactyly, Patau syndrome, hydrops fetalis, the lung in cystic fibrosis, sacrococcygeal teratoma, Wilms tumor, neuroblastoma and necrotizing enterocolitis;
- describe microscopic slides of hyaline membrane disease, chorioamnionitis, extramedullary hematopoiesis in the liver in hydrops, the pancreas in cystic fibrosis, neuroblastoma and Wilms tumor;
- determine the cause of death of a newborn on the basis of clinical and anatomic data.

11.1. Features of childhood and the perinatal period

A child's organism is not a scaled-down copy of an adult's: at each age the structure and course of diseases are distinctive. From the standpoint of pathological anatomy, the following periods are important:

- **antenatal (prenatal) period** — from fertilization to birth;
- **perinatal period** — from the **22nd week** of pregnancy (fetal weight 500 g) to **7 days** after birth (168 hours): it includes the antenatal, intranatal (during delivery) and early neonatal periods;
- **neonatal period** — up to **28 days** after birth;
- **infancy** — up to 1 year; then the periods of childhood. The causes of death in children change with age: **under 1 year** — congenital malformations, prematurity and low birth weight, respiratory distress syndrome, perinatal infections and sepsis, sudden infant death syndrome, asphyxia; **1–4 years** — congenital malformations, accidents, infections (pneumonia, diarrhea), tumors; **5–14 years** — accidents, malignant tumors (leukemias, brain tumors), congenital malformations.

11.2. Prenatal pathology

Prenatal pathology comprises diseases of the period of intrauterine development (**kyematopathies**, Greek kyema — embryo). Their type and outcome depend on the period of ontogenesis at which the damage occurred:

Table 11.1. *Periods of prenatal pathology*

Stage	Time	Pathology	Outcomes
Gametopathies	Maturation of germ cells (gametes) and before fertilization	Mutations of gametes — hereditary diseases and malformations	Infertility, spontaneous abortion, hereditary diseases, chromosomal diseases
Blastopathies	The first 15 days after fertilization	Damage to the blastocyst	Death and loss of the embryo (often unnoticed), twin anomalies (conjoined twins), cyclopia
Embryopathies	From day 16 to day 75 (weeks 8–10) — the period of organogenesis	Congenital malformations caused by teratogens — the most sensitive period	Congenital malformations, death of the embryo
Fetopathies	From day 76 to birth (early fetal — up to day 180; late fetal)	The organs are formed; damage causes diseases rather than malformations: infections (fetal hepatitis, pneumonia), hemolytic disease, endocrine disorders (diabetic fetopathy), hypoxia	Intrauterine growth restriction, prematurity, congenital diseases, organ hypoplasia, signs of dysontogenesis

REMEMBER

The period of organogenesis (**weeks 3–8**) is the period **most sensitive** to teratogens: an agent acting at this time disrupts the formation of organ primordia and causes a **congenital malformation**. The time of teratogen exposure determines the **type** of malformation — each organ has its own “critical period” (for example, the heart — weeks 3–6, the palate — weeks 6–9). An agent acting in the fetal period causes not a malformation but a **disease** (infection, inflammation).

11.3. Congenital malformations

Congenital malformations are permanent changes in the shape, structure and function of an organ or body part present at birth. They occur in about **3%** of newborns and are one of the main causes of infant mortality.

Types of anomalies (by the disturbance of morphogenesis):

- **Malformation** — abnormal formation of an organ as a result of a disturbance of its intrinsic genetic program (heart defects, polydactyly, cleft lip and palate);
- **Disruption** — secondary destruction of a previously normally formed organ by an external factor (for example, **amniotic bands** — amputation of fingers or limbs);
- **Deformation** — a change of shape due to mechanical pressure (a narrow uterine cavity, **oligohydramnios** — clubfoot);
- **Sequence** — a chain of secondary changes arising from one primary defect: the **Potter sequence** — renal agenesis → absence of urine → **oligohydramnios**

→ compression of the fetus → flat face, limb deformities, **pulmonary hypoplasia** (the cause of death); • **Syndrome** — a set of several defects caused by one cause (Down syndrome). **Morphologic**

terms (Topic 1): **agenesis** (complete absence of an organ), **aplasia** (the organ as a primordium), **hypoplasia** (incompletely developed), **atresia** (closure of an orifice or canal — esophageal, anal atresia), **stenosis** (narrowing), **ectopia** (an organ in an unusual location — the testis remaining in the abdominal cavity — cryptorchidism), **heterotopia** (tissue elements in another organ), **persistence** (survival of embryonic structures — patent ductus arteriosus), **dysraphia** (failure of closure of embryonic clefts — spina bifida with meningocele, cleft lip), **atavism** (appearance of structures typical of phylogenetic ancestors), **gigantism**, **polydactyly** (extra fingers), **syndactyly** (fused fingers).

Causes:

1. **Genetic factors** (about 25%): chromosomal diseases (Down, Edwards and Patau syndromes), monogenic mutations;
2. **Environmental factors — teratogens** (about 10%): – **infections** — **TORCH**: toxoplasmosis, others (syphilis, VZV, parvovirus B19, Zika), **rubella** (cataract, heart defects, deafness — the Gregg triad), **cytomegalovirus** (microcephaly, periventricular calcification), **herpes**; – **drugs and chemicals** — **thalidomide** (phocomelia — shortening of the limbs), warfarin, antiepileptic drugs (valproate — neural tube defects), retinoids (isotretinoin), ACE inhibitors, **alcohol**; – **maternal diseases** — **diabetes mellitus** (heart defects, neural tube defects, macrosomia — **diabetic fetopathy**), phenylketonuria; – **radiation**, nutritional deficiency (**folic acid** deficiency — **neural tube**

defects: anencephaly, spina bifida); 3. **Multifactorial** (genetic predisposition + environment) and **unknown** causes — the most common (about 50–65%): cleft lip and palate, neural tube defects, congenital heart defects.

CLINICAL SIGNIFICANCE

Fetal alcohol syndrome is the most severe consequence of alcohol consumption during pregnancy: growth restriction, microcephaly, intellectual disability, facial anomalies (short palpebral fissures, thin upper lip, smooth philtrum). There is no safe dose of alcohol. Taking **folic acid** before pregnancy and in the first months significantly reduces the risk of neural tube defects.



Figure 11.2. *Patau syndrome (trisomy 13): cleft lip and palate, small eyes (TMA systemic pathology study guide, 2018)*



Figure 11.1. *Polydactyly: an extra 6th finger on the left hand (TMA systemic pathology study guide, 2018)*

11.4. Hereditary diseases

Chromosomal diseases

Chromosomal diseases develop as a result of changes in the **number** (aneuploidy — trisomy, monosomy) or **structure** (deletion, translocation, inversion, duplication) of chromosomes. Most

chromosomal anomalies lead to early death of the embryo (they are found in half of spontaneously aborted conceptuses).

Table 11.2. *Main chromosomal diseases*

Disease	Karyotype	Main features
Down syndrome	Trisomy 21 (95% — meiotic nondisjunction; related to maternal age), translocation, mosaic	The most common chromosomal disease (about 1:700); intellectual disability, flat face, oblique (upslanting) palpebral fissures, epicanthus, large tongue, transverse palmar crease; congenital heart defects (40%; atrioventricular canal), duodenal atresia, risk of acute leukemias, early Alzheimer disease
Edwards syndrome	Trisomy 18	Micrognathia, prominent occiput, low-set ears, flexed and overlapping fingers, “rocker-bottom” feet, heart and kidney defects; most die in the first year
Patau syndrome	Trisomy 13	Cleft lip and palate, microphthalmia, holoprosencephaly, polydactyly, heart defects; severe, early death
Klinefelter syndrome	47,XXY	Males; tall stature, testicular atrophy, infertility (hypogonadism), gynecomastia, eunuchoid body habitus
Shereshevsky–Turner syndrome	45,X (monosomy X)	Females; short stature, webbed neck (in the fetus — cystic hygroma), streak ovaries (primary amenorrhea, infertility), coarctation of the aorta, lymphedema
22q11.2 deletion	Microdeletion	DiGeorge syndrome (Topic 8): hypoplasia of the thymus and parathyroid glands, heart defects

Monogenic (Mendelian) diseases

Diseases caused by mutation of a single gene; by mode of inheritance:

- **Autosomal dominant** — **one** mutant allele is enough for the disease; it occurs in every generation; 50% of the patient's children are affected. Most often defects of **structural proteins** and receptors: **Marfan syndrome** (fibrillin-1 — tall stature, arachnodactyly, lens dislocation, **aortic aneurysm and dissection**), **familial hypercholesterolemia** (LDL receptor — early atherosclerosis), **neurofibromatosis** (NF1), polycystic kidney disease (adult type), achondroplasia, Huntington disease.
- **Autosomal recessive** — when **both** alleles are mutated; the parents are healthy carriers (heterozygotes); each child has a 25% chance of being affected; more common in **consanguineous marriages**. Most often **enzyme** defects: **phenylketonuria**, **galactosemia**, **cystic fibrosis**, **lysosomal storage diseases** (Gaucher, Niemann–Pick, Tay–Sachs — Topic 3), **glycogenoses**, hemochromatosis (Topic 4), Wilson disease, sickle cell anemia, thalassemia, albinism.
- **X-linked (recessive)** — mainly **boys** are affected, the mother is a carrier: **hemophilia A and B**, **Duchenne muscular dystrophy** (dystrophin), glucose-6-phosphate dehydrogenase deficiency, Bruton agammaglobulinemia, Lesch–Nyhan syndrome. **Phenylketonuria** — deficiency of the enzyme phenylalanine hydroxylase: phenylalanine and its products accumulate and **damage the brain** (intellectual disability, seizures, demyelination); melanin is low (fair skin, light hair); it is prevented by neonatal **screening** and a phenylalanine-free diet.

Galactosemia — deficiency of galactose-1-phosphate uridyltransferase: after milk intake galactose-1-phosphate accumulates — **fatty degeneration of the liver** and cirrhosis, **cataract**, brain damage, renal tubular damage; it is prevented by a milk-free diet.

Cystic fibrosis (mucoviscidosis) — mutation of the **CFTR** gene (most often deltaF508) — chloride channel function is impaired: exocrine glands secrete **thick, viscous mucus**, while the sweat glands excrete much sodium chloride (the **sweat test** is the basis of diagnosis). It is the most common lethal autosomal recessive disease in white populations. **Morphology:**

- **pancreas** (85–90%) — the ducts are plugged with mucus and dilated (**cysts**), with atrophy and fibrosis of the acini — exocrine insufficiency (fatty stools, weight loss, deficiency of fat-soluble vitamins);
- **lungs** — the most severe involvement and the main cause of death: the bronchi are plugged with mucus, **chronic bacterial infections** (staphylococcus, **Pseudomonas aeruginosa**), **bronchiectasis**, abscesses, pulmonary fibrosis, cor pulmonale;
- **intestine** — **meconium ileus** in newborns;
- **liver** — focal biliary cirrhosis; **reproductive system** — congenital absence of the vas deferens in men (infertility); sweat glands — no morphologic changes.

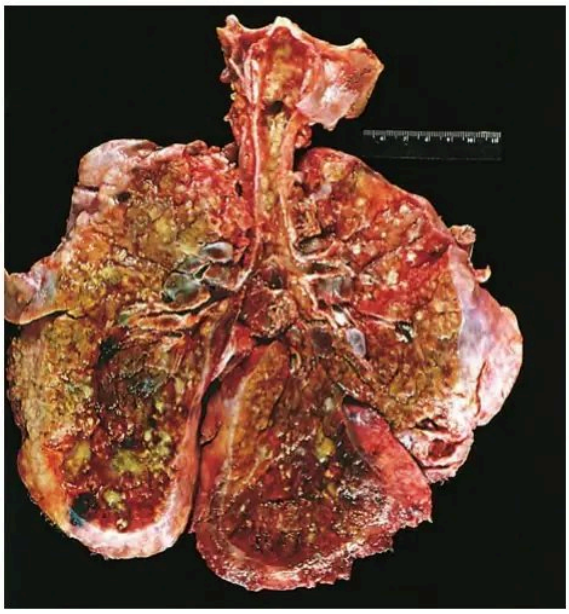


Figure 11.4. *The lung of a patient who died of cystic fibrosis: extensive mucus plugging and dilatation of the tracheobronchial tree; the parenchyma is consolidated by secretions and pneumonia (Robbins, 9th ed., Fig. 10-21)*

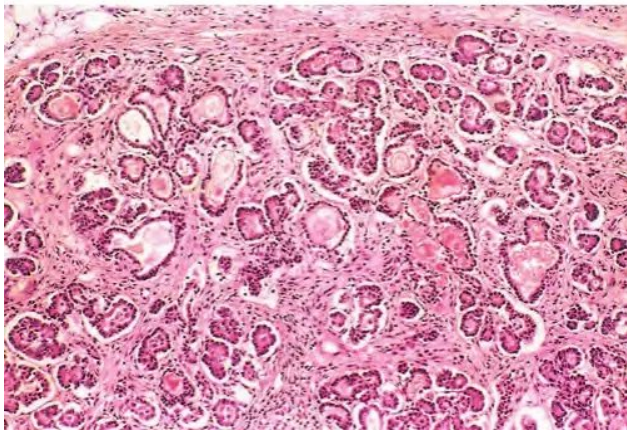


Figure 11.3. *The pancreas in cystic fibrosis: the ducts are dilated and plugged with eosinophilic mucus, and the glands are atrophic and replaced by fibrous tissue (Robbins, 9th ed., Fig. 10-20)*

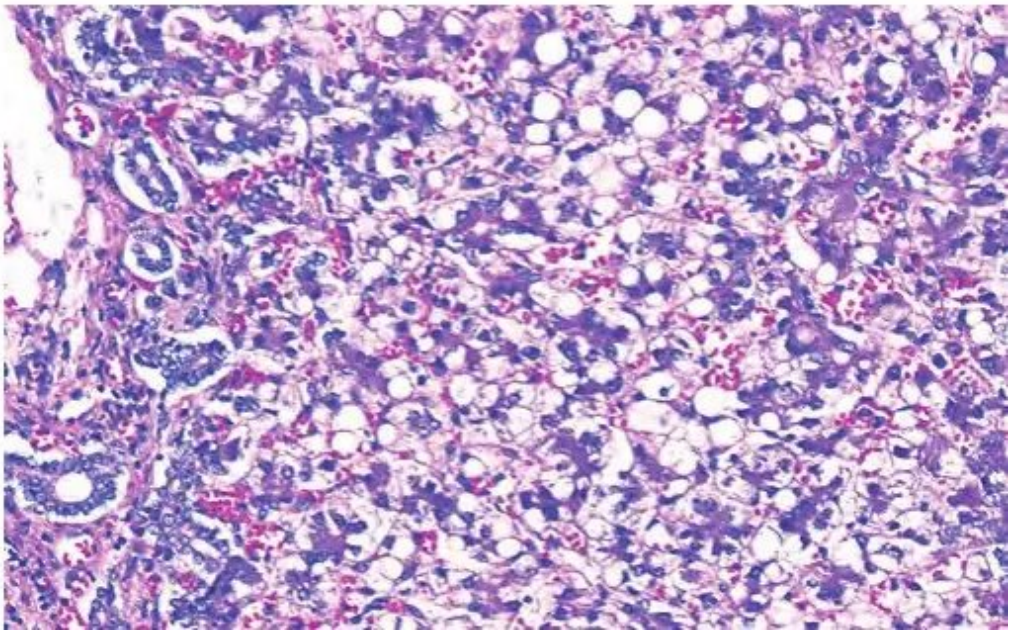


Figure 11.5. *Galactosemia: extensive fatty degeneration and fibrosis in the liver (Robbins, 9th ed., Fig. 10-16)*

11.5. Prematurity and fetal growth restriction

A **premature (preterm)** infant is born **before 37 weeks** of pregnancy. A **low-birth-weight** infant weighs **less than 2500 g** at birth. Prematurity is the most important cause of neonatal death (respiratory distress syndrome, necrotizing enterocolitis, sepsis, intracerebral hemorrhages).

Risk factors for prematurity: **premature rupture of membranes**, **intrauterine infection** (chorioamnionitis — one of the most important causes), anomalies of the uterus, cervix and placenta (placenta previa, premature placental separation), multiple pregnancy.

Intrauterine growth restriction (small-for-gestational-age fetus) — causes: **fetal factors** (chromosomal diseases, congenital malformations, infections — symmetric restriction), **placental factors** (placental infarcts, vascular insufficiency — asymmetric: the brain relatively spared), **maternal factors** (preeclampsia, arterial hypertension, malnutrition, smoking, alcohol, drug abuse).

Signs of prematurity and immaturity (at autopsy): low weight and length, scant subcutaneous fat, fine hairs on the skin (lanugo), soft ear and nasal cartilages, nails not reaching the fingertips, undescended testes in boys, labia majora not covering the labia minora in girls; immature internal organs (embryonic glomeruli in the lungs and kidneys, few sulci and gyri in the brain).

11.6. Diseases of the perinatal period

Birth injury

Birth injury is mechanical damage to the infant's tissues during delivery. Risk factors: a large fetus, a narrow pelvis, abnormal fetal presentation (breech presentation), precipitate or prolonged labor, use of obstetric forceps and vacuum extractor, prematurity.

- **Caput succedaneum** — edema and hemorrhage of the soft tissues of the presenting part of the head (resolves quickly);
- **Cephalohematoma** — hemorrhage beneath the periosteum of the skull; its border does not cross the bone sutures;

- **Intracranial hemorrhages** — **subdural hematoma** (tears of the falx cerebri and tentorium cerebelli, rupture of veins — marked molding of the head), **subarachnoid** hemorrhage; **in premature infants — intraventricular and periventricular (germinal matrix) hemorrhages** (immature vessels, hypoxia, fluctuations in blood pressure);
- **spinal cord** injury, **brachial plexus** injury (Erb palsy), clavicle fracture, hemorrhage into the liver and adrenal glands.

Asphyxia (hypoxia) — oxygen deficiency in utero or during delivery: umbilical cord entanglement, placental separation, abnormal labor. Morphology: venous congestion, **petechial hemorrhages** (pleura, pericardium, thymus, internal organs), **aspiration of amniotic fluid** (meconium — in the lungs), **hypoxic-ischemic encephalopathy** in the brain (later **cerebral palsy** — CP).

Respiratory distress syndrome (hyaline membrane disease)

Neonatal respiratory distress syndrome is the main disease of premature infants and one of the causes of their death. The main cause is **lung immaturity** and **surfactant deficiency**. Surfactant (phospholipids, mainly lecithin) is produced by type II alveolar cells and reduces surface tension in the alveoli; its adequate synthesis begins after the **35th week** of pregnancy.

Risk factors: prematurity (in 60% of infants born before 28 weeks), **maternal diabetes mellitus** (insulin suppresses surfactant synthesis), cesarean section (absence of labor stress and glucocorticoids), male sex, twins.

Pathogenesis: surfactant deficiency → the alveoli collapse with each expiration (**atelectasis**) → hypoxia and hypercapnia → acidosis → pulmonary vasoconstriction and damage to the endothelium and epithelium → leakage of plasma proteins (fibrinogen) into the alveoli → **hyaline membranes form from fibrin and necrotic cell debris** → gas exchange worsens further (a vicious circle).

Morphology: the lungs are of normal size but **firm, airless, red-violet and liver-like**, and **sink** in water. Microscopically: alternating areas of **atelectasis** and dilated alveoli; **eosinophilic, homogeneous hyaline membranes** lining the respiratory bronchioles, alveolar ducts and alveoli; little inflammation. Reparative changes in infants who survived more than 48 hours.

Complications: bronchopulmonary dysplasia (prolonged mechanical ventilation and oxygen toxicity — chronic lung damage), **retinopathy of prematurity**, patent ductus arteriosus, intraventricular hemorrhages, necrotizing enterocolitis.

> **CLINICAL SIGNIFICANCE**
> Giving **glucocorticoids** (dexamethasone, betamethasone) before delivery to a pregnant woman at risk of preterm birth accelerates surfactant synthesis in the fetal lungs; **exogenous surfactant** is given to the infant through the trachea, and breathing is supported. These measures have sharply reduced deaths from respiratory distress syndrome. Lung maturity can be assessed by the **lecithin/sphingomyelin ratio** in amniotic fluid (above 2:1).

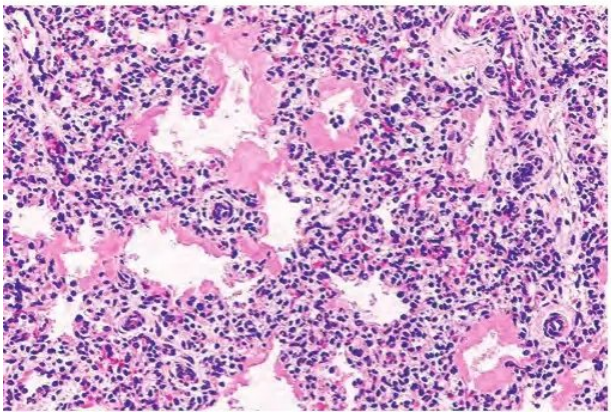


Figure 11.6. *Hyaline membrane disease: alternating atelectasis and dilatation of alveoli; thick eosinophilic hyaline membranes lining the dilated alveoli (Robbins, 9th ed., Fig. 10-7)*

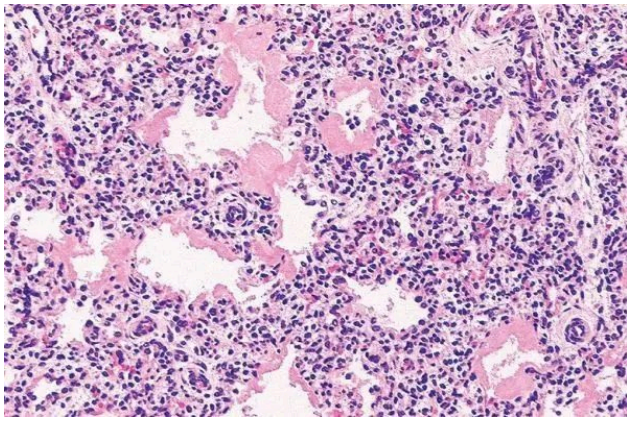


Figure 11.7. *Hyaline membrane disease of the newborn (microscopic slide) (TMA systemic pathology study guide, 2018)*

Necrotizing enterocolitis

Necrotizing enterocolitis is inflammation and necrosis of the small and large intestine, mainly in **premature infants** (especially after formula feeding is started). Intestinal **ischemia**, **bacterial colonization**, immaturity of the intestinal barrier and enteral feeding are important in its pathogenesis. The **terminal ileum, cecum and ascending colon** are most often affected: the intestine is dilated, its wall thinned and dark red-black (hemorrhagic infarction and necrosis), with **gas bubbles in the submucosa** (**pneumatosis intestinalis** — a diagnostic sign on X-ray), perforation and peritonitis. As a consequence — intestinal strictures.



Figure 11.8. *Necrotizing enterocolitis: A — in a severe case the entire small intestine is markedly dilated, with a perilously thin wall (preperforation); B — the congested portion of the ileum corresponds to areas of hemorrhagic infarction and transmural necrosis, with gas bubbles in the submucosa (arrows) (Robbins, 9th ed., Fig. 10-8)*

Perinatal infections

Infection of the fetus and newborn is transmitted by two routes:

1. **Transcervical (ascending)** — the vaginal and cervical microflora (group B streptococcus, Escherichia coli, mycoplasmas) infect the amniotic fluid and membranes — **chorioamnionitis** (neutrophilic infiltrate in the membranes) and **funisitis** (inflammation of the umbilical cord); the fetus swallows or aspirates the infected amniotic fluid — **congenital pneumonia**, sepsis. Chorioamnionitis is an important cause of premature rupture of membranes and **preterm birth**.
2. **Transplacental (hematogenous)** — agents in the maternal blood cross the placenta: **TORCH** — toxoplasmosis, others (syphilis, listeriosis, parvovirus B19, HIV, hepatitis B), rubella,

cytomegalovirus, herpes. Their clinicomorphologic manifestations are common: fever, encephalitis, chorioretinitis, hepatosplenomegaly, pneumonitis, myocarditis, hemolytic anemia, rashes; depending on the period of pregnancy at which infection occurred — congenital malformations or fetal disease. **Neonatal sepsis: early** (in the first 7 days of life, most often in the first 4–5 days) — agents transmitted from the mother (**group B streptococcus**, Escherichia coli, Listeria) — pneumonia, sepsis, meningitis; **late** (from 7 days to 3 months) — hospital infections (Listeria, Candida, staphylococci), **umbilical sepsis** (the umbilical wound is the portal of entry: purulent omphalitis, purulent thrombophlebitis and arteritis of the umbilical vessels).

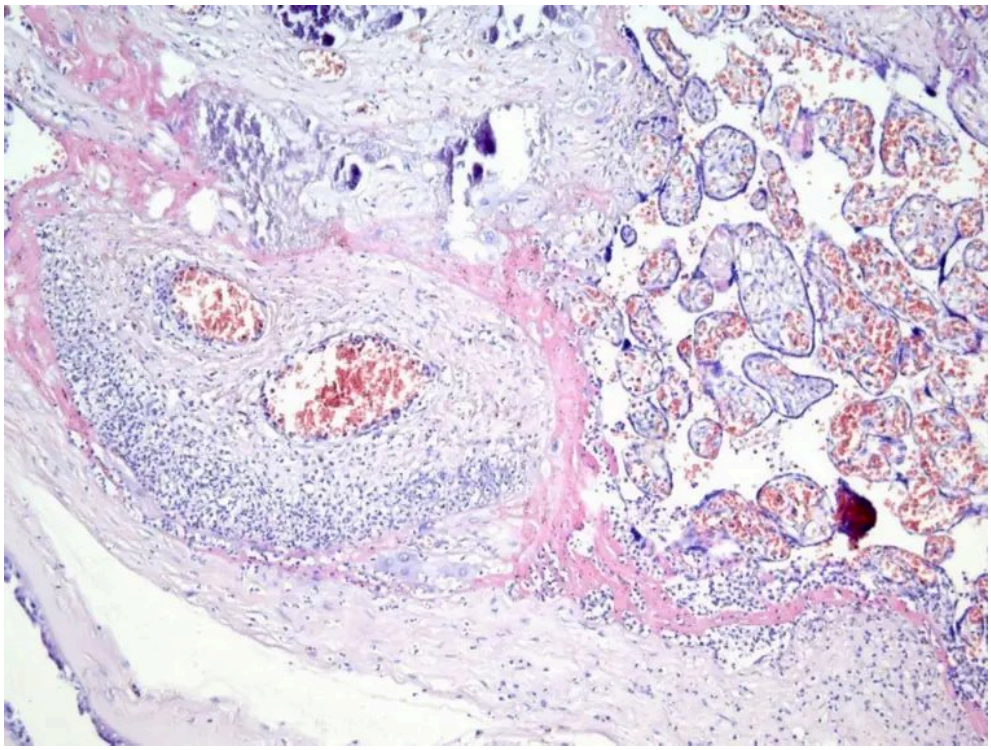


Figure 11.9. *Chorioamnionitis: diffuse leukocytic infiltration in the subchorionic part of the chorionic plate (TMA systemic pathology study guide, 2018)*

Hemolytic disease of the newborn and hydrops fetalis

Hydrops fetalis is **generalized edema** of the fetal body and accumulation of fluid in body cavities during intrauterine development. There are **immune** and **nonimmune** types.

Immune hydrops — hemolytic disease of the newborn — **antigenic incompatibility** between maternal and fetal blood: an Rh-positive fetus of an Rh-negative mother (most often the **D antigen**), less often ABO incompatibility. During a previous pregnancy, abortion or delivery, fetal red cells enter the maternal blood and sensitize her; in the next pregnancy maternal **IgG antibodies** cross the placenta and destroy fetal red cells (**type II hypersensitivity**, Topic 8). Therefore the first pregnancy is usually not affected.

Pathogenesis and morphology:

- **hemolysis** → **anemia** → hypoxia → **heart failure** and liver damage (decreased protein synthesis — hypoproteinemia) → **generalized edema** (hydrops): subcutaneous edema, ascites, hydrothorax, hydropericardium;
- **extramedullary hematopoiesis** in response to anemia — the liver and spleen enlarge (**hepatosplenomegaly**), with foci of erythropoiesis in them; many young red cells in the blood — **erythroblastosis** (erythroblastosis fetalis);
- the placenta is enlarged, edematous and pale;

- **hemolytic jaundice**: unconjugated bilirubin increases, crosses the blood–brain barrier and stains yellow and damages the **subcortical nuclei of the brain** (basal ganglia, thalamus, cerebellum, brainstem nuclei) — **kernicterus (nuclear jaundice)**: severe neurological sequelae in survivors (CP, deafness, intellectual disability). **Clinical forms:** intrauterine death; **generalized hydrops** (the most severe); **hemolytic jaundice** (risk of kernicterus); the **anemic** form (mild).

Causes of **nonimmune hydrops**: **cardiovascular defects** and arrhythmias, **chromosomal diseases** (Turner — cystic hygroma, trisomies 21 and 18), fetal anemia (**thalassemia**, **parvovirus B19** — intranuclear inclusions in erythroid precursors), infections, twin-to-twin transfusion syndrome.

CLINICAL SIGNIFICANCE

Prevention of Rh hemolytic disease — giving the Rh-negative mother **anti-D immunoglobulin within 72 hours** after the birth of each Rh-positive child (or after abortion, and at the 28th week of pregnancy): it destroys fetal red cells that have entered the maternal blood before sensitization. In severe cases, intrauterine transfusion and, after birth, exchange transfusion and phototherapy are used.

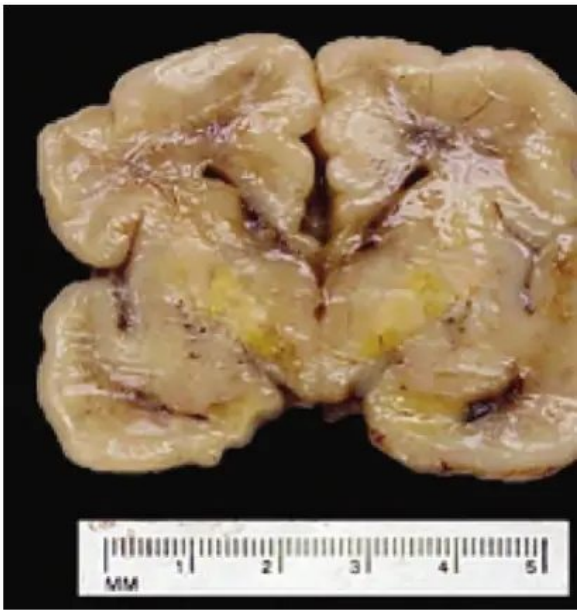


Figure 11.11. *Kernicterus (nuclear jaundice): yellow staining of the brain parenchyma, especially in the basal ganglia next to the ventricles (Robbins, 9th ed., Fig. 10-13)*



Figure 11.10. *Hydrops fetalis: A — generalized accumulation of fluid; B — fluid is especially abundant in the soft tissues of the neck (cystic hygroma) (Robbins, 9th ed., Fig. 10-11)*

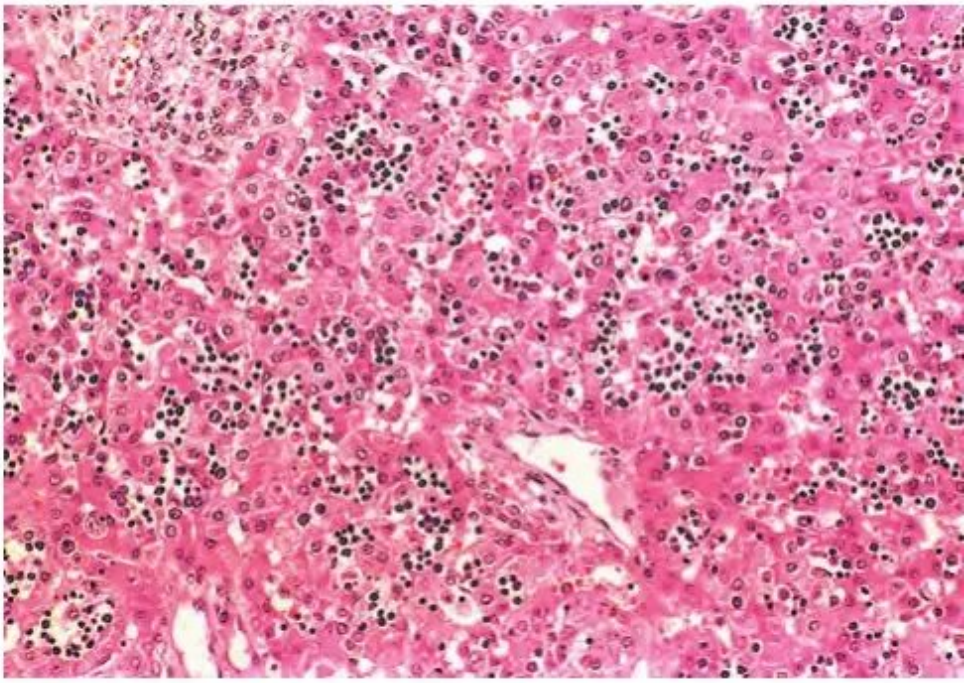


Figure 11.12. *Extramedullary hematopoiesis in the liver in hydrops fetalis: numerous islands of small blue cells among the mature hepatocytes (Robbins, 9th ed., Fig. 10-12)*

Sudden infant death syndrome

Sudden infant death syndrome is the **sudden, unexpected** death of a child **under 1 year of age** that remains unexplained even after a complete autopsy, examination of the death scene and review of the clinical history. It occurs most often in infants **2–4 months** old, **during sleep** (at night). **Risk factors:** putting the infant to sleep **prone (on the stomach)**, a soft mattress, overheating, maternal smoking and drug abuse during pregnancy, prematurity, low birth weight, male sex, not breastfeeding, bed sharing with parents. Mechanism — immaturity of the brainstem (respiratory and arousal centers), a defect of the serotonergic system. Nonspecific changes at autopsy: **petechiae** in the pleura, pericardium and thymus, pulmonary edema and congestion, gliosis in the brainstem.

> **CLINICAL SIGNIFICANCE**
> The "**Back to Sleep**" campaign (putting the infant to sleep only on the back, on a firm surface, with no soft objects nearby) reduced the frequency of this syndrome several-fold.

11.7. Childhood tumors

Tumors in children differ from those in adults: **benign tumors and tumor-like lesions** are more common; among malignant tumors, **leukemias**, tumors of the **central nervous system**, **lymphomas** and tumors of the **embryonal (blastoma)** type predominate; carcinomas are rare. Most malignant tumors appear microscopically as **small, round, blue cell** tumors (neuroblastoma, Wilms tumor, retinoblastoma, rhabdomyosarcoma, Ewing sarcoma, lymphoma); they are associated with **genetic** factors and respond well to modern treatment.

Benign tumors and tumor-like lesions

- **Hamartomas** and **choristomas** (Topic 9).
- **Hemangiomas** — the most common tumor of children; most often in the skin (face, head), of the **capillary** type; most **regress spontaneously**. **Lymphangiomas** — neck, axilla (cystic hygroma).

- **Teratomas** — most often in the **sacrococcygeal** region (**sacrococcygeal teratoma**, more often in girls), and also in the ovary, testis and mediastinum; most are mature (benign), but they may contain immature (malignant) components.

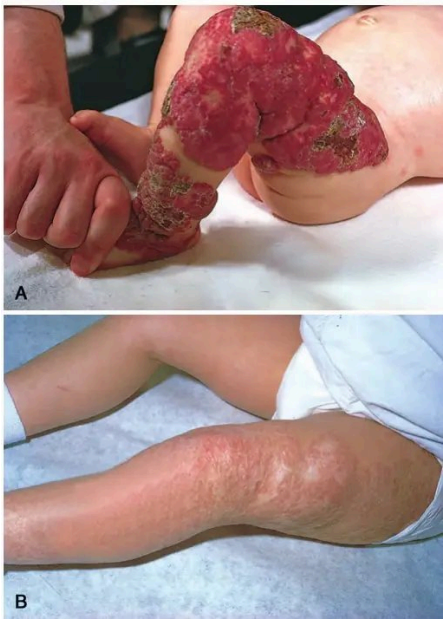


Figure 11.13. *Congenital capillary hemangioma: A — at birth; B — at 2 years of age, after spontaneous regression (Robbins, 9th ed., Fig. 10-22)*



Figure 11.14. *Sacrococcygeal teratoma: the size of the tumor compared with that of the stillborn fetus (Robbins, 9th ed., Fig. 10-23)*

Malignant tumors

Neuroblastoma — a tumor arising from neural crest cells of the sympathetic nervous system and **adrenal medulla**; the most common **extracranial solid tumor** of early childhood (median age 18 months). In 40% of cases it arises **in the adrenal gland**, the rest along the sympathetic chain (abdomen, mediastinum). **Grossly:** a soft, gray-pink nodule of variable size with foci of hemorrhage, necrosis and **calcification**. **Microscopically: sheets of small, round cells with dark nuclei**, a delicate fibrillar matrix (neuropil) and **Homer Wright pseudorosettes** (cells arranged in a ring around central neuropil); the degree of maturation varies — ganglioneuroblastoma, **ganglioneuroma** (mature, benign). **Urinary catecholamine metabolites** (vanillylmandelic and homovanillic acids) are increased — a diagnostic sign. Prognosis depends on age (good under 1.5 years), stage and **amplification of the MYCN oncogene** (unfavorable); spontaneous regression or maturation is possible in infants.

Wilms tumor (nephroblastoma) — the most common **primary renal tumor** of children (at 2–5 years of age). In some cases it is associated with syndromes (**WAGR** — Wilms tumor, aniridia, genital anomalies, intellectual disability — deletion of the WT1 gene; Denys–Drash and Beckwith–Wiedemann syndromes). **Grossly:** a large, solitary, sharply demarcated nodule, soft and uniform on the cut surface, **gray-yellowish (tan)**, with foci of hemorrhage and cysts. **Microscopically — triphasic structure: blastema** (sheets of small blue cells), an **epithelial** component (embryonic tubules and glomeruli) and **stroma** (fibroblast-like or myxoid, sometimes skeletal muscle). **Anaplasia** (large, hyperchromatic nuclei, atypical mitoses; TP53 mutation) — unfavorable prognosis. Clinically — a large abdominal mass. With modern treatment **more than 90%** of patients are cured.

Retinoblastoma — the most common malignant tumor of the retina; **RB** gene mutation (“two hits”, Topic 9): familial (40%; often bilateral, early, with a later risk of osteosarcoma) and sporadic forms. Clinically — leukocoria (“cat's eye” — the pupil shines white), strabismus. Microscopically — small blue cells, Flexner–Wintersteiner rosettes.



Figure 11.15. *Adrenal neuroblastoma in a 6-month-old child: a hemorrhagic, partially encapsulated tumor displacing the left kidney (Robbins, 9th ed., Fig. 10-24)*

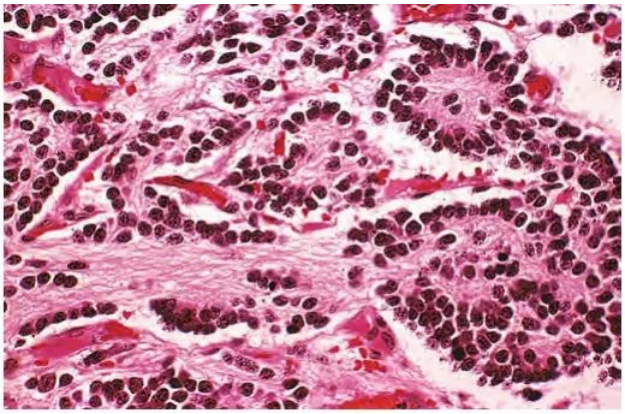


Figure 11.16. *Neuroblastoma (microscopic slide): small cells embedded in a delicate fibrillar matrix (Robbins, 9th ed., Fig. 10-25)*

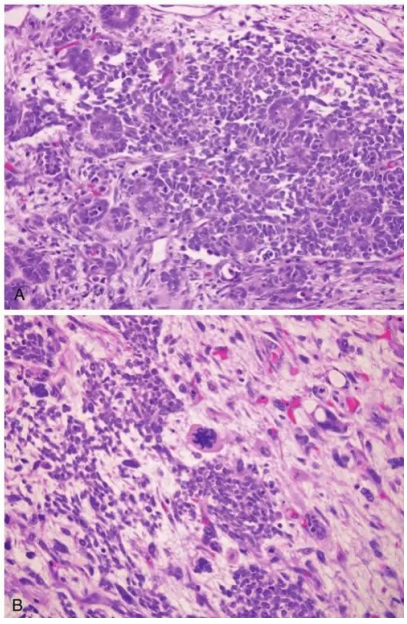


Figure 11.18. *Wilms tumor: A — tightly packed blue cells (blastema) with primitive tubules between them (epithelial component); B — focal anaplasia: large hyperchromatic nuclei, atypical mitoses (Robbins, 9th ed., Fig. 10-29)*

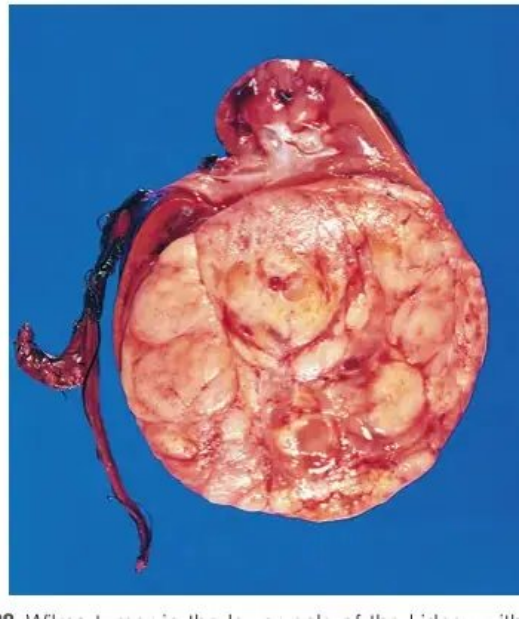


Figure 11.17. *Wilms tumor in the lower pole of the kidney: characteristic tan color and sharp borders (Robbins, 9th ed., Fig. 10-28)*

11.8. Clinicopathological section

Determining the cause of perinatal death

In perinatal death the autopsy is performed together with examination of the **placenta**, because placental changes (chorioamnionitis, infarcts, abruption, umbilical cord anomalies) often indicate the cause of fetal death. The pathological diagnosis states **maternal** (preeclampsia, diabetes, infection), **placental** and **fetal** factors. The time of fetal death is estimated from the degree of **maceration** (skin slippage, softening of tissues — postmortem autolysis in utero).

Example: “Underlying disease: neonatal respiratory distress syndrome — hyaline membrane disease of the lungs (gestational age 29 weeks, weight 1200 g). Complications: hemorrhage into the lateral ventricles of the brain. Background condition: maternal diabetes mellitus; acute chorioamnionitis.”

CLINICAL SIGNIFICANCE

Neonatal screening — in all newborns, blood is taken from the heel in the first days after birth to detect **phenylketonuria**, **congenital hypothyroidism**, cystic fibrosis, galactosemia and other treatable hereditary diseases. Early diagnosis and diet or hormone therapy prevent intellectual disability.

Practical skills

Skill 1. Gross specimens

1. **“Polydactyly”.** An extra finger (toe) on the hand (foot). Conclusion: congenital malformation.
2. **“Patau (or Down) syndrome”.** Cleft lip and palate, microphthalmia, polydactyly (Patau); flat face, upslanting (“mongoloid”) palpebral fissures, large tongue, transverse palmar crease (Down). Conclusion: chromosomal disease.
3. **“Hydrops fetalis”.** Generalized edema of the fetal body, especially of the neck and head; ascites, hydrothorax; the liver and spleen are enlarged; the placenta is edematous. Conclusion: hydrops fetalis (immune or nonimmune).
4. **“The lung in cystic fibrosis”.** The bronchi are dilated (bronchiectasis) and filled with thick mucus and pus; foci of pneumonia and abscesses in the lung tissue. Conclusion: cystic fibrosis.
5. **“Necrotizing enterocolitis”.** The intestine is dilated, its wall thin and dark red-black, with perforation in places. Conclusion: necrotizing enterocolitis.
6. **“Sacrococcygeal teratoma”.** A large cystic-solid tumor in the sacral region. Conclusion: teratoma.
7. **“Wilms tumor”.** A large, sharply demarcated nodule in the kidney, gray-yellowish on the cut surface. Conclusion: nephroblastoma.
8. **“Neuroblastoma”.** A soft tumor in the region of the adrenal gland with foci of hemorrhage and calcification. Conclusion: neuroblastoma.

Skill 2. Microscopic slides

1. **“Hyaline membrane disease”** (hematoxylin and eosin). Areas of atelectasis; eosinophilic hyaline membranes lining the walls of dilated alveolar ducts and alveoli; the capillaries are congested. Conclusion: neonatal respiratory distress syndrome.
2. **“Chorioamnionitis”** (hematoxylin and eosin). A diffuse neutrophilic infiltrate in the chorionic and amniotic membranes and in the subchorionic area. Conclusion: ascending intrauterine infection.
3. **“The liver in hydrops”** (hematoxylin and eosin). Foci of extramedullary hematopoiesis (islands of erythroid cells) among the hepatocytes and in the sinusoids. Conclusion: extramedullary hematopoiesis (hemolytic disease).
4. **“The pancreas in cystic fibrosis”** (hematoxylin and eosin). The excretory ducts are dilated and plugged with eosinophilic mucus; atrophy of the acini, interstitial fibrosis. Conclusion: cystic fibrosis.
5. **“Neuroblastoma”** (hematoxylin and eosin). Sheets of small, round cells with dark nuclei, a fibrillar matrix, Homer Wright pseudorosettes. Conclusion: neuroblastoma.

6. **“Wilms tumor”** (hematoxylin and eosin). A triphasic structure consisting of blastema, primitive tubules and stroma. Conclusion: nephroblastoma.

Review questions

1. Features of the perinatal period and childhood from the standpoint of pathological anatomy. The main causes of death in children by age.
2. Periods of prenatal pathology: gameto-, blasto-, embryo- and fetopathies.
3. Types of congenital anomalies (malformation, disruption, deformation, sequence). The Potter sequence.
4. Causes of congenital malformations. Teratogens and the TORCH complex.
5. Chromosomal diseases: Down, Edwards, Patau, Klinefelter and Turner syndromes.
6. Modes of inheritance of monogenic diseases and examples.
7. Morphology of phenylketonuria, galactosemia and cystic fibrosis.
8. Prematurity and fetal growth restriction; signs of immaturity.
9. Morphology of birth injury and asphyxia.
10. Pathogenesis and morphology of respiratory distress syndrome.
11. Necrotizing enterocolitis.
12. Routes of transmission of perinatal infections; neonatal sepsis.
13. Pathogenesis and morphology of hemolytic disease of the newborn and hydrops fetalis; kernicterus.
14. Sudden infant death syndrome.
15. Features of childhood tumors; neuroblastoma, Wilms tumor, retinoblastoma.

Test questions

1. The period most sensitive to teratogens:
   A) Blastogenesis (days 1–15)
   B) Gametogenesis
   C) Late fetal period
   D) Embryogenesis (days 16–75, organogenesis)
   E) Neonatal period

2. The chain renal agenesis → oligohydramnios → pulmonary hypoplasia:
   A) Deformation
   B) Syndrome
   C) Disruption
   D) Malformation
   E) Potter sequence

3. Characteristic defects resulting from the effect of the rubella virus on the fetus:
   A) Phocomelia
   B) Neural tube defects
   C) Cleft lip
   D) Cataract, heart defects, deafness
   E) Polydactyly

4. The main cause of Down syndrome:
   A) 45,X
   B) Trisomy 13
   C) Trisomy 21
   D) Trisomy 18
   E) 47,XXY

5. Cleft lip and palate, holoprosencephaly and polydactyly are characteristic of:
   A) Klinefelter syndrome
   B) Patau syndrome
   C) Down syndrome
   D) Turner syndrome
   E) Edwards syndrome

6. Characteristic of Shereshevsky–Turner syndrome:
   A) Short stature, webbed neck, coarctation of the aorta
   B) Intellectual disability, large tongue
   C) Tall stature, gynecomastia
   D) Testicular atrophy
   E) Cleft lip

7. Cystic fibrosis is associated with mutation of which gene?
   A) HFE
   B) WT1
   C) RB
   D) CFTR
   E) FBN1

8. An X-linked recessive disease:
   A) Marfan syndrome
   B) Neurofibromatosis
   C) Familial hypercholesterolemia
   D) Phenylketonuria
   E) Hemophilia A

9. The main cause of neonatal respiratory distress syndrome:
   A) Heart defect
   B) Surfactant deficiency
   C) Meconium aspiration
   D) Diaphragmatic hernia
   E) Pneumonia

10. Increases the risk of hyaline membrane disease:
   A) Female sex
   B) Maternal glucocorticoid treatment
   C) Maternal diabetes mellitus and prematurity
   D) Labor stress
   E) Term birth

11. The antibodies that destroy fetal red cells in Rh hemolytic disease of the newborn:
    A) Fetal IgA
    B) Maternal IgM
    C) Maternal IgG
    D) Fetal IgG
    E) Maternal IgE

12. The substance that causes kernicterus (nuclear jaundice):
    A) Urobilinogen
    B) Unconjugated bilirubin
    C) Hemosiderin
    D) Biliverdin
    E) Conjugated bilirubin

13. The main risk factor for sudden infant death syndrome:
    A) Prone sleeping (on the stomach)
    B) Breastfeeding
    C) Female sex
    D) Supine sleeping (on the back)
    E) A firm mattress

14. The most common primary renal tumor of children:
    A) Angiomyolipoma
    B) Clear cell carcinoma
    C) Neuroblastoma
    D) Oncocytoma
    E) Wilms tumor (nephroblastoma)

15. Increased in the urine as a diagnostic sign of neuroblastoma:
    A) Vanillylmandelic acid (catecholamine metabolites)
    B) Galactose
    C) Bilirubin
    D) 5-hydroxyindoleacetic acid
    E) Phenylpyruvate

Clinical cases

Case 1

An infant born at 29 weeks of pregnancy and weighing 1250 g developed shortness of breath, grunting, chest retractions and cyanosis a few hours after birth. The mother has diabetes mellitus and did not receive glucocorticoids before delivery. The infant died on day 3. At autopsy the lungs are firm, airless, red-violet and sink in water.

Questions: 1) Your diagnosis? 2) What changes are expected microscopically? 3) Pathogenesis of the disease and risk factors.

Answer: 1) Neonatal respiratory distress syndrome — hyaline membrane disease. 2) Alternating atelectasis and dilated alveoli; eosinophilic hyaline membranes (fibrin + necrotic epithelial debris) lining the respiratory bronchioles, alveolar ducts and alveoli; little inflammation. 3) Surfactant deficiency due to lung immaturity → collapse of the alveoli (atelectasis) → hypoxia, acidosis → damage to the endothelium and epithelium → leakage of plasma proteins → hyaline membranes.

Risk factors: prematurity, maternal diabetes mellitus (insulin suppresses surfactant synthesis), absence of antenatal glucocorticoids.

Case 2

The third pregnancy of an Rh-negative woman. She did not receive anti-D immunoglobulin after the previous pregnancies. The fetus died at 32 weeks. At autopsy: generalized edema of the fetal body, ascites, hydrothorax; the liver and spleen are markedly enlarged; the placenta is large, edematous and pale. Microscopically, numerous islands of erythroid cells in the liver.

Questions: 1) Your diagnosis? 2) The mechanism of the disease. 3) If the fetus had been born alive, what dangerous complication could have developed? What is the prevention?

Answer: 1) Hemolytic disease of the newborn, the generalized hydrops form (immune hydrops fetalis). 2) During previous pregnancies, Rh-positive fetal red cells entered the maternal blood and sensitized her; the mother's anti-D IgG antibodies crossed the placenta and destroyed the fetal red cells (type II hypersensitivity) → severe anemia → hypoxia, heart failure and liver damage (hypoproteinemia) → generalized edema; extramedullary hematopoiesis in the liver and spleen in response to anemia (hepatosplenomegaly, erythroblastosis). 3) Kernicterus (nuclear jaundice) — unconjugated bilirubin damages the subcortical nuclei of the brain, leading to severe neurological sequelae. Prevention — giving the Rh-negative mother anti-D immunoglobulin within 72 hours after the birth of an Rh-positive child, after abortion, and at the 28th week.

Case 3

The mother of a 3-year-old child felt a large hard mass in the child's abdomen while bathing him. Computed tomography: a sharply demarcated tumor 10 cm in size in the left kidney. The child has no iris in the eyes (aniridia). After nephrectomy, the tumor on the cut surface is uniform, gray-yellowish and sharply demarcated; microscopically — sheets of small blue cells, primitive tubules and stroma.

Questions: 1) Your diagnosis? 2) Explain the association with aniridia. 3) What determines the prognosis?

Answer: 1) Wilms tumor (nephroblastoma) — triphasic structure: blastema, an epithelial component (primitive tubules) and stroma. 2) WAGR syndrome — Wilms tumor, aniridia, genital anomalies, intellectual disability; caused by a deletion at the 11p13 locus that includes the WT1 (tumor suppressor) and PAX6 genes. 3) The prognosis depends on the stage and the presence or absence of anaplasia (large hyperchromatic nuclei, atypical mitoses — TP53 mutation; unfavorable); with modern combined treatment (surgery, chemotherapy, radiotherapy), more than 90% of patients are cured.

Quick review

Table 11.3. *Key facts on Topic 11*

Question	Answer
Perinatal period	From the 22nd week to 7 days after birth
Period of embryopathies	Days 16–75 (organogenesis) — congenital malformations
Fetopathy	From day 76 — diseases, not malformations
Potter sequence	Renal agenesis → oligohydramnios → pulmonary hypoplasia
TORCH	Toxoplasma, others, rubella, CMV, herpes

Question	Answer
Thalidomide	Phocomelia
Folic acid deficiency	Neural tube defects
Down / Edwards / Patau	Trisomy 21 / 18 / 13
Turner / Klinefelter	45,X / 47,XXY
Autosomal dominant	Marfan, familial hypercholesterolemia, NF1
Autosomal recessive	PKU, galactosemia, cystic fibrosis, storage diseases
X-linked	Hemophilia, Duchenne dystrophy
Cystic fibrosis	CFTR; thick mucus; the lung — the main cause of death; sweat test
Prematurity	Before 37 weeks; low birth weight — less than 2500 g
Hyaline membrane disease	Surfactant ↓; prematurity, maternal diabetes; eosinophilic membranes
Necrotizing enterocolitis	Premature infants; pneumatosis intestinalis; perforation
Chorioamnionitis	Ascending infection; a cause of preterm birth
Rh hemolytic disease	Maternal anti-D IgG; hydrops, erythroblastosis, kernicterus
Sudden infant death	2–4 months, during sleep; prone sleeping
Neuroblastoma	Adrenal gland; small blue cells, pseudorosettes; VMA ↑; MYCN
Wilms tumor	2–5 years; triphasic; WT1; anaplasia — unfavorable
Retinoblastoma	RB gene; leukocoria

Glossary of terms

Terms of heredity and genetics

- **Gene** — a sequence of nucleotides in DNA that encodes a particular RNA molecule.
- **Genome** — the totality of genetic information in a cell of an organism.
- **Genotype** — the genetic makeup of an organism (the set of alleles at one or several loci).
- **Phenotype** — the set of external and internal traits of an organism formed under the influence of the genotype and the environment.
- **Allele** — one of the alternative forms of a gene; located at the same locus of homologous chromosomes.
- **Locus** — a region of a chromosome with a particular genetic determinant.
- **Homozygote** — a cell or organism with two identical alleles at a particular locus of homologous chromosomes.
- **Heterozygote** — a cell or organism with two different alleles at a particular locus of homologous chromosomes.
- **Hemizygosity** — presence of a gene in only one copy (for example, genes on the X chromosome in males).
- **Dominance** — expression in a heterozygote of the trait of only one allele.
- **Recessiveness** — expression of the trait of an allele only in the homozygous state.
- **Autosome** — any chromosome that is not a sex chromosome; humans have 22 pairs of autosomes.

- **Autosomal dominant inheritance** — a mode of inheritance in which one mutant allele on an autosome is enough for the trait (disease) to be expressed.
- **Autosomal recessive inheritance** — a mode of inheritance in which the mutant allele must be inherited from both parents for the trait (disease) to be expressed.
- **Holandric inheritance** — Y-linked inheritance (only from father to son).
- **Homologous chromosomes** — paired chromosomes with the same set of genes.
- **Centromere** — the part of a chromosome to which the spindle fiber attaches and which ensures the distribution of chromosomes to the daughter cells.
- **Chromatin** — the complex of DNA and associated proteins in the nucleus; **heterochromatin** — its dense, transcriptionally inactive part.
- **Gamete** — a mature germ cell; **haploid** — a cell with a single set of chromosomes; **somatic cells** — cells of non-germ tissues.
- **Mutation** — a heritable change in genetic information; **mutagenesis** — the process of formation of mutations; **mutagens** — physical, chemical and biological agents that cause mutations.
- **Chromosomal aberration (anomaly)** — the general name of any chromosomal mutation: deletion, translocation, inversion, duplication and others.
- **Deletion** — loss of part of a chromosome; **duplication** — doubling of part of a chromosome.
- **Aneuploidy** — a chromosome set with one or more chromosomes fewer or more than normal (trisomy, monosomy).
- **Amplification** — an increase in the number of copies of a gene (for example, amplification of oncogenes in tumors).
- **Crossing over** — exchange of segments between homologous chromosomes in meiosis.
- **Replication** — doubling of DNA; **transcription** — synthesis of RNA on a DNA template; **translation** — synthesis of a protein polypeptide chain on ribosomes.
- **Genetic code** — the system for recording the sequence of amino acids in a protein by nucleotide triplets (codons) in DNA (RNA); **codon** — a nucleotide triplet that specifies one amino acid or the end of translation.
- **Exon** — the coding part of a gene retained in mature mRNA; **intron** — the noncoding part removed during splicing; **splicing** — the process of removing introns from RNA.
- **Promoter** — the region of DNA to which RNA polymerase binds to begin transcription; **enhancer** — a region of DNA that increases transcription.
- **DNA repair** — correction of a damaged DNA molecule.
- **Reverse transcriptase** — an enzyme that catalyzes DNA synthesis on an RNA template (in retroviruses).
- **Retroviruses** — RNA viruses that encode reverse transcriptase and integrate into the chromosome (provirus).
- **Proto-oncogenes** — normal genes that, as a result of mutation or activation, confer malignant tumor properties on a cell; **oncogenes** — activated genes capable of transforming a cell into a tumor cell.
- **Clone** — a set of genetically identical cells derived asexually from a single ancestral cell.
- **Monoclonal antibodies** — antibodies specific for a single antigen produced by hybridomas; **hybridoma** — a cell formed by fusion of a myeloma cell with an immunized lymphocyte.
- **Microsatellites** — short tandemly repeated sequences in DNA; used in genetic fingerprinting (personal identification) and in tumor diagnosis.

- **Sequencing** — determination of the sequence of monomers in a nucleic acid or protein molecule.
- **Blotting** — a method of transferring DNA, RNA or protein molecules to a membrane after electrophoresis.
- **Vaccine** — an antigenic preparation used to induce immunity against an infection.
- **Vector** — a DNA molecule (plasmid, virus) used to introduce genetic information into a cell.
- **Plasmid** — a circular DNA molecule that replicates independently of the cell chromosome.

Terms of environmental pathology

- **Adaptation** — adjustment of an organism to particular environmental conditions by means of a set of morphologic, physiological and behavioral features.
- **Allergy** — hypersensitivity of the organism to allergens (chemicals polluting the air and water, food, drugs, plant pollen); one of the most widespread diseases.
- **Benzo(a)pyrene** — a potent carcinogen of the group of polycyclic aromatic hydrocarbons; found in industrial gases, vehicle exhaust and tobacco smoke.
- **Carcinogenic substances** — chemicals that cause cancer.
- **Dioxins** — among the most toxic substances known; formed during combustion of chlorine-containing substances; they have carcinogenic and teratogenic effects.
- **Suspended dust (particulate matter)** — fine solid particles in the air; they accumulate in the lungs (pneumoconioses) and carry heavy metals and carcinogens.
- **Heavy metals** — metals with a density greater than 4.5 g/cm^3 (mercury, lead, cadmium, arsenic and others); they accumulate in the body and damage the nervous system, kidneys and hematopoiesis.
- **Mercury** — a heavy metal; its organic compounds (methylmercury) are especially dangerous; entering the body through fish, they damage the nervous system (Minamata disease).
- **Lead** — a heavy metal; especially dangerous for children and pregnant women: anemia, damage to the nervous system and kidneys, delayed mental development in children.
- **Cadmium** — a heavy metal; damages the kidneys and bones.
- **Carbon monoxide** — a colorless, odorless, highly toxic gas; it forms carboxyhemoglobin with hemoglobin and causes tissue hypoxia.
- **Nitrates and nitrites** — salts of nitric and nitrous acids; in excess they cause methemoglobinemia and lead to the formation of nitrosamines (carcinogens).
- **Nicotine** — a highly toxic alkaloid of tobacco smoke; it constricts vessels and harms fetal development; damage to the leg vessels in heavy smokers (“smoker's leg”).
- **Tobacco smoke** — a mixture of gases and aerosols consisting of several thousand substances, including dozens of carcinogens.
- **Pesticides** — chemicals used to protect plants (insecticides, herbicides, fungicides and others); **DDT** — one of the most dangerous and persistent insecticides, it accumulates in the body.
- **Radioactivity** — decay of isotopes of some chemical elements with emission of ionizing radiation (alpha and beta particles, gamma rays); **radon** — a radioactive inert gas that accumulates in the lower floors of buildings and increases the risk of lung cancer.
- **Ultraviolet rays** — electromagnetic radiation with a wavelength of 180–400 nm; in large doses it causes skin cancer and cataract.

- **Smog** — a toxic haze consisting of a mixture of gases, smoke and dust over industrial plants and cities.
- **Toxicity** — the property of a substance to have a harmful effect on the organism and health.
- **Maximum allowable concentration (MAC)** — an environmental standard defining the highest amount of a substance that does not harm human health.

Answers to the test questions

Topic 1: 1-A; 2-D; 3-E; 4-E; 5-C; 6-A; 7-B; 8-C; 9-B; 10-D; 11-B; 12-D; 13-E; 14-A; 15-C.

Topic 2: 1-B; 2-C; 3-D; 4-A; 5-E; 6-A; 7-B; 8-C; 9-E; 10-A; 11-C; 12-D; 13-B; 14-E; 15-D.

Topic 3: 1-B; 2-E; 3-B; 4-C; 5-C; 6-A; 7-B; 8-E; 9-D; 10-A; 11-E; 12-D; 13-A; 14-D; 15-C.

Topic 4: 1-E; 2-C; 3-E; 4-D; 5-A; 6-A; 7-C; 8-C; 9-D; 10-A; 11-D; 12-E; 13-B; 14-B; 15-B.

Topic 5: 1-C; 2-A; 3-A; 4-D; 5-B; 6-E; 7-B; 8-A; 9-E; 10-E; 11-D; 12-D; 13-B; 14-C; 15-C.

Topic 6: 1-D; 2-B; 3-E; 4-C; 5-E; 6-B; 7-B; 8-A; 9-E; 10-C; 11-D; 12-C; 13-D; 14-A; 15-A.

Topic 7: 1-C; 2-D; 3-B; 4-A; 5-B; 6-E; 7-B; 8-D; 9-A; 10-E; 11-C; 12-E; 13-C; 14-A; 15-D.

Topic 8: 1-D; 2-B; 3-D; 4-C; 5-E; 6-E; 7-E; 8-A; 9-C; 10-A; 11-B; 12-B; 13-A; 14-C; 15-D.

Topic 9: 1-B; 2-A; 3-E; 4-E; 5-D; 6-A; 7-B; 8-D; 9-E; 10-C; 11-D; 12-B; 13-C; 14-A; 15-C.

Topic 10: 1-C; 2-E; 3-A; 4-A; 5-C; 6-C; 7-D; 8-B; 9-D; 10-B; 11-B; 12-D; 13-D; 14-E; 15-A.

Topic 11: 1-D; 2-E; 3-D; 4-C; 5-B; 6-A; 7-D; 8-E; 9-B; 10-C; 11-C; 12-B; 13-A; 14-E; 15-A.

References

1. Abdullakhodjaeva M.S. Patologik anatomiya. 1-qism. Umumiy patologik anatomiya [Pathological anatomy. Part 1. General pathological anatomy]. — Tashkent: Tafakkur-Bo‘stoni, 2012.
2. Abdullakhodjaeva M.S. Patologik anatomiya. 2-qism. Xususiy patologik anatomiya [Pathological anatomy. Part 2. Special pathological anatomy]. — Tashkent: Tafakkur-Bo‘stoni, 2012.
3. Strukov A.I., Serov V.V. Патологическая анатомия [Pathological anatomy]. 6th ed. — Moscow: GEOTAR-Media, 2015.
4. Strukov A.I. Патологик анатомия [Pathological anatomy] (translated from the revised 2nd edition). — Tashkent: Meditsina, 1973.
5. Умумий патологик анатомия: ўқув қўлланма [General pathological anatomy: a study manual]. — Tashkent: Tashkent Medical Academy, 2018.
6. Хусусий патологик анатомия: ўқув қўлланма [Special pathological anatomy: a study manual]. — Tashkent: Tashkent Medical Academy, 2018.
7. Kumar V., Abbas A.K., Aster J.C. Robbins and Cotran Pathologic Basis of Disease. 9th ed. — Philadelphia: Elsevier Saunders, 2015.
8. Klatt E.C., Kumar V. Robbins and Cotran Review of Pathology. 3rd ed. — Philadelphia: Saunders Elsevier, 2010.
9. Sattar H.A. Fundamentals of Pathology: Medical Course and Step 1 Review. — Chicago: Pathoma LLC, 2011.
10. Kaplan USMLE Step 1 Lecture Notes: Pathology. — New York: Kaplan, 2016.
11. Mishra D. Pathology (Notespaedia). — New Delhi.